

Higher-order structures and parasitism in grammar

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It is a truth universally acknowledged that doing a PhD is a challenging time. Adding a pandemic into the mix (as well as the odd geopolitical event for good measure) does not exactly make it easier. Needless to say, this thesis happened despite a lot of things. This is all the more reason to acknowledge what lead to its completion. To start from the beginning:

When I started my PhD in 2019, the two courses IGRA01 and 02 that introduced the overall topic of the Graduiertenkolleg provided an amazing bird's-eye view of the core areas of grammatical theory. Especially the lectures by Ezer Rasin left a lasting impression. Never before have I met a person that is so gifted at breaking down any topic into its most fundamental assumptions and then, starting from zero, putting these pieces together into a well-structured whole. I had little to no background in phonology before, yet in two semesters I got a great overview of that field, complete with remarks on its history, sociology and epistemology. Meeting Gereon Müller was inspiring due to his unwavering commitment to always find the deepest and most beautiful theory – we had a number of discussions where our opinions differed in that regard (as well as about the way beauty should play a role in science).

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at that, which is quite different from the version that can be found in this thesis. My attempts to make this operation work fully the way I intended did not succeed for a couple of months and it was quite crucial to me that Mike continued to be supportive during that pretty difficult time.

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1 Introduction

The leitmotif of this thesis is parasitism. It also contains variations on this theme that at first sight might seem unrelated, but they turn out to be connected nonetheless. To understand why one might call a syntactic relation parasitic, compare the following examples:

- (1) a. Which article₁ did you file t_1 ?
b. *Which article₁ did you file a book without reading t_1 ?
c. ?Which article₁ did you file t_1 without reading t_1 ?
- (2) Mary doesn't know WHEN she will come,
a. but Peter not even WHETHER he will come.
b. *but Peter doesn't even know WHETHER.
c. ?but Peter not even WHETHER.

In (1a), there is *wh*-movement of the object to a sentence initial position. In (1b), the attempted extraction out of the adjunct [*without reading*] results in ungrammaticality; it therefore appears that a long-distance dependency between a *wh*-phrase and a position in an adjunct is not possible. Unexpectedly, such a dependency is possible in (1c), although the sentence is slightly marked. The crucial factor here is that this sentence also contains the acceptable long-distance dependency in (1a). It therefore appears that the (1b)-type dependency is only possible if the (1a)-type dependency is present. In other words, the former is *parasitic on* the presence of the latter. This phenomenon is called parasitic gaps (Ross 1967, Engdahl 1983 and many others) since the creation of an otherwise ungrammatical gap becomes grammatical in the presence of another licit gap.

(2a) is an example of gapping, a process where the finite verb of a sentence (*know*) is elided. It can be recovered from the preceding clause, so that the meaning of (2a) is understood as '*but Peter doesn't even know whether he will come*'. Not every element in a sentence can be left out even when they can be recovered from context. This is the case in (2b), a sentence that speakers of many dialects of English reject. It appears to be a property of English that ellipsis after *whether* is not possible. Yet unexpectedly, (2c) is an acceptable sentence. As above, (2c) combines the processes of both (2a) and (2b): it contains both the acceptable gapping of the finite verb and the previously ungrammatical ellipsis after *whether*. The latter ellipsis becomes acceptable in the presence of the former; it is therefore an instance of parasitic ellipsis (Fitzgibbons 2014).

Parasitic ellipsis was discovered in Fitzgibbons (2014) who described it as a process in Russian that can occur after certain prepositions if the respective

PP is the remnant of gapping. One of the contributions of this thesis is that I show that parasitic ellipsis is a much more general process: not only can it occur in other languages such as German, English and Estonian, it can also occur after many other categories such as C, A, N and V:

- (3) The dog stole the sausage WITH Masha's help
 - a. ?*but the cat stole the sausage WITHOUT.
 - b. but the cat WITHOUT.
- (4) Peter is happy ALTHOUGH he was fired
 - a. *but Mary is happy BECAUSE.
 - b. ?but Mary BECAUSE.
- (5) Paul was HAPPY with his son,
 - a. #but Ethan was UPSET. *(generic/'strict')*
 - b. but Ethan UPSET. *(strict/sloppy)*
- (6) Mary has a PICTURE of her grandparents,
 - a. #and Peter has a PAINTING. *(generic/'strict')*
 - b. and Peter a PAINTING. *(strict/sloppy)*

I arrive at the following generalization about parasitic ellipsis:

Parasitic ellipsis generalization (PEG)

For every possible remnant of gapping, the complement of its head can optionally be deleted if the head is contrastively focussed and its complement given.

The first half of this thesis deals with parasitic ellipsis, the second half with parasitic gaps. The core questions I attempt to answer are what parasitism in syntax is and how it should be dealt with in a theory of grammar.

The answer that I provide subverts an implicit assumption in the text above, namely that the (c)-examples contain two separate processes to begin with. In fact, I argue that parasitic dependencies in the literal sense are illusory, and moreover, should not be part of a theory of grammar since they could generate pathological dependencies resulting in unattested phenomena. Parasitic phenomena, in the empirical sense, do of course exist, as demonstrated above. In 4.3 I compare parasitic gaps and parasitic ellipsis and show that they share a number of distinctive properties, some of which are unexpected if one views them simply as a subtype of feeding. I conclude from that that parasitism is a genuine notion of grammar in its own right that cannot be reduced to other more well-known rule interactions such as feeding. What I propose in this thesis instead is that parasitic phenomena

are the result of a *single* but more powerful operation. In other words, (1c) is not the outcome of two separate movements with a mysterious connection that ensures that both gaps refer to the same element; instead it is the outcome of a single movement of the same *wh*-phrase, but over a more complex structure. (2c) is not the outcome of two separate deletions with a communication channel established between them that ensures the dependency; it is rather the outcome of a *single*, higher-order ellipsis operation that elides the missing material all at once. In this thesis I compare two views of parasitism with each other: the *separate-but-dependent processes* view vs. the *single-but-more-powerful process* view. I provide a number of arguments that the latter approach is more appropriate and that the former cannot explain core properties of parasitism while relying on conspiracies to make the purported parasitic dependency between the processes work. Each half of the dissertation is devoted to developing a syntactic operation that realizes the vision of the *single-process* view: first, I use a 3rd-order ellipsis operation to generate host and parasitic ellipsis all together, then I define a new structure-building operation, *3rd-merge*, that creates structural configurations where a single phrase leads to two traces if it is extracted out of it.

The framework that I use throughout this dissertation are Minimalist Grammars (Stabler 1997), a syntax formalism from computational linguistics that formalizes core aspects of minimalist syntax. Both the ellipsis framework that I use as well as the new structure-building operation that I define to derive parasitic gaps are couched within Minimalist Grammars (MGs). MGs differ from current minimalist syntax in a few aspects, most of which are notational. I will introduce MGs in section 2.1 in a way so that hopefully someone who has never heard of this formalism can follow anything that is said in this thesis. Wherever possible I use a notation that is as close to current minimalist syntax as possible in order to make the content of this thesis as accessible to as many people as possible.

I introduce the ellipsis framework couched in MGs that I use in Part I of this thesis in section 2.2. There I also explain the term *higher-order structure* that occurs in the title of this thesis. In this approach an ellipsis operation is a pro-form of a specific syntactic type. This pro-form is then anaphoric to a phrase of the same type in the preceding context. As an example, say you have a VP-ellipsis:

- (7) Someone will [_{VP} praise Joscha]. Anton will **e**_{VP}

In this approach, you have a silent ellipsis pro-form **e**_{VP} in the second sentence that has type VP. This pro-form is anaphoric to the VP in the preceding sentence [_{VP} praise Joscha]. Since this ellipsis pro-form is anaphoric to a full

constituent, it is a *first-order* ellipsis operation.

In the case where there is a phrase that moves out of the ellipsis site, something else needs to be said, for reasons that I explain in detail in the respective section. The ellipsis pro-form needs to make room for the item that moves out later, as e.g. in a passive VP-ellipsis:

- (8) Someone will be [praised *t*]. Joscha will be $e_{NP \rightarrow VP}$

The pro-form needs to stand in for a VP, *minus* the logical object that moves out ($e_{NP \rightarrow VP}$). It is therefore anaphoric to a constituent missing a subconstituent, a *2nd-order constituent*: $\lambda x.[\text{praised } x]$.

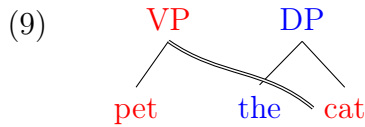
The original article on that framework, Kobele (2015), restricted itself to 2nd-order ellipsis operations. However, I make the argument that a *3rd-order* constituent is precisely the structure of the material that is elided in parasitic ellipsis: you elide the phrase containing the finite verb, say, CP (1st-order); but you exempt at least one phrase contained in it from deletion, such as the clause ‘*whether he will come*’ (2nd-order); to go to 3rd order would mean not to rescue the whole CP, but continue eliding a subconstituent of it, e.g. everything below *whether*, leaving this element as the sole survivor. Such a 3rd-order ellipsis operation ($e_{(TP \rightarrow CP) \rightarrow CP}$) describes the elided material as a single, albeit complex, structure ($\lambda f.[_{CP} \text{ knows } [_{CP} f(\text{he will come})]]$).

It is not necessary to immediately grasp all these notions in an unfamiliar framework via an admittedly short description. All these concepts and structures will be explained and visualized in detail in the respective sections. For now it is enough to understand that a 3rd-order operation can elide the missing material in parasitic ellipsis in a *single* although more powerful process. Under such a view, parasitic ellipsis is the outcome of one process, not two separate but somehow dependent ones. I want to stress that not much hinges on the idiosyncrasies of the specific framework used here. The empirical generalizations made here are independent of the theoretical framework, and the observation that the elided material corresponds to a 3rd-order structure is true regardless of whether ellipsis is viewed as a pro-form or not. The actual analysis proposed here implements this insight transparently, but a *one-process* analysis can also be stated in other approaches.

In the second part of this dissertation I apply the same kind of reasoning to parasitic gaps to investigate whether this phenomenon can also be generated in one fell swoop. If a higher-order ellipsis operation leads to parasitic ellipsis, would a more powerful, ‘higher-order’ structure-building operation lead to parasitic gaps? In a loose analogy, let us say that *merge* combines (1st-order) constituents (*hugged, him*) to yield another constituent; *move* leads to constituents missing a subconstituent (*who, $\lambda x.$ did you hug x*). Following

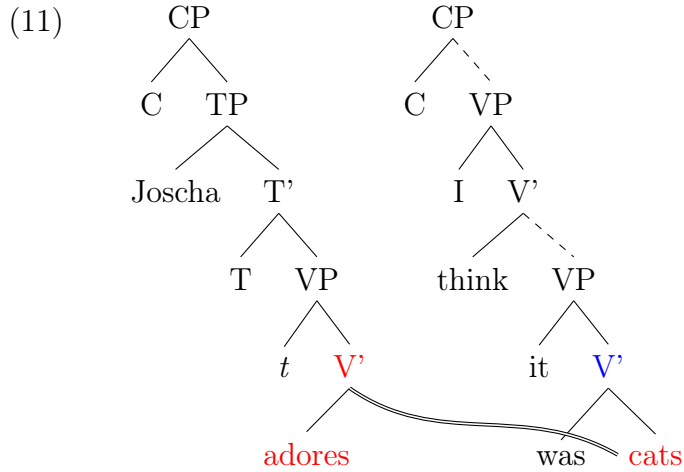
this analogy, what would a structure-building operation look like that leads to structures of the following type: $\lambda f.(\alpha, f(\beta))$ - creating a structure in which α and β are one unit, but across intervening material (f).

Although the analogy is rather informal, it is possible to assign a meaningful function to such a ‘*3rd-order merge*’ operation: long-distance in-situ selection of α for β , across a structure f that also merges with β but not selected by and hence unconnected to α . I argue that such an operation can be proposed as a potential solution to the DP vs. NP issue, i.e. the debate whether the noun or the determiner is the head of the nominal domain (see e.g. Salzmann 2020). While there are arguments for DP as the highest projection, the fact that verbs select for types of nouns but never for types of determiners can be seen as evidence for a direct connection between verb and NP (Bruening 2009). With the new operation, a verb can long-distance select for an NP (*pet*, $f(\textit{cat})$), while the determiner can at the same time select for that NP as well, making it the highest project in the nominal domain ($f=\lambda x.\textit{the}(x)$). I propose that this new type of long-distance dependency always holds between verbs and their nominal arguments. As mentioned above, there is no selection relationship between α and f (*pet* and *the*) which can be modelled by granting the D-layer the status of a second root. I formalize the new operation, called *3rd-merge*, in such a way that it leads to the growth of such a secondary root:



Once such structures are available, it is possible to grow secondary roots even further, beyond just the small D-layer. There are at least two options: one is that the root grows further into a full clause and remains independent. I claim that this corresponds to the phenomenon of Horn amalgams (Lakoff 1974, trees below are simplified):

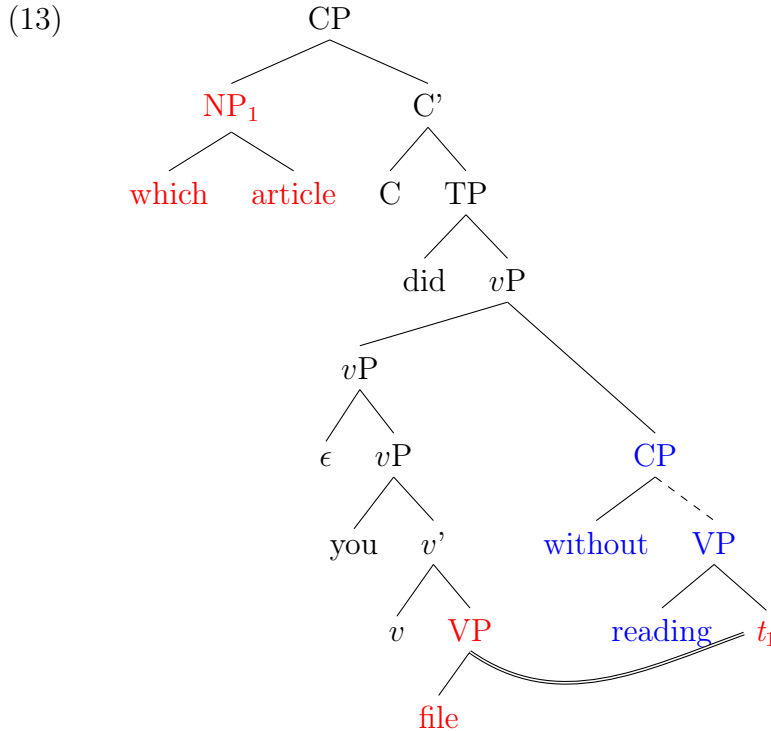
(10) Joscha adores [I think it was cats].



The pivot *cats* is the argument of both *adore* and *was*, yet the clauses headed by both verbs are unconnected: no c-command relationship holds between any of their elements other than the pivot (Kluck 2011).

Another option is to grow the secondary root further but reconnect it to the main root at some point during the derivation, resulting in a cyclic structure. If the connecting element is moved out, this leads to the appearance of two distinct gaps in a 1D-string connected to the same element after linearization. I argue that this corresponds to parasitic gaps:

(12) Which article₁ did you file *t*₁ without reading *t*₁?



Under this analysis, a single movement leads to the creation of two gaps at the same time because it extracts an element out of a more complex structure generated by a more elaborate operation. Here, the connection between such an operation and the parasitic phenomenon is more indirect than in the parasitic ellipsis case; yet it is still in accordance with the view proposed here that parasitism is one single phenomenon, not two separate but dependent ones.

This approach provides a unified perspective on three apparently unrelated phenomena: the structure of the nominal domain, Horn amalgams and parasitic gaps. The structure of the DP is the seed that enables other multi-root phenomena. I take the fact that both Horn amalgams and parasitic gaps appear to be (at least prototypically) restricted to NPs as evidence that proposing such a connection is justified.

However, the analyses of these phenomena are logically independent from each other in that parasitic gaps and Horn amalgams can be derived with *3rd-merge* without using this operation for the structure of the DP. I therefore start with the weaker claim that verbs simply select for the complete nominal domain, not some subpart of it, optionally allowing the growth of secondary roots and therefore Horn amalgams (chapter 7) and parasitic gaps (chapter 8). It is only in the last chapter (9) that I make the stronger claim that one can view the connection between verb and nominal argument as

the sole purpose of *3rd-merge*, with Horn amalgams and parasitic gaps as possible consequences.

This thesis is an attempt to combine various ways of doing research in linguistics. It is a thesis in computational linguistics that uses MGs to provide clear formalizations of the analyses proposed here; it documents in some detail (to my knowledge) genuine new data in a number of languages; and it also is a work in theoretical syntax in that it attempts to provide insightful analyses and to uncover new connections between a number of phenomena. However, this thesis also runs into the danger of leaving specialists from each strand wanting: there are no sweeping proofs of generative capacity; there could be a much more extensive elicitation of data, both with respect to the number of speakers and the languages shown here; and use of a different notation from a less widespread syntax framework might make the analyses more difficult to understand and connect to relevant current topics.

However, I do think that all these approaches can usefully be combined to provide a deeper understanding to the phenomenon of parasitism in syntax than each could have done on its own. Plenty of new data on an understudied parasitic process makes it possible to compare it to already known phenomena; from this new generalizations about parasitism as a general phenomenon can emerge that any theoretical account has to derive. Minimalist Grammars from computational linguistics provide formal rigour which makes it necessary to spell out explicitly the mechanisms of one's theory. This is especially important when defining a new structure-building operation that leads to unusual structures – avoiding hand-waving is rarely as crucial as in such a case. Finally, providing a specific analysis means not just making abstract statements about a phenomenon but making an attempt to derive core aspects of the phenomenon at hand, ideally with clear predictions that can be checked. My hope is that this thesis meets these requirements.

The structure of this thesis is as follows. Chapter 2 provides the necessary theoretical background. First, I introduce the grammar formalism Minimalist Grammar (section 2.1) and the ellipsis approach used here (section 2.2), explaining and visualizing the core ideas of 1st- and 2nd-order ellipsis operations. Then starts part I of this dissertation, at the heart of which is the phenomenon of parasitic ellipsis. First, however, I show that a restriction to 2nd-order operations is not possible in the ellipsis framework if one considers ellipsis remnants that have undergone remnant movement (section 3.1). This serves as segue to the main part of this half of the dissertation, parasitic ellipsis (3.2). I present this phenomenon as originally described in Fitzgibbons (2014) and then show that parasitic ellipsis occurs in other languages and after different categories as well in section 3.3. I also discuss the generalizations and restrictions that emerge from an in-depth look at parasitic ellipsis

(section 3.4).

Turning to the theoretical part of this half of the dissertation in chapter 4, I make the case that parasitic ellipsis cannot be reduced to an interplay of already existing processes such as scrambling or head movement (4.1). After a critical look at Fitzgibbons' own account of parasitic ellipsis (4.2), I discuss the phenomenon of parasitism in general. I compare parasitic gaps and parasitic ellipsis and show that they exhibit a number of similarities constant across both constructions (4.3). I then discuss that there are various properties that are specific to parasitism and conclude that parasitism is a genuine notion of grammar in its own right. I argue that parasitism is distinct from and cannot be reduced to other process interactions like feeding, for a number of reasons (4.4). First, parasitic processes appear to refer to the presence of processes, not contexts; this persists in some theoretical implementations - a device that I argue is too powerful. Second is the **parasitism of likes** generalization: parasitic processes are always dependent on processes of the same type, an unexpected conspiracy under approaches that view parasitism as a simple process interaction. Third, if the parasitic process truly is a separate operation and part of grammar, one would expect that the same type of restrictions hold as for the process it is parasitic on. Yet this is markedly not what one finds: the putative separate parasitic process enjoys both much more freedom, but is at the same time subject to apparently ad-hoc restrictions.

I propose that a *single-process* approach to parasitism avoids a number of issues that arise under a *separate-but-dependent* view. It treats parasitism as a genuine phenomenon with its own mechanisms, not as the accidental outcome of various, allegedly unrelated, processes that conspire to reliably produce it. Yet it also avoids pathological dependencies that are possible under approaches where operations can sense each others' presence.

I provide such an analysis in the ellipsis framework at hand in section 4.5, using the 3rd-order ellipsis operations proposed earlier. A simple restriction in terms of a locality condition for going from 2nd- to 3rd-order ellipsis operations ensures that such operations are not too powerful and generate the desired data.

I conclude this part of the dissertation with remarks on various further issues in chapter 5: I discuss whether 4th-order ellipsis is possible and how it would look like empirically. I also mention a number of issues that some phenomena present for this ellipsis framework that someone working on it in the future would need to solve. I then compare parasitic ellipsis to other cases that might be construed as additional deletion that is enabled by the presence of another ellipsis, such as relative clause deletion in Collins (2015). I also compare parasitic ellipsis to other phenomena in the literature that

have received the epithet ‘parasitic’, such as parasitic participles (e.g. den Dikken & Hoekstra 1997) or parasitic scope (Barker 2007).

The second part of this thesis further pursues the hint of a connection between parasitism and higher-order structures. Via the reasoning described above I arrive at a new structure-building operation *3rd-merge* that can be viewed as a form of long-distance in-situ merge leading to the creation of secondary roots in a syntactic tree (6.1). I then formalize this operation in MGs and apply it to two different phenomena (6.2). The following chapters then spell out in detail the application of this new operation to three phenomena.

In chapter 7, I show that growth of a secondary root that stays independent corresponds to Horn amalgams, with a number of desired properties. Chapter 8 is the core of this half of the thesis where I show that a reconnected secondary root with subsequent movement of the connecting element leads to the creation of two gaps at the same time, corresponding to the same element, in other words parasitic gaps. I discuss in 8.4 how this approach fares with respect to the desiderata for a theory of parasitic process put forward in the first half. Finally, I propose in chapter 9 that the dependency created by *3rd-merge* holds between verbs and their nominal arguments, in the way described above. Horn amalgams and parasitic gaps are then simple consequences of this approach to the structure of the nominal domain. In all three chapters I first provide a short overview over relevant empirical generalizations about the respective phenomenon. After detailing the specific analysis with step-by-step derivations, I also discuss a number of further issues for these proposals.

Chapter 10 concludes this dissertation and provides an outlook of future research avenues to pursue. Specifically, I bring up the point that proposing a new structure-building operation means proposing a new (long-distance) dependency. I argue that this might be justified, based on the fact that this dependency as well as the constructions that can emerge from it share properties such as a categorial restriction and a number of other restrictions. This is reminiscent of a classification of a certain set of dependencies proposed in Postal (1998) and further developed in Stanton (2016). Other phenomena falling into that class such as *tough*-movement might also be amenable to a multi-root analysis similar to the ones above. Future research needs to determine whether it is justified to propose a third long-distance dependency next to A- and $\bar{A}$ -dependencies by searching for more common properties and more fine-grained diagnostics and by investigating which other phenomena might be related to the ones addressed above.

2 The frameworks

This section introduces the grammar formalism that I use throughout the dissertation: Minimalist Grammars by Stabler (1997). Both the approach to ellipsis in the first half of the dissertation, and the new structure building operation in the second half are couched in Minimalist Grammars (MGs). After the core formalism is introduced (Section 2.1), I introduce the ellipsis approach as well (Section 2.2). The new structure building operation, as an extension of the original formalism, will only be defined in the second half of the dissertation. This section is intended as an introduction for someone who has never encountered Minimalist Grammars. People familiar with this framework can skip section 2.1 and can concentrate on the presentation of the ellipsis approach (2.2).

2.1 Minimalist Grammars

Minimalist Grammars (Stabler 1997) can be seen as a formalization of core aspects of current minimalist syntax. Minimalist Grammars (MG) are a strongly feature-driven formalism. Once the structure-building operations *merge* and *move* are defined, what is left to provide an analysis is to specify the feature sequence on the lexical items that occur in a sentence. These features regulate the application of *merge* and *move*. To illustrate the make-up of a lexical item, take a transitive verb like *hug* which looks as follows:

$$(14) \quad hug :: =n.=n.V$$

It consists of its phonological realization *hug*, a string of phonemes taken from an alphabet Σ , the phoneme inventory of a language. Its syntactic features after $::$ determine the syntactic processes it triggers as well as its category. In this case, *hug* demands to select, i.e. *merge* two nouns. After this is done it is a complete phrase of type V (i.e. a VP). Below are lexical entries of two nouns that *hug* could merge with:

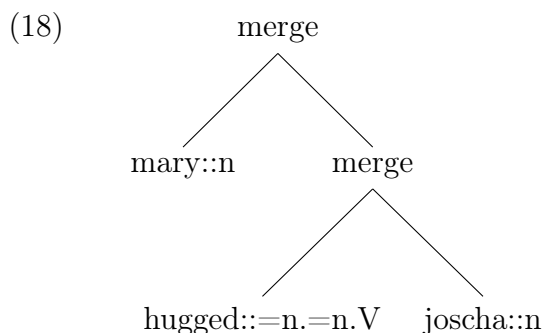
$$(15) \quad joscha :: n \qquad \qquad \qquad mary :: n$$

In MGs, both syntactic operations *merge* and *move* have their associated positive/negative feature pair. For *merge*, the ‘negative’ feature is the category symbol, say *n*. The positive merge feature is indicated by the equal sign, as in $=n$, indicating ‘I want something of type *n*’. In MGs, once an operation has applied, the two features involved are erased. Let us look at a tree for a simplistic syntactic analysis of the following sentence; indicated below are the lexical items that I propose as an analysis for this sentence:

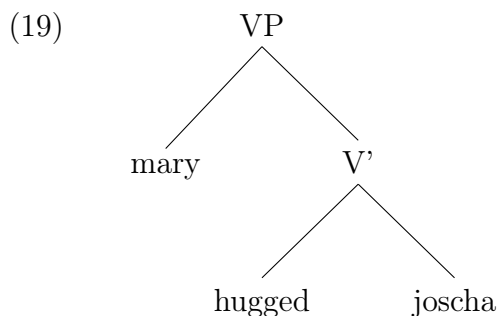
(16) Mary hugged Joscha.

(17) $hugged :: =n.=n.V$ $joscha :: n$ $mary :: n$

In MGs, the corresponding tree would look like this:



This is what is called a **derivation tree**: it is a tree that details the steps that were taken during the derivation. First, *hugged* and *joscha* were combined by an application of *merge*, and then the resulting structure was combined with *mary* by an application of *merge* as well. This is notated on the node dominating the items participating in the operation. However, nothing much hinges on this notational convention. It is equally possible to write the more familiar phrasal information on nodes. The tree below is entirely equivalent to the one above.



Throughout this dissertation I will mostly write phrasal information on nodes to increase readability.¹

When it comes to *move*, MG derivation trees differ in more than just

¹As a side note, MG trees do not per se encode the linear sequence of two sisters. Sometimes linear sequence of sisters is used to encode head and argument status where the sister containing the head of the constituent (i.e. containing the feature that triggered the operation) is to the left, and the phrase that was selected is to the right (i.e. specifiers to the right). I will not use this encoding here and instead use whatever ordering is most intuitive without making a claim about headedness/linearity.

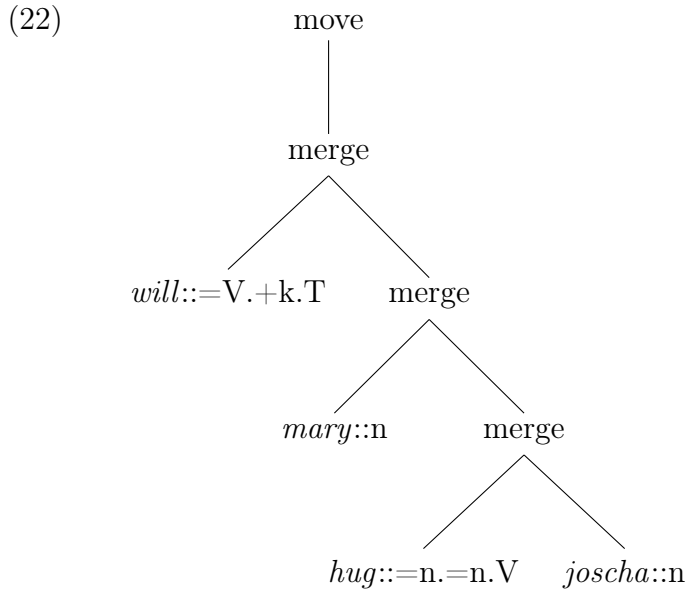
aesthetic preferences. But first, the relevant features: since MGs are a feature-driven formalism, there is also a negative/positive feature pair for items involved in a *move*-operation. The head that triggers movement carries the positive feature, say $+wh$, and the item that needs to move carries the negative movement feature, $-wh$. Let us take a sentence that involves subject movement, and the associated simplistic analysis I propose in form of a selection of lexical items with their feature specification:

(20) Mary will hug Joscha.

(21) $will :: =V.+k.T$ $joscha :: n$
 $hug :: =n.=n.V$ $mary :: n.-k$

The analysis for this sentence now contains subject movement. *Mary* moves to SpecTP, attracted by *will*. The movement feature pair involved is $+k/-k$. Since *mary* needs to move, it carries the $-k$ feature, following the category symbol. Negative movement features are in general ordered after the category symbol: first a head can select its arguments, then it can be selected itself; if it moves, then it does so after it has been merged.

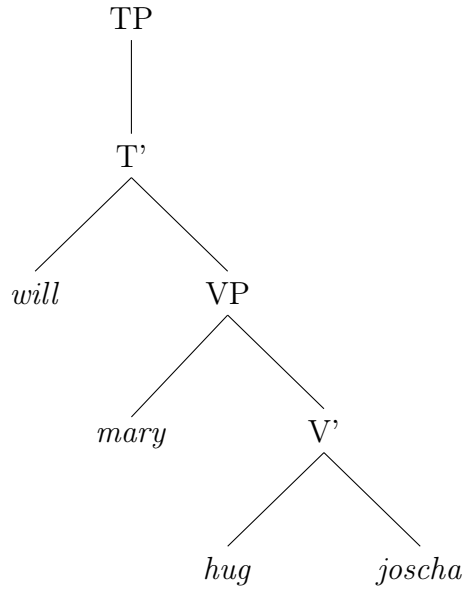
In the structure below, the T-head *will* first merges with its complement VP and then attracts an element into its specifier via $+k$. The corresponding derivation tree for this analysis looks as follows:



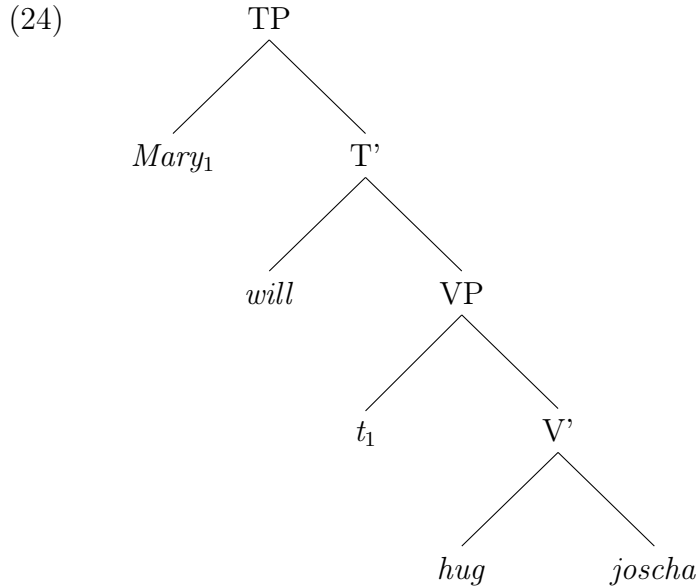
Switching to phrasal information on the nodes and leaving out the feature sequence on the leaves does not change the crucial difference between MG *derivation* trees and the type of trees more commonly used in minimalist

praxis:

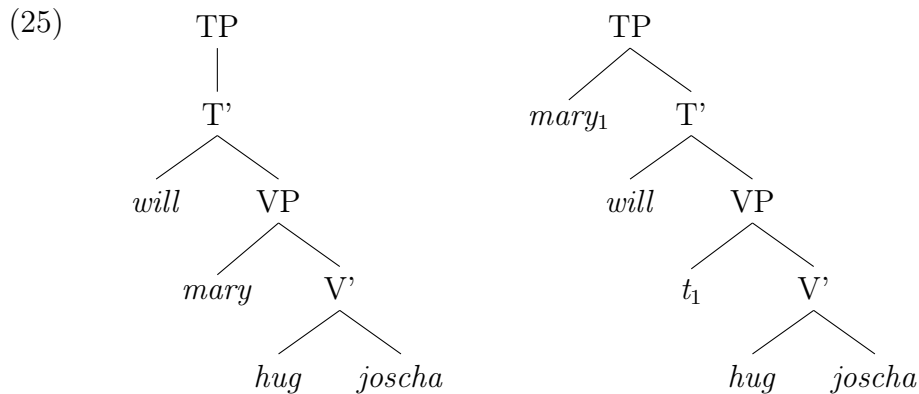
(23)



In MGs, the node that indicates the application of a move operation is a unary branching node. This makes sense when recalling that MGs use *derivation trees*: a structure that indicates the steps of a derivation, similar to a recipe (‘apply X to the result of the preceding step’). So after a *merge*-step takes place triggered by $=n$, a *move*-operation takes place $(+k)$, faithfully indicated in the node above. This is different from a tree in which one also notates the result of the operation. This is what you can know as a **derived tree** which shows the structure that is derived when executing all operations.

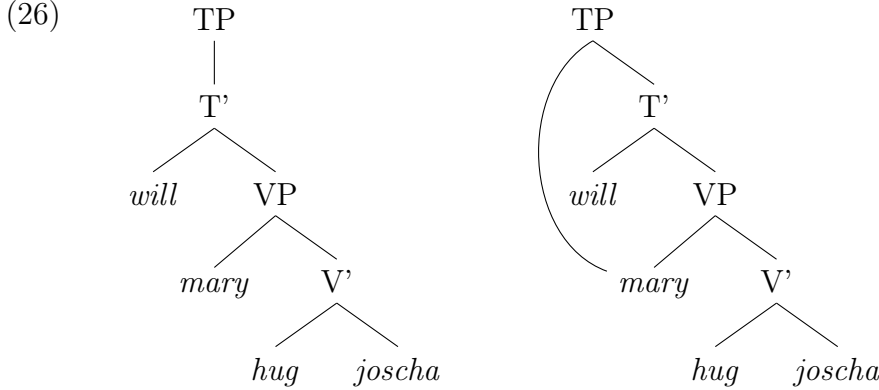


However, it is still the case that both types of trees impart the same information: derivation trees indicate that movement happens, and due to the way MGs are formalized (more about this later) the element that moves is always uniquely identifiable, as well as the locus of its realization. That is, there is a deterministic procedure to get from one tree to the other:



MGs view *derivation trees* as the more central data structure that the grammar refers to. This will have an effect when talking about the interaction of ellipsis and movement in the next section.

One way to make MG derivation trees more familiar is to visualize them with multidominance notation (*without*, however, committing to this specific theoretical account of movement). Throughout this dissertation I will often draw an arc between a moving item and its final landing position in order to make the tree more readable and to avoid the unfamiliar unary branching.



For the purposes of this dissertation, all tree types - *derived trees* which execute movement and leave behind a trace, or *derivation trees* with in-situ movers either with or without additional arc - are to be understood simply as different notational conventions for the same underlying MG calculus that is presented below. This concludes the informal presentation of Minimalist Grammars. MGs consist solely of the operations *merge* and *move* whose application is precisely regulated by features on lexical items. Nothing else so far is part of Minimalist Grammars. Other aspects of syntactic analysis (e.g. case) can be added, but require a formal implementation. I ignore these issues here since they are mostly not relevant for the arguments made in this thesis. I also abstract away from head movement. Head movement has been formalized in MGs (Stabler 2001), but for the present purposes I simply write verbs and their potential affixes in their base position, or as one unanalysed whole. This, too, does not matter for the arguments made here.

I now turn to the precise formalization of MGs, specifically the algebraic definition of MGs by Stabler & Keenan (2003). Let me start with a block of text; afterwards I go through each statement step by step and explain it in prose.

A Minimalist Grammar $G = \langle \Sigma, F, Types, Lex, \mathcal{F} \rangle$, with a non-empty alphabet Σ , the set of Features F consisting of *base* features $(n, v, c, \dots)$, the corresponding selection features, as well as licenser and licensee features for movement, i.e. $F = base \cup \{=f | f \in base\} \cup \{+f | f \in base\} \cup \{-f | f \in base\}$, the $Types = \{::, :\}$. There is a special start symbol $c \in F$. Stabler & Keenan (2003) call Chains $C = \Sigma^* \times Types \times F^*$, and Expressions $E = C^+$, with the lexicon $Lex \subseteq C^+$ as a finite subset of $\Sigma^* \times \{::, :\} \times F^*$. There are two operations $\mathcal{F} = \{merge, move\}$, defined as partial functions. I use $s, t, u, v, w \in \Sigma^*$; $\beta, \gamma \in F^*$; $\delta, \varepsilon, \zeta \in F^+$, chains $\alpha_1, \dots, \alpha_k$ or $\iota_1, \dots, \iota_\ell$ or $\mu_1, \dots, \mu_m$ with $k, l, m \geq 0$.

Let us unpack these statements one by one. To define a Minimalist Grammar G , we need five things:

We have an alphabet Σ containing the phoneme inventory of the language ($a, i, u, p, l, m, \dots$). Out of those we can construct the phonological form of our lexical items, i.e. the part that appears before $::$ in the above examples. Then there is a set of Features F which contains the category features, such as n or V . Using those category features, which are at the same time the ‘negative’ features in a *merge* operation, we can construct the positive merge features that indicate that an element wants to merge with something of that category. This is indicated by ‘=’. x and $=x$ are a feature pair with opposite polarity. They can participate in a merge operation after which both features are erased.

The same can also be done to generate positive and negative *move* features, written as $+f$ and $-f$. The complementizer of an interrogative sentence, e.g., might trigger *wh*-movement into its specifier. For that it has to carry a $+wh$ -feature, attracting a *wh*-word with $-wh$. Note that there is some redundancy in the original definition because in practice there are no base features that are used both as categories and in movement features - there is e.g. no *wh* category. One might as well say that movement features are generated from a different set distinct from categories ($wh, top, k, \dots$). This would not change anything of substance, however.

Next comes a description how one can construct lexicon entries and more complex entities. Let us start with the simple case, a lexicon entry such as *hug*. For that one needs the set $Types = \{::, :\}$. The element ‘ $::$ ’ indicates that an object is a simple lexicon entry which has not participated in any syntactic operation yet. Conversely, ‘ $:$ ’ indicates that something is a syntactic object which has already participated in an operation. $\cdot \in \{::, :\}$ as is used as shorthand when the distinction between the two does not matter. F^* means zero or more instances of an element out of F , i.e. a sequence of features. A chain is defined as $C = \Sigma^* \times Types \times F^*$. In other words, to construct a chain means to take a phonological string, a type and a sequence of features.

$$(27) \quad hug :: =n.=n.V$$

The above is an instance of a chain: it is a phonological string $hug \times$ a type ‘ $::$ ’ $\times$ a sequence of three features, two positive and one negative *merge* feature. It is also a special form of chain, namely a lexical item (‘ $::$ ’). The lexicon Lex is the subset of chains that have the type ‘ $::$ ’.

Expressions can consist of either one or several chains. To understand how a complex expression can come about, it is necessary to introduce the precise definitions of the two operations *merge* and *move*. In these abstract definitions, $s, t, u, v, w \in \Sigma^*$ will be used as stand-ins for more realistic phonological forms; s could as well be *cat* or *will hug Joscha* or any other combination

of lexical items. $\beta, \gamma \in F^*$ denote zero or more features; $\delta, \varepsilon, \zeta \in F^+$ one or more features; in expressions, which are combinations of chains, some chains can be abbreviated as $\alpha_1, \dots, \alpha_k$ or $\iota_1, \dots, \iota_\ell$ or $\mu_1, \dots, \mu_m$, with $k, \ell, m \geq 0$.

The two syntactic operations are defined using the format of an inference rule. In practice this means listing the items participating in an operation above the line - the inference line - as premises (e.g. selector and selectee); below the line is the conclusion, the result of applying the operation, specifying the changes that occur as a consequence. The inference rule therefore defines the effect of the operation.

$$\frac{\text{premise1} \quad \text{premise2}}{\text{conclusion}} \text{rule}$$

There is an intuitive correspondence between this format and the more familiar tree representation as I illustrate below.

The operation *merge* is defined by three subcases. *Merge* takes two expressions and combines them into one ($\text{merge} : (E \times E) \rightarrow E$).

$$\begin{aligned} & \frac{s :: = f\delta \quad t \cdot f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{merge1} \\ & \frac{s : = f\delta, \alpha_1, \dots, \alpha_k \quad t \cdot f, \iota_1, \dots, \iota_\ell}{ts : \delta, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{merge2} \\ & \frac{s \cdot = f\delta, \alpha_1, \dots, \alpha_k \quad t \cdot f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{merge3} \end{aligned}$$

merge1 describes the case where an element is merged into complement position, *merge2* where an element is merged into specifier position and *merge3* where a phrase is merged that contains a movement feature and hence cannot be linearized directly.

I illustrate each of those with an example. In *merge1*, the element triggering *merge* is a head directly out of the lexicon ($::$). Since by definition this head has not participated in any syntactic operation yet, its argument is its first to be merged. This argument therefore becomes the complement of this head and - at least in this specific definition - is linearized to its right. Note that the *Type* of the argument does not matter (therefore $\cdot$). It can itself be a lexicon item like a proper name or a more complex syntactic object.

$$\frac{hug :: = n. = n.V \quad joscha :: n}{hug \text{ } joscha : = n.V} \text{merge1}$$

$$\frac{will :: =V.+k.T \quad hug\ joscha : V, mary : -k}{will\ hug\ joscha : +k.T, mary : -k} \text{merge1}$$

As can be seen in the above examples, the positive and negative merge features are erased in the outcome and the phonologies of selector and selectee are combined. The result is in any case a derived expression (‘:’). In case the selectee contained any elements that need to move further - as is the case in the more complex second example where the subject *mary* carries a movement feature - they are simply ‘carried’ over and notated after the head of the expression, separated by commata (I return to these more complex expressions when discussing *merge3*). Note that the inference-rule format seen above is in a sense an upside down tree:

$$(28) \quad \begin{array}{c} \text{merge1} \\ \swarrow \quad \searrow \\ hug::=n.=n.V \quad joscha::n \end{array} \quad \begin{array}{c} hug\ joscha:=n.V \\ \swarrow \quad \searrow \\ hug::=n.=n.V \quad joscha::n \end{array}$$

All these various representations are equivalent. One can either write an application of *merge* as an inference rule, or as an upside down derivation tree. The derivation tree indicates that a *merge1* step occurred, equivalent to the annotation to the right of the line in the proof format. The featural make-up of the top node of the tree is the same as the bottom line in the inference rule format. It is possible to annotate that directly into the *merge1*-node in the tree. I omit this in most cases for readability, with the implicit understanding that both representations carry the same amount of information. I usually impoverish the trees even further, by writing phrasal information on the nodes, not rule applications, as well as not writing out the feature sequence on leaves. This representation, too, is meant to be equivalent to both the enriched tree format as well as the inference rule format.

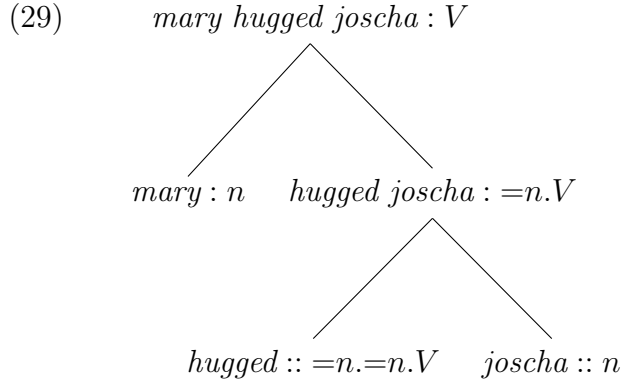
Turning back to the definition of *merge*, rule *merge2* describes the case where the selector is already a derived expression. Therefore it contains at a minimum already a complement, but possibly already one or more specifiers. Any item it selects will therefore become an (additional) specifier. Since the selector is a derived expression, it might or might not contain one or more moving items ($\alpha_1, \dots, \alpha_k$), just as its selectee ($\iota_1, \dots, \iota_\ell$). In the result, both items combine, their positive/negative merge-features are erased and in this formalization the selectee is linearized to the left.

$$\frac{\frac{hugged\ joscha : =n.V \quad mary : n}{mary\ hugged\ joscha : V} \text{merge2}}$$

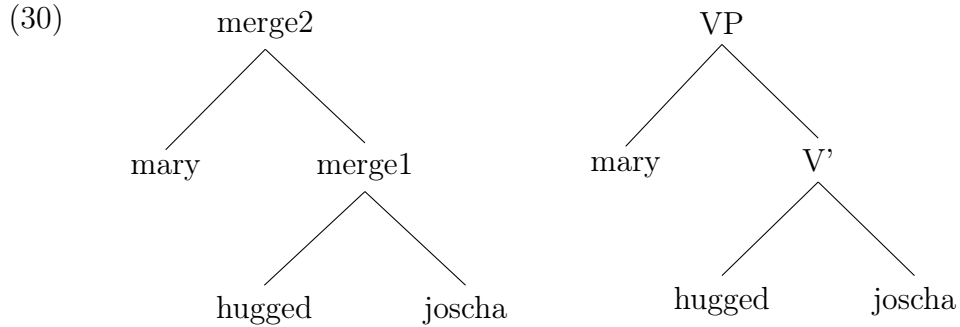
This inference rule starts out with verb and object already combined. It is possible to write out the construction of the full VP from the start. Multiple inference rules get stacked, just like a tree of growing depth:

$$\frac{\frac{\frac{hugged :: =n.=n.V \quad joscha :: n}{hugged\ joscha : =n.V} \text{merge1} \quad mary :: n}{mary\ hugged\ joscha : V} \text{merge2}$$

This brings out the branching, tree-like nature of the inference rule format even more. The corresponding tree version of the above is:



which, again, is understood as entirely equivalent to:

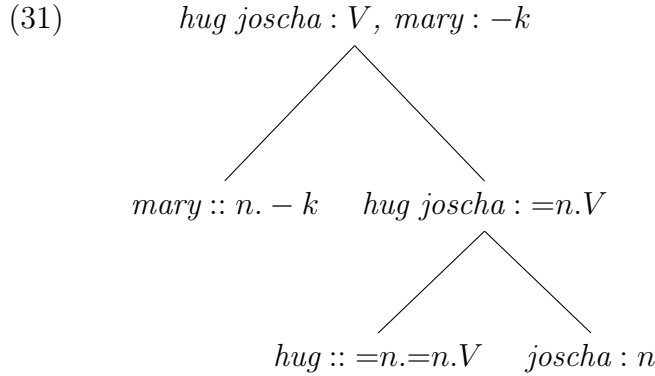


As a last merge rule, there is *merge3* which defines the case where the selectee itself is a mover, i.e. it has a movement feature after its category symbol ($f\varepsilon$). The selectee will therefore still play a role in the derivation and can not yet be linearized. For this reason the Type of the selector, i.e. whether it selects

for a complement or specifier, does not matter ($\cdot$).² Merging a moving item leads to an expression consisting of multiple chains. In the simple cases above we deal only with trivial expressions consisting each of a single chain. The outcome of *merge3* is an expression consisting of multiple chains, with the head of the expression written first and all other chains following.

$$\frac{hug\ joscha : =n.V \quad mary :: n. - k}{hug\ joscha : V, mary : -k} \text{merge3}$$

This inference rule corresponds to the upper portion of the equivalent tree.



The verb *hug*, after selecting its object, also selects for its subject (*mary*) which in this example carries a movement feature after the category symbol. Selector and selectee feature ($=n, n$) are erased, as usual. Since *mary* has a movement feature, it cannot be linearized together with the V' constituent yet. It is therefore written as one chain of the larger expression.

The head of the expression is the chain whose head carries the positive merge feature triggering *merge3* and is written as the leftmost chain. Any elements that are part of this expression but are still syntactically active in that they contain a movement feature that needs to be saturated are listed after the head of the expression (their order does not matter). This concludes all subcases of *merge*.

We now turn to the operation *move*. *Move* consists of two subcases; in each case, it takes an expression and turns it into another expression ($move : E \rightarrow E$).

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, t : -f, \alpha_{i+1}, \dots, \alpha_k}{ts : \delta, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{move1}$$

²In case $\cdot$ is for $::$, the selector itself can of course not yet contain movers. The number of movers k can only be greater than 0 in the $:$ case. This rule is therefore an implicit abbreviation that glosses over this distinction.

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, t : -f\varepsilon, \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \alpha_1, \dots, \alpha_{i-1}, t : \varepsilon, \alpha_{i+1}, \dots, \alpha_k} \text{ move2}$$

Move applies to expressions that consist of more than one chain, executing movement of one of its chains which was triggered by the head of the expression. *Move1* describes the case where a mover reaches its final landing site. *Move2* describes the case where an element moves to a position that is not its final landing site because it contains yet another movement feature ($f\varepsilon$); this could e.g. be an item that first moves for case and is then $\bar{A}$ -moved. I illustrate with examples:

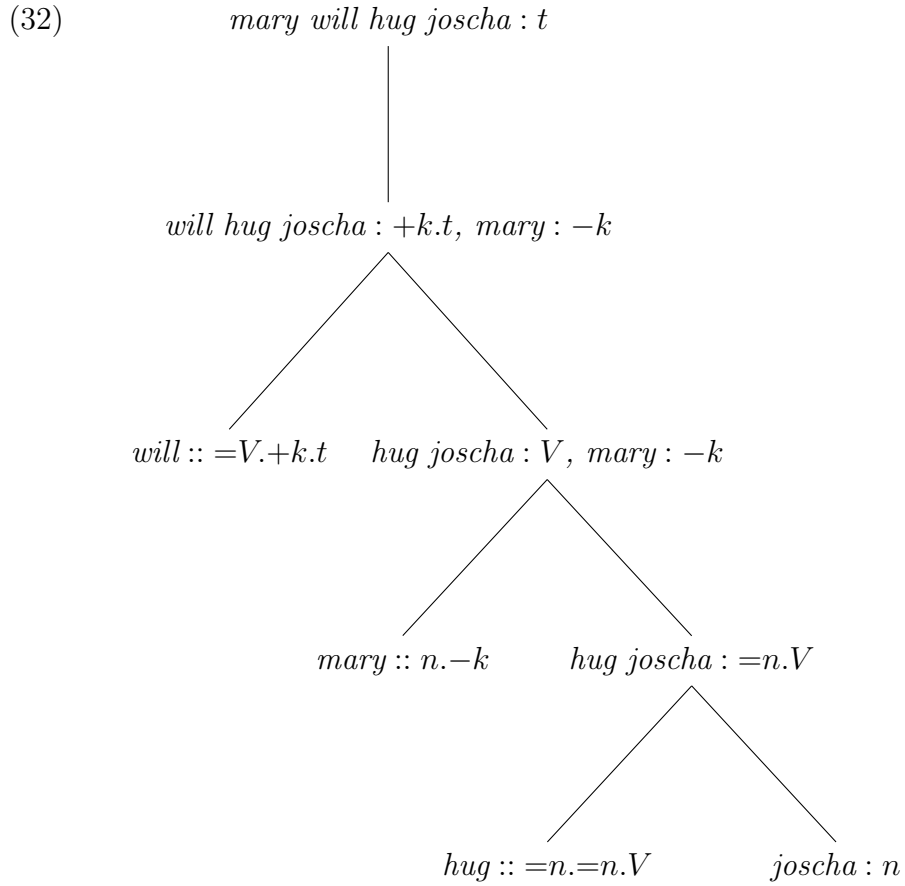
$$\frac{\text{will hug joscha} : +k.T, \text{mary} : -k}{\text{mary will hug joscha} : T} \text{ move1}$$

$$\frac{\text{will hug joscha} : +k.T, \text{who} : -k.-wh}{\text{will hug joscha} : T, \text{who} : -wh} \text{ move2}$$

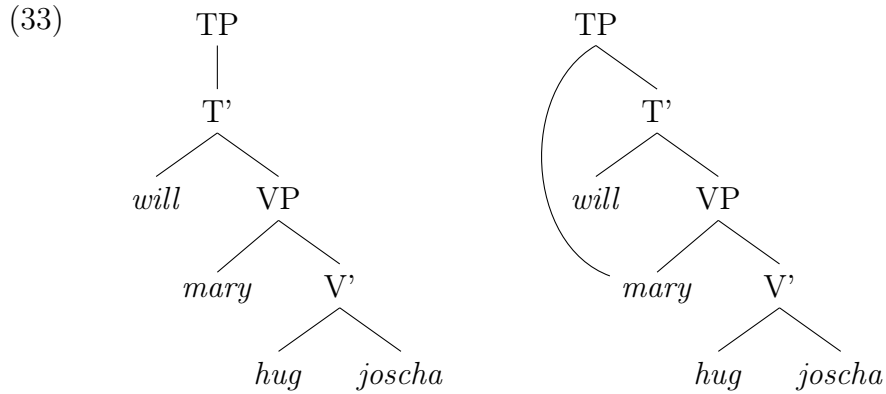
In both cases, the first feature on the head of the whole expression is a positive movement feature $+f$. Following the head of the expression are one or more chains. One of those chains has as its first feature the corresponding negative movement feature $-f$. If this movement feature is the only feature left on this chain, then the mover is linearized to the left as a specifier of the head of the expression. This happens in the exemplary *move1* case above, where *mary* moves to subject position and by doing so has all its features checked and can be linearized. Positive and negative movement features are erased, just as with *merge*.

Move2 is illustrated in the subsequent rule where the *wh*-word first needs to move to subject position ($-k$) but is also *wh*-moved afterwards. What happens in this specific instance is that only the subject-movement features are erased, but *who* is still listed as a chain of the more complex expression, as an element that still needs to move. Note that in this rule we use ε after f since we have at least one or more movers ($\varepsilon \in F^+$).

It now becomes clear why *move* is indicated via a unary branching node in a derivation tree. *Merge* is an operation that takes two expressions and combines them into one ($\text{merge} : (E \times E) \rightarrow E$), hence it is a binary branching node. *Move* takes a single expression and administers a change to a member of that single expression ($\text{move} : E \rightarrow E$). There is a single expression above and below the line in *move1* and *move2*. Hence a *derivation tree* representation for the *move1* case above would be:



Therefore, the *derivation tree* equivalent is a unary branching node that indicates an effect on the expression that the daughter node represents. Since no outside element is involved, no binary branching is necessary. It is of course possible to indicate which element in the expression undergoes the change. This information is added via the outgoing movement arc in one of the equivalent notations below:



A very important constraint on Minimalist Grammars comes into play here: the **Shortest Move Constraint (SMC)**. It says that in the expression in the *move1/2* rule, only the mover in question ($t : -f(\varepsilon)$) has the relevant movement feature $-f$, but no other member of $\alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k$ can carry it.

In other words, what the SMC says is that there can never be a competition between movers of the same movement type. It is forbidden that the head of an expression carries an attraction feature, say $+wh$, and there are two chains whose first feature is $-wh$. For this reason it is not necessary to indicate which element is actually the mover corresponding to a *move* node in a derivation tree because this element can per definition always be uniquely identified.

The SMC as defined here is probably the biggest difference to standard minimalist practice where such a configuration would not lead to a crash but would instead be decided e.g. by principles of relativized locality (Rizzi 1990). However, this difference does not play a role for any of the core points made in this thesis. The same goes for another difference, namely that $\bar{A}$ -movement in MGs does not proceed successive-cyclically through phase edges. This equally does not affect the main analyses in this thesis which are consistent with either of those options. I will therefore gloss over these difference from now on.

As a last point, we have defined c as the start category or start symbol. This tells us when a syntactic object is complete, namely when we have an expression (or the corresponding tree) whose only unchecked feature is the category c on its head. There must not be any moving pieces left over; nor does any other category have this special status (so far). This means that none of the trees above are technically complete syntactic objects. I have done so for reasons of space and readability; I continue to do so throughout this thesis, with the implicit understanding that each sentence is headed by a (possibly silent and not visually indicated) C . The below expression is complete since the only feature left unchecked is c :

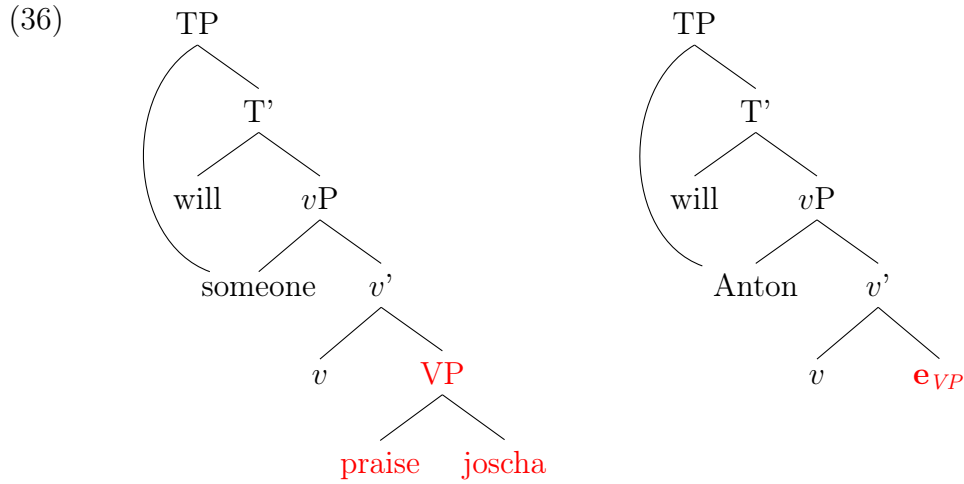
(34) *mary will hug joscha : c*

This concludes the overview of Minimalist Grammars.

2.2 Ellipsis in Minimalist Grammars

I now turn to the ellipsis framework that I use in the first half of the dissertation. Kobele (2015) is a derivational copying approach to ellipsis that is implemented in Minimalist Grammars. Ellipsis is conceived of as a pro-form of a specific syntactic type. “[E]llipsis sites are resolved by replacing them with their antecedent semantically, but antecedents are delimited syntactically” (2015:244). This is best illustrated with an example. Say that you have a sentence with VP-ellipsis:

(35) Someone will [_{VP} praise Joscha]. Anton will e_{VP}



This approach postulates that in the second elliptical sentence, there is an ellipsis pro-form present that has the correct syntactic type. In the context of this ellipsis framework, *syntactic type* is defined as follows: the type of an expression is $type(t) = \langle \alpha, A \rangle$, with α the feature sequence on the head of the expression, and A the feature sequences on the heads of other chains if they are present. As an example, the type of the expression *praise joscha* : V is $\langle V \rangle$; if *joscha* were a mover in the expression *praise* : V , *joscha* : $-k$, then the type of this expression would be $\langle V, -k \rangle$. For simplicity, I refer to an ellipsis pro-form of type $\langle V \rangle$ simply as a VP pro-form, often indicated as subscript on the ellipsis (e_{VP}). Visually, I indicate an ellipsis pro-form - as well as the material that serves as its antecedent - in red.

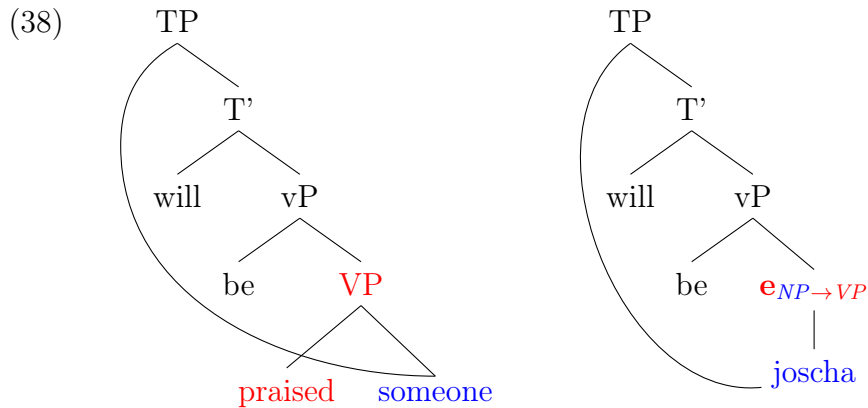
In this approach, to say that a language has VP-ellipsis means to say that the language has e_{VP} in its lexicon. English would be an example for such a language, whereas German which lacks VP-ellipsis correspondingly also lacks e_{VP}. German, however, would have e_{NP} for NP-ellipsis and both languages would possess e_{TP} for sluicing. Phonologically, these ellipsis pro-forms are spelled out as silent elements. Syntactically, their behaviour is determined

by their syntactic type. The type of $\mathbf{e}_{VP}$ is $\langle V \rangle$. In the above sentence, the silent v -head selects for a VP-complement ($\epsilon ::= V.n.v$). Since $\mathbf{e}_{VP}$ is of category V, both elements can be merged and the merge features can be erased.

The crucial work happens in the “copying” process, in other words in determining how the ellipsis is interpreted in a specific context. An ellipsis pro-form $\mathbf{e}_{VP}$ of type $\langle V \rangle$ is anaphoric to an expression (or corresponding equivalent derivation tree) that has the exact same type. The short context in (35) happens to contain such an expression: *praise joscha* : V (or equivalently, $[_{VP} \text{praise Joscha}]$). In this specific context, therefore, the meaning of $\mathbf{e}_{VP}$ is the meaning of $[_{VP} \text{praise Joscha}]$. Importantly, this framework assumes that ellipsis pro-forms are anaphoric to portions of a *derivation tree*. This fact becomes relevant below.

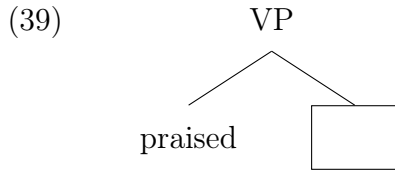
The above case describes an instance where an ellipsis pro-form is a stand-in for a full constituent. Anticipating terminology a bit, I call these ellipsis pro-forms **1st-order ellipsis operations**. However, it is well-known that phrases whose base position is within a constituent that is elided can survive this deletion if they are moved to a position that is higher than the ellipsis. In this case the fact that MGs refer to *derivation trees* becomes a relevant factor. Let me illustrate with an example. In the (simplified analysis of) passive, the logical object of the verb, in the example below *someone*, moves to subject position. In case a speaker chooses to apply VP-ellipsis in a similar enough sentence, the simple picture above does not work any more in this framework:

(37) *Someone* will be [praised t]. *Joscha* will be $\mathbf{e}_{NP \rightarrow VP}$



Since moving items are notated in their base position we cannot just claim that the VP-ellipsis in the second sentence can copy the VP from the first

sentence ‘as is’.³ We do not want to copy [praise someone] into the sentence ‘*Joscha will be*’. This leads to a clash between the logical object of the first sentence, and the logical object of the second sentence. Despite containing a VP-ellipsis, the second sentence still provides an overt version of the object of the verb that is elided. This is because the object moves to SpecTP, a position higher than the VP. Instead, what we want to say is that the ellipsis copies the VP from the first sentence, but *without* the corresponding object *someone*, since the elliptical sentence can supply an object itself. In a sense, what needs to be copied is the VP with a hole:



The ellipsis approach used here postulates ellipsis pro-forms that can do exactly this. In addition to full constituent pro-forms, there are also ellipsis pro-forms that can stand in for a constituent while ignoring a subpart of this constituent. An ellipsis operation that wants to copy a VP, minus an NP contained in it is written $\mathbf{e}_{NP \rightarrow VP}$; in the above case it has the type $\langle n, -k \rangle \rightarrow \langle V, -k \rangle$ ⁴ or as shorthand just $\mathbf{NP} \rightarrow \mathbf{VP}$. As before, I use **red** to indicate ellipsis as well as the material it is anaphoric to. I now use **blue** to indicate the element that is base-generated inside the ellipsis but not elided, leading to the more complex pro-form, as well as the corresponding item within the antecedent that therefore needs to be ignored/cut out.

In order to describe these structures, I now introduce important terminology that is highly relevant throughout this thesis. The ellipsis $\mathbf{e}_{\mathbf{NP} \rightarrow \mathbf{VP}}$ is anaphoric to a **higher-order** antecedent. It is not anaphoric to a full constituent, a **first-order** antecedent/constituent, like $\mathbf{e}_{\mathbf{VP}}$; instead, it is anaphoric to a **second-order** antecedent, a constituent-minus-subconstituent. A second-order constituent is therefore a constituent, out of which a constituent has been cut out. I have represented this above as a tree with a hole

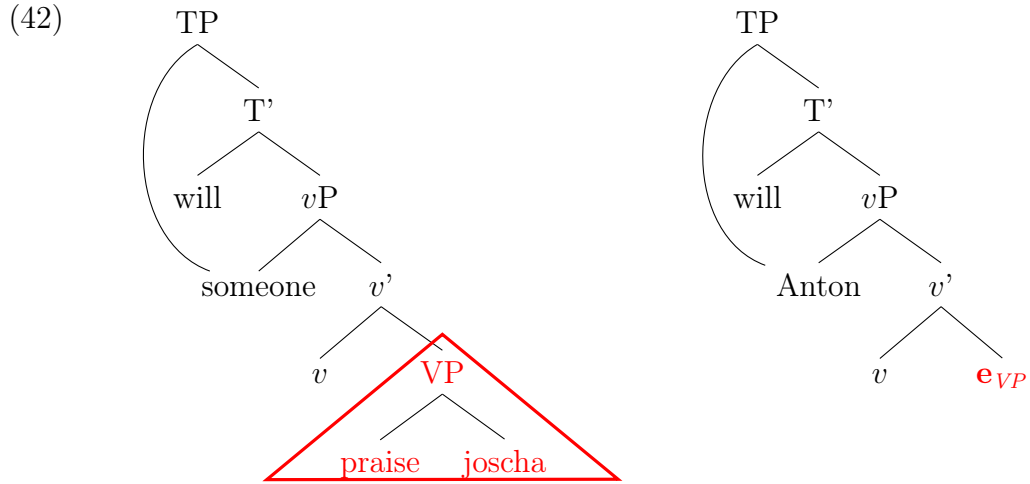
³Note that this is not ‘just’ a quirk of the movement notation chosen. Remember that the trees above are equivalent to the more elaborate formal expressions in MGs. The expression for the VP node is $\textit{praised} : V, \textit{someone} : -k$ which is what would be copied when instructed to copy the VP node.

⁴ $n, -k$ since it is the feature sequence on *joscha*, and $V, -k$ since V is the feature that a VP node would have, with a $-k$ after the comma because this specific VP contains a mover whose only feature left after it would be merged with a ‘copied in’ verb would be its movement feature.

(also called ‘*tree context*’). A way to talk about second-order constituents without always showing 2D-trees is to make use of λ -notation. Whereas $[_{VP} \text{praise Joscha}]$ is the antecedent for e_{VP} , I now say that $\lambda x.[_{VP} \text{praise } x]$ is the antecedent for $e_{NP \rightarrow VP}$. A completely equivalent way of expressing it is to use a representation more similar to the original MG derivation trees: one can write a mother node *merge* with daughters α and β as $merge(\alpha, \beta)$. A second-order antecedent then would be $\lambda x.merge(\text{praise}, x)$. To summarize this in one place:⁵

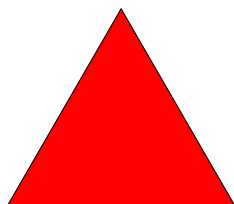
- (40) 1st-order ellipsis / antecedent
- a. e_{VP}
 - b. $[_{VP} \text{praise Joscha}] / merge(\text{praise}, \text{Joscha})$
- (41) 2nd-order ellipsis / antecedent
- a. $e_{NP \rightarrow VP}$
 - b. $\lambda x.[_{VP} \text{praise } x] / \lambda x.merge(\text{praise}, x)$

To represent this as trees, I repeat the trees from above. In the first-order case, ellipsis is a simple node, corresponding to a complete subtree (i.e. a node and all elements it dominates). Abstractly, one could visualize the antecedent as a triangle that encompasses this complete phrase:



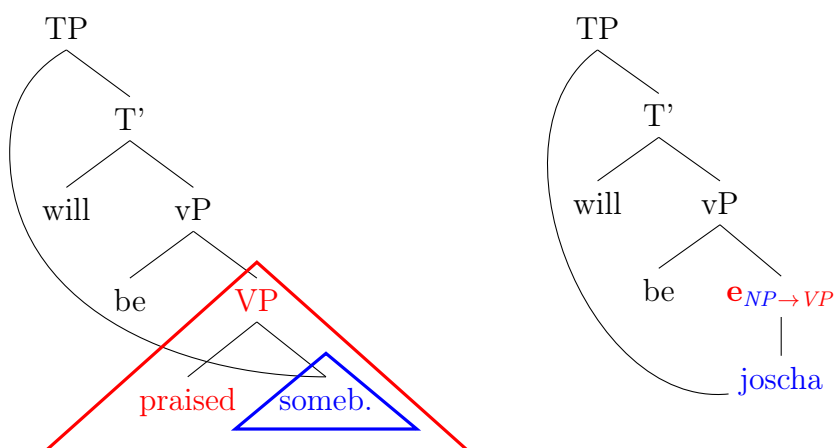
⁵To preempt a potential confusion, use of λ -notation does not mean that the semantics module is invoked here. λ -calculus is a notation in which any function can be written, so it can in principle be used in any subfield of linguistics or beyond and is not tied to a specific module.

(43)

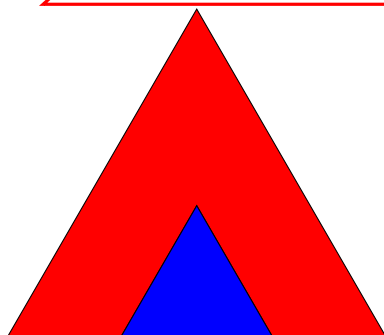


In the second-order case, ellipsis is now a node with a daughter (more about this below), the unelided subpart of the incompletely elided constituent. Since non-leaf-nodes in MG derivation trees denote operations, one can also call this ellipsis pro-form an operation. I use ellipsis pro-form or operation interchangeably here. This ellipsis operation corresponds to a incomplete constituent where that part is ignored that is present in the elliptical sentence. Visually, one could represent this as the triangle that covers this subtree, with the ignored subpart cut out in form of a smaller (blue) triangle:

(44)



(45)

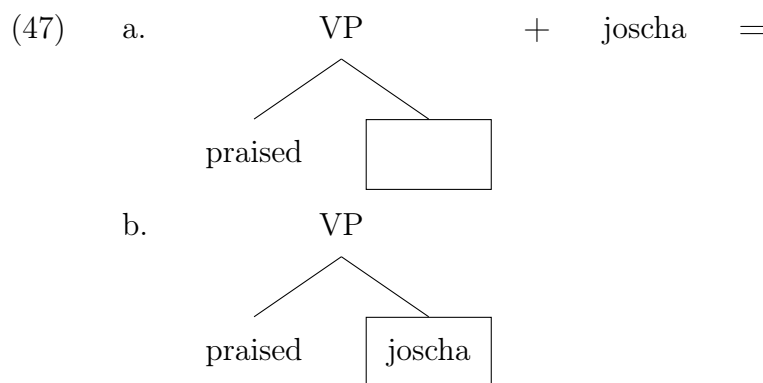


After the introduction of important terminology, let us return to the question how ellipsis is resolved in the passive VP-ellipsis example above. The way one wants to resolve the ellipsis $e_{NP \rightarrow VP}$ is to copy a VP minus an NP. $\lambda x.[_{VP} \text{praise } x]$ is a suitable candidate. This object is of type $\langle n, -k \rangle \rightarrow \langle V, -k \rangle$, in other words: if provided with a noun that moves for case, it

would become a VP containing a case mover. If this structure is copied into the ellipsis site, it is a perfect match for the structural configuration there: a VP with a hole for an NP is needed, and an NP with a movement feature can be provided. The ellipsis can therefore be resolved like this:

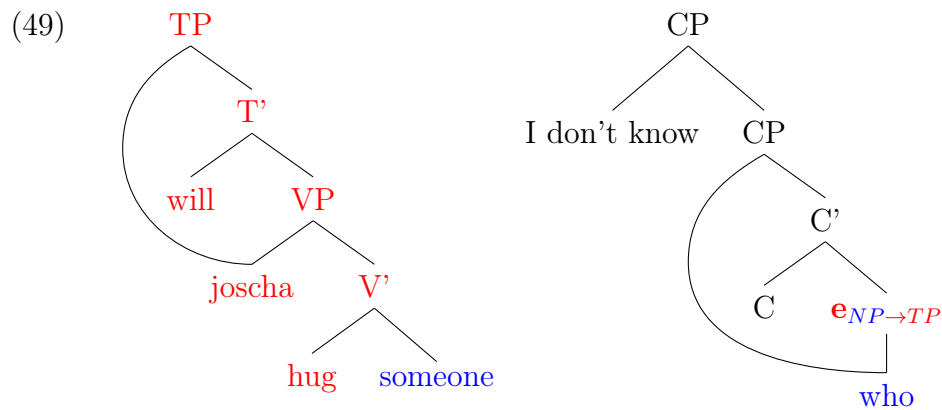
$$(46) \quad \lambda x. [\text{VP praise } x] (\text{joscha}) = [\text{praise joscha}]$$

This is exactly the derived outcome. A way of representing this visually might be:



Let me illustrate with yet another (simplified) example:

(48) Joscha will hug someone. But I don't know who.



If we say that sluicing here is ellipsis of TP, then what is missing in the elliptical sentence is a TP except for the object NP that moves to SpecCP in the elliptical sentence. What we therefore need to look for is a TP in the preceding context, cut out an NP in object position, and then paste the TP-minus-NP into the ellipsis site:

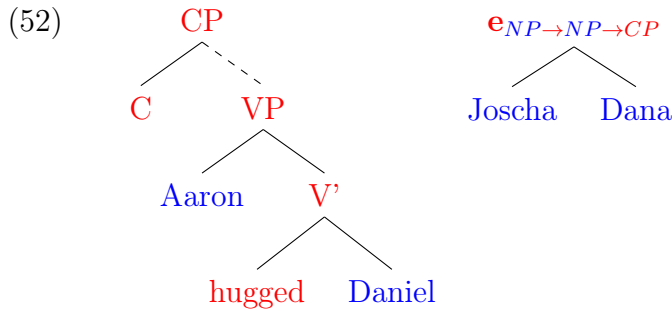
$$(50) \quad \lambda x.[_{VP} \text{joscha will hug } x] (\text{who}) = [\text{joscha will hug who}]^6$$

Note that this approach is not all too different from deciding the identity requirement of ellipsis over derived trees. Antecedents for ellipses with movement survivors also do not come for free. The trace/copy of the moved item in the antecedent is not understood as referring to said moved item but instead needs to be converted into an open variable that the moved item in the elliptical sentence can then refer to (see Fox 2002). This has the same effect of creating an antecedent of the form $\lambda x.[_{VP} \text{praise } x]$.

I also want to address a potential confusion that might arise due to the position of the remnant of ellipsis. The logical object in the passive VP-ellipsis sentences above is written as the daughter of the 2nd-order ellipsis operation. It is not ‘selected’ by the ellipsis operation in the sense of the word where ‘select’ means taking part in a *merge* operation. It fulfils the type requirement of the ellipsis operation which requires an NP to become a complete VP. The NP is in a sense understood to be selected by the structure that the ellipsis is anaphoric to. In this case, the VP-minus-NP contains such an element, namely the verb *praise* :: =*n.V* which can select elements of the form *joscha* : *n*.

As a final point, I want to mention that increasing the number of ‘daughters’ of an ellipsis operation, and therefore the number of holes in the antecedent, does not increase the *order* of the ellipsis operation. Instead, it increases its *arity*. Ellipses can have more than one remnant, especially in gapping, the form of ellipsis discussed in the next few sections. Consider the following sentence and assume that gapping elides full CPs (in this massively simplified analysis).

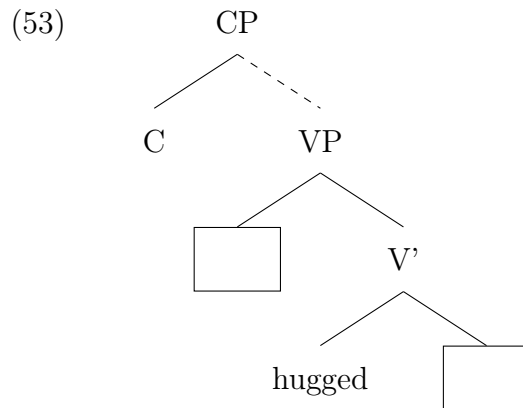
(51) Aaron hugged Daniel, and Joscha Dana.



Here, the ellipsis operation needs to make room for two remnants. Its type is therefore $e_{NP \rightarrow NP \rightarrow CP}$, or shorter $e_{NP, NP \rightarrow CP}$. Its antecedent consequently

⁶For simplicity I abstract away from the inversion of auxiliary and subject in the embedded question.

will be a constituents with two holes: $\lambda x.\lambda y.[y \text{ hugged } x]$. Each hole, however, corresponds to a complete (first-order) constituent, so the highest-order variable in the expression describing this structure is still first-order, making it overall a 2nd-order term.



This concludes the overview of the approach to ellipsis used in this thesis. With the next chapter the main content of this thesis begins. I first show shortly that 1st- and 2nd-order ellipsis operations are not enough to satisfactorily describe remnant movement remnants in this approach; instead, even 3rd-order ellipsis operations are needed.

I then describe the phenomenon of parasitic ellipsis, the main focus of the first half of this thesis. I discuss properties of parasitic constructions in general and come to the conclusion that they are best analysed as a single process. I show that in parasitic ellipsis the elided material also corresponds to a 3rd-order antecedent and one can use 3rd-order ellipsis operations to derive parasitic ellipsis in one fell swoop.

As a last note, the choice of the ellipsis approach is not crucial for the broader points I make about parasitic ellipsis here. It just happens to be the framework that is implemented in Minimalist Grammars, the formalism I use throughout this thesis. Even though the insight of a deletion of a 3rd-order structure can transparently be encoded in this framework, the more general remarks about how parasitism should be dealt with in grammar can also be transported to other ellipsis approaches and other grammar formalisms.

3 3rd-order ellipsis operations

3.1 Remnant movement - going beyond 2nd-order ellipsis

Kobele (2015) considers only ellipsis of 1st-order and 2nd-order constituents (also called *tree contexts*): “[r]estricting our attention to *contexts* amounts to, in the more general setting of the λ -calculus, limiting our attention to terms of order at most two”⁷ (Kobele 2015:fn.18).

The goal of this section is to show that a restriction to terms of order two or lower does not suffice to describe all elliptical phenomena in natural language. To reiterate the basic picture from the last chapter, one can elide constituents, giving rise to a 1st-order ellipsis operation. We have seen that a phrase that originates within an elided constituent but that moves to a position higher than the ellipsis site leads to a 2nd-order ellipsis operation that is anaphoric to a constituent missing the moved constituent. Multiple phrases that independently move out of an ellipsis site do not increase the order of the ellipsis operation, only the number of holes in the tree context that such an ellipsis operation is anaphoric to.

Things get more interesting when movers are nested as is the case in remnant movement. Take the following German sentence:

- (54) $[t_1 \text{ Lesen}]_2$ wird $[\text{das Buch}]_1$ jeder t_2 , $[t \text{ rezensieren}]$ keiner.
read will the book everyone, review noone.
 $\approx$ ‘Everyone will READ the book, but noone will REVIEW it.’

The traditional analysis for such a sentence (see e.g. Müller 1998) is that *das Buch* scrambles out of the VP; subsequently, the VP containing the trace of the object is topicalized into SpecCP. In the corresponding elliptical sentence, an instance of gapping, one of the remnants is not a complete constituent any more, but a constituent missing a constituent. Such incomplete remnants are not restricted to a particular quirk of the syntax of German; remnant movement in English (as discussed in Abels 2007) leads to analogous results:

- (55) $[\text{How likely to } t_1 \text{ win the lottery}]_2$ is he₁ t_2 and how likely to t win the race?

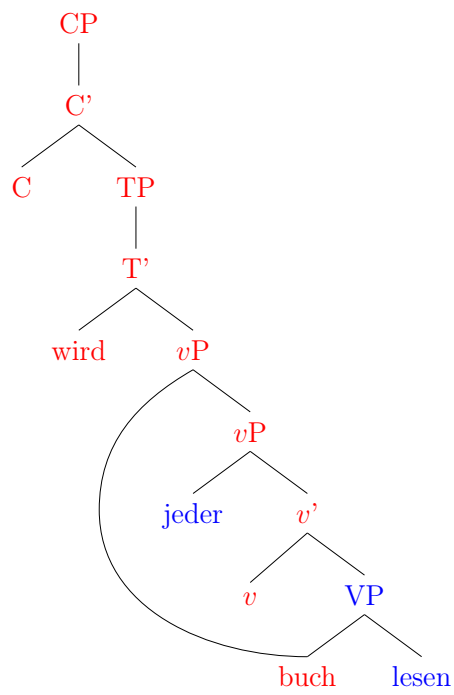
Here, A-movement of the subject out of the AP is followed by wh-movement of said AP, leading to a sluicing remnant containing a trace in the second conjunct.

⁷Where a λ -term of order one would be a simple constituent, and a term of order two a constituent missing a constituent, as in $\lambda x. [\text{VP praise } x]$.

The question is how such sentences are best analysed in the current approach. Below I show the derivation trees for the antecedent clause in (54) and (55), each directly followed by two possible analyses for the subsequent elliptical clause.⁸

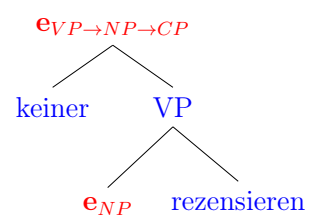
⁸A note regarding the derivation of contrastive ellipses such as gapping: here and throughout the dissertation I do not commit to a specific stance towards either the size of the constituent elided in gapping (CP, TP, *v*P), simply choosing CP for convenience, or the mechanism by which gapping remnants survive ($\bar{A}$ -movement, exceptional movement, in-situ focus marking,...). Whether movement or not occurs does not change the base position of the mover since the data structures used here are derivation trees. It would only add additional move nodes on top of the ellipsis site which does not change anything about the points made within the current framework. Similarly for the size of the elided constituent: the status of a phrase as ellipsis remnant does not change as long as it is contained within the phrase that is elided. Whether that phrase is a CP, TP or *v*P is irrelevant as long as it contains the remnant. I take this to hold for all the examples discussed here. As long as the assumption holds that all ellipsis remnants discussed in this thesis are in fact ellipsis remnants, all points made here hold and are not affected by the manner of survival or size of ellipsis. I therefore do not indicate further structure above the ellipsis operation in contrastive ellipsis for readability.

(56)

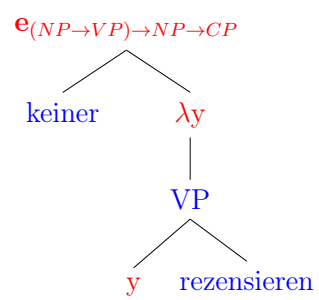


(57)

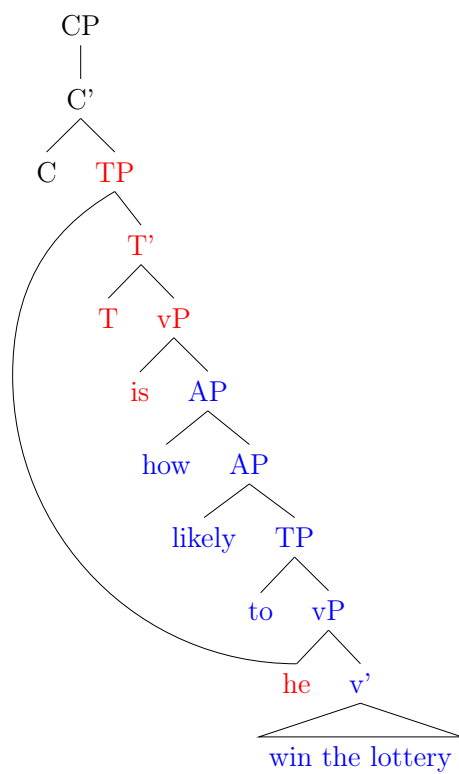
a.



b.

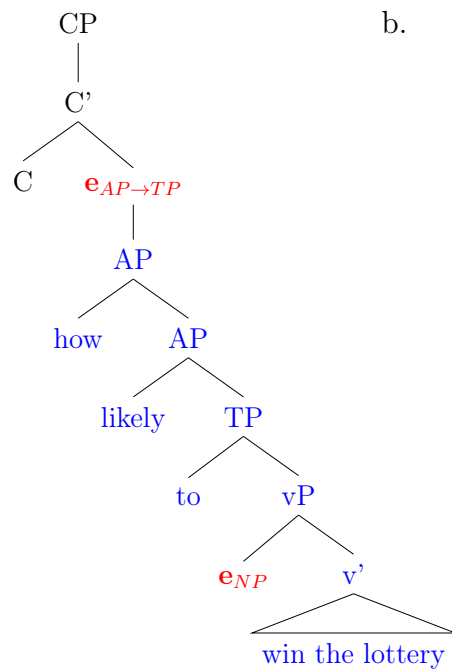


(58)

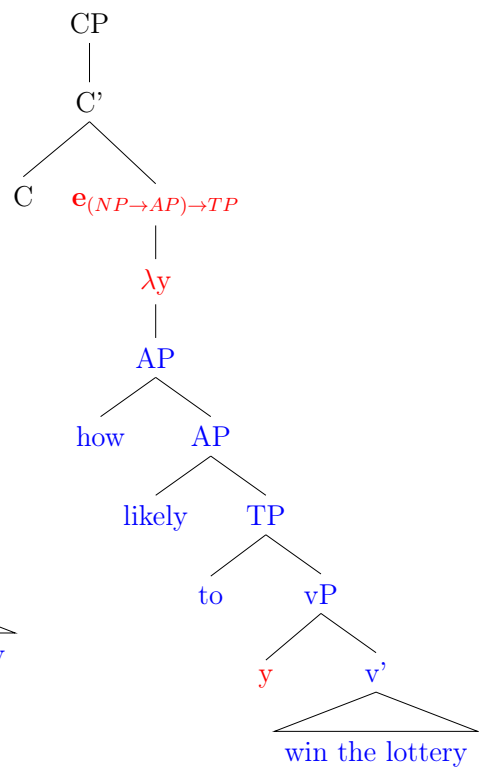


(59)

a.



b.



Take (54) as an example: what survives ellipsis is *jeder* and *lesen*. For *jeder* this is business as usual: a full (1st-order) constituent survives which requires a 2nd-order ellipsis operation with a hole in the corresponding position. The more interesting case is *lesen*. Intuitively one wants to say that the whole VP is an ellipsis remnant ($[_{VP} t \text{ lesen}]$ or $\lambda x.[_{VP} x \text{ lesen}]$), not just the verb, and the object is missing since it has been scrambled out. Usually it is not single heads but full phrases that are ellipsis remnants. This becomes much more unambiguous in the English remnant movement example where there is more material surrounding the missing element. The most natural assumption is that the fully spelled-out version of the elliptical sentence would be:

(60) [How likely to t_1 win the race] is he_1 ?

Here the phrase headed by *how likely* is the ellipsis remnant, minus the subject of *win* which moves to a higher position in the normal course of events:

(61) He_1 is [very likely to t_1 win the race].

The reason that the object of *lesen* and the subject of *win* are not pronounced in these elliptical sentences is that they move out of the phrase that survives ellipsis to a position that is *included* in the ellipsis site.

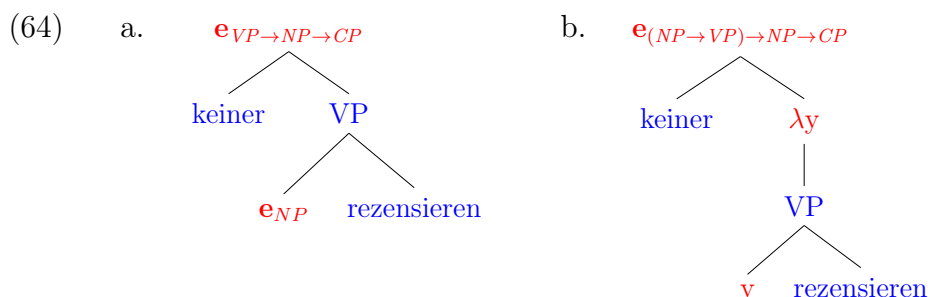
One has to account for the non-pronunciation of the missing constituent in these remnants somehow in the current approach. We know from the interpretation of the above elliptical sentences that the additional missing constituent (in italics) is recovered from the antecedent, in other words, their meaning is:

(62) ... *das Buch* rezensieren wird keiner.

(63) ... how likely to win the lottery is *he*.

Not only the finite verb is recovered, but also the object of *lesen* / the subject of *win*. Since, again, the data structure used here is *derivation trees*, one cannot simply not represent the missing phrase, or alternatively represent it with a trace. We need to express that something that fulfils the role of the missing NP is present, and that it is part of the ellipsis site.

There are two possibilities to represent this: either as a separate, additional ellipsis operation of the item in question ((57a), (59a)), or as a single, discontinuous and therefore higher-order ellipsis operation ((57b), (59b)). I want to argue against the former and for the latter option (illustrated here with the two versions of the German sentence above):



The argument against a separate and *independent* ellipsis operation e_{NP} is that such ellipsis operations are not independently available in the grammar of the two languages discussed here. A truly independent ellipsis would look as follows:

- (65) Mary drank two beer,
 a. and Susan three beer.
 b. and Susan three.
 c. and Susan drank three.

(65a) is missing a finite verb, an instance of gapping; in (65b) one can see that it is possible to elide *beer* as well. Yet this NP-ellipsis is completely independent from gapping, as (65c) illustrates.

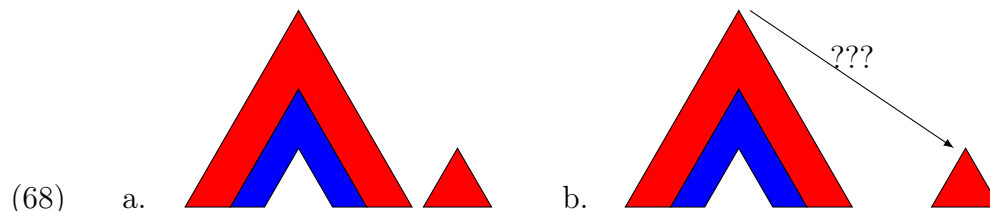
This contrasts with the remnant movement examples above. German does not generally allow something like object drop in this environment, nor does English have a universally available rule of subject ellipsis. Take the non-gapped counterparts to the two sentences discussed above:

- (66) *Lesen wird das Buch jeder, rezensieren wird keiner.
 read will the book everyone, review will noone.
 ‘Everyone will read the book, noone will review.’

- (67) *How likely to win the lottery is?

The only reading that is marginally available for the German sentence is one of a generic reviewing activity which is distinct from the meaning of the gapped clause in (54). The English sentence is ungrammatical. The only way to make the *separate ellipsis* option viable is to make it a fully dependent operation that can only be grammatical if another ellipsis operation is present. In the previous chapter, I represent ellipsis operation abstractly as triangles, where a single red triangle is for an ellipsis of a (1st-order) constituent; a 2nd-order ellipsis where one elides a constituent except for a subconstituent is represented as a triangle (red) with a triangle-shaped hole (blue). The choice between the two different analyses can therefore be represented as

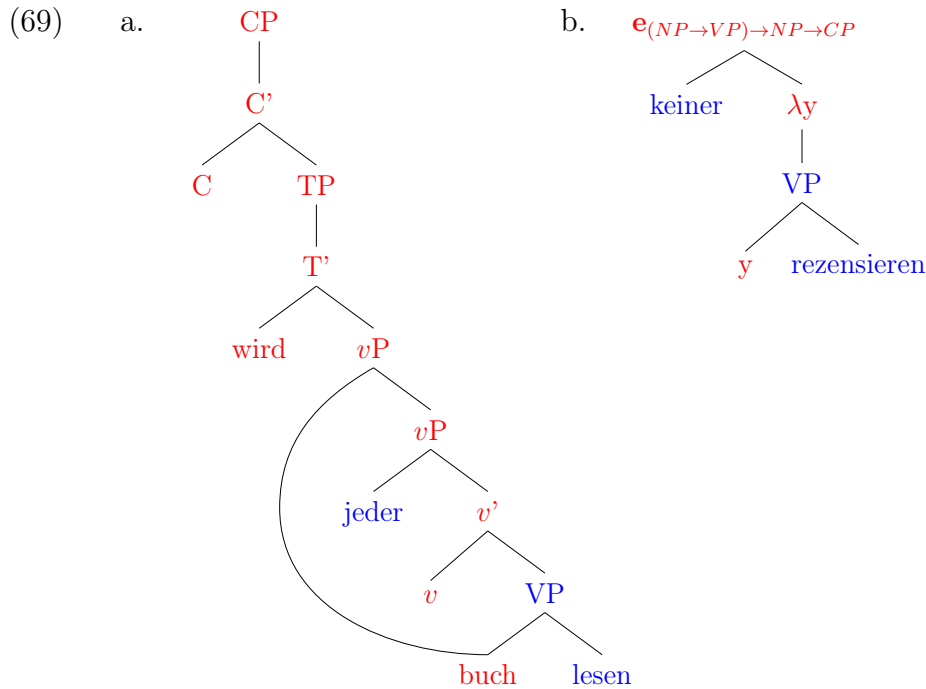
follows:



In (68a) there is an ellipsis (big red triangle) with a surviving constituent (blue triangle); yet it so happens that there is an additional ellipsis within the surviving constituent (small triangle, to the side). This would describe a case where there is NP-ellipsis in a gapping remnant as in (65). A separate but dependent ellipsis is represented in (68b), where the arrow indicates a yet to be described dependency between the two ellipses.

To foreshadow, whether and how the presence of one ellipsis operation might enable further ellipsis operations, and whether the possibility of operations that depend not on the context that other operations provide but on their mere presence, will be discussed in the sections on parasitic ellipsis. However, at least for the current phenomenon I do not want to propose such a potentially powerful mechanism. The reason for this is that the cause for the shape of these remnants is well-understood, namely (remnant) movement. In fact, in post-syntactic approaches to ellipsis, nothing at all needs to be said about these examples. If there is syntactic structure present in the ellipsis site and only after all syntactic movements have applied the (non-)pronunciation of a constituent is decided due to the presence or absence of an E-feature, any phrase that is part of that constituent at that point, whether base merged or moved, will not be pronounced. This means that there is only a single ellipsis for sentences (54) and (55). I view it as conceptually desirable to analyse the ellipses in (54) and (55) as a single ellipsis in the current framework as well, for the reasons indicated above: the putative second ellipsis is not available in other contexts and appears to magically coincide with remnant movement constructions in all cases.

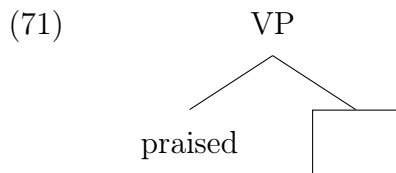
In order to express that all elided material in (54) and (55) is part of a single ellipsis operation, one has to go above 2nd-order ellipsis to (at least) 3rd-order ellipsis operations. This option is represented in the *b*) sentences of (57) and (59), repeated here for the German sentence:



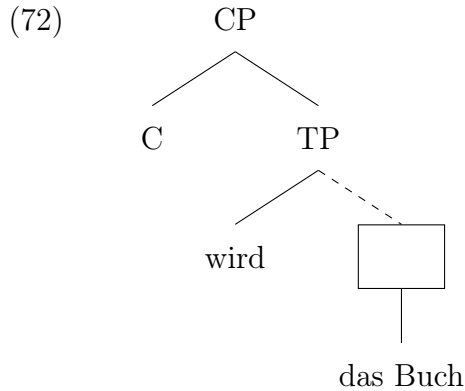
Ignoring the subject *keiner* for now, what is elided in (69b) and copied from (69a) is the whole CP minus the VP, but *plus* the object *das Buch*, i.e. the following term:

$$(70) \quad \lambda f.[_{CP} \dots [_{vP} v f(\text{das Buch})]]$$

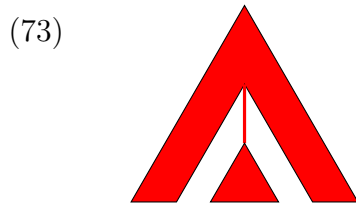
The expression above is just a formal way of referring to all the material in red at once. This was easiest in 1st-order ellipsis where one describes just a constituent. In a 2nd-order antecedent, one needs to exclude the remnants; the resulting ‘holes’ are described like this: $\lambda x.[_{VP} \text{praise } x]$.



In a 3rd-order antecedent, one does not just cut out a whole constituent but only a part of it - the ‘hole’ now has substructure, so to speak. We ignore the VP but keep the object. For this reason, when describing the antecedent a higher-order variable occurs which expresses that we *exclude* some material (f), but *include* material structurally contained within it ((*das Buch*)).



An abstract way to represent the structure $(\lambda f. [\text{CP} \dots [\text{vP } v \text{ f}(\text{das Buch})]])$ that needs to be copied into the elliptical sentence (69b) is the triangle visualization used before:



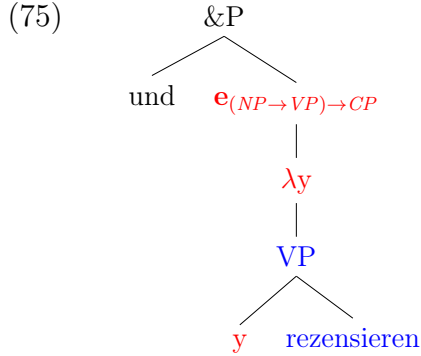
The antecedent is a constituent (big triangle) missing a subconstituent (the white cut-out part), but not the whole subconstituent is missing, some subpart of *it* is yet again part of the antecedent (the smaller triangle). But they are not two different structures - they are a single but discontinuous unit⁹.

Since the ellipsis in (69b) elides a CP, except for a VP remnant, of which the object is elided yet again, it has the type $\mathbf{e}_{(NP \rightarrow VP) \rightarrow CP}$ (ignoring the subject remnant again for simplicity). It has one more level of embedding than a 2nd-order ellipsis operation, so it is a 3rd-order ellipsis operation.

We can now combine the elliptical sentence with its antecedent. Let me illustrate this with the simplified example where the crucial VP-remnant is the only remnant:

- (74) Lesen wird das Buch jeder, und rezensieren (auch)
 read will the book everyone and review (too)
 Everyone will read the book, and review it, too.

⁹I represent the unity of the whole structure via the small red connecting line.



Combining the antecedent that the ellipsis operation is anaphoric with the structure of the elliptical sentence that is overtly present yields the following:

$$\begin{aligned}
 (76) \quad & \lambda f. [_{CP} \dots [_{vP} v \text{ f(das Buch)}]] (\lambda y. y \text{ rezensieren}) = \\
 & [_{CP} \dots [_{vP} v [(\lambda y. y \text{ rezensieren})(\text{das Buch})]] = \\
 & [_{CP} \dots [_{vP} v [_{VP} \text{ das Buch rezensieren}]]]
 \end{aligned}$$

The left expression in the first line is the structure that the ellipsis operation is anaphoric to, the CP of the previous sentence, minus its VP except for the object. The right expression is the overt part of the elliptical sentence, the VP-remnant which additionally is missing an object. The remnant is not a complete constituent which is why there is a variable in the place of its object. The two terms have matching types so they can be combined. The left-hand term requires a 2nd-order argument which is then inserted in the position of the variable; the argument is itself unsaturated, requiring a (1st-order) constituent, it becomes a function in the next step, taking the object as its argument. This results in the desired outcome, namely that the second clause is understood to mean¹⁰:

$$(77) \quad \text{und das Buch rezensieren wird jeder.}$$

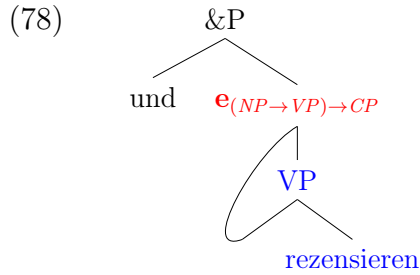
A note on the appearance of λy in the syntax trees above: it might seem unusual that a λ appears in a place where phrasal information is usually located. I use this representation simply as a notational clutch to express

¹⁰For completeness, here is also how the version with the initial gapping sentence with two remnants would work:

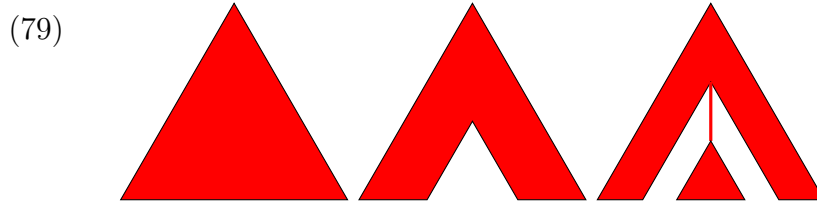
$$\begin{aligned}
 (i) \quad & \lambda f. \lambda x. [_{CP} \dots [_{vP} x v \text{ f(das Buch)}]] (\lambda y. y \text{ rezensieren}) (\text{keiner}) = \\
 & [_{CP} \dots [_{vP} \text{ keiner } v [_{VP} \text{ das Buch rezensieren}]]]
 \end{aligned}$$

As remarked above, a 1st-order remnant such as *keiner* is business as usual.

in a tree the same idea that is expressed in the bracketed notation in (76), namely the presence of a higher-order and therefore discontinuous elliptical structure. It is difficult to represent a single 3rd-order ellipsis continuously in a 2D-tree which is why I resort to λ -notation. Another representation might be a multidominance notation, expressing the exact same idea:



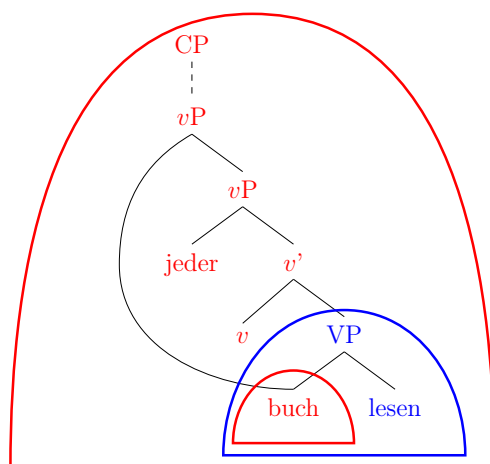
Let us summarize the theoretical findings of this section: in order to express that in an ellipsis with a remnant movement remnant there is a single, coherent antecedent, it is necessary in the current framework to go to 3rd-order ellipsis operations. This is forced due to the choice of viewing *derivation trees* as the central data structure in which movement steps are not executed. There is a succession of higher-order ellipsis types depending on whether there is no movement, movement, or nested movers:



Whereas one elides constituents in 1st-order ellipsis, and 2nd-order constituents with 1st-order remnants in 2nd-order ellipsis, 3rd-order ellipsis has 2nd-order remnants where the highest elided constituent and the missing constituent in the 2nd-order remnant constitute a *single but discontinuous* 3rd-order ellipsis to the exclusion of the 2nd-order remnant. This is illustrated again below with an actual example:¹¹

¹¹I have smoothed out the triangles for reasons of space.

(80)



As noted above, 3rd-order ellipsis operations also cannot be avoided by claiming that in the German example only a single head survives, not the whole phrase, or that these constructions should not be analysed as the result of remnant movement (see Fanselow 2002 for such a discussion). The English example above, repeated here, contains material surrounding the trace on both sides, making a remnant movement analysis the most likely:

(81) How likely to win the lottery is he, and how likely to win the race?

The purpose of this chapter, however, is not to argue for the correct analyses of these sentences but simply to point out that if there are remnant movement remnants in ellipsis, the approach to ellipsis used here cannot restrict itself to 2nd-order ellipsis operations. It needs at least 3rd-order ellipsis operations if it wants to maintain certain assumptions about the ellipsis processes discussed, first and foremost the assumption that only a single gapping process takes place. This leads to the question whether derivation trees are the most suitable data structure to talk about ellipsis since derived trees derive all the facts discussed here without any further stipulation. This might suggest that derived trees, i.e. the shape of the syntactic structure *after* all operations have been executed, are the more appropriate data structure over which the properties of ellipsis are computed (strengthening the case for ellipsis as a post-syntactic operation). However, I ultimately leave this question open here.

I also did not pursue the option of dependent ellipsis operations since it would postulate ellipsis of phrases that are otherwise not deletable. A compositional analysis can of course be made to work if it is ensured that these new ellipses *always* depend on another ellipsis and perfectly coincide with the presence of remnant movement at which point it becomes indistinguishable from a single ellipsis for all practical purposes.

The next section discusses cases where the initial intuition that there might be an ellipsis that is dependent on another ellipsis is more plausible, a phenomenon called parasitic ellipsis. In this case the discussion is also independent from the choice between derived or derivation trees. Nonetheless, I will come to the conclusion that even in this case the phenomenon at hand is better analysed as a single ellipsis - and more generally that grammatical processes should not be dependent on other processes, only on the contexts that they create.

3.2 Parasitic ellipsis in Russian

We have seen in the last section that not all elliptical phenomena in natural language can be described with 1st or 2nd-order terms. Remnant movement remnants require either an additional, always dependent ellipsis operation that is potentially ungrammatical on its own, or going from 2nd to 3rd-order ellipsis operations to be able to describe such an ellipsis as a single process. I have argued for the latter approach. These issues arise in a framework that takes derivation trees as the relevant data structure; post-syntactic approaches to ellipsis that operate over derived trees accommodate these data more easily.

In this section I want to discuss a phenomenon that superficially looks quite distinct from the constructions of the last section. *Theoretically*, however, very similar issues arise with respect to the question how this phenomenon is best analysed - in this case independent of whether one looks at derivation or derived trees. The phenomenon in case is parasitic ellipsis, first described by Fitzgibbons (2014). Fitzgibbons shows that ellipsis of the complement of some prepositions is impossible, even if this complement is recoverable (her exx. 2b, 3, 4a; all examples in this section are from Fitzgibbons 2014, with minor adaptations):

- (82) **Koška ukrala kolbasu s Mašinoj pomošč'ju, a sobaka ukrala*
cat stole sausage with Masha's help, but dog stole
kolbasu bez.
sausage without.
The cat stole the sausage with Masha's help, but the dog stole the
sausage without.

However, when there is gapping in the same clause (what Fitzgibbons calls *vP* ellipsis) the formerly ungrammatical ellipsis becomes (marginally) possible:

- (83) ?*Koška ukrala kolbasu s Mašinoj pomošč'ju, a sobaka*
cat stole sausage with Masha's help, but dog
bez.
without.
The cat stole the sausage with Masha's help, but the dog without.

Besides *bez* 'without', only two other prepositions *pod* 'under' and *nad* 'above' are judged acceptable by some of Fitzgibbon's consultants (fn. 2, exx. (iii)c, (iv)b, ex. 1d based on Gribanova 2009):

- (84) ?*Maša sprjatala podarok za divanom, a Vasja pod.*
Masha hid gift behind couch, but Vasja under.

- Masha hid the gift behind the couch, but Vasja under it.
- (85) Samolët proletel pod mostom, a vertolët nad.
plane flew under bridge, but helicopter above.
The plane flew under the bridge, but the helicopter above.
- (86) Katja krasila nad oknom, a Vasja pod.
Katja painted above window, but Vasja under.
Katja was painting above the window, but Vasja under (it).

Fitzgibbons, citing Gribanova (2009), points out that there are additional factors that determine acceptability: the preposition whose complement is elided needs to contain a vowel, or rather have a CV structure (fn. 2 (ii)a,b):

- (87) *V pervye tri mesjaca Pëtr žil po sosedstvu s
in first three months Petr lived on neighbourhood with
Sergeem, a Ivan u.
Sergej, but Ivan at.
During the first three months, Petr lived next to Sergei, but Ivan with Sergej.
- (88) *Inna položila korobku pod knižnyj škaf, a Tanja v.
Inna put box under book case, but Tanja in.
Inna put the box under the book case, but Tanja into the book case.

Fitzgibbons also remarks that only clausal ellipsis seems to allow for this additional ellipsis but not nominal ellipsis (exx. 16a, 17a)¹²:

- (89) ?V dožd' lučše xodit' s zontom, čem bez.
in rain better walk with umbrella than without
When it rains, it is better to go with an umbrella than without.
- (90) ???Vorkug stojali vrači v xalatax i bez.
around stood doctors in coats and without.
Doctors in lab coats and without stood around.

This appears to indicate that only clausal ellipses such as stripping and gapping (presumably the more general term for her *vP* ellipsis) license this phenomenon.

Fitzgibbons calls this phenomenon parasitic ellipsis and proposes a theoretical account in terms of E-feature valuation dependence. A contrastively

¹²I am not sure that these examples and others in the paper conclusively show this point. They all involve coordination, except for the comparative above, and it is not always clear which structure is coordinated and if it requires verbal ellipsis at all. However, I deal with the question which ellipsis licenses parasitic ellipsis more extensively in 3.4.

focussed preposition such as *bez* can optionally carry an E-feature, causing ellipsis of its complement. This feature is an interpretable but unvalued E-feature; it gets valued by the c-commanding E-feature of the *vP*-ellipsis that enables the additional deletion. This models the core property of parasitic ellipsis: an otherwise impossible ellipsis that becomes grammatical in the presence of another ellipsis.

I now turn to the main empirical contribution in this dissertation: I show that the phenomenon of parasitic ellipsis is neither confined to Russian and can be found in other languages as well; nor that it is confined to complements of prepositions but can instead be found in the complements of (at least) complementizers, predicative adjectives and nouns.

3.3 Widespread parasitism

To demonstrate that a language allows for parasitic ellipsis in a specific environment one needs to show at least two things:

- 1) the non-pronunciation of an XP in a sentence leads to ungrammaticality (or less/different meanings) in the absence of an ellipsis that contains the base position of XP
- 2) the presence of such an ellipsis makes the sentence grammatical (or leads to more/different meanings)

The purpose of this section is to demonstrate that this occurs in quite a number of contexts. Before I start, a word on the results presented below. In her seminal paper on the better known parasitic phenomenon, parasitic gaps, Engdahl (1983) remarks that there is a lot of variation in speaker judgements. She provides an accessibility hierarchy of constructions in which speakers accept parasitic gaps. Parasitic gaps in constructions at the top of the hierarchy are accepted even by very restrictive speakers while the lowest constructions are only accepted by permissive speakers. Judgements regarding parasitic ellipsis are similar in a couple of respects. Judgements are sometimes subtle and parasitic ellipsis is often judged as slightly marked even though there are examples where licensed parasitic ellipsis strongly increases acceptability compared to its unlicensed counterpart.¹³ There is also inter-speaker variation in parasitic ellipsis; most variation is due to different preferences regarding the specific heads whose complements are elided. Not every speaker likes every parasitic ellipsis after every head of a certain category equally. One speaker might prefer parasitic ellipsis after *for* over parasitic ellipsis after

¹³I presuppose that the reader is familiar with the usual caveats about grammaticality and acceptability.

against, for another speaker it is the other way around. I have not been able to discern which general properties make a head of some category a more suitable head for parasitic ellipsis than others; factors like frequency, semantic plausibility in their contrastive usage and phonological shape might be at play here. While there is no exact hierarchy, there are definitely lexical items after which most speakers allow parasitic ellipsis. In what follows I focus mostly on those elements for which there was the most agreement between speaker judgements.

The most important finding, however, is that most speakers confirm a difference between an unlicensed and licensed parasitic ellipsis. This persists even for some speakers for who the unlicensed baseline is already somewhat acceptable or for who a licensed parasitic ellipsis is only marginally acceptable.¹⁴ The overall generalization is this: there is an acceptability difference between unlicensed and licensed parasitic ellipsis, and the difference is always in the same direction. I take this to mean that this is a real effect and that a containing ellipsis can license an otherwise unavailable parasitic ellipsis. The subtleties of speaker judgements is a property that parasitic ellipsis shares with the parasitic gap construction.

3.3.1 Prepositions

The purpose of these sections is to demonstrate that parasitic ellipsis of the complements of prepositions is not restricted to Russian and occurs in German and English (Indo-European) as well. Though the focus here is on English and German, I have also been able to gather basic data on parasitic ellipsis in Estonian (Finno-Ugric).¹⁵

Most English and German speakers report a difference in acceptability in the translated equivalents of Fitzgibbons' original parasitic gap example:¹⁶

- (91) a. ??The dog stole the sausage WITH Masha's help, but the cat stole
the sausage WITHOUT.

¹⁴It is interesting to note here that some people are relatively permissive in the kinds of *independent* ellipses that they allow, sometimes counter to generalizations put forward in the literature (e.g. in Lobeck 1995, see discussion below). As such, those speakers are irrelevant for discussions of *parasitic* ellipsis since they can independently elide an XP without fulfilling any preconditions. Relevant for the current purposes are speakers who judge an ellipsis ungrammatical on its own but grammatical under a containing ellipsis.

¹⁵Data are from 12 German native speakers (2 linguists) plus my own judgements, 4 English native speakers (all linguists) and one Estonian native speaker (not a linguist).

¹⁶I only indicate contrastive focus on the relevant second remnant and its antecedent, not the first one, simply to reduce clutter.

- b. ?The dog stole the sausage WITH Masha’s help, but the cat WITHOUT.
- (92) a. *Der Hund hat die Wurst MIT Mashas Hilfe gestohlen, aber
the dog has the sausage with Masha’s help stolen, but
die Katze hat die Wurst OHNE gestohlen.
the cat has the sausage without stolen
‘The dog stole the sausage with Masha’s help, but the cat stole
the sausage without.’
- b. ?Der Hund hat die Wurst MIT Mashas Hilfe gestohlen, aber die
Katze OHNE.
‘The dog stole the sausage with Masha’s help, but the cat with-
out.’

Note also that the different case requirements of *mit* (dative) and *ohne* (accusative) do not affect the grammaticality of the ellipsis, as expected. Whether an unlicensed parasitic ellipsis¹⁷ occurs sentence-medially or sentence-finally makes a difference for some speakers. Unlicensed sentence-medial ellipsis is generally judged to be bad whereas this improves for some speakers in sentence-final position.

- (93) ?*Der Hund stahl die Wurst MIT Mashas Hilfe, aber die Katze
the dog stole the sausage with Masha’s help but the cat
stahl die Wurst OHNE.
stole the sausage without.

To exclude this additional factor I restrict myself to sentence-final parasitic ellipsis where possible.

Not all English and German speakers completely reject ellipsis after *with* or *without*. German e.g. does not generally forbid ellipsis after *mit* ‘with’ or *ohne* ‘without’ in all cases; the slogan of a well-known condom advertisement is *Mach’s mit!* ‘Do it with!’ and there is an idiomatic phrase *oben ohne* (lit. ‘above without’) meaning ‘topless’. This does not mean, however, that this option is completely productive in all cases; most speakers do report a difference in acceptability for less idiomatic expressions, as indicated above.

A similar gradient can be found in other pairs of prepositions, such as *for* and *against*:

¹⁷For speakers who do allow the ellipsis in question independently this ellipsis is of course not parasitic any more. For ease of exposition, however, I continue to refer to the ellipsis whose parasitic status is tested as *parasitic ellipsis*, independent of its actual status, to distinguish it from the containing *licensing ellipsis*. Parasitic ellipsis where the latter is missing I refer to as *unlicensed (parasitic) ellipsis*.

- (94) a. ??Ørjan ist gegen den Vorschlag und Darko ist für.
 Ørjan is against the proposal and Darko is for.
 ‘Ørjan is against the proposal and Darko is for.’
 b. ?Ørjan ist gegen den Vorschlag und Darko für.
 ‘Ørjan is against the proposal and Darko for.’

- (95) a. ?Peter is against the proposal but Mary is for.
 b. Peter is against the proposal but Mary for.

While (94a) is not completely out for most German speakers, the sentence improves in the licensed condition. The difference is less pronounced for the English version of that sentence.

Data similar to those above have already been observed in Yoshida, Nakao & Ortega-Santos (2014) in the context of *why*-stripping or stripping in general. They observe that e.g. bare verbs or prepositions can be remnants of ellipsis in these constructions (their exx. 102 & 125):

- (96) a. A: John should sell his banana boat.
 B: Why SELL?
 b. A: Veterans are honored after death, but not before death.
 B: Why AFTER?
 (97) a. John should LEND his car, not SELL (his car).
 b. Veterans are honoured AFTER death, not BEFORE (death).

They argue that in this case there are actual bare P-remnants (and therefore do not consider an approach in terms of parasitism), an approach that I argue against below in 4.1. For now I want to point out that the apparently single head remnants are not restricted to (*why*-)stripping and can occur in gapping as well:

- (98) Pop stars are honoured BEFORE death, politicians AFTER.

In the above examples, there is a difference in acceptability without, however, the unlicensed condition necessarily being fully unacceptable. It is possible to produce examples where the contrast is much more pronounced:

- (99) a. *Wine is after the lecture, but coffee is during.
 b. ?Wine is after the lecture, but coffee during.
 (100) a. *John drank wine after the lecture, but Mary drank wine during.
 b. ?John drank wine after the lecture, but Mary during.

Unlicensed ellipsis after *during* is strongly unacceptable, but improves sig-

nificantly in the presence of a containing licensing ellipsis.

- (101) a. *Die meisten Studenten schlafen NACH der Vorlesung, aber
the most students sleep after the lecture but
Peter schläft schon WÄHREND.
Peter sleeps already during
'Most students sleep after the lecture, but Peter already sleeps
during.'
b. ?Die meisten Studenten schlafen NACH der Vorlesung, aber Pe-
ter schon WÄHREND.
'Most students sleep after the lecture but Peter already during.'

Consider Maria, a contrarian child that usually does the exact opposite of what she is told:

- (102) a. *Maria erledigte die Aufgabe TROTZ der versprochenen
Maria completed the task despite the promised
Belohnung, aber Peter erledigte die Aufgabe WEGEN.
reward but Peter completed the task because.of
'Maria did her chores despite the promised reward but Peter
completed the task because of.'
b. ?Maria erledigte die Aufgabe TROTZ der versprochenen Beloh-
nung, aber Peter WEGEN.
'Maria did her chores despite the promised reward but Peter
because of (it).'

There are a couple of preposition pairs in German where the presence of a licensing ellipsis improves acceptability only for some speakers while most still reject it:

- (103) a. *Marie steht VOR dem Baum und Peter steht
Marie stands in.front.of the tree and Peter stands
HINTER.
behind.
'Marie is standing in front of the tree and Peter is standing
behind it.'
b. ?Marie steht VOR dem Baum und Peter HINTER.
'Marie is standing in front of the tree and Peter behind.'
- (104) a. *Marie steht VOR den Stühlen, aber Peter steht
Marie stands in.front.of the chairs but Peter stands
ZWISCHEN.
between

- ‘Marie is standing in front of the chairs but Peter is standing between.’
- b. ?Marie steht VOR den Stühlen, aber Peter ZWISCHEN.
 ‘Marie is standing in front of the chairs but Peter between.’
- (105) a. *Maria ist UNTER dem Durchschnitt, und Sophie ist ÜBER.
 Maria is under the average and Sophie is above
 ‘Maria is below average and Sophie is above.’
- b. Maria ist UNTER dem Durchschnitt, und Sophie ÜBER.
 ‘Maria is below average and Sophie above.’
- (106) a. *Peter steht AN der Wanne, und Marie steht IN.
 Peter stands at the bathtub and Marie stands in
 ‘Peter is standing at the bathtub and Marie is standing in.’
- b. ?Peter steht AN der Wanne und Marie IN.
 ‘Peter is standing at the bathtub and Marie in.’

The reason why many speakers reject these examples might be due to competition with lexical items that are combinations of the preposition plus *da* ‘there, that’, e.g. *dazwischen*, *dahinter*, *darin* (‘between that’, ‘behind that’, ‘in that’). The existence of a very similar and completely acceptable alternative might lead to a stronger degradation than in examples where such a minimally different alternative is not at hand.¹⁸

In general, however, all these examples become markedly more acceptable as single stripping remnants even for restrictive speakers:

- (107) Maria steht VOR den Stühlen, nicht ZWISCHEN.
 Maria stands in.front.of the chairs not between.
 ‘Maria is standing in front of the chairs, not between.’
- (108) Maria steht VOR dem Baum, nicht HINTER.
 Maria stands in.front.of the tree not behind.
 ‘Maria is standing in front of the tree, not behind.’

English differs slightly from German in this respect since similar examples do not necessarily lead to unacceptability but rather to different readings:

- (109) a. #Mary is standing in front of the tree, and Peter is standing behind.
 b. Mary is standing in front of the tree, and Peter behind.

If the sentence without gapping means anything, it talks about a generic

¹⁸Some speakers specifically state that they would want items like *dahinter* to occur and therefore reject the example.

‘standing behind’ activity by Peter, he could e.g. be standing behind Mary. In the version with gapping, however, Peter is standing behind a tree as well.

- (110) a. #Peter ran up the hill, and Mary ran down.
b. Peter ran up the hill, and Mary down.

A similar situation can sometimes be found in German as well:

- (111) a. #Peter ist links vom Stuhl und Maria ist rechts.
Peter is left of.the chair and Mary is right
‘Peter is to the right of the chair and Mary.’
b. Peter ist links vom Stuhl und Maria rechts.
‘Peter is to the left of the chair and Maria to the right.’

The unelided version only means either that Maria is to the right of some unspecified reference frame or that she is politically right-wing. I take this to mean that the presence of a licensing ellipsis opens up the possibility of parasitic ellipsis that is otherwise unavailable in the a)-versions of the above sentences - hence the absence of certain readings.¹⁹

Evidence for parasitic ellipsis after prepositions can also be found in Estonian.²⁰

- (112) a. *Jaan värvis akna kohal, ja Marie värvis all.
Jaan painted window above and Marie painted under
‘Jaan painted above the window and Marie painted below.’
b. ?Jaan värvis akna kohal, ja Marie all.
‘Jaan painted above the window and Marie below.’

To conclude, English, German and Estonian also exhibit the phenomenon that an otherwise unacceptable ellipsis noticeably improves when it is contained in another ellipsis. Several factors interact with its acceptability such as whether parasitic ellipsis is sentence-internal or final, and whether there are idiomatic uses of prepositions without complement. The presence of a competing ‘rescuing’ alternative like the existence of combinations with *da* ‘there, that’ for German *hinter*, *in*, *unter* etc. can also interfere. While there is some variation, the overall trend is clear and it is possible to construct examples where the unlicensed version is strongly unacceptable and the licensed version significantly improves. I take that to mean that the phenomenon of parasitic ellipsis also exists in these languages.

¹⁹I discuss this property in more detail in the section on parasitic ellipsis after adjectives.

²⁰All Estonian data in this thesis are due to Indrek K., p.c.

3.3.2 Complementizers

The previous section demonstrated that parasitic ellipsis in PPs is not restricted to Russian. I now want to demonstrate that this operation is not only not language-specific, but also not restricted to any specific category. For this we turn to complementizers.

Lobeck (1995:46) claims that a sluiced phrase cannot be the complement of *for*, *that*, *whether* and *if* (her ex. 26, slightly adapted).

- (113) *Although *whether/if* is unclear, Sue thinks John made it to work on time.

Most English speakers I consulted strongly reject ellipsis after *whether*, even if it is not cataphoric as in the example above:²¹

- (114) *Mary doesn't know WHEN she will come, but Peter doesn't even know WHETHER.

Those speakers, however, report a strong increase in acceptability for the following:

- (115) Mary doesn't know WHEN she will come, but Peter not even WHETHER.

The same goes for ellipsis after *if* or *because*:

- (116) a. *Mary doesn't know WHEN her salary arrives, Susan doesn't even know IF.
b. ?Mary doesn't know WHEN her salary arrives, Susan not even IF.
- (117) *Peter loved his job but quickly found a new one; Mary hated her job and is glad she doesn't have to work there any more.*
a. *Peter is happy ALTHOUGH he was fired, but Mary is happy BECAUSE.
b. ?Peter is happy ALTHOUGH he was fired, but Mary BECAUSE.

The situation is similar for German:

- (118) a. *Marco schläft schlecht, OBWOHL er jemanden beleidigt hat,
Marco sleeps badly although he someone insulted has
aber Morten schläft schlecht, WEIL.
but Morten sleeps badly because

²¹Though this is by no means universal, there seem to be some English speakers who find this sentence acceptable.

- ‘Marco sleeps badly although he insulted someone, but Morten sleeps badly because.’
- b. ?Marco schläft schlecht, OBWOHL er jemanden beleidigt hat, aber Morten WEIL.
Marco sleeps badly although he insulted someone, but Morten because.
- (119) a. *Peter kriegt seine Belohnung BEVOR er seine Aufgaben
Peter gets his reward before he his tasks
erledigt hat, aber Paul kriegt seine Belohnung NACHDEM
completed has but Paul gets his reward after
‘Peter gets his reward before he completed his task but Paul after.’
- b. ?Peter kriegt seine Belohnung BEVOR er seine Aufgaben erledigt
hat, aber Paul NACHDEM.
‘Peter gets his reward before he completed his task but Paul after.’

A similar example can be found in Estonian:

- (120) a. *Jaan oli õnnelik kuigi ta vallandati, ja Artur oli
Jaan was happy although he fired.was and Artur was
õnnelik sest.
happy because
‘Jaan was happy although he was fired, and Artur was happy because.’
- b. ?Jaan oli õnnelik kuigi ta vallandati, ja Artur sest.
‘Jaan was happy although he was fired, and Artur because.’

To conclude, parasitic ellipsis exists not only after prepositions, but after complementizers as well.

3.3.3 Adjectives

Predicative adjectives also exhibit a form of parasitic ellipsis, though here there is an additional complication that makes the judgements slightly more subtle. Lobeck (1995:48) claims that “ellipted complements of the lexical heads V, P and A are ungrammatical in English”, citing the following example (her ex. 37a):

- (121) *Mary is crazy about the idea, and Bill is also [happy [_{PP} e]].

The sentence is ungrammatical in the intended meaning. However, since

the complements of predicative adjectives tend to be optional, the sentence without the elided verb is not necessarily ungrammatical in general but rather lacks certain readings or is pragmatically marked.

I claim that the presence of parasitic ellipsis can nonetheless be detected due to the presence or absence of these readings. Licensed parasitic ellipses allow for the recovery of the complement, both in the strict and sloppy reading when said complement contains a pronominal, in contrast to ungapped sentences:

- (122) Marie ist BEGEISTERT von ihrem Idol, aber Peter ist
 Mary is excited by her idol but Peter is
 ENTTÄUSCHT.
 disappointed.
 ‘Mary is amazed by her idol but Peter is disappointed.’

The above sentence means that Marie is excited about a specific idol of hers, but that Peter is disappointed in general. Via pragmatic reasoning one could infer that - since they are both mentioned in the same sentence - what Peter is disappointed about is the same issue, e.g. the performance of the already mentioned pop star, namely Marie’s idol.

The same sentence with the verb elided, however, has additional meanings:

- (123) Marie ist BEGEISTERT von ihrem Idol, aber Peter ENTTÄUSCHT.
 Marie is excited by her idol but Peter disappointed.
 ‘Marie is amazed by her idol, but Peter disappointed.’

This sentence means that Peter is disappointed about an idol, in either the strict or the sloppy reading, i.e. it can be both Marie’s or his own idol. The existence of strict and sloppy readings is completely expected under an analysis where the complement of the adjective has been subject to ellipsis. The same goes for the fact that the gender features of the pronominal are irrelevant for the antecedent recovery. These two properties are hallmarks of ellipsis.

My own and my informants’ intuition is that the recovery of the adjectival complement (and therefore the parasitic ellipsis reading) is immediate in the gapped sentence and is often even preferred over the simple, non-parasitic reading without additional ellipsis after the adjective. (122) lacks these readings since in the absence of gapping parasitic ellipsis is not licensed.²² In what

²²To be more precise, this sentence can, as mentioned, have a pragmatic quasi-strict meaning since this is often the only way to make sense of the conjunction in a discourse. But Peter might as well be disappointed about Mary liking this specific pop star. In addi-

follows I mark such sentences with # to indicate it lacks the intended reading.
 English exhibits the same pattern as German:

- (124) a. #Peter is CONTENT with his idol, and Paul is DISAPPOINTED.
 b. Peter is CONTENT with his idol, and Paul DISAPPOINTED.
- (125) a. #Peter ist GLÜCKLICH über das Bild von seinen Eltern,
 Peter is happy about the picture of his parents,
 und Paul ist traurig.
 and Paul is sad
 ‘Peter is happy about the picture of his parents, and Paul is sad.’
 b. Peter ist glücklich über das Bild von seinen Eltern, und Paul traurig.
 ‘Peter is happy about the picture of his parents, and Paul sad.’
- (126) a. #Peter is happy about the picture of his parents, and Paul is sad.
 b. Peter is happy about the picture of his parents, and Paul sad.
- (127) a. #Peter is AFRAID of his students and Mary is proud.
 b. Peter is afraid of his students and Mary proud.

Estonian shows the same phenomenon:

- (128) a. #Marie oli oma poja üle uhke, ja Susan oli pettunud.
 Marie was own son over proud and Susan was disappointed
 ‘Marie was proud of her son and Susan was disappointed.’
 b. Marie oli oma poja üle uhke, ja Susan pettunud.
 ‘Marie was proud of her son and Susan disappointed.’

In conclusion, although the diagnostics are slightly different, the phenomenon is arguably the same. Here, the diagnostic is not pure syntactic unacceptability but the presence or absence of certain readings. This diagnostic has also been used in prepositions as well, as shown above in (111), to the same effect. I view the examples presented in this section as good evidence that parasitic ellipsis exists in predicative adjectives as well.

tion, if Peter is generally disappointed, nothing in principle precludes the logical possibility that he so happens to be disappointed about someone that is his idol. That, however, is not its *semantic* meaning but a (quite remote) inference from the utterance. My claim is that the additional readings in the parasitic case are part of the underlying semantics of the sentence. So when I claim that certain readings are absent, this is a shorthand for the above more precise claim.

3.3.4 Nouns

The diagnostic from the previous section can also be used to detect parasitic ellipsis in the nominal domain:

- (129) a. #Peter besitzt ein GEMÄLDE von seinen Großeltern, aber
 Peter possesses a painting of his grandparents but
 Maria besitzt ein PHOTO.
 Maria possesses a picture.
 ‘Peter has a painting of his grandparents, but Maria has a picture.’
 b. Peter besitzt ein GEMÄLDE von seinen Großeltern, aber Maria
 ein PHOTO.
 ‘Peter has a painting of his grandparents, but Maria a picture.’

(129a) means that Maria possesses some generic picture; a common inference is that it has to be a picture of Peter’s grandparents. (129b), on the other hand, means that Mary has a picture of either Peter’s grandparents (strict) or her own (sloppy). As is the case with adjectives, these readings are immediate and strongly preferred over a simple gapping analysis without additional ellipsis. The same holds for English:

- (130) a. #Peter has a PAINTING of his grandparents, but Mary has a
 PICTURE.
 b. Peter has a PAINTING of his grandparents, but Mary a PIC-
 TURE.

The German examples also work with genitive complements (anticipating a later discussion of a potential evacuation analysis):

- (131) a. #Peter besitzt ein GEMÄLDE seiner Großeltern,
 Peter possesses a painting his.GEN grandparents.GEN
 aber Maria besitzt ein PHOTO.
 but Maria possesses a picture.
 ‘Peter possesses a painting of his grandparents but Maria pos-
 sesses a picture.’
 b. Peter besitzt ein GEMÄLDE seiner Großeltern, aber Maria ein
 Bild.
 ‘Peter possesses a painting of his grandparents, but Maria a
 picture.’

It works with other nouns as well:

- (132) a. #Maria las einen ARTIKEL über ihr Forschungsthema, und
 Maria read an article about her research.topic and
 Peter las ein BUCH.
 Peter read a book.
 ‘Maria read an article about her research topic and Peter read
 a book.’
- b. Maria las einen ARTIKEL über ihr Forschungsthema, und Peter
 ein BUCH.
 ‘Maria read an article about her research topic and Peter a
 book.’
- (133) a. #Mary read an ARTICLE about her research topic and Peter read
 a BOOK.
- b. Mary read an ARTICLE about her research topic and Peter a
 BOOK.
- (134) a. #Mary met the CEO of her favourite company, and Peter met
 the president.
- b. Mary met the CEO of her favourite company, and Peter the
 president.

In Estonian, the sentence that still has an overt verb can receive a strict interpretation, but the gapped sentence marginally allows for a sloppy interpretation which is not possible in the ungapped version:

- (135) a. #Artur luges ARTIKLI oma kangelase kohta, ja Jaan luges
 Artur read article own hero about and Jaan read
 RAAMATU.
 book
 ‘Artur read an article about his hero and Jaan read a book.’
- b. Artur luges ARTIKLI oma kangelase kohta, ja Jaan
 Artur read article own hero about and Jaan
 RAAMATU.
 book
 ‘Artur read an article about his hero and Jaan a book.’

To conclude, gapping creates the possibility of recovering a nominal complement in the strict and sloppy readings if there is an appropriate antecedent. Nouns pattern with adjectives in that the unlicensed sentences are not ungrammatical per se, but lack at least the sloppy readings. In the licensed versions, however, the parasitic ellipsis reading, with its strict and sloppy version, becomes available.

3.3.5 Verbs

The last category I want to investigate are verbs. The status of parasitic ellipsis after verbal elements is not as straightforward as with other categories. As mentioned above, Yoshida, Nakao & Ortega-Santos (2014) already discuss data like the following:

- (136) Mary should LEND her car, not SELL.

In English, parasitic ellipsis is generally judged to be quite degraded in the context of gapping:

- (137) ??Peter should LEND his car, but Mary SELL.

German allows for the following:

- (138) a. *Peter wird einen Kredit aufnehmen KÖNNEN, aber Marie
Peter will a loan take.out can but Marie
wird MÜSSEN.
will must.
'Peter will be able to take out a loan, but Marie will have to.'
b. ?Peter wird einen Kredit aufnehmen KÖNNEN, aber Marie MÜSSEN.
'Peter will be able to take out a loan, but Marie will have to.'
- (139) a. *Peter hat sein Segelboot VERKAUFT, aber Paul hat
Peter has his sailing.boat sold but Paul has
VERSCHENKT.
given.away
'Peter has sold his sailing boat but Paul has given it away.'
b. ?Peter hat sein Segelboot VERKAUFT, aber Paul VERSCHENKT.
'Peter has sold his sailing boat but Paul has given it away.'²³

German does not generally allow VP-ellipsis as can be seen in the *a*)-versions of these examples. Yet in the presence of gapping these examples improve. Modal complement ellipsis (as proposed for Dutch by Aelbrecht 2012), i.e. ellipsis after an inflected modal is marginally possible:

- (140) ?Peter MUSS in die Schule gehen, aber Maria KANN.
Peter must in the school go but Maria can.
'Peter has to go to school but Maria can.'

²³It has come to my attention late in the writing process that data such as these have been discussed in a term paper by Schmidt (2014). He proposes that the infinite verb is a proper subconstituent of the VP in German and can be fronted, even without prior scrambling. I argue against both this and the scrambling approach below.

But the modal in (138) is not finite since gapping deletes the finite verb, and ellipsis after a non-finite modal is not possible. On top of that, this phenomenon is not restricted to modals:

- (141) ?Peter wird das Buch LESEN, Marie REZENSIEREN.
 Peter will the book read Maria review.
 ‘Peter will read the book, and Maria review it.’

I conclude therefore that this phenomenon, however analysed, is related to gapping and not another ellipsis process. The question then is how sentences as (141) are best analysed.

One might propose that in the elliptical sentence, the object of *rezensieren* scrambled out of the VP, in other words that *rezensieren* is a remnant movement remnant, as seen in the last section. This is the analysis given to equivalent examples in Dutch by Boone (2014). In contrast to the sentences discussed in 3.1, however, that scrambling operation does not have a parallel movement step in the antecedent, as it has in the example below:

- (142) Lesen wird das Buch jeder, rezensieren keiner.
 read will the book everyone review noone
 ‘Everyone will read the book, but noone will review it.’

Parallelism between both sentences makes a scrambling analysis in (142) the default option but it would have to be stipulated for (141). Additionally, parasitic ellipsis-like sentences also work in environments where scrambling of the verbal complement is quite degraded:

- (143) a. #Maria wird ANFANGEN ihren Brief ans Ministerium zu
 Maria will start her letter to.the ministry to
 schreiben, aber Sophie wird AUFHÖREN.
 write but Sophie will stop.
 ‘Maria will start writing her letter to the ministry but Sophie
 will stop.’
 b. ?Maria wird ANFANGEN ihren Brief ans Ministerium zu schreiben,
 aber Sophie AUFHÖREN.
 ‘Maria will start writing her letter to the ministry but Sophie
 will stop doing so.’
- (144) ?*Sophie wird ihren Brief ans Ministerium zu schreiben wohl
 Sophie will her letter to.the ministry to write probably
 aufhören.
 stop.

≈ ‘Writing her letter to the ministry, Sophie will stop.’²⁴

(143a) is semantically odd in that the second conjunct describes a situation where Sophie stops an unspecified activity; the recovery of Sophie’s activity is clearer in (143b). Another example is the following:

- (145) a. #Marie wird verneinen, ihre Mutter angerufen zu haben, aber
 Marie will deny her mother called to have but
 Sophie wird bejahen.
 Sophie will confirm
 ‘Marie will deny having called her mother but Sophie will confirm.’
 b. ?Marie wird verneinen, ihre Mutter angerufen zu haben, aber
 Sophie bejahen.
 ‘Marie will deny having called her mother but Sophie will confirm.’

Another diagnostic whether the phenomenon at hand is parasitic ellipsis is the size of the remnant. In the next section I discuss that the locality condition on parasitic ellipsis appears to be that only a single head can survive whose immediate complement needs to be elided; a larger remnant destroys the possibility of parasitic ellipsis. The following sentence lacks the relevant strict and sloppy readings even though parasitic ellipsis is in principle possible after N:

- (146) #Maria ist BEEINDRUCKT von einem Bild von ihren
 Maria is impressed by a picture of her
 Großeltern, aber Sophie ENTTÄUSCHT von einem Bild.
 grandparents but Sophie disappointed by a picture.
 ‘Maria is impressed by a picture of her grandparents but Sophie disappointed by a picture.’

If incomplete predicate fronting or fronting of the predicate (complex) is at work in the above examples then the following example should be acceptable, contrary to fact:

- (147) #Marie wird ANFANGEN ihren Brief ans Ministerium zu
 Marie will start her letter to.the ministry to

²⁴A reviewer (Fabian Heck) notes that this example independently appears to be bad if one drops the *vP*-boundary particle *wohl*, weakening the argument against scrambling. If this is the case, and no more clear-cut examples can be found, this would void this specific argument against a non-parasitic analysis. Establishing the existence of VP parasitic ellipsis for English might then be more clear-cut due to the absence of scrambling.

schreiben, aber Sophie AUFHÖREN zu schreiben.
 write but Sophie stop to write
 ‘Marie will start writing her letter to the ministry but Sophie stop
 writing.’

This sentence lacks the same readings that (143a) lacks. Yet fronting such a predicate seems somewhat acceptable:

- (148) ?Aufhören zu schreiben wird Peter seine Lieder auch unter
 stop to write will Peter his songs even under
 Gewaltandrohung nicht.
 force.threat not
 ‘Peter will not stop writing his songs, even under threat of force.’

However, a more in-depth investigation of these constructions is necessary to conclusively rule out alternative analyses for German. Furthermore, judgments regarding parasitic ellipsis in the verbal domain are subtle and it is unclear why English appears more restrictive with respect to parasitic ellipsis after verbs. The situation for Estonian is also unclear:

- (149) a. Artur peaks oma auto välja laenama, aga Jaan peaks
 Artur should own car out lend but Jaan should
 müüma.
 sell.
 ‘Artur should lend his car but Jaan should sell.’
 b. Artur peaks oma auto välja laenama, aga Jaan müüma.
 Artur should own car out lend but Jaan sell
 ‘Artur should lend his car but Jaan sell.’

(149a) appears to somewhat allow already for the recovery of *oma auto* ‘his car’ and as such disqualifies it as testing ground for further parasitic ellipsis, even though the preferred reading for (149b) is the sloppy one.

For now, I tentatively conclude that parasitic ellipsis after verbs is a possibility but further research is necessary to demonstrate this conclusively.

The main conclusion of the empirical part of this half of the thesis is the following: the phenomenon of parasitic ellipsis exists not only in Russian for complements of certain prepositions. It occurs at least in English, German and Estonian, as well, and, most importantly, occurs in the complement of a variety of categories, namely at least P, C, A and N.

3.4 The parasitic ellipsis generalization

The following appear to be common properties of parasitic ellipsis across different categories: when a phrase is a remnant of gapping and its head is focussed, one can optionally elide its complement.

- (150) Peter bestieg den Berg MIT seinem besten Freund, aber
Peter climbed the mountain with his best friend but
Marie OHNE.
Marie without.
'Peter climbed the mountain with his best friend but Marie with-
out.'
- (151) Jordan knows WHEN he gets his salary, but Sven not even WHETHER.
- (152) Jaan oli õnnelik kuigi ta vallandati, ja Artur sest.
Jaan was happy although he fired.was and Artur because
'Jaan was happy although he was fired, and Artur because.'
- (153) Susan is AFRAID of her students, but Peter PROUD.
- (154) Paul has a DRAWING of his grandparents, and Peter a PICTURE.

As expected under an ellipsis analysis, even phrases that do not match in features such as case (150) and gender (153) can serve as antecedents. Another expected property of ellipsis is the availability of strict and sloppy readings. I propose the following generalization:

Parasitic ellipsis generalization (PEG)

For every possible remnant of gapping, the complement of its head can optionally be deleted if the head is contrastively focussed and its complement given.

Let us unpack the claims made in this statement. The first thing to point out is that this generalization undoubtedly overgenerates. Not every preposition, complementizer, adjective or noun is equally suited for every speaker to allow subsequent parasitic ellipsis. There are of course trivial reasons like the fact that e.g. not every noun selects a complement. Since the head preceding parasitic ellipsis must be contrastively focussed, it needs a non-empty set of semantically reasonable alternatives to produce a felicitous sentence. There might be syntactic or lexical reasons that severely restrict the available contrastive alternatives. PP-complements of adjectives are headed by different prepositions and mismatched prepositions decrease acceptability:

- (155) ??Peter is PROUD of his son, and Paul CONTENT.

That is, not every head might have other heads of the same category that is sufficiently parallel. A head also needs reasonable semantic alternatives for the contrast to make sense. Not every preposition has strong lexical content, and contrastively focussing prepositions like *of* might sound artificial. The contexts in which such a contrast is possible might be contrived, further reducing acceptability. Phonological factors also play a role. As pointed out by Fitzgibbons (citing Gribanova 2009), parasitic ellipsis does not work with prepositions consisting only of a consonant. To reformulate, the relevant head needs to be able to overtly express contrastive focus. All these factors conspire to often significantly reduce the size of the set of *completely parallel* alternatives even in the absence of parasitic ellipsis. Even taking all these lexical, syntactic, semantic and phonological factors into account, there are further factors interfering with the acceptability of parasitic ellipsis. One such factor is the existence of existing minimally different constructions that compete with a parasitic ellipsis construction. This is the case for some German prepositions that have an anaphoric counterpart, like *hinter* or *zwischen* ('behind', 'between') and *dahinter* or *dazwischen* ('behind it', 'between it'). Parasitic ellipsis is more marginal after the former due to the readily available latter option. Something similar might be the reason why verbal parasitic ellipsis is often marginal; a minimally different *sell it* instead of *sell* as sole survivor increases acceptability. Speakers also show individual preferences for certain items over others; sometimes switching the contrasted heads in the conjunction leads to an increase or decrease in acceptability and I have not been able to discover general trends that hold for all speakers. In short, there are intervening factors that have nothing to do with the core syntactic licensing condition of parasitic ellipsis. For this reason I ignore this variation in the generalization and attribute it to the factors discussed above. There are enough items of each category where parasitic ellipsis *is* possible and it seems unreasonable to me that a theory of syntax should make reference to specific lexical entries (especially to ones belonging to an open class) to which an operation can apply. The generalization above simply states the structural conditions that have to be met for parasitic ellipsis to apply.

One important claim in the parasitic ellipsis generalization is about the structural **locality conditions** that have to be met. This concerns how much material must or must not survive in the ellipsis remnant. The generalization states that the **head** of the remnant must be present and additionally be contrastively focussed. Consider the cases where material survives that the head of the ellipsis remnant selects for:

- (156) *Peter is CONTENT with his son, but Paul DISAPPOINTED with.

- (157) *Peter ist TRAURIG über das Ergebnis, aber Maria FROH über.
 Peter is sad about the result but Maria happy about.
 ‘Peter is sad about the result but Maria happy about.’

Consider CP-remnants where parasitic ellipsis only applies after T:

- (158) *Peter ist glücklich, OBWOHL sein Boss ihn gefeuert hat, und
 Peter is happy although his boss him fired has and
 Maria WEIL hat.
 Maria because has.
 ≈‘Peter is happy although his boss fired him and Maria because
 did.’

Non-complements such as specifiers/adjuncts, however, do not cause ungrammaticality:

- (159) Peter ist ziemlich TRAURIG über das Ergebnis, aber Maria
 Peter is quite sad about the result but Maria
 ziemlich FROH.
 quite happy
 ‘Peter is quite sad about the result but Maria quite happy.’

Parasitic ellipsis can only apply after the head of the remnant but not after a more deeply embedded element, even if that element is a potential head for parasitic ellipsis in other circumstances:

- (160) #Maria ist BEEINDRUCKT von einem Bild von ihren
 Maria is impressed by a picture of her
 Großeltern, aber Sophie ENTTÄUSCHT von einem Bild.
 grandparents but Sophie disappointed by a picture.
 ‘Maria is impressed by a picture of her grandparents but Sophie
 disappointed by a picture.’
- (161) #Mary was HAPPY with the picture of her son, and Susan DISAP-
 POINTED with the picture.

The above sentences lack a sloppy reading where Sophie is disappointed with a picture of her own grandparents. This does not change when focus is switched to the more deeply embedded element:

- (162) #Maria ist beeindruckt von einem GEMÄLDE von ihren
 Maria is impressed by a painting of her
 Großeltern, und Sophie beeindruckt von einem BILD.
 grandparents and Sophie impressed by a picture.

‘Maria is impressed by a painting of her grandparents and Sophie impressed by a picture.’

- (163) #Mary was happy with the PAINTING of her son, and Susan happy with the PICTURE.

This means that parasitic ellipsis, or the ‘dependency’ between parasitic ellipsis and its enabling ellipsis is subject to a very local selection relationship: among the elided elements of gapping there is an elided head who has an argument that survives said ellipsis.²⁵ The *first* argument (i.e. complement) of the head of that phrase can optionally be elided as well.²⁶ Parasitic ellipsis does not work in an environment that is less local than the minimal ‘complement of remnant head’ relationship. More removed relationships like ‘complement of complement of remnant head’ are ruled out.

The attentive reader will have noticed a potential exception to the generalization stated above: nouns behave differently from every other category with respect to this locality condition if one assumes the **DP-hypothesis**. If D is the head of a nominal expression, it is neither focussed nor is its complement elided but rather the complement of its complement. But as we have seen, such constellations prevent licensing of parasitic ellipsis (162). This means that the domain where there is arguably the most debate about the status of its head is the one that is the only potential exception to the above generalization. However, the assumption that D is the head of the nominal domain has recently been criticized (e.g. Bruening, Dinh & Kim 2018, Bruening 2020) and is by no means universal. If one assumes that D

²⁵A reviewer (Fabian Heck) notes (162) allows for a parasitic reading for him. My own judgement and that of my informants diverge from this. He also notes a version of (163) allows a parasitic reading for him:

- (i) ?Maria ist beeindruckt von einem GEMÄLDE von ihren Großeltern, und Sophie
 Maria is impressed by a painting of her grandparents and Sophie
 von einem BILD.
 by a picture

I am unsure about my judgements in this case, so a more careful empirical investigation is necessary. If true, one might have to change the locality condition so that e.g. what is parasitically elided is the complement of the most deeply embedded focussed head. Another option would be that PPs work similarly to DPs as I touch on in the next paragraphs. To anticipate results from the second half of the thesis, not only D but, on top of that, P might also be part of the secondary root, leaving the complement-head relation between verb and noun intact.

²⁶Another possibility would be that of parasitic ellipsis of the *specifiers* of the remnant head. I will not explore this possibility in this dissertation and restrict myself to the more basic *complement-of* relation. However, I briefly mention potentially applications for relaxing that restriction in 5.3.

is the head selecting for N, one might expect that parasitic ellipsis would be possible following it. This is decidedly not the case. In English, NP-ellipsis is not possible after some determiner-like elements like *the*, *a*, and *every* (see e.g. Lobeck 1995:44). Parasitic ellipsis interestingly does not remedy this situation which is unexpected given that it reliably does so in many other categories:

- (164) a. *Mary likes some professors, but Peter likes every.
 b. *Mary likes some professors, but Peter every.

Again, if D were the head of the nominal domain it behaves differently from all the other categories that I have tested so far. It would be the only one that does not allow parasitic ellipsis of its complement but the only one that allows non-local application of that process. Under the assumption that N is the head of the nominal domain, the above generalization holds fully.

This shows that the DP/NP-issue is relevant for the validity of the above generalization; vice versa, the data in parasitic ellipsis might also serve as circumstantial evidence in that debate. Since I do not want to stipulate an exception from both the locality condition and the ability to host parasitic ellipsis solely for D, I assume the NP-hypothesis for now like the authors above. In the second half of this dissertation, I develop an analysis that makes it possible to both say that the verb selects its nominal argument and that the determiner is the highest projection of the nominal domain as a secondary root. Under such an account, no exception to the above generalization needs to be stipulated, even under the DP-hypothesis.

Finally, there is the claim about the type of ellipsis that phrases that host parasitic ellipsis have survived. The examples so far appeared in gapping constructions, i.e. clausal ellipsis with (at least) two remnants. Fitzgibbons (2014) notes that other clausal ellipses but not nominal ellipsis can license parasitic gaps as well.

Stripping appears to be much more permissive with respect to parasitic ellipsis. Otherwise marked gapping examples often become significantly more acceptable, and successful parasitic ellipsis cases work just as well:

- (165) a. ?*Peter steht AN der Wanne, Maria IN.
 Peter stands at the bathtub Maria in
 ‘Peter is standing at the bathtub, Maria in.’
 b. Peter steht AN der Wanne, nicht IN.
 Peter stands at the bathtub not in.
 ‘Peter is standing at the bathtub, not in.’
 (166) a. ?*Mary went out to play BECAUSE it was hot outside, but Susan

ALTHOUGH.

- b. Mary went out to play BECAUSE it was hot outside, not ALTHOUGH.²⁷

- (167) Peter ist ENTTÄUSCHT von seinem Idol, nicht BEEINDRUCKT.
Peter is disappointed by his idol, not impressed.
'Peter is disappointed by his idol, not impressed.'

This also works in Estonian:

- (168) ?Jaan oli õnnelik KUIGI ta vallandati, mitte SEST.
Jaan was happy although he fired.was, not because.
'Jaan was happy although he was fired, not because.'

As mentioned above, this was already discovered for APs and VPs in stripping by Yoshida, Nakao & Ortega-Santos (2014:363):

- (169) a. John should LEND his car, not SELL.
b. Veterans are honoured AFTER death, not BEFORE.

The situation is similar in fragment answers:

- (170) A: Do you want the dish WITH or WITHOUT spices?
B: WITHOUT.
- (171) A: Did she kill her ALTHOUGH or BECAUSE she loved her?
B: BECAUSE.
- (172) a. A: War Johanna BEEINDRUCKT von ihrem Idol?
Was Johanna impressed by her idol
'Was Johanna impressed by her idol?'
b. B: Ne, ENTTÄUSCHT.
Nope, disappointed.
'Nope, disappointed.'

Similar data are presented in Griffiths (2019:26, ex. 78a&b)

- (173) a. A: Should he REVOLVE or TILT the gyroscope?
B: REVOLVE, of course.
b. A: Is IN or UNDER the bed the best hiding place?
B: UNDER, I reckon.

although he claims that in reactions to preceding assertions such constructions are not possible any more because assertions do not invoke the necessary

²⁷While parasitic ellipsis after *because*, with *although* as antecedent, is acceptable the reverse order is less acceptable.

licensing conditions (implicit *wh*-questions) (Griffiths 2019:29, ex. 85b):

- (174) A: UNDER the bed is the best hiding place.
B: *No, IN.

This is not true at least for German:

- (175) a. A: Peter wollte seinen Döner MIT scharf.
Peter wanted his döner with spicy.
'Peter wanted his döner with spices.'
b. B: Nee, OHNE.
nope without.
'Nope, without.'
- (176) a. A: Peter hat mit seinem Freund schlussgemacht,
Peter has with his boyfriend broke.up
OBWOHL er berühmt wurde.
although he famous became
'Peter broke up with his boyfriend although he became famous.'
b. ?B: Ne, WEIL.
Nope because
'Nope, because.'

One English speaker I consulted accepts the following sentence:

- (177) A: (Believe me,) Peter wanted his fries WITH mayonnaise!
B: ?No, WITHOUT.

Three English speakers I consulted find the following sentence acceptable:

- (178) A: Peter broke up with his girlfriend ALTHOUGH she became famous.
B: No, BECAUSE.

To conclude, data that show strong parallels to the construction observed in gapping occur in other clausal ellipses as well; this indicates that parasitic ellipsis is not an idiosyncratic property of gapping alone. However, the focus of this thesis is parasitic ellipsis in gapping and more work is necessary to investigate whether the parallelism between parasitic ellipsis in gapping and in other clausal ellipses is exact. Relevant questions are whether stripping/fragment answers can generate at least all the parasitic ellipsis remnants that occur in gapping, i.e. if they are a full superset of the gapping data, and why at least stripping seems to be more permissive than gapping. One also needs to determine whether they share all properties demonstrated above.

This is not always easy to test: sloppy identity, e.g., is difficult to test for with a single remnant. On the theoretical side, the relevant question is whether the same ellipsis mechanism is behind the three cited ellipsis types. While I leave these issues for future research, the data above warrant the stronger conjecture that parasitic ellipsis is not too specific to gapping.

Things get murkier when considering whether parasitic ellipsis can occur in all clausal and/or contrastive ellipses. Sluicing appears to behave differently with respect to the availability of parasitic ellipsis. Griffiths & Lipták (2014) group sluicing together with other clausal ellipses and argue that it can be contrastive as well (though Merchant 2004 provides a different analysis for sluicing in contrast to fragments).

However, German and English speakers report that they only get generic readings for sluicing remnants, or an inference to a strict reading in sluicing examples but that a sloppy reading is unavailable:

- (179) #Paul wusste, welches BUCH über seinen Helden ausgezeichnet
 Paul knew which book about his hero honoured
 wurde, und Peter wusste, welcher ARTIKEL.
 was and Peter knew which article
 ‘Paul knew which book about his hero was given an award, and
 Peter knew which article.’
- (180) #Paul knew which BOOK about his hero was given an award, and
 Peter knew which ARTICLE.

Similarly, in Estonian it is possible to get a strict interpretation, but a sloppy reading is impossible:

- (181) #Artur teadis, millist ARTIKLIT tema kangelase kohta tunnustati,
 Artur knew which article his hero about awarded.was
 ja Jaan teadis, millist RAAMATUT.
 and Jaan knew which book
 ‘Artur knew which article about his hero was awarded and Jaan
 knew which book.’

Regarding other categories, with sluicing it is difficult to test for anything besides nominal remnants; pied-piped PPs with elided complements are ungrammatical but probably for reasons that have nothing to do with parasitic ellipsis since what is elided here is the *wh*-word responsible for fronting:

- (182) *Peter doesn’t know WITH who he wants to spend his life, he only
 knows WITHOUT.

- (183) *Peter weiß nicht, MIT wem er sein Leben verbringen will, er
 Peter knows not with who he his life spend wants he
 weiß nur, OHNE.
 only knows without who
 ‘Peter doesn’t know with who he wants to spend his life, he just
 knows without who.’

It appears that the generalization that *all* clausal contrastive ellipses allow for parasitic ellipsis is too strong if one includes sluicing. A more in-depth empirical and theoretical investigation into the interaction of parasitic ellipsis and sluicing is necessary which I leave for further research.

The negative generalization that parasitic ellipsis cannot be licensed by any non-clausal ellipsis is still possible. Turning to VP-ellipsis in English the situation appears to confirm to this generalization at first. Speakers definitely reject the sloppy reading in topicalized VP remnants and in A-moved remnants:

- (184) #The BOOK about his idol, Peter could buy, but the ARTICLE, John
 couldn’t.
 (185) #Mary said that the PAINTING of her grandson will be stolen, but
 Susan said that the PICTURE might be.

This also shows that the relevant factor is *not* solely whether an ellipsis remnant is contrastively focussed. Testing for other categories, parasitic ellipsis in CP remnants is also not possible:

- (186) *WHEN he would arrive, Peter knew, but WHETHER, Mary didn’t.

Turning to *wh*-remnants of VP-ellipsis, however, the picture - both strict and sloppy - becomes more complicated. Some English speakers report to get a sloppy reading in *wh*-remnants of VP-ellipsis:

- (187) ?I know which ARTICLE about his hero Bill should read and which
 BOOK John shouldn’t.

I do not have a full account of this puzzling fact. However, there is a difference worth noting between the property of *wh*-movement and topicalization with respect to extraction out of VP-ellipsis which I want to point out before ultimately leaving this issue for further research. Schuyler (2001) notes that the two movement types have different focus requirements when they evacuate out of a VP-ellipsis. Simplifying her proposal a bit, she states the following generalization: “For *wh*-movement out of the site of VPE to be licensed, there must be a contrastively focused expression in the C-command

domain of the VPE-extracted *wh*-phrase.” (2001:14)²⁸. She demonstrates this with the following examples (2001:7, her ex. 27&35, slightly adapted):

- (188) *I think you should adopt ONE of these puppies, but I don’t know WHICH one you should.
- (189) I think YOU should ride the TALLEST camel, but I don’t know which one PHIL should.

She notes that topicalization does not behave like that since it does not need intervening contrastive foci and it suffices for the remnant phrase to be focussed itself, in contrast to (188) (2001:13, ex. 88):

- (190) The BLUE papers I think Pete signed, and the GREEN ones I think he did, too.

Wh-remnants of VP-ellipsis appear to be the only counterexample to the generalization that only some contrastive clausal ellipses can license parasitic ellipsis; but in precisely this counterexample contrastive focus (on an element other than the remnant) plays a crucial role: a domain containing the ellipsis site requires the presence of a contrastive focus (on an element other than the remnant). This is a condition that is otherwise irrelevant for *wh*-movement and is only at play for *wh*-remnants of VP-ellipsis. This suggests that there might be a unifying property that is shared by these examples and contrastive ellipsis and that the same factors are at play that license parasitic ellipsis in both cases. To formulate a unifying account of these phenomena, or to determine whether the factors in Schuyler (2001) are relevant for this point at all, is beyond the scope of this thesis, however.

Roughly speaking, clausal ellipses tend to license parasitic ellipsis while non-clausal ellipses do not. Sluicing and *wh*-remnants of VP-ellipsis are exceptions on both sides. The safer claim is that parasitic ellipsis is licensed at least in the prototypical clausal ellipsis types gapping, stripping and fragment answers. Future work is necessary to determine whether the stronger claim that only clausal contrastive ellipses license parasitic gaps can be defended and whether apparent counterexamples can be made sense of.

To conclude, parasitic ellipsis is at least licensed in gapping, though other ellipses might also license the same phenomenon. In gapping, parasitic ellipsis is possible after at least C, A, N and P. There is a strict locality condition in that only the complement of the head of an ellipsis remnant can be elided

²⁸This is not her final generalization. She ultimately proposes a unification for her generalizations about *wh*-movement and topicalization based on assumptions she has to make about the focus status of *wh*-words. This is immaterial for present purposes since for now I am only interested in the empirical part of the generalizations.

parasitically. The parasitic ellipsis generalization, repeated here, encapsulates the results of this section.

Parasitic ellipsis generalization (PEG)

For every possible remnant of gapping, the complement of its head can optionally be deleted if the head is contrastively focussed and its complement given.

4 Theoretical approaches to parasitic ellipsis

4.1 Ruling out alternative analyses

The last section demonstrated the properties of the parasitic ellipsis that an account of this construction has to derive. Before I present my own proposal, I want to argue against certain types of analyses with which one could derive the data. Those are approaches that try to derive the parasitic ellipsis phenomenon through independently available mechanisms so that the parasitism is only apparent.

One way to analyse the data would be to claim that what survives is not an XP with additional ellipsis, but actually just a single head X. For approaches that assume that remnants of contrastive ellipses survive via (exceptional) movement, this is hardly tenable. Such a head movement is strongly ungrammatical in overt syntax.

(191) *WITH₁, the dog stole the sausage *t*₁ Masha's help.

(192) *WEIL ist Peter froh [_{CP} *t*₁ er gefeuert wurde].
because is Peter happy he fired was
'Because Peter is happy he was fired.'²⁹

Fitzgibbons (2014) makes the same argument for Russian, showing that extraction of bare P is ungrammatical (2014:174, fn. 6, adapted):

(193) *Ivan₁ bez nam pomog *t*₁ vsjakiz voprosov.
Ivan without us helped any questions
'Ivan helped us without asking any questions.'

Such extractions would also violate locality restrictions on head movement since it does not move a head to the head of its selector. Head movement would also be cross-clausal in the case of C-remnants in (192). We also know that more than a simple head can survive, as is the case with adjuncts, or specifiers in nominal expressions:

(194) Kofi ist ziemlich FROH über sein Ergebnis, und Anna ziemlich TRAU-
RIG.
Kofi is quite happy about his result and Anna quite sad
'Kofi is quite happy about his result and Anna quite sad.'

A more reasonable alternative analysis would therefore be one that postulates

²⁹The intended (and completely unavailable) reading is of course that the firing is the cause of happiness, not the other way round. German does not have the ambiguity of the English translation since the subordinated clause is verb-final in this example.

prior evacuating movement of the unrealized complement. Such an approach could among other things be used to reanalyse parasitic ellipsis in the nominal domain as scrambling of the complement of the noun. At least for German, this is an independently available process:

- (195) Peter besaß [von seinen Eltern]₁ nur [Gemälde *t*₁].
 Peter possessed of his parents only paintings
 ≈‘Of his parents, Peter had only paintings.’

One might therefore postulate that the complement is scrambled into the Mittelfeld where it is deleted by normal gapping, giving rise to an apparent parasitic process:

- (196) Jordan hat ein GEMÄLDE von seinen Eltern, und Paul [ein
 Jordan has a painting of his parents and paul a
 BILD *t*].
 picture
 ‘Jordan has a painting of his parents and Paul a picture.’

It has been noted in Sag et al. (1985) that antecedent and elliptical sentence can differ in their word order and the movement operations that have applied (1985:158, ex. 106):

- (197) A policeman walked in at 11, and at 12, a fireman.

This example, however, differs from the parasitic ellipsis construction in a number of relevant ways. First, the PP *at 12* receives a topic interpretation. In contrast, the parasitic ellipsis sentence does not have an interpretation that differs e.g. in its information structure from its antecedent. That is, the interpretive effects of scrambling are absent. Second, the additional movement in (197) is overt while the one in (196) would be covert. Movement approaches to gapping therefore would need to not only stipulate *exceptional rescuing movement*, of elements that need to survive, but also *exceptional destructive movement*, of elements that unexpectedly get deleted. However, such a device faces a number of issues, beyond the issues mentioned above. German has scrambling, but English does not, yet it has parasitic ellipsis. Even if English could be said to have scrambling solely in elliptical contexts, this would still not explain why parasitic ellipsis is possible in German even in contexts where scrambling is forbidden (the following data, except for the parasitic ellipsis examples, are discussed in Müller, Englisch & Opitz 2022; for a more in-depth discussion, consider this article and references therein). In the examples below, extraction from the NP headed by *book* is blocked when its verb is not a ‘natural predicate’ such as *read*:

- (198) a. Worüber hat Peter ein Buch gelesen?
 about.what has Peter a book read
 ‘What did Peter read a book about?’
 b. *Worüber hat Peter ein Buch zerstört?
 about.what has Peter a book destroyed
 ‘About what did Peter destroy a book?’
 c. *Peter hat darüber wohl ein Buch zerstört.
 Peter has about.that presumably a book destroyed
 ‘Peter presumably destroyed a book about that.’

But parasitic ellipsis after *book* or *picture* is acceptable in German in such a context:

- (199) Peter zerstörte einen ARTIKEL über sein Lieblingsthema, und
 Peter destroyed an article about his favourite.topic and
 Paul ein BUCH.
 Paul a book
 ‘Peter destroyed an article about his favourite topic and Paul a
 book.’
 (200) Peter zerstörte das GEMÄLDE von seinem Großvater, und Paul
 Peter destroyed the painting of his grandfather and Paul
 das BILD.
 the picture
 ‘Peter destroyed the painting of his grandfather and Paul the pic-
 ture.’

These sentences have both a strict and a sloppy reading. This is similar in English where parasitic ellipsis is possible where extraction is not:

- (201) a. *About which actress did you destroy a book?
 b. ?Mary destroyed an ARTICLE about her favourite actress and
 Susan a BOOK.

German does not allow for extraction from definite NPs and indirect dative objects (Müller, Englisch & Opitz 2022:1621, ex. 3a&4, adapted):

- (202) a. *Worüber₁ hat man [einem Buch *t*₁] einen Preis verliehen?
 about.what has one a book an award given
 ‘About what was a book given an award?’
 b. ?*Worüber₁ hat der Fritz [das Buch *t*₁] gelesen?
 About.what has the Fritz the book read?
 ‘About what did Fritz read the book?’

Scrambled versions are similarly degraded:

- (203) a. *Peter hat darüber gestern einem Buch einen Preis
 Peter has about.that yesterday a book an award
 verliehen.
 given.
 ‘About that, Peter gave a book an award yesterday.’
 b. ?*Peter hat darüber gestern das Buch gelesen.
 Peter has about.that yesterday the book read.
 ‘Peter read a book about that yesterday.’

Yet parasitic ellipsis in these contexts is acceptable:

- (204) Maria hat einem BUCH über ihr Lieblingsthema einen Preis
 Maria has a book about her favourite.topic an award
 verliehen, und Sophie einem ARTIKEL.
 given and Sophie an article
 ‘Maria gave a book about her favourite topic an award and Sophie
 an article.’
 (205) Sarah hat den ARTIKEL über ihren Helden gelesen und Lena
 Sarah has the article about her hero read and Lena
 das BUCH.
 the book
 ‘Sarah read the article about her hero and Lena the book.’

What is more, extraction of genitive arguments is strongly unacceptable, yet parasitic ellipsis of genitive arguments is fine:

- (206) *Peter hat [seines Vaters]₁ wohl ein Bild _{t₁}.
 Peter has his.GEN father.GEN presumably a picture
 ‘Peter has a picture of his father.’
 (207) Maria hat ein GEMÄLDE ihres Vaters und Laura ein
 Maria has a painting her.GEN father.GEN and Laura a
 BILD.
 picture
 ‘Maria has a painting of her father and Laura a picture.’

If *exceptional destructive movement* is somehow more permissive than its un-exceptional, overt version, one could expect that such a movement operation could also function as *exceptional rescuing movement*, i.e. also lead to the survival of these elements. This is also not the case:

- (208) *Peter besitzt ein Bild seines Sohnes und Paul seiner
 Peter possesses a picture his.GEN son.GEN and Paul his.GEN
 Tochter.
 daughter.GEN
 ‘Peter has a picture of his son, and Paul of his daughter.’

Under an analysis with additional movement operations, genitive arguments and other similar elements can apparently only be moved to their demise, but not their salvation. Such a movement could also occur counter to any restriction that is imposed on its overt version.

This means that non-pronunciation of NP-complements cannot be reanalysed as prior evacuation and that this phenomenon is to be treated on par with parasitic ellipsis in other categories; such an analysis is therefore also not a way to avoid the DP/NP-issue in the generalization.

Turning to other categories besides NPs, they do not fare better with respect to potential prior evacuating movement. German generally does not allow extraction out of PPs, but parasitic ellipsis can elide the argument of prepositions:

- (209) *[Mashas Hilfe]₁ hat der Hund die Wurst gestohlen mit *t*₁.
 Masha’s help has the dog the sausage stolen with
 ‘The dog stole the sausage with Masha’s help.’
- (210) *Der Hund hat [Mashas Hilfe]₁ gestern die Wurst gestohlen
 the dog has Masha’s help yesterday the sausage stolen
 mit *t*₁.
 with
 ‘The dog stole the sausage with Masha’s help.’

Not all English prepositional elements allow extraction. Extraction from *during* is quite degraded, yet the corresponding parasitic ellipsis is acceptable:

- (211) a. *What did Mary drink wine during?
 b. ?Susan drank wine AFTER dinner but Mary DURING.

In APs, subextraction is generally allowed in English and German:

- (212) Peter ist [auf seine Tochter]₁ wohl ziemlich [stolz *t*₁].
 Peter is of his daughter presumably quite proud
 ‘Peter is presumably quite proud of his daughter.’
- (213) Which student is Mary proud of?

Here, however, the issue still remains that no effect on information structure

is perceivable in German parasitic ellipsis and that English does not possess a scrambling-like movement operation in overt syntax (*exceptional destructive movement* would presumably not be an $\bar{A}$ -type movement because the landing site must *not* be high enough to ensure rescuing). Additionally, one might expect at least English to maintain its p-stranding property under exceptional destructive movement. This is also not the case, as we have seen in the discussion of the locality conditions on parasitic ellipsis:

- (214) #Susan is HAPPY about the result, and Mary SAD about.

Note also that stranded prepositions in ellipsis remnants are not forbidden in general, they are only excluded in parasitic ellipsis:

- (215) Which child is Mary PROUD of, and which child DISAPPOINTED with?

To continue with other categories, subextraction of TP from a subordinated CP is a grotesque claim; it leads to strong ungrammaticality in overt syntax:

- (216) *[Er von seinem Boss gefeuert wurde]₁ ist Peter glücklich, weil
 he by his boss fired was is peter happy because
 *t*₁

‘Peter is happy because he was fired by his boss.’

- (217) *Peter ist [er von seinem Boss gefeuert wurde]₁ wohl
 Peter is he by his boss fired was presumably
 glücklich, weil *t*₁
 happy because
 ‘Peter is happy because he was fired by his boss.’

- (218) *[He was fired by his boss]₁, Peter is happy because *t*₁

An equivalent exceptional *evacuating* movement operation that rescues TPs from gapping is also not attested:

- (219) *Peter ist traurig weil er gefeuert werden WOLLTE, und Paul
 Peter is sad because he fired will want and Paul
 er gefeuert WURDE.
 he fired was
 ‘Peter is sad because he wanted to be fired, and Paul he was fired.’

- (220) *Mary is sad because she WAS fired and Susan she WASN’T fired.

If all this could be made to work, there is still no explanation for the locality

restrictions on parasitic ellipsis. Exceptional destructive movement would need to be a movement type that can only move phrases if those phrases are the complement of a contrastively focussed head. Even slightly more embedded phrases would be impossible to move. This movement type would also need to distinguish between the above cases and the constellation where the phrase to be moved is the complement of a contrastive focussed head but that head is not the head of the ellipsis remnant (as in (162)). As we have seen above, those cases also do not allow parasitic ellipsis. That is, to derive the locality conditions in total, this movement type would need access to the information ‘*The head of my selector is the focussed head of an ellipsis remnant*’ to avoid over/undergeneralization. This would be quite a powerful non-modular approach to the capabilities of movement features - on top of the unexplained violations of known restrictions and otherwise non-availability in overt syntax.

In conclusion: an analysis that assumes prior evacuation of the complement would need to postulate movement that does not behave like its overt counterpart in languages that possess it, postulate it for languages that do not exhibit it at all and allow it to violate otherwise strong restrictions on this kind of movement while at the same time being sometimes more restrictive than other movements as is the case with p-stranding. Such an exceptional destructive movement could lead to non-pronunciation of elements that can *not* be moved by its exceptional rescuing counterpart. Such a movement would also be sensitive to locality and focus conditions that other types of movement are not subject to. In short, exceptional destructive movement would be an operation that is both difficult to motivate or falsify. The empirical generalization, on the other hand, is quite simple to state. The data in this section demonstrated that an independently available movement operation (even stretching the meaning of ‘independent’ to include exceptional movement) cannot be responsible for all the data in the preceding sections.

This is also a reason to reject the non-parasitic approach in Yoshida, Nakao & Ortega-Santos (2014). As mentioned, they note that otherwise immovable elements like V and P heads can survive (*why*-)stripping. Their approach that relies (next to other factors) on focus-movement of the apparently ‘incomplete’ remnant. This faces the issues discussed in this section: more than a single head can survive, such as modifiers and specifiers. Therefore one cannot just move a single head. Its complement would need to be evacuated prior to focus movement, leading to a plethora of theoretical issues.

Parasitic ellipsis is a genuine phenomenon in its own right and requires an analysis that takes this into account. The next sections discuss Fitzgibbons’ own parasitic analysis, properties of parasitism in general and how an analysis can treat parasitic ellipsis as a genuine phenomenon but avoid some of the

issues of previous approaches to parasitism.

4.2 Parasitic operations and their issues

Fitzgibbons (2014) provides a genuinely parasitic analysis of the ellipsis of prepositional complements in gapping remnants. In her account, prepositions like *bez* ‘without’ can optionally carry an E-feature which triggers ellipsis of its complement if this preposition is contrastively focussed. She models the ‘dependency’ of this ellipsis on gapping by proposing that the E-feature on the remnant head needs to enter into a relationship with the E-feature on the head that triggers gapping. This is achieved by adopting the four-way classification of possible features from Pesetsky & Torrego (2007) as used in Chou (2013) where features can be (un)interpretable and (un)valued. A prepositions such as *bez* carries $iE[-]$, an interpretable but unvalued E-feature, that is valued by the $iE[+]$ feature on a Foc head under c-command.

Additionally, the PP needs to vacate the ellipsis site before gapping can take place. The responsible focus movement is triggered by the same head Foc with an $iFoc$ -feature that checks a $uFoc$ -feature on *bez*.

As an example, a derivation of the original parasitic ellipsis sentence would precede as follows: *bez* carries $iE[-]$ and $uFoc$, and the Foc head responsible for gapping carries $iE[+]$ and $iFoc$. When Foc is merged, it values $iE[-]$ on *bez* as $iE[+]$. Ellipsis of the complement of *bez* can already take place, but not of the complement of Foc since it still contains the contrastively focussed remnant. Movement of the remnant is triggered by $iFoc$ on Foc checking $uFoc$ on *bez*, after which gapping can apply.

In my view, the main issue with this proposal is the interpretability of the E-feature. An interpretable feature is a feature that is relevant for LF/semantics. Fitzgibbons (2014) claims that parasitic ellipsis of the complement of a contrastively focussed head is “always possible [...] but its interpretation will not be accessible if the E-feature is not valued” and that the ungrammaticality of unlicensed parasitic ellipsis is “due to a crash at the interface with semantics” (2014:175). However, it is unclear what the meaning of an E-feature is compared to e.g. tense or number. The purpose of an E-feature is to trigger a syntactic operation or an instruction to PF/phonology for non-pronunciation.

In the feature system used here, Pesetsky & Torrego (2007) propose that all four combinatorial possibilities of (un)interpretable and (un)valued features exist; e.g. the T head carries $iTense[-]$, probes for the c-commanded verb carrying $uTense[+past]$ and finds its value there. Here, the c-command relation is the reverse compared to the one in the ellipsis case. Chou (2013), on the other hand, uses an $iTopic[-]$ on an NP that undergoes A-movement

to the specifier of T which carries $uTopic[+]$ to c-command the goal; in this approach, it is the *moving* item that triggers movement to receive the relevant value. Once the NP is in SpecTP it is valued for the topic feature and receives a topic interpretation. Fitzgibbons adopts this approach; in her system, it is the *lower* E-feature $iE[]$ as well that needs to receive a value (while focus movement on the other hand is still triggered by a higher iF feature on the attracting head).

A major difference to Chou’s proposal where the A-moved phrase gets its topic value and therefore a topic *interpretation*, is that no additional semantic effect arises from an interpretable E-feature, in contrast to how interpretable features are used in the above accounts. The way interpretable features are used here is simply to trigger the application of an operation. To my understanding, ‘apply an operation’ is not a semantic meaning. Nor is ellipsis an operation that can be executed in the semantics module - this is exactly the wrong interface to which this information should be sent. Another difficulty arises when one considers in which circumstances heads can carry $iE[]$ -features. Fitzgibbons claims that *bez* (and other functional prepositions such as *under* and *above*) can optionally carry this feature if it is contrastively focussed. Yet we have seen above that there are locality restrictions on the heads whose complements can parasitically be elided. This means not every head can optionally carry this feature if it is focussed. The putative mechanism that assigns $iE[]$ to heads would therefore also have to check how deeply inside the remnant the head is in order to not overgenerate. A contrastively focused element can only get an $iE[]$ -feature if it is the head of a remnant but not the head of its complement (or any other element). To assign this feature, structural information needs to be taken into account, i.e. it can only happen after the structure is already built sufficiently far.

To conclude, the underlying idea in Fitzgibbons (2014) is to let operations communicate with each other via features to express the ‘dependence’ of one operation on the other. But I am unconvinced that using the combinatorial possibilities of the above feature system is a fruitful implementation since the claims one makes about the status of the operation (such as a potential semantic ‘meaning’ of an E-feature) are doubtful and one potentially predicts interactions with the already existing machinery of the grammar that go far beyond the phenomenon that one wants to derive. In principle all features can then be allowed to communicate with each other, leading to probably unwanted empirical and theoretical consequences.

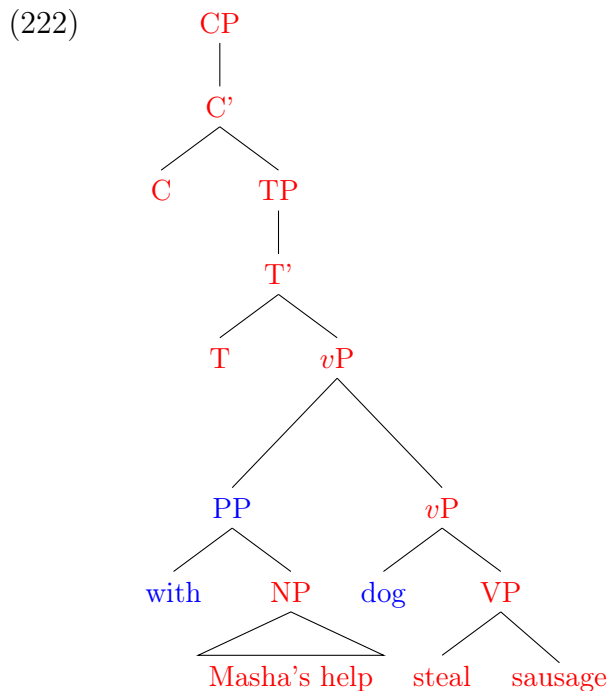
I now want to sketch the two main options how parasitic ellipsis can be modelled in the ellipsis framework I have been using so far. In the next section I discuss which of those options is the most appropriate. After that

I discuss the solution that I commit to in more detail.

Consider again the classic parasitic ellipsis sentence:

- (221) The dog stole the sausage WITH Masha's help, and the cat WITHOUT.

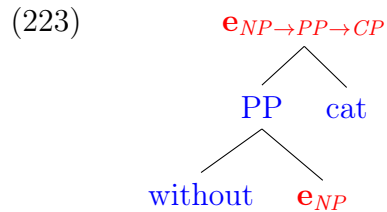
Below is the structure of the antecedent sentence (not indicating movers for now):



As in the previous sections, we have in red the structure that the ellipsis operation(s) in the following clause will be anaphoric to, and in blue the elements that must be ignored because the elliptical sentence provides these elements overtly; as I put it previously, the elements in blue are the ‘holes’ in the antecedent context. To repeat a point made in the section about remnant movement, I abstract away from specific accounts with respect to the size of the gapped constituent or potential additional (exceptional) movements of the remnants. The only relevant point for the analysis is that the elided constituent contains the base positions of the remnants. Any other variation in its size does not affect the arguments made here. The same goes for the assumption of additional movement steps. While different approaches to the *cause* of a remnant’s survival might provide a more restrictive theory as to what element can or cannot be a gapping remnant, they do not change the fact that when viewed over *derivation trees* the elements in question still

occur in their base positions. Movement steps only lead to additional movement nodes higher up in the tree. These questions are therefore orthogonal to the approach I want to argue for.

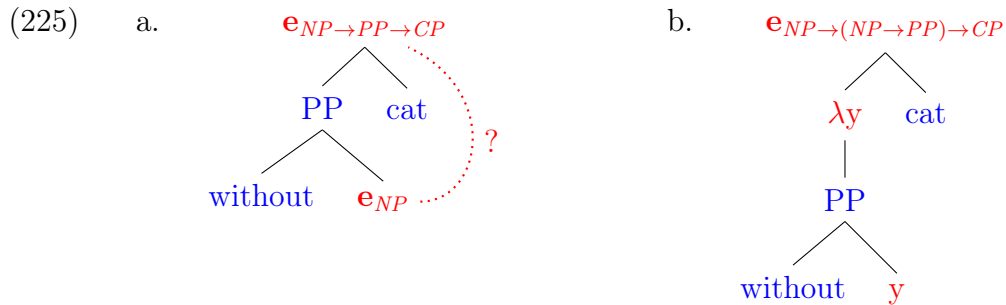
We now turn to the potential structure of the elliptical sentence. A simple analysis as the one below does not work:



Tree (223) represents the option of treating the ellipsis of the nominal complement of *without* as a *separate* and *independent* process of the grammar. We know that this is not the case. An analysis as in (223) would only work for something like NP-ellipsis inside a gapping remnant:

- (224)
- a. Peter bought five beer and Mary six (beer).
 - b. Peter bought five beer and Mary bought six (beer).

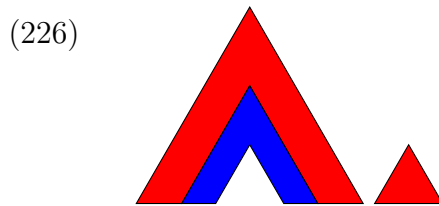
Both gapping and NP-ellipsis can independently occur (or not occur) and do not affect the felicitous occurrence of the other operation. Parasitic ellipsis is markedly different. This leaves the following two options:



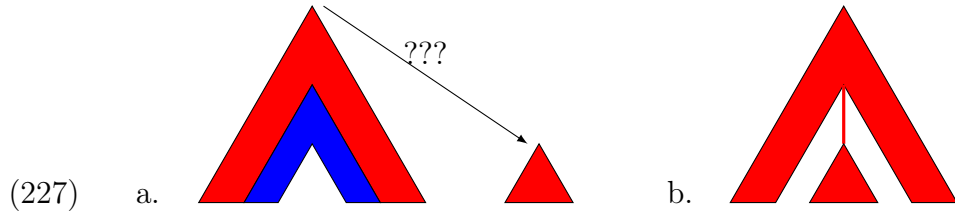
One could either have a *separate* but *dependent* ellipsis operation (225a). This is a pre-theoretic, intuitive understanding of parasitism: there is an operation whose existence depends in some way on another operation of the same type, i.e. in some way these operations can ‘talk to each other’ (graphically represented by the dotted line). The details of the dependence is what an account of the licensing conditions for the parasitic process has to work out.

But there is a third option, at least in the current framework: one can do away with the assumption that parasitic ellipsis is an operation that is *separate* from the operation that licensed it. This way, one can describe parasitic ellipsis as a *single* but *higher-order* operation (225b).

Another way to visually represent these options are the familiar triangles from the last section where different approaches to remnant movement remnants were discussed: we have ruled out the case of a second-order ellipsis operation (red triangle containing a blue triangle) and a separate, independent first-order ellipsis:



Options (225a) and (225b) are represented below where we have the same structural constellation as above but additionally some form of dependency between both ellipses; or the alternative where the elided structure forms a single but higher-order and therefore discontinuous ellipsis:

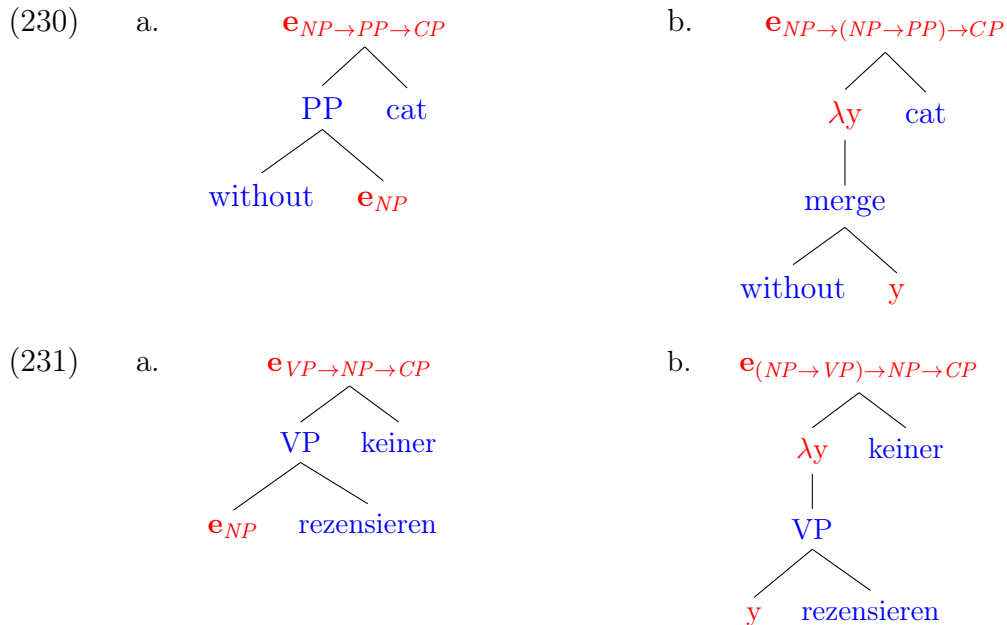


Notice that the structural configuration and the analytical possibilities for parasitic ellipsis in this framework are very similar to what we have encountered in remnant movement (when viewed over derivation trees).

(228) The dog stole the sausage WITH Masha's help and the cat WITHOUT.

(229) [Lesen t_2]₁ wird [das Buch]₂ jeder, [rezensieren t_3] keiner.

I repeat here the trees corresponding to the two approaches to parasitic ellipsis and remnant movement, respectively. The *a*)-versions represent the approach of a separate ellipsis with a so far unspecified dependency on the main ellipsis, the *b*)-versions represent the single but higher-order ellipsis approach:



As a short reminder, remnant movement remnants are non-trivial for ellipsis frameworks that take derivation trees as the central data structure. What is elided is not a full constituent, e.g. a VP (e_{VP}), or a constituent that

is missing a constituent like sluicing which elides TP with a surviving NP ($e_{NP \rightarrow TP}$); what is elided is a constituent to the partial exclusion of a remnant where not the whole remnant survives, e.g. because the object scrambled out of the surviving VP prior to those VP's topicalization and therefore becomes part of the ellipsis. This, however, was framework dependent since an approach over derived trees did not need additional machinery. We see that with parasitic ellipsis, the same question arises: **Is it one process or two processes?**

The initial intuition is that e.g. sluicing or VP-ellipsis is a single process, no matter whether they have a remnant or not, or whether there are additional movements out of the remnant. The additionally elided constituents do not form a natural class and cannot be elided independently; and even a separate, dependent ellipsis that can communicate with the initial ellipsis does not capture the theoretical intuition that derived tree-approaches derive effortlessly: there is still only a single ellipsis process. To capture this, I proposed that one needs to allow for third-order ellipsis operations for remnant movement remnants. With this one can express that an ellipsis operation is a single operation, no matter whether it elides a complete constituent, whether it has simple remnants (2nd-order) or incomplete remnants (3rd-order).

In parasitic ellipsis, the same issues arise again, but now it is independent from the choice of data structure. No matter whether one views derived trees or derivation trees as the more central data structure, the same structural configuration arises: unpronounced elements inside the remnant of an ellipsis operation. This time, the unpronounced element is not inevitable a mover, as in the case of remnant movement remnants. I have show that a movement analysis to parasitic ellipsis is not tenable. Therefore, reference to derived trees does not change the basic picture inside the remnant, of additionally elided material. So the question remains: is it one process or two processes?

For parasitic ellipsis, one might tend to the initial intuition to treat the observed deletion as two processes with some form of parasitic dependency between the them. In this thesis I want to make the case that this is as unwarranted as it would have been in the remnant movement case. However, intuitions are at best treated with caution and are at worst simply prejudices of the researcher. For this reason, I want to discuss the phenomenon of parasitism in more depth and to what extent parasitism is comparable to other relationships between operations like feeding. The aim is to show that parasitism is markedly distinct from feeding and that one should carefully distinguish between the claims that an operation is ‘parasitic on’ vs. ‘depends on the output of’ another operation.³⁰ As a consequence, I want to

³⁰Until now I have usually used the word ‘depend(ent)’ in scare quotes so as to make

claim that parasitic phenomena are better analysed as a single phenomenon, not two parasitic ones, or in other words that true parasitism does not exist in grammar. The arguments I present are largely conceptual (and should therefore be treated with caution as well) but some are concerned with the issue of potential overgeneration. That is, allowing operations to communicate with each other leads us to expect phenomena that we never find whereas a more powerful operation that explains the phenomenon as a *single* process might turn out to be the lesser of two evils (unless one restricts the former so strongly so as to become a notational variant of the latter).

4.3 Parasitism compared

To talk about parasitism in grammar it is important to discuss the more well-known parasitic phenomenon, parasitic gaps, and compare it with parasitic ellipsis. I provide a short overview over parasitic gaps. First discovered by Ross (1967), parasitic gaps were first described in-depth in the seminal work by Engdahl (1983),³¹ including the by now classic sentence used to demonstrate the phenomenon (1983:5, ex. 1):

- (232) a. ?Which article did you file _ without reading _*pg*?
 b. Which article did you file _ without reading *it*?
 c. *Which article did you file *it* without reading _?

In (232b), there is regular *wh*-extraction of the object of the matrix clause verb; such extraction is ungrammatical out of adjuncts since they are considered islands (232c). Interestingly, in a sentence where the object of the matrix verb has been *wh*-extracted, the object of the verb in the adjunct can optionally go unpronounced (232a). The interpretation of (232a) is invariably that the object of the matrix and adjunct verb is the same expression *which article*, i.e. it is the same as the interpretation of (232b) when *it* is coreferent with *which article*. Therefore one can say that the gap in the adjunct is *parasitic* on the ‘normal’ or ‘real’ gap, or on the movement step that induced this gap. There is speaker variation with respect to the acceptability of parasitic gaps in general and in the type of constructions in which parasitic gaps are allowed. They are usually slightly degraded but are acceptable to a significant number of speakers. The acceptability of parasitic gaps depends on a number of properties that the literature has converged on (for an overview, see e.g. Culicover 2001). I discuss most of the central properties

clear that this is not a claim about the nature of these operations but a pre-theoretic use of this term.

³¹But see also Taraldsen 1980, Kayne 1983, Pesetsky 1982, Chomsky 1982

here. First is the structural configuration in which real gap, parasitic gap and extracted element have to appear. The true gap must not c-command the parasitic gap. This is called the *anti-c-command requirement* (Engdahl 1983:20, ex. 54, adapted).

(233) *Who₁ *t*₁ sent a picture of *pg*₁?

Another property of parasitic gaps is *island insensitivity*, as can be seen in (232a). The relationship between the *wh*-expression and the parasitic gap crosses an island barrier (in this case an adjunct). However, parasitic gap constructions get significantly more degraded if they cross more than one island:

(234) *Which article did you file *t* without complimenting the guy who read *pg*?

Parasitic gaps are restricted to DPs/NPs:

(235) *How much did he donate *t* without you demanding *pg*?

(236) *To which place did George go *t* after sending his friends *pg*?

Parasitic gaps are said to be *optional* in the sense that in most cases the parasitic gap can be plugged with a coreferent pronoun without any loss of meaning, as can be seen in (232) above, repeated here:

(237) a. ?Which article did you file *t* without reading *pg*?
b. Which article did you file *t* without reading it?

Engdahl (1983) calls obligatory parasitic gaps those gaps that precede the true gap because a pronoun in that position leads to unacceptability (1983:27):

(238) Which city do the people from *pg*/*it always talk about *t*?

This, however, might be due to independent factors since movement of the *wh*-element induces a cross-over violation.

Another important property is that only $\bar{A}$ -movement but not A-movement seems to license parasitic gaps (Engdahl 1983:13, ex. 31, adapted):

(239) *John was killed *t* by a tree falling on *pg*.

In what follows I want to list a couple of pre-theoretic observations and expectations about parasitic gaps and parasitic ellipsis, compare their properties and then discuss the potential theoretical conclusions to be drawn from them.

- 1) In both cases, an operation appears to ‘depend’ in some way on the presence of another operation *of the same type*. In parasitic gaps, the creation of one gap depends on the creation of another; in parasitic ellipsis, one ellipsis depends on the presence of another ellipsis.
- 2) Note that purely descriptively speaking we seem to refer to the presence of the licensing operation, not necessarily the context it creates (parasitic gaps are licensed by $\bar{A}$ -movement, parasitic ellipsis is licensed by another ellipsis). The question is to which extent this persists in the theoretical implementation (as discussed for the feature-valuation approach in Fitzgibbons 2014) or if we at least need to extend our view of what can fall under a context that operations in grammar can refer to. Put differently, a potential question is whether parasitic ‘dependencies’ are distinct from rule interactions like feeding.
- 3) In both cases, the parasitic process applies in the domain defined for its host process. To define $\bar{A}$ -movement and formulate potential restrictions, one needs to refer to the base position of the moved item, its landing site and the span it travels; a parasitic gap must occur in a constituent attached to that span (i.e. path containment in the sense of Pesetsky 1982, see also Kayne 1983), but e.g. not lower than the base position of the licensing mover. Parasitic ellipsis must occur in an item whose base position is contained in the licensing ellipsis.
- 4) Both parasitic gaps and parasitic ellipsis appear to be optional.
- 5) parasitic gaps and parasitic ellipsis are often judged as marked by many speakers

Parasitic relations appear to hold not between any two operations but only between operations of the same type. The parasitic instance of the process occurs not just anywhere, but in that part of the structure that is relevant for the licensing process. Parasitic phenomena also seem to possess a relatively marginal status.

A side remark on the optionality of parasitic phenomena: whether a process is optional or obligatory appears to alter whether it is perceived as parasitic or not. The two phenomena discussed so far are optional. Another parasitic process that has recently been described is parasitic agreement (Corver 2021). I cannot do this phenomenon full justice here and a more complete comparison with the present phenomena and the question of its actual parasitism has to be explored in future research, but the basic pattern is as follows: in Dutch, an adjectival degree word modifying an inflected

attributive adjective can optionally exhibit the same suffix (Corver 2021:186, ex. 3):

- (240) een erg(-e) leuk-e auto
 a very(-e) nice(-e) car
 ‘a very nice car’

Compare this to a process where agreement of multiple elements with respect to one feature is not optional but obligatory, like agreement in the German NP (i.e. concord):

- (241) d-em klug-en stark-en Kind-e
 the-DAT.SG smart-DAT.SG strong-DAT.SG kid-DAT.SG
 ‘the smart kid’

In contrast to the above examples, concord is usually not presented as a prime example of parasitism.³² However, I see no a priori reason why the status of its optionality should affect whether a phenomenon should be classified as parasitic or not (or whether a one process or two process analysis is deemed more appropriate). The crucial factor should be the status of the relationship between grammatical entities. To conclude this side remark, more phenomena might fall under the umbrella of parasitism if the optionality of a process is factored out. For now, however, I want to restrict myself to the two phenomena at hand.

Let us assume (less pre-theoretically now) that what is observed in these phenomena is the result of two separate processes. This leads to the initial expectation that grammar enforces the same restrictions for operations of the same type. One would expect that the parasitic process underlies the same restrictions as its licensing process (obviously minus the additional dependency requirement). This is decidedly not the case, in both directions.

- 6) The parasitic operation is *more permissive* than the licensing operation.

The dependency between mover and parasitic gap can cross one of the most fundamental restrictions on movement, namely an island boundary. Interestingly, however, it is not ‘anything goes’, only a single island boundary can be crossed.

In parasitic ellipsis, one ellipsis appears to license additional ellipsis across focussed material (or in the 3rd-order analysis sketched above, ellipsis skips over a focussed head), otherwise a similarly fundamental

³²Henderson 2006 is an exception who describes the agreement morphology of an element in a concord relation as ‘parasitic’ on the features of another element.

barrier for deletion. Yet again, not ‘anything goes’, only a single focussed head under strict locality conditions can be skipped. Another place where parasitic ellipsis is much more permissive are the kinds of different heads that apparently license it. It can occur after P, N, A, C and possibly V and affects NP/DP, PP and TP; whereas other ellipsis types are usually *defined* by the head that licenses it (or alternatively the complement of that head that is elided). NP-ellipsis, sluicing or VP-ellipsis exhibit distinct profiles, whereas parasitic ellipsis could not care less about the constituent it elides.

- 7) The parasitic operation is *more restrictive* than the licensing operation.

Parasitic gaps are subject to categorial restriction whereas standard $\bar{A}$ -movement can also displace PPs, APs and more.

In parasitic ellipsis, the apparent licensing head needs to be contrastively focussed (as well as being able to express focus overtly and possessing reasonable semantic alternatives). This is in stark contrast to other ellipsis types like sluicing where the licensing C-head must *not* be pronounced and, at least for German, some mechanism needs to ensure that the verb is not pronounced in C where it obligatorily moves to in normal circumstances. It is in contrast especially to the ellipses that license parasitic ellipsis in the first place, like gapping and stripping, where no licensing head (if there is any) is overtly present.³³

One conclusion from these observations is that parasitic ellipsis and parasitic gaps are remarkably similar in several ways even though there are some differences that are due to the nature of the respective process. Viewed abstractly enough, however, many parallels emerge. I take this to mean that parasitism is a genuine notion of grammar in its own right.

On the other hand, however, parasitic phenomena exhibit at first sight unexpected properties if we assume that the parasitic process is a separate operation of its own. The parasitic operation can both violate fundamental restrictions that hold for its licensing operation while on the other hand being subject to apparently ad-hoc restrictions that do not hold for operations of the same type.

³³Note that not all these points can be reduced to the fact that *some* focus is necessary since we are talking about a contrastive ellipsis remnant. If it were not for the strict locality conditions, parasitic ellipsis might occur somewhere deeper inside the remnant on the complement of a given or silent head.

	parasitic gaps	parasitic ellipsis
type of process	gap creation parasitic on gap creation	ellipsis parasitic on ellipsis
domain	c-commanded by landing site, within span attractor-target	base-position of remnant contained in ellipsis
optionality	yes (except for some weak-crossover configurations)	yes (except for some sloppy readings)
acceptability	degraded	degraded
more freedom	can cross islands (once)	can cross focussed heads (once), categorical blindness
less freedom	restriction to DPs	contrastive focus requirement

Table 1: Properties of parasitic phenomena

4.4 Parasitism $\not\subset$ feeding

As we have seen above, parasitic phenomena exhibit unexpected properties if viewed as two separate operations. It is insightful to investigate to what extent parasitism is similar to or distinct from other rule interactions like feeding. One way to define feeding is as follows:

“Two rules A and B stand in a potentially feeding relation if the application of A creates *new* input to B.” (Kenstowicz 1994:94, emphasis mine)

Taking this definition literally would mean that parasitism is not a form of feeding since the host movement or host ellipsis is the *only* rule/ process/ operation that provides ‘input’ or a ‘context’ for the parasitic process to apply. Under a more careful definition that does not presuppose already existing inputs for a parasitic process, parasitism could be viewed as a special subset of feeding, namely those instances where it so happens that the only case where the context specification for Rule B is met is when Rule A applied. This might then simply be an accident of a specific grammar; were the language to change in such a way that it does provide a ‘non-derived’ context for Rule B, then that rule would change from a parasitic to a standard one without a change in its definition. The definition for this conception of parasitism as a special subcase of feeding would be as follows (where, staying close to the biological metaphor, ‘host’ is the term for providing the context

for a parasitic process):

Parasitism (wrong):

Process A hosts Process B iff A is the *only* process providing the input for the application of B.

Standard feeding, then, would be when A provides *additional* contexts to B, and parasitism if A provides the *only* context.

In what follows I want to argue that the equation of parasitism with (accidentally) fully-dependent feeding is misguided. There are several ways in which process interactions like feeding differ from parasitic interactions and there are generalizations or properties that such a classification does not capture.

Property #1: contexts vs. rules

A *fully dependent* process still only talks about the contexts in which it can apply, not to the presence of another operation. Imagine the following toy phonological grammar: there is a rule that resolves hiatus by deleting one of the vowels; yet the morpheme structure constraints of the language are by some accident in such a way that this context never arises by just concatenating morphemes (e.g. there are only CV-syllables). The only way this context comes about is through another process of deletion of intervocalic voiceless stops. Despite depending completely on the outputs of other rules, the hiatus rule is no different than any other rule in the grammar and might find new applications, e.g. through language change or lone words. Only the researcher knows that the input to the fully dependent rule already has a derivational history, the rule itself is blind to that information.

Compare this to parasitic ellipsis. Here, in order to decide whether to apply, an ellipsis process needs to sense whether there is material in a certain structural configuration that has been deleted. Crucially, it is also necessary to know the information *which ellipsis* is responsible for that non-pronunciation: deletion due to gapping fulfils this requirement, deletion due to VP-ellipsis does not. This leads to the question how to model this theoretically. In Fitzgibbons (2014), E-features communicate via feature valuation. Note that here the parasitic E-feature does *not* refer to an empirical property of a syntactic element such as its non-pronunciation; it cannot since in her proposal, parasitic ellipsis is licensed before the host ellipsis is applied. Instead, the E-feature refers to a theoretical diacritic, a notational device that implements a specific proposal. If one allows for this, one considerably extends what it means to be a context. This proposal differs from a standard case of Agree where e.g. a property of a noun such as number or person is

reflected in the inflection of a verb. Here the theory refers to the abstract properties that it postulates for a language, not to itself.

It is non-trivial to implement parasitic ellipsis theoretically without referring to the actual licensing operation - contrast this with the fully dependent hiatus resolution rule above. It would be necessary for a probably syntactic operation to sense whether certain material is phonologically realized and what operation caused it. This also leads to questions in which module/interface parasitic ellipsis should be located and to timing questions: the higher licensing ellipsis needs to occur prior to the lower parasitic ellipsis if it is only the outcome of the former ellipsis that matters.

In parasitic gaps, a gap, in order to know that it is parasitically licensed apparently needs to scan upwards across an island barrier, and then probe downwards to see whether there exists another gap that corresponds to a moving item (scanning upwards again) that c-commands both gaps. Besides sensing not only for a gap but rather the fact that a movement dependency has been established, it is crucial to know that this movement dependency is due to $\bar{A}$ -movement.

An approach like Nissenbaum (2000) relies on a number of conspiring factors that all need to come together to derive this: empty operators exist (postulating that which should ideally be derived, the existence of additional gaps) that move, e.g. to the edge of the containing adjunct (overt counterparts of this movement are often strongly ungrammatical). The licensing movement proceeds successive-cyclically through vP exactly where all adjuncts that can contain parasitic gaps merge via late adjunction, tucking in below the landing site. At this point, the correct configuration for the semantics module is achieved so that predicate modification can combine the structures ensuring that both empty operator and trace refer to the same element. Here, a putative syntactic process is ultimately solved in semantics. Other approaches like sideways movement (Nunes 1995) derive parasitic gaps in a less accidental way but need to resort to the definition of a more powerful operation: a phrase can move between two structures that are not yet conjoined at the top node, leading to the creation of parasitic gaps. In either case, some heavy strings need to be pulled to derive this construction.

It appears to be difficult to get rid of reference to specific operations without resorting to additional machinery when viewing parasitism as the result of two separate processes. Implementing such reference directly as in Fitzgibbons (2014) also harbours the danger of overgeneration. If such mechanisms are allowed it could in principle generate syntactic phenomena like the following: ‘Agree with element A iff element B is A-moved which is only possible after C was $\bar{A}$ -moved to topic position’. Such possibilities, and the accompanying ‘grammar self-awareness’, should probably be treated

with extreme caution.

In conclusion, parasitic phenomena seem to differ substantially from simply fully dependent ones. So for terminological clarity, I will use the term **dependent** only for this specific subset of feeding and use the term **parasitic** for the stronger claim about the type of phenomena discussed here. A dependent process only talks about the contexts in which it can apply and is blind to the fact whether this context is ‘base-generated’ or not. Descriptively, in parasitism a process appears to refer to the presence of another operation and the question is to what extent this persists in theoretical implementations, and what alternatives there are to derive parasitism.

Property #2: *primus et ultimus inter pares*

Another property of parasitic processes is that they are simultaneously more permissive and more restrictive than their host processes (modulo the parasitism requirement). This is unexpected under the belief that those are two separate processes of the same type. Roughly the same restrictions should hold for the same type of operations. Additional restrictions are, of course, partly expected, given that one process has the additional requirement of ‘depending’ on another process. Restrictions such as anti-c-command or the focus requirement are by no means insurmountable for any theory, single process or more; e.g. the second process has to stand in *some* structural relationship to the second process. Depending on the theory, those properties might be derived from basic assumptions (though a restriction to DPs in parasitic gaps does not at first sight follow inevitably from the structural location of the parasitic process). Even more surprising is the fact that operations with an additional parasitism requirement exhibit *more* freedom than their host counterparts. Both parasitic gaps and parasitic ellipsis apply across what are otherwise strong ‘barriers’ for their host processes: islands and focused phrases; but both can do so only once.³⁴ Parasitic ellipsis, in contrast to other ellipses, appears to be blind towards the types of phrases it elides. Approaches to parasitic phenomena should ideally account for why a process that at first sight appears so similar exhibits markedly different behaviour from its host process. If parasitism were simply a subset of feeding, these properties are unexpected.

³⁴In that, parasitism is similar to certain harmony processes in phonology: an element that could block spreading since it does not share a certain feature does not do so and instead is transparent for the process; yet the skipping cannot occur across an unbounded number of such elements. This is e.g. the case in Hungarian vowel harmony (see e.g. Ringen 1988), where a single transparent vowel can be skipped but not more.

Property #3: the *same process* conspiracy

A third and crucial property of parasitism is that the parasitic phenomenon always appears to be an instance of the same process that enabled it. Or, to not presuppose separate processes, one might say that there appears to be an overapplication of a process. We find an additional gap in a gap creation process and additional ellipsis in ellipsis. The phenomenon of parasitic Agree mentioned above follows the same pattern. I put forward the following generalization:

Parasitism of Likes:

A parasitic phenomenon is always an instance of the same process that enables it.

Corver (2021) proposes a similar generalization about the phenomenon of parasitism: “the presence of a symbol of type α in a syntactic representation is dependent (i.e., parasitic) on the presence of another symbol of type α in that same representation” (2001:185).³⁵

As the generalization proposes, there is only ever parasitism of likes. What we do not find are all other combinatorial possibilities in case one assumes that parasitism is actually a relationship that holds between two distinct processes: ellipsis parasitic on movement, ellipsis parasitic on anaphor binding, quantifier raising parasitic on agree, etc.³⁶ If distinct operations can be parasitic on another, with n operations one naively expects n^2 potential parasitisms. Yet it just so happens that out of all the potential interactions, we only find parasitism between processes of the same type (i.e. (less than) n out of n^2). This is a conspiracy or an accident under a simple rule reference approach. It is of course possible to restrict communication only to similar operations. Note, however, that a parasitic dependency between two different overt $\bar{A}$ -movements is still distinct from the parasitic gap phenomenon, leading to wrong predictions even under this restricted view.

This concludes my attempt at a comparison between parasitism and other process interactions. As one can see, parasitism differs from simple feeding. The type of information that it needs to have access to is unusual compared

³⁵It might be difficult to fruitfully describe all parasitic processes in terms of symbols. Gaps are represented quite differently in different theories and it is unclear to what extent ellipsis is a symbol. I therefore prefer talking about processes.

³⁶A reviewer (Fabian Heck) notes that multiple sluicing in otherwise single *wh*-movement languages might be a counterexample. There are, however, a number of non-parasitic analyses that can be explored (such as e.g. in-situ approaches to ellipsis) before positing the need for a parasitic relationship, in contrast to the parasitic phenomena noted above where such a possibility is highly unlikely.

to other processes. For a putative process of the same type, it also shows markedly different behaviour: it is at the same time more restricted than its host process, yet can lift otherwise fundamental restrictions on such processes. Parasitism can also only hold between similar processes. From this I conclude that parasitism is distinct from feeding-like rule interactions, i.e. parasitism is *not* another name for fully dependent feeding but a distinct phenomenon of its own. Mnemonically, I state this conclusion as follows:

parasitism $\not\subset$ feeding

The question then arises what place in grammatical theory such a phenomenon should have and how one accounts for it. One option is to attempt to coordinate and fine-tune already existing mechanisms in such a way that they give rise to the parasitic phenomenon. This does not do justice to the fact how different parasitism appears from other process interactions, as discussed above; it relegates parasitism to an almost accidental interplay of grammatical machinery and ignores the distinct identity of parasitism.

The other option is to directly encode the parasitic relationship in grammar. This runs into overgeneration issues that need to be reigned in. An issue that separate-but-dependent approaches to parasitic processes share, no matter which of those two options are chosen, is the unexpectedly different behaviour of the parasitic process and the apparently accidental parasitism of likes generalizations.

In this thesis I want to pursue a different approach. I do not want to view parasitism as a conspiracy of sorts, a forced accident of pre-existing operations; instead I want to assume that parasitism is a genuine phenomenon in its own right. Yet I also do not want to resort to actual rule reference, to open up a Pandora's box of communicating operations, only to then close the lid almost completely again by restricting the emerging possibilities heavily. Instead, I want to call into question that the parasitic process is a separate process to begin with. The theoretical approach I want to pursue is to claim that parasitic phenomena are the outcome of a *single* but more powerful operation than the one that leads to the basic, non-parasitic process. These new operations are intended to be a natural continuation of their base version and generate both the host and the parasitic process in one fell swoop. Under such a view, there is no parasitism per se, as a type of relationship between grammatical entities, any more. It is only a superficial appearance that there are two separate processes. The approach pursued here, namely that parasitism in the literal sense does not exist, could be put into the following slogan:

No parasitism conjecture

Every instance of parasitism is the outcome of a single but more powerful operation than the operation that gives rise to the non-parasitic phenomenon.

This thesis is devoted to the attempt of developing such operations and to address the properties of parasitism described in the last sections. This half of the dissertation proposes a higher-order ellipsis operation that describes the parasitic ellipsis and its licensing ellipsis as a single phenomenon, the second half of the dissertation proposes a more powerful structure-building operation that can give rise to multiple gaps.

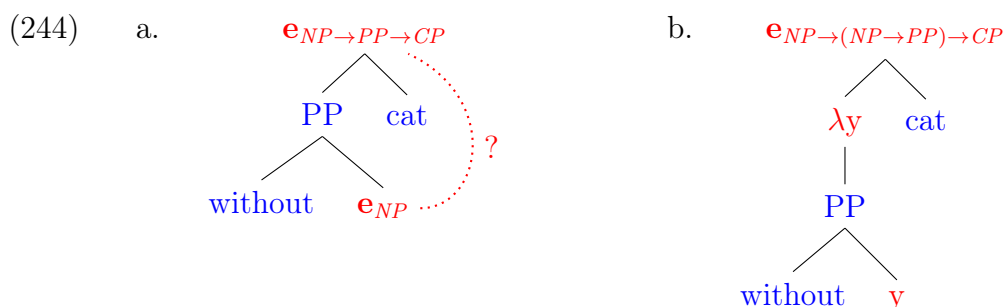
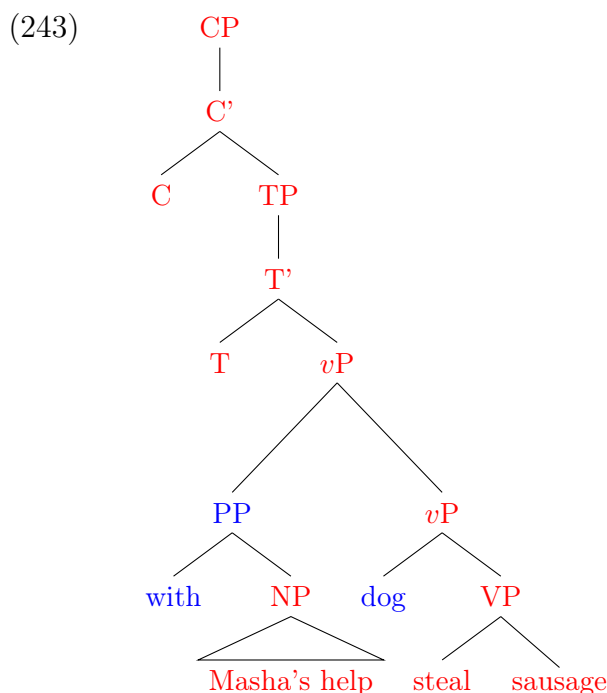
Note that two of the issues discussed above are immediately addressed under a *single process* approach: ‘rule reference’ and the same process-conspiracy. There is no communication between separate operations if there is only a single operation. There is additional application of a process without giving rise to pathological process interactions that have to be ruled out afterwards. Under the assumption that parasitism is the outcome of a more powerful version of the operation that triggered the host process, it is expected that both operation talk about similar phenomena. When going from a 2nd- to a 3rd-order ellipsis operation, one does not arrive at parasitic quantifier raising but at another ellipsis operation.

I also attempt to show that under such an approach at least some of the unexpected behaviours of the parasitic process can be derived, such as the categorial blindness of parasitic ellipsis. Later, in the second part of this dissertation I develop a new structure-building operation and discuss how this derives the idiosyncrasies of parasitic gaps. In the next section, however, I demonstrate my approach to parasitic ellipsis.

4.5 Parasitic ellipsis as higher-order operation

After an in-depth discussion of parasitism I now return to the actual analyses of parasitic ellipsis that I sketched in 4.2. The question that triggered this discussion was what the preferred analysis of parasitic ellipsis is: as two separate but dependent processes, with a yet to be defined way of expressing their dependency, or a single but higher-order operation. I repeat here the prototypical parasitic ellipsis sentence, the structure of the antecedent sentence and the two possible analyses:

- (242) The dog stole the sausage WITH Masha’s help, and the cat WITHOUT.



In the last section I discussed the issues with separate-but-dependent approaches to parasitism as in (244a). Instead, I aim to explore an approach to parasitism that views it as a single process that is the outcome of a more powerful operation. (244b) is a suitable candidate.

The ellipsis used in (244b) is a 3rd-order ellipsis operation familiar from the discussion in 3.1. 3rd-order ellipsis operations are a natural progression of the already available options in framework used here. As a brief reminder, ellipsis of constituents as in VP-ellipsis without remnants corresponds to a 1st-order ellipsis operation such as e_{VP} with its antecedent [_{VP} verb object]. Ellipsis remnants necessitate 2nd-order ellipsis operations that express that a constituent was elided but a subpart of it was spared. This is the case e.g. in passive VP-ellipsis with a surviving logical object: $e_{NP \rightarrow VP}$ and its antecedent $\lambda x.[_{VP} \text{ verb } x]$.

There is nothing inherent about 2nd-order contexts that make them a natural stopping point. The choice is ‘arbitrary’ in the sense that higher-order contexts initially did not appear to be necessary to explain all empirical phenomena. Remnant movement remnants forced us to go to higher orders, but this might be an artefact of how the evacuation movement is represented in derivation trees. This is different in parasitic ellipsis where less depends on such choices. In parasitic ellipsis, the non-pronunciation of the complement of *without* is not due to well-understood independent factors such as scrambling that are non-trivial to account for in derivation trees. I showed in 4.1 that a ‘destructive movement’ analysis for parasitic ellipsis is not tenable. In other words, the structural configurations indicated above also hold for derived trees where movements are already executed; the additional deletion needs to be accounted for by other means.

It is noteworthy that if one assumes 3rd-order ellipsis operations to exist, one actually *predicts* a phenomenon with a profile similar to that of parasitic ellipsis. A 3rd-order ellipsis operation is of the type $\mathbf{e}_{(XP \rightarrow YP) \rightarrow ZP}$: eliding a ZP with a YP remnant of which some elements (XP) are still part of the ellipsis. Its antecedent is of the abstract form $\lambda f. [\mathbf{ZP} \dots \mathbf{f}(\mathbf{XP})]$, i.e. what needs to be copied into the ellipsis is a phrase of type ZP from a suitable antecedent, minus the constituent YP that is still present in the elliptical sentence (therefore $\mathbf{f}$ as a hole in the antecedent). But the ellipsis site provides YP only partially, the XP part is also elided and needs to be copied from the antecedent again. This is exactly the profile of parasitic ellipsis: we have an ellipsis with a remnant in which the ellipsis continues. A 3rd-order constituent is discontinuous, therefore a 3rd-order ellipsis elides discontinuous material, skipping over some intervening elements. In the specific case above, there is ellipsis of an entire CP, with two remnants: the subject *cat* and the incomplete PP remnant *without* whose complement is elided as well. What this ellipsis operation is therefore anaphoric to is a CP, cutting out an NP and part of a PP (*with*) but including the complement of the preposition. If this is provided as antecedent for the elliptical sentence, the structures can compose to yield the correct interpretation:

$$\begin{aligned}
 (245) \quad & \lambda x. \lambda f. [\mathbf{CP} \dots [\mathbf{vP} [\mathbf{f}(\text{Masha's help})] [\mathbf{vP} \mathbf{x} \mathbf{v} [\mathbf{VP} \text{ steal sausage}]]]] \\
 & (\lambda y. \text{without } y) (\text{cat}) = \\
 & \lambda f. [\mathbf{CP} \dots [\mathbf{vP} [\mathbf{f}(\text{Masha's help})] [\mathbf{vP} \text{ cat } v [\mathbf{VP} \text{ steal sausage}]]]] \\
 & (\lambda y. \text{without } y) = \\
 & [\mathbf{CP} \dots [\mathbf{vP} [\text{without Masha's help}] [\mathbf{vP} \text{ cat } v [\mathbf{VP} \text{ steal sausage}]]]]
 \end{aligned}$$

As the above discussion shows, structures like these are readily available in

this framework by simply allowing 3rd-order ellipsis operations as well, lifting a restriction to 2nd-order operations. This fulfils one of the goals I mentioned in the last section: the parasitic phenomenon is the outcome of a *single* but more powerful operation that is in some sense a natural progression from the operation that - descriptively speaking - license it in the first place.

While it is true that a 3rd-order ellipsis operation already corresponds to the general shape of parasitic ellipsis, the actual locality conditions of the construction are stronger. As the parasitic ellipsis generalization states, parasitic ellipsis occurs after the head of a remnant of ellipsis, not just somewhere within the remnant. Translated into the general shape of 3rd-order ellipses, $\mathbf{e}_{(XP \rightarrow YP) \rightarrow ZP}$, this means there is a locality restriction between the second-most (YP) and most deeply (XP) embedded constituent in that formula:³⁷

$$(246) \quad [\underbrace{XP \rightarrow YP}_{\text{complement head[foc]}}] \rightarrow ZP$$

To derive the correct profile of parasitic ellipsis, one needs to state the requirement that XP and YP stand in a head-complement relation, a basic, fundamental relationship in syntax.

There is nothing unusual about stating such restrictions. There are already locality conditions in place that restrict the relationship between the two elements in a 2nd-order ellipsis operation:

$$(247) \quad [\underbrace{XP \rightarrow YP}_{\text{MCC}}]$$

Gapping is usually subject to the clause mate condition (e.g. Lasnik 2013), i.e. the requirement that the remnants are not separated by a CP-boundary from the root node of the ellipsis:

$$(248) \quad ?*\text{MARY thought that Peter loves JOHN, and SUSAN PAUL.}$$

Another requirement is the Major Constituent Constraint (MCC) proposed by Hankamer (1973): “A “major constituent” of a given sentence S_0 is a constituent either immediately dominated by S_0 or immediately dominated by VP, which is immediately dominated by S_0 ” (1973:18). A more recent version can be found in Wilder (1997), as discussed by Osborne (2008): “any head belonging to the Extended Projection that constitutes a conjunct, or

³⁷Put differently, the *f* in the antecedent needs to be minimal in the sense that it is selected by an elided head and itself immediately selects as its first argument an elided constituent

any phrasal sister of such a head, counts as a major constituent” (1997:104). Another way to phrase the MCC then would be to require that remnants of gapping need to be phrases that have been selected by heads of the clausal spine. If gapping remnants that are complements of prepositions are possible this needs to be amended:

(249) ?Mary talked to Joscha and Anne Susan.

In this case one has to allow not only for elements selected by heads of the clausal spine, but also elements that have been selected by those elements. The relationship is ‘once removed’, i.e. daughters and granddaughters of the clausal spine are possible remnants of gapping.³⁸

The relationship between ellipsis and remnant in an ellipsis type is therefore much more involved and less local than the locality condition that holds between the remnant and the parasitically elided phrase.

A naive 3rd-order ellipsis operation already generates parasitic ellipsis-like patterns without any additional stipulations; the stricter locality requirements that have to be stated are as simple as it gets in syntax. By allowing 3rd-order gapping operations that follow this restriction one generates exactly the patterns described in the parasitic ellipsis generalization repeated here:

Parasitic ellipsis generalization (PEG)

For every possible remnant of gapping, the complement of its head can optionally be deleted if the head is contrastively focussed and its complement given.

This generalization states that the complement of a head of a gapping remnant can additionally be elided; stating that there are 3rd-order gapping operations where the most deeply embedded phrase is the complement of the second-highest phrase is a direct translation of this statement into this framework. At this point I also want to address worries about overgeneration if one allows grammar to refer to 3rd-order structures. One of the arguments for exceptional movement to ellipsis is that otherwise one would have to resort to ‘non-constituent deletion’ (see e.g. Merchant 2004), i.e. it is a mechanism partly invented for preventing talking about 2nd-order ellipsis. The conceptual argument is that syntax should refer to constituents, with the often implicit understanding that if one allows for 2nd-order constituents, anything goes. Of course, any theory of gapping needs to specify

³⁸The actual precise relationship between gapping and its remnants is presumably much more intricate and beyond the scope of this thesis. I assume that most proposals for this relationship can be ported over to the current approach.

what its possible remnants are. Both in-situ and exceptional focus movement accounts need to specify under which conditions both mechanisms can rescue a phrase from deletion, neither of them come for free. It is beyond the scope of this thesis to adjudicate between these approaches. Both can in principle be implemented in the current framework. Whatever properly restricted approach is in place, one either elides 2nd-order constituents directly in the theory, or at least descriptively on the surface. One can either exceptionally move out remnants before deleting a constituent, or delete a constituent and determine conditions under which elements are exempt from deletion, e.g. by virtue of being focussed (as in a ‘radical deaccenting’ approach as in Tancredi 1992). In a sentence with parasitic ellipsis, the material that is elided is a 3rd-order constituent. This is a fact independent of the mechanisms that ultimately determine what is elided or not. The approach used here refers to this structure directly. The only thing necessary to avoid overgeneration is the locality condition used here. With the simple recipe of ‘when going from 2nd- to 3rd-order, respect a simple head-complement locality restriction’, one generates ellipses of the form ‘ellipsis with a remnant in which ellipsis continues’.³⁹ As we have seen, many of these structures in fact do exist, so we do want to be able to refer to them. As long as independently necessary restrictions on 2nd-order ellipses are in place, the proposed account generates them, but nothing more.

Let us take stock of how this approach fares with respect to the properties of parasitism discussed in the last sections.

First, we have a single but higher-order operation that generates parasitic ellipsis all at once. By going from a 2nd- to a 3rd-order ellipsis operation, one cannot but arrive at another ellipsis operation. The fact that on the surface it appears that one ellipsis is parasitic on another ellipsis is a simple consequence from allowing 3rd-order ellipsis operations. This derives the *parasitism of likes* generalization for ellipsis.

Second, since there is only a single operation there is no literal communication between processes in the theory. We are talking about a 3rd-order constituent all at once and are restricting its shape according to certain locality conditions. Those conditions are stated in terms of head-complement relationships, a simple and fundamental relationship in syntax. No further restrictions are necessary to reign in the power of communicating operations.

³⁹Deaccenting or in-situ approaches to ellipsis can easily emulate this approach by stipulating that while usually a deletion process is halted by focus, it can skip focussed material and continue deleting afterwards if the skipped material consists only of a single focussed head. This would capture the spirit of the analysis here. Exceptional destructive movement of the additionally elided material, however, is hardly a tenable analysis as shown above.

The appearance of two separate processes is simply a relict of empirical description in this approach; here, parasitic ellipsis is much more similar to a phonological spreading process that can optionally skip one blocker than to two processes that communicate across a barrier. An approach that views parasitic phenomena as the outcome of a single, more powerful operation derives both these points.

As a last point, this approach can also account for some of the idiosyncrasies of the apparently separate parasitic operation. One is its blindness towards the category it elides. As mentioned previously, what constituent an ellipsis elides usually defines its identity: VP-ellipsis has a different profile than sluicing. Parasitic ellipsis deletes a smörgåsbord of categories without any apparent difference in behaviour. This falls out naturally under the current account which cares only about the structural configuration in which ellipsis can continue in a remnant. I have shown that parasitic ellipsis occurs in the categories C, N, A, P and marginally V. The choice of potential categories participating in parasitic ellipsis is constrained by the possible remnants of gapping. If a CP can be a gapping remnant, then its complement can be parasitically elided; if a PP can be a gapping remnant, its complement can also be elided and so on. Yet the deletion is still a version of the same initial gapping process, one that can optionally skip a focussed head. It is not a separate ellipsis that needs to be licensed by a plethora of heads.

Another observation is if not derived at least accounted for, namely the ‘licensing’ or spreading of ellipsis over focussed material. If one postulates that 3rd-order ellipsis operations exist, then they necessarily skip intervening material (otherwise they would be 2nd-order ellipsis operations). What does not come for free is the precise locality requirement of parasitic ellipsis which has to be postulated. This restriction in turn forces the locus of contrastive focus to be on the surviving intervening material between the ‘two’ ellipses. Focussed elements cannot be elided, and a remnant of contrastive ellipsis has to carry focus. Only the fact that it has to be on the head is an additional restriction that needs to be postulated. Since there is no triggering of ellipsis by intervening heads, this account is not faced by the problem that parasitic ellipsis differs strongly from the type of ellipsis that enables it in the first place, namely contrastive ellipsis, in that it can apparently only be triggered by focused heads, in stark contrast to completely silent structure except for the remnants in the latter. This account also avoids the need to postulate that all heads in the lexicon can optionally carry E-features when contrastively focussed as in Fitzgibbons (2014).

To conclude the theoretical account of parasitic ellipsis: postulating 3rd-order ellipsis operations and restricting them with a simple locality condition

generates a phenomenon that on the surface looks like two ellipses in which one ‘depends’ on the presence of the other. Yet in the underlying theory there is only a single operation, an approach which avoids potentially pathological dependencies between various processes. The parasitism of likes generalization falls out as a natural consequence in an approach in which the parasitic phenomenon is a single process and it avoids unexpected behaviour that a putative separate parasitic process would exhibit.

Before closing this half of the dissertation and moving on to parasitic gaps, I want to touch on a number of topics, such as issues for the ellipsis framework used here, how potential 4th-order ellipses would look like and other places where one might find parasitic phenomena, elliptical or not, that would need to be compared in-depth to the two discussed here.

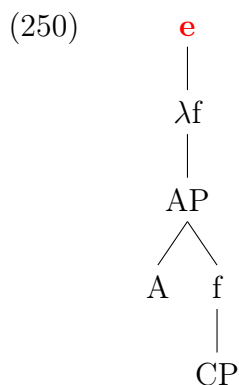
5 Further topics

5.1 4th-order ellipsis?

When arguing that a restriction to 2nd-order ellipsis operations must be lifted to allow for 3rd-order operations, the question naturally arises whether there are sentences that require going beyond 3rd-order ellipsis operations. The question is difficult to answer conclusively, due to at least two factors: on the one hand, potential examples for 4th-order ellipsis are increasingly contrived and reliable judgements difficult to come by; it is unclear to which extent performance issues affect acceptability ratings (similar to three or more center embeddings). On the other hand, much depends on the specific analysis assigned to these sentences, and that the necessary interplay of operations is available and not excluded for other reasons. Nonetheless, I want to showcase how such examples would have to look like, leaving it to future work to find such examples and to determine the empirical and theoretical validity of these constructions.

Nested movers

We have encountered two causes for 3rd-order ellipsis operations: one is due to how remnant movement is represented in derivation trees, the other one is parasitic ellipsis which is not due to movement. Since the former is rather an artefact of a specific representation, I will not delve deeply into the matter of 4th-order ellipsis operations due to movement. A 3rd-order remnant that causes a 4th-order ellipsis would look as follows,



This means, an ellipsis operation that has as argument not just an incomplete constituent in which ellipsis continues; now it is an incomplete constituent in which ellipsis continues and then *stops* again at some point, due to another remnant that is nested within the remnant. To get to 4th-order ellipsis (or

equivalently, to a 3rd-order ellipsis remnant) via movement, one would need constructions with three nested movers. There are $3! = 6$ possible timings for such movements. The one that is needed here is a sequence where the complement of A moves out of AP into its destruction. Then its subpart CP needs to move outside of the ellipsis site surfing-type movement (Sauerland 1996, Abels 2007, i.e. one that is potentially subject to freezing); the AP also needs to move to a point higher than the ellipsis site, a case of classical remnant movement.

Putting aside for now any restrictions that hold for such nesting movers (see Abels 2007), how would such an example in principle look like? Imagine a conversation about literature students and the speaker wants to contrast which kinds of books they like to read for fun and which books they prefer to write essays about:

- (251) ??[t_2 lesen]₁ wollen [Bücher t_3]₂ alle nur über die Liebe t_1 ,
 [t_2 read]₁ want [books t_3]₂ everyone only about the love t_1 ,
 [rezensieren t_2]₁ nur [über den Tod]₃.
 [review t_2]₁ only [about the death]₃
 $\approx$ ‘As for reading, everyone wants to do that with books about love,
 as for reviewing, only about death.’

I am unable to provide a reliable judgement for such sentences, regarding both syntactic and information structural properties. In informal elicitations, speaker judgements vary from ‘quite bad’ and ‘hard to get’ to ‘If I know the context and if it is said with very prominent pauses and intonation it is kind of okay’. Assuming for the sake of the argument that such an example or a similar one could be found, we would have the desired remnant (ignoring head-finality for readability):

- (252) $\lambda f. [\text{rezensieren } f \text{ ([über den Tod])}]$

What the ellipsis would be anaphoric to is a CP minus a VP plus an NP minus a PP: $\mathbf{e}_{((PP \rightarrow NP) \rightarrow VP) \rightarrow CP}$.

As has become clear, it is difficult to construct the necessary examples while making sure that the sequence of syntactic operations and the information structure of the sentence are licit. The sentences are hard to judge and might only be relevant for approaches using derivation trees as the data structure over which to analyse ellipsis.

Multiple parasitic ellipsis

Another way to get 3rd-order ellipsis is parasitic ellipsis, as we have seen. A 4th-order parasitic ellipsis would be one where we have e.g. a gapping construction with a remnant with a focused head; in contrast to 3rd-order ellipsis, the complement of the focused head would not be elided completely but a subconstituent of it is pronounced again. Imagine a conversation between employees of a shop that talk about the fact that their customers differ in how entitled they are.

- (253) Peter comes into the shop WITH the question WHEN the new model will arrive, but Paul comes into the shop WITHOUT even the question WHETHER it will arrive.

As we know, parasitic ellipsis is possible after *whether* or *without*.

- (254) Peter comes into the shop WITH the question when the new model will arrive, but Paul WITHOUT:

The question then is how acceptable nested parasitic deletion is for (253), where one elides the complement of WITHOUT (i.e. *the question*) but now keeps as *parasitic remnant* its complement ('WHETHER it will arrive'):

- (255) ??Peter comes into the shop WITH the question WHEN the new model will arrive, but Paul WITHOUT even WHETHER it will arrive.

Such a sentence is quite marked, leading to similar issues as above. A careful, in-depth elicitation of such constructions is necessary to make substantiated claims about the existence of 4th-order parasitic ellipsis.

5.2 Issues for the framework

There are a couple of issues for the current ellipsis framework that need to be addressed in potential further work. One issue is that in contrast to non-contrastive ellipsis, the number of remnants in contrastive ellipsis is potentially unbounded. To see this, consider the following examples:

- (256) Mary seems like an idiot to me, and Susan like a genius to you.

For those speakers of English that accept more than two gapping remnants, this is an acceptable sentence. We can then use verbs that select for infinitival clauses to increase the number of potential gapping remnants:

- (257) ?Mary seems to me to want to seem like a hero to her husband, and

[Peter] [to you] [like a tough guy] [to his wife].

This sentence is in accord with the Major Constituent constraint and no remnants are separated by a finite clause boundary, just like (256). (257) might sound more strained than (256) which might be due to the slightly artificial meaning and the increased depth of embedding. Yet (257) does not violate any additional constraints (known to me) that (256) does not violate. To see that gapping *seem to want* is not an issue, consider the following sentences:

(258) Peter seems to want to kiss his daughter, and Paul his son.

That is, as far as grammar is concerned, both ellipses are on equal footing, differing due to external factors, like cases of multiple center embedding. In principle, there is no syntactic bound for the number of times a verb selecting infinitival clauses happens to select such a clause headed by a verb of the same type, i.e. *seem to want to seem to...* ad infinitum. We can therefore increase the number of gapping remnants each time by at least one e.g. by making the ‘experiencer’ of each *seem* (something like *to you* or *to Mary*) another remnant. In this case, the number of potential gapping remnants is unbounded. Neijt already claimed “in principle, an infinite number of gaps is possible. This number is presumably restricted only by the fixed capacity of human working memory” (Neijt 1979:94). Her examples, however, involve modifiers of the same verbal domain for which one could find a workaround by saying that they survive in bulk as one unit. This is not possible with different arguments. This can lead to a number of problems. The first would occur if one would adopt a movement approach to contrastive ellipsis remnants. This is an issue for Minimalist Grammars which assume the Shortest Move Constraint which, in short, says that there can never be more than one potential target for an attractor feature. As a quick reminder, a derivation crashes in which a probing $+wh$ -feature on a head finds two elements carrying a $-wh$ -feature. In a movement approach to contrastive ellipsis remnants one would have to assume a different feature for each element which then would lead to a non-finite set of features and therefore a non-finite lexicon.

Yet even a naive in-situ approach leads to problems of its own. In the current approach ellipsis types need to be lexically stored to express which language possesses which ellipsis type (VP-ellipsis, sluicing, etc...). Additionally, one also needs to account for ellipses with remnants. Phrases can have different means of survival, e.g. via case movement or *wh*-movement. Therefore, one would have various *vP*-ellipsis types such as $\langle d, -k, \emptyset \rangle \rightarrow \langle v, -k \rangle$ and $\langle d, -wh, \emptyset \rangle \rightarrow \langle v, -wh \rangle$ which is “not yet entirely satisfactory” (Kobelev 2015:248, fn. 19). What one would want to state is that “for any α , it

has the category” $\langle d.\alpha, \emptyset \rangle \rightarrow \langle V, \alpha \rangle$ (ibid.). The sentences discussed above worsen this issue since in contrast to different movement types their number is not bounded. In the absence of a general procedure to determine ellipsis types, one would have to assume that such a *lexicon of operations* is non-finite, containing e.g. an unbounded number of ellipsis operations with various combinations of NP, AP and PP remnants. If one wants to avoid this, a generalization over possible ellipsis types is necessary that derives them. Since the issue of unbounded remnants only occurs in contrastive ellipsis, one would need a statement that contrastive ellipsis operations (such as e_{CP}) can have an arbitrary number of arguments with the condition that they carry a feature [focus] in an appropriate theory of information structure.

Another issue is a certain sub-kind of antecedent-contained deletion (ACD). ACD is the phenomenon where an ellipsis occurs whose antecedent appears to contain the ellipsis site:

- (259) a. *Mary [_{VP} said that John did [_{VP} e]] =
b. Mary said that John did say that John did say that John...

where (b) is the meaning reconstructed for (a). However, some *prima facie* antecedent containment contexts are acceptable, as in this relative clause:

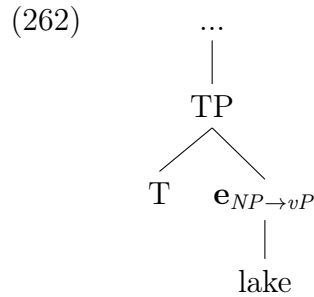
- (260) a. Mary [_{VP} investigated every case that her rival did [_{VP} e]] =
b. Mary investigated every case that her rival investigated.

As mentioned in the original article, the approach in Kobele (2015) deals easily with ACD since no issue of infinite regress arises in the first place. No resort to quantifier raising as rescuing mechanism is necessary. Under the standard assumption that the head of the relative clause moves, whether it is the relative clause pronoun, an empty operator or the actual element appearing in the higher clause, the VP-ellipsis will be a 2nd-order context, i.e. crucially missing the moving element. The antecedent that is recovered is therefore a VP minus the relevant NP, completely ignoring the offending NP. There is however a case of ACD where the identity of the moved element *matters*, namely in so-called doubly-headed ACD, as discussed e.g. in Kennedy (2001), Sauerland (2004) (Sauerland 2004:64, ex. 2, adapted):

- (261) a. *Polly visited every town that is near the lake Erik did.
b. Polly visited every town that is near the one Erik did.

If the judgements above are correct, then the ellipsis in the relative clause is sensitive to the content of its remnant, i.e. the head noun *lake*. Such *lexical content effects* cannot straightforwardly be accounted for in this approach

since it simply ignores the remnant. Standard cases of ACD work because the 2nd-order ellipsis does *not* refer to the evacuated mover, yet in the example above this is exactly what one would need.



A naive approach where what is compared for identity is the ellipsis plus all its arguments also does not work since in normal VP-ellipsis such an identity requirement would make the wrong predictions (see the well-formed ‘*Mary was kissed, and Peter was, too.*’). What is more, the lexical content effect vanishes under passivization, i.e. there is a difference between whether the evacuating movement is an instance of A or $\bar{A}$ -movement (Sauerland 2004:113, ex. 127a, adapted):

- (263) The town near the lake that was visited by vandals seems to have been ~~visited by vandals~~, as well.

In the matrix clause, the town has been visited but in the relative clause it was the lake. Sauerland explains this contrast by the differences between both movement types, depending on their reconstruction properties and the types of traces they leave behind. So far, the properties of different movement types did not play a role in this approach to ellipsis. Such a distinction would need to be implemented, leading to the question whether the general idea can be upheld that the remnant is not at least initially part of the ellipsis site.

5.3 Other potential cases of parasitic ellipsis

The cases of parasitic ellipsis that I have considered so far were of the following form: in gapping, the complement of the head of the remnant can optionally be elided as well. For the sake of completeness, I want to discuss other cases of ellipsis that appear to be ‘parasitic on’ another ellipsis, but that might differ in their behaviour from parasitic ellipsis proper as discussed in the last sections.

To stay close to the original, there might be other potential structural constellations than the head-complement relation mentioned above that allow for parasitic ellipsis. One might be that if the (highest) *specifier* of a remnant is focussed any lower element can be elided. One construction that fits this profile is left-branch extraction (LBE) under sluicing in languages that do not allow this extraction outside of ellipsis (see Ross 1969, Merchant 2001). An example is the following:

- (264) a. I know she dated someone’s sister, but I don’t know whose.
b. *I know she dated someone’s sister, but I don’t know whose she dated sister.

Next to analyses of LBE as an ‘exceptional’ movement only enabled or ameliorated by ellipsis (see Merchant 2001) there are also analysis of this phenomenon as a form of additional deletion (in the spirit of ‘distributed deletion’ in Fanselow & Ćavar (2002) for split-NP constructions in German and other languages). Such an analysis has recently been proposed in Bondarenko & Davis (2023) for Russian, a language in which this extraction is also possible in the absence of ellipsis. The sentence above would then have an analysis as sketched below:

- (265) ... but I don’t know [[whose sister]₁ ~~she dated *t*₁~~]

What appears as left-branch extraction would then be additional ellipsis within the standardly moved NP that is in some way enabled by the fact that the NP originates within a sluicing remnant. Unification with parasitic ellipsis proper, however, is unlikely, even when the different structural configuration is ignored. This is due to the fact that sluicing does license parasitic ellipsis, as discussed in 3.4.

Data that are similar to the above phenomenon are remnant degree phrases. Neijt (1979) discusses these examples but claims that the crucial ellipsis is independent of gapping (1979:112, exx. 10& 11, adapted):

- (266) The first house is 50 feet deep and the second house 60 feet.

(267) The second house is 60 feet

The same, however, is not true for German:

- (268) a. Das erste Haus ist 50 Meter hoch und das zweite Haus 60
the first house is 50 meter high and the second house 60
Meter.
meter
'The first house is 50 meters high and the second house 60
meters.'
- b. #Das erste Haus ist 50 Meter hoch und das zweite Haus ist
the first house is 50 meter high and the second house is
60 Meter.
60 meter
'The first house is 50 meters high and the second house is 60
meters.'

The ungapped sentence can only have one quite degraded meaning: it equates the house with the abstract entity that is described by the phrase *60 Meter*, but not as describing a specific property of its spatial extension.

Interestingly, in measure phrases there are even further ellipsis options. In a gapped sentence, one can also elide the unit of measurement:

- (269) a. #Das erste Haus ist 50 Meter hoch und das zweite ist 60.
the first house is 50 meter high and the second is 60
'The first house is 50 meters high and the second is 60 years
old.'
- b. Das erste Haus ist 50 Meter hoch und das zweite 60.
the first house is 50 meter high and the second 60
'The first house is 50 meters high and the second 60 meters.'
- (270) a. *Peter arbeitet 50 Meter unter dem Boden, und Paul
Peter works 50 meter under the ground, and Paul
arbeitet 60.
works 60.
'Peter works 50 meter underground, and Paul 60.'
- b. Peter arbeitet 50 Meter unter dem Boden, und Paul 60.
Peter works 50 meter under the ground, and Paul 60
'Peter works 50 meters underground, and Paul 60.'

Without gapping, such deletion only works with a few measurement units such as years for which expression without unit is idiomatic:

- (271) Maria ist 5.
 Maria is 5
 ‘Maria is five years old.’

As such, the only available but unintended reading for (269a) is that the second, anthropomorphized house is 60 years old. Structurally, this is surprising, since an element like *60* might not be a specifier of the remnant, but a specifier of its specifier. However, the structure of degree and measure phrases (and similar structures like partitives etc.) is beyond the scope of both my expertise and this thesis (for a discussion of LBE and degree phrases see Corver 1990). I list these examples simply to indicate that there are other cases where gapping appears to enable additional non-pronunciation. The survival of otherwise immovable modifiers might also receive an analysis of additional deletion under gapping:

- (272) *Sehr ist das Buch *t* interessant.
 very is the book interesting
 ‘Very, the book is interesting.’
- (273) Peter ist ein BISSCHEN verliebt, aber Paul (*ist) SEHR.
 Peter is a bit enamoured but Paul is very
 ‘Peter is a little bit in love but Paul (is) very much.’

However, whether the above examples are best analysed as exceptional extraction or additional deletion, and whether such a deletion is truly parasitic, is left for further research.

Another possible structural relation for parasitic ellipsis would be the case where the head of the remnant is focused but what is elided is an element *higher* than the head, i.e. some or all of its modifiers or specifiers.⁴⁰ One such case might be Comparative Subdeletion (e.g. Kennedy & Merchant 2000). In comparatives with attributive adjectives, comparative deletion cannot target the latter as discussed in Pinkham (1985) (Kennedy & Merchant 2000:92, ex. 7a):

- (274) *Pico wrote a more interesting novel than Brio wrote a __ play.

Yet when the phrase containing the adjective is an ellipsis remnant, i.e. if that phrase is contained in an ellipsis site, this additional deletion becomes possible (Kennedy & Merchant 2000:98, fn.7):

- (275) Pico wrote a more interesting novel than Brio a __ play.

⁴⁰Under the NP-hypothesis, that is.

This sentence means that Pico wrote a more interesting novel than Brio wrote an interesting play. In essence, the analysis in Kennedy & Merchant (2000) boils down to exceptional left-branch extraction of the AP (*interesting* in the above case) out of the containing NP so that it is outside of the later NP remnant which leads to its deletion. This adjective extraction is ungrammatical in overt syntax so they propose an amelioration effect by ellipsis, i.e. ellipsis removes the offending property of these constructions. Implementing this analysis in the current ellipsis framework would lead in any case to a 3rd-order ellipsis operation similar to the remnant movement cases. The difference to those cases, however, is that the additional movement is not independently motivated. This *exceptional destructive movement* is ungrammatical in overt syntax and has no detectable difference in meaning. That is, this analysis runs into similar problems as the destructive movement approach to parasitic ellipsis that I argued against in 4.1. As an alternative to an ellipsis amelioration account one could therefore pursue an approach in terms of parasitic deletion of attributive modifiers. Such an account, however, would have to be properly constrained to avoid unrestricted deletion of subparts of the nominal domain.

Another example of additional deletion of nominal modifiers is relative clause deletion as described in Collins (2015:57):

(276) At the party, I saw three boys who I know and one girl.

This sentence can have an interpretation where the speaker saw a girl who she knows, i.e. the relative clause modifier *who I know* from the first conjunct is recovered. The NP head of the deleted relative clause needs to be contrastively focussed. He also notes that such an ellipsis can receive a sloppy reading (2015:63):

(277) I met three freshmen who liked their professors and two sophomores.

These examples are similar to parasitic ellipsis in that recovery of unpronounced material appears to ‘depend’ on another ellipsis. Without e.g. stripping, the intended reading is very difficult or impossible to get:

(278) #I saw three boys who I know, and I saw one girl ~~that I know~~.

As mentioned above, if relative clause deletion is to be subsumed under some general phenomenon of parasitic nominal modifier deletion, research into its restrictions are necessary. In this case, a parasitic ellipsis is possible for a specific construction. But it is not the case that any modifier of a noun can be elided:

(279) #Peter saw a happy boy, and a ~~happy~~ girl, too.

As a last instance of additional deletion in the nominal domain, there is the most well-known instance of a ‘dependent’ ellipsis called determiner sharing (McCawley 1993):

- (280) a. Every musician likes Beethoven and ~~every~~ painter KO Götz.
 b. *Every musician likes Beethoven and ~~every~~ painter likes KO Götz.

(280a) is understood in a way that every painter likes KO Götz. The ellipsis of the determiner/quantifier is only possible if gapping takes place as well; without gapping, ungrammaticality ensues (280b). What the structural profile of determiner sharing depends again crucially on the assumptions one makes about the structure of the NP/DP. If D is analysed as the highest specifier of NP then it would be another instance of a nominal modifier deletion.⁴¹

If verbs select DPs, then the structural profile of determiner sharing is similar to that of p-stranding remnants. In other words, the ellipsis remnant is not a daughter of a verbal projection but a granddaughter, i.e. in a movement approach the NP ‘strands’ the D. Such a movement is not possible in overt syntax at least in English, though it is possible in certain contexts in e.g. German. For such an analysis of determiner sharing see Schwarzer (2021). Under an analysis that proposes that the cause of ellipsis survival is an independently available process there is of course nothing parasitic about determiner sharing, i.e. it would be a case where under closer inspection an apparently parasitic phenomenon is the result of the interplay of already existing operations in the grammar.

A different approach to determiner sharing is found in Ackema & Szendrői (2002). They write that determiner sharing involves a “combination of coordinate ellipsis plus a second process which we will term ‘*dependent* ellipsis’, a term meant to indicate both that the process is *parasitic* on coordinate ellipsis and targets dependents of the elided head” (2002:191, emphasis mine).⁴² They build their analysis on Williams (1997). To present a simplified version of the analysis: in gapping, the head of the second conjunct can be null. This

⁴¹In the second half of my dissertation I present a proposal where it is both the case that verbs select for NPs and D selects for that same NP as well. This would then still constitute a sort of ‘upward licensing’ in that the path proceeds downwards from (deleted) verbal spine to the N-head of its NP argument, but the path from nominal head to D is upwards.

⁴²The terms *dependent* and *parasitic* are used interchangeably here. I have argued above that there is a distinction between dependent and parasitic and use these terms differently.

head then licenses further ellipsis: the head of the argument (or the whole argument) that this null head selects can also be null which in turn enables further ellipsis in the same way (Ackema & Szendrői 2002:194, ex. 10):

- (281)
- a. John wants to decapitate Fred and Bill wants to hamstring Pierre
 - b. John wants to decapitate Fred and Bill 0 to hamstring Pierre.
 - c. John wants to decapitate Fred and Bill 0 0 hamstring Pierre.
 - d. John wants to decapitate Fred and Bill 0 0 0 Pierre.

They also note that ellipsis cannot skip heads. Ackema & Szendrői (2002) then analyse determiner sharing as an instance of one additional further ellipsis. They ultimately give up the iterative or ‘transitive’ licensing of dependent ellipsis to avoid overgeneration so that a null head that is the result of dependent ellipsis cannot itself license further ellipsis (and derive the examples above in a different way).

I present their analysis especially to highlight certain similarities to the approach used here, and how the same insights from the analysis in this thesis could be ported over other approaches.⁴³ Note that ellipsis here works similar to the iterative spreading of vowel harmony in e.g. rule-based phonology. One vowel triggers assimilation of the following vowel (under a suitable locality definition). This vowel then iteratively triggers the same process for any following vowel. In ellipsis, the spreading then proceeds from head to argument head, not over strings but over trees. In this system, parasitic ellipsis proper would be similar to spreading processes where an element that is expected to be a blocker for harmony surprisingly becomes transparent for such a process without itself undergoing harmony. Contrastively focussed heads usually block ellipsis, yet here ellipsis continues in the complement of the head while the head remains pronounced. In other words, there are special conditions under which their ‘Don’t skip heads’ rule can be circumvented, namely the locality condition of parasitic ellipsis. Yet parasitic ellipsis would still be one further instance of the underlying spreading process which is close in spirit to the ‘single ellipsis operation’ approach pursued here. Their analysis for determiner sharing itself then also shares the same spirit even though it occurs under different structural configurations (depending again on the assumed structure of the nominal domain).

⁴³Another interesting approach is the one pursued by Osborne (2008) in dependency grammars. He proposes the concept of *catena* which subsumes constituents and what is in effect 2nd-order constituents, i.e. continuous collection of nodes with potential holes. He proposes that what gapping elides is always a catena. As we have seen, under an analysis of parasitic ellipsis as a single unit one would need to go beyond even catenae, i.e. beyond 2nd-order constituents.

To summarize, there are a number of deletion processes that exhibit superficial similarity to parasitic ellipsis in that they appear to ‘depend’ on the presence of another ellipsis. Such kinds of ellipsis appear to be found especially in the nominal domain. It is unclear to which extent unification is possible or desirable. I am least convinced that LBE is similar to parasitic ellipsis. It occurs as a genuine phenomenon in languages like Russian in the absence of a licensing ellipsis on which it could be parasitic in the first place. Comparative subdeletion, relative clause deletion and determiner sharing at least share the property that they occur in a phrase that is an ellipsis remnant with a contrastively focussed head. Their locality profile differs in that the additional ellipsis appears ‘upwards’, i.e. higher than the contrastively focussed noun.

These processes differ from parasitic ellipsis in that they specifically target a specific element/construction unlike parasitic ellipsis which is blind to the category that it elides as long as it is the complement of a certain head. Determiner sharing and relative clause deletion target the elements from which they derive their name, and comparative subdeletion specifically targets attributive adjectives. Each of these phenomena has its own distinctive profile, depending to a great part on the nature of its target. It therefore appears that for now the constructions just listed should be treated as distinct from parasitic ellipsis. Further research into the mechanisms of ‘dependence’ in each phenomenon, or to what extent such a ‘dependence’ is actually necessary, is required to enable a more precise comparison.

5.4 Parasitism elsewhere

There are other phenomena in the literature that carry the label ‘parasitic’ besides parasitic gaps and parasitic ellipsis. I cannot do those phenomena full justice here. An in-depth discussion is necessary to determine how similar they are to the two constructions discussed here, whether they are truly parasitic or dependent and how this property is implemented in the theoretical approach. I want to list them nonetheless to provide a rough overview over potentially similar constructions to indicate avenues for further research.

A third parasitic process that was already mentioned above is Parasitic Agree (Corver 2021). In Dutch, degree modifiers on attributive adjectives can optionally carry inflectional morphology, depending on whether the adjective carries this inflection as well. Adjectives modifying indefinite, singular, neuter nouns do not exhibit this inflection and as a consequence, the additional inflection on the degree word is not possible (Corver 2021:188, ex. 7, 8, adapted):

- (282) a. een erg(-e) leuk-e auto
 a very-e nice-AGR car
 b. een erg(*-e) leuk huis
 a very-e nice house

Corver discusses further restrictions on this construction, such as the type of modifiers that allow this form of parasitic agree (e.g. it cannot occur on temporal, modal or evaluative modifiers). This process also appears to be iterative (Corver 2021:192, ex. 18, adapted) but it cannot skip modifiers:

- (283) a. een hel-e erg-e dur-e fiets
 a very-e much-e expensive-AGR bike
 b. *een hel-e erg dur-e auto
 a very-e much expensive-AGR car

Corver argues that this is not a PF-phenomenon based on string adjacency but a process for which a structural configuration with a specific feature constellation that is relevant for its occurrence. He proposes that Parasitic Agree happens in a Spec-Head relation. As for the meaning of this construction, he suggests that the additional *-e*-inflection is a ‘marker of expressive emphasis’. In this point parasitic agree seems to differ from parasitic gaps and parasitic ellipsis which tend to not differ from their non-parasitic counterparts in meaning (i.e. unelided sentences or sentences with pronouns instead of gaps⁴⁴).

Another phenomenon that carries the label *parasitic* is parasitic participles (see e.g. den Dikken & Hoekstra 1997, Wurmbrand 2012). In Frisian for example, a perfect auxiliary selects for a verbal element in participle form but it is optionally possible that an additional verbal element that usually appears in the infinitive is realized as a participle as well (den Dikken & Hoekstra 1997:1058, ex.3):

- (284) a. hy soe it dwaan wollen ha
 he would it do-INF want-PTC have-INF
 b. hy soe it *dien* wollen ha
 he would it do-PTC want-PTC have-INF
 ‘He would have liked to do it.’

In both cases, *ha* requires *wollen* to be a participle; the verb *dwaan* of the VP selected by *wollen* usually appears as an infinitive but can optionally

⁴⁴Modulo finer details such as the availability of sloppy readings and the impossibility of pronouns in certain contexts that are orthogonal to parasitic gaps and have to do with separate binding principles.

be realized as a participle.⁴⁵ Whether this element appears in infinitive or participle form has no effect on the semantic interpretation, i.e. this parasitic participle is semantically vacuous. It appears therefore that one auxiliary can ultimately be responsible for a participle realization of more than one verbal element. The more deeply embedded participle *dwaan* is therefore ‘parasitic’ on the higher ‘host’ participle *wollen*. Triple parasitic participles are also possible but parasitic participles cannot skip over non-participial forms which they call the connectedness requirement (den Dikken & Hoekstra 1997:1068):

- (285) a. hy soe it *dien* *kinnen* wollen ha
 he would it do-PTC have can-PTC want-PTC
 b. ??hy soe it *dien* kinne wollen ha
 he would it do-PTC can-INF want-PTC have
 He would like to be able to have done it.

Wurmbrand (2012) provides an overview of different types of parasitic participles in various languages (including the phenomenon introduced above) and a discussion of the major theoretical proposals. While all of these approaches differ in various respects what most of them have in common is the basic idea that what underlies parasitic participles is an extension of a one-to-one agreement relation to a one-to-many relation between an auxiliary and other verbal elements.

It is unclear whether a ‘dependent’ one-to-many agree/licensing relationship always justifies classifying such a phenomenon as parasitic; more fine-grained diagnostics are needed that can abstract away from the specific construction to compare across such distinct operations as ellipsis, movement and agree.

Richards (1998) mentions a number of phenomena where an otherwise ill-formed dependency is licit in the presence of a dependency of the same type. Three of them are binding phenomena (I use the binding possibilities of Dutch *zich* here to illustrate), one are parasitic gaps and the other is an amelioration of subjacency (Richards 1998:599–600, ex. 1, 4):

- (286) a. *Henk_i wees zich_i aan mij toe.
 Henk assigned self to me
 ‘Henk assigned himself to me.’
 b. Henk_i wees mij aan zichzelf_i toe.
 Henk assigned me to self-self
 ‘Henk assigned me to himself.’

⁴⁵The hierarchical structure is assumed to be as follows: [soe [ha [wollen [dien]]]].

- c. Henk_i wees zich_i aan zichzelf_i toe.
Henk assigned self to self-self
'Henk assigned himself to himself.'
- (287)
- a. *What_j do you wonder who bought t_j?
 - b. Who_i t_i wonders who bought the book?
 - c. Who_i t_i wonders who bought what_j?

In Dutch, the realization of an anaphor that is bound in its own domain is *zichzelf*, not *zich*. Yet if a clause already contains a dependency between a binder and *zichzelf*, a *zich* can also be bound bind the same binder, contrary to expectation. (287) shows that in English, overt movement cannot escape a *wh*-island, yet a covert LF-movement of *what* to achieve matrix scope does not. Richards proposes to re-envision subjacency as not being about overt versus covert movement. Instead, a similar principle as the one above is at work: the presence of a successful *wh*-extraction in (287b) (*who*) enables the otherwise illicit *wh*-extraction of the embedded *what*.

Richards proposes the principle of minimal compliance (PMC) which states roughly that if a dependency D that obeys a constraint C is present then under specific circumstances elements that are relevant to determine whether D obeys C can be ignored for another such dependency D' (Richards 1998:601).

The main issue with the PMC is that it is rather a meta-principle than a specific theoretical proposal. In other words, it can neither predict or derive for a specific phenomenon which elements can ignored and which cannot; nor is there a formal implementation of this principle. For example, parasitic gaps need to be contained in a path from host gap to mover - yet in the subjacency examples above, *what* is lower than entire span of the 'licensing' movement. Yet both are $\bar{A}$ -movement phenomena, and the PMC seems able to ignore different aspects of their restrictions for different phenomena. It is also unclear why the PMC only allows an overt movement to license a covert movement in (287b), but another over movement would still be ungrammatical. Furthermore, the question arises what the power of a mechanism like the PMC is. A mechanism that can sense certain dependencies and connect them to others needs to be properly constrained or it runs into the issues of 'rule reference' approaches discussed in the previous sections. There are alternative approaches to the data in (287) and parasitic gaps can be derived with a *single operation* approach as I demonstrate in the next half of this thesis. The binding data, however, appear to belong to the family of one-to-many relationships like the ones discussed in the preceding paragraphs.

Yet another parasitic relationship is parasitic NPI-licensing of the Dutch polarity item *heel* ('whole') (den Dikken 2002, den Dikken 2006, Hoeksema

2007). Under normal (non-parasitic) circumstances, *heel* is a universal quantifier that needs to be clausemate with a negation but - in contrast to other NPIs - it must *not* be in the scope of that negation (den Dikken 2002:39, 40, 59, ex. 6, 7&20).

- (288)
- a. dat ik die (hele) vent met niets gelukkig ka maken.
that I the whole bloke with nothing happy can make
'I can't make that bloke happy with anything at all.'
 - b. *Ik wil niet met die hele vent praten.
I want not with that whole bloke talk.
'I don't want to talk to the bloke at all.'
 - c. *Ik geloof niet dat die hele vent me kent.
I believe not that the whole bloke me knows.
'I don't believe that bloke knows me at all.'

heel is licensed by negation but not in other non-veridical contexts like questions or conditionals. There is also a parasitically licensed version of *heel* where it occurs in a configuration in which it would not be licensed on its own (*heele* is usually banned in subject position) but where it is nonetheless grammatical if another successfully licensed version of *heel* or another polarity item (*überhaupt*, 'at all') is present (den Dikken 2002:51, ex. 23)

- (289)
- a. Hij wil die student die (hele) constructie niet
he wants that student that whole construction not
uitleggen.
explain
 - b. Hij wil die (?*hele) student die constructie niet uitleggen.
 - c. Hij wil die (hele) student die hele constructie niet uitleggen.

den Dikken (2002) claims that this 'parasitic licensing' has "all the properties which one would expect of parasitism [...], including the anti-c-command condition, the connectedness requirement, and the need for parasitic licensing to be established prior to Spell-Out." (2002:54). Parasitic *heel* needs to be contained in a path from true polarity item to its licenser. It cannot occur in a position lower than the polarity item it wants to parasitize on (den Dikken 2002:54, ex. 30):

- (290)
- Ik geloof niet dat ook maar iemand die (*hele) vent kent.
I believe not that also but anyone that whole bloke knows

den Dikken compares this to the connectedness of Kayne (1983) which requires that parasitic gaps occur on the path between the real gap and the moved item.

Parasitic *heel* is less restrictive than standard *heel* in that it can also be licensed in e.g. conditionals, as long as its host polarity item is standardly licensed in such a context. For parasitic *heel* the clausemate restriction can also be obviated (den Dikken 2002:53, exx. 20&28):

- (291) a. *Ik geloof niet dat ik die hele vent ken.
 I believe not that I that whole bloke know
 ‘I don’t believe I know that bloke at all.’
 b. Ik geloof niet dat ik die hele vent ooit gezien heb.
 I believe not that I the whole bloke ever seen have
 ‘I don’t believe I have ever seen that bloke at all.’

den Dikken (2006) descriptively states the configuration in which parasitic *heel* is licensed as such: *heel* can ‘link up’ to the dependency between the negation and the host polarity item, given that the parasite is in the correct structural configuration. While he gives a formal account of the licensing and movement of standard *heele* (which I gloss over for reasons of space), there is no explicit implementation of the probing mechanism that parasitic *heel* has to carry out. It is similar to the two parasitic constructions discussed in the main text of this thesis so far in that what parasitic *heel* appears to probe for is not a specific feature but, descriptively speaking, a *dependency having been established*. In his own words: “all that [polar *heel*] cares about is that there be an S-structure dependency relationship involving polarity item licensing *that it can structurally connect to*.” (2006:168, emphasis mine).⁴⁶

Allowing to probe for the existence of such dependencies leads exactly to same issues that were discussed in the previous sections. On the other hand, it is exactly this overlap of properties between parasitic *heel* and parasitic gaps such as anti-c-command, connectedness and the unexpected obviation of otherwise strong restrictions on the licensing of *heel* that increases the confidence that this is a proper parasitic phenomenon belonging in the same class as parasitic gaps and parasitic ellipsis.

Another phenomenon that might fit the bill of genuine parasitism is parasitic scope (Barker 2007). Barker provides a compositional analysis of *same*

⁴⁶This might lead to timing issues or counter-cyclicity. Dikken assumes that “in order for a dependency between a [NEG]-specified head and a [...] PI to be able to parasitically license polar *heel*, that dependency itself must have been fully established, with all uninterpretable features erased.” (den Dikken 2006:164). This means that at a point in the derivation where probing/licensing is executed, and in a bottom-up fashion has reached a position already higher than the parasitic item, that same mechanism needs to start over again to probe from a position that is lower than the root of the subtree it has already reached. This might or might not be a problem depending on the theoretical account or definition of cyclicity.

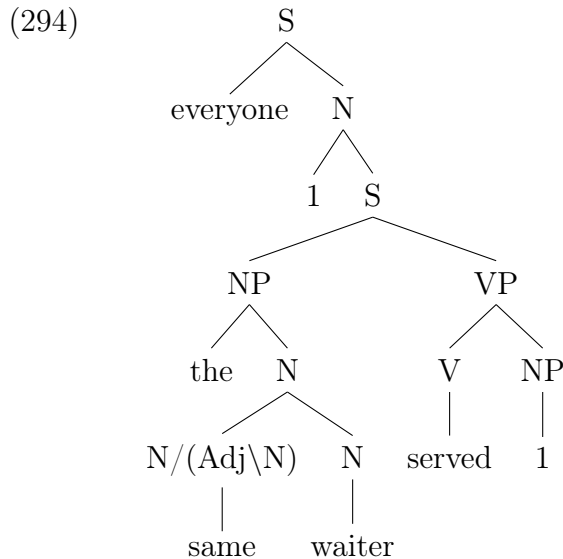
as used in the following sentences (2004:409, ex. 2):

- (292) a. The same waiter served John.
b. The same waiter served everyone.

(292a) has a deictic reading in which *the same waiter* refers to a specific, contextually salient waiter. (292b), on the other hand, has an additional, so-called ‘internal’ reading that is understood distributively. A paraphrase might be that there is a waiter for who it is the case that this waiter served every person. There is also an NP-internal use of *same* (2007:418, ex. 20):

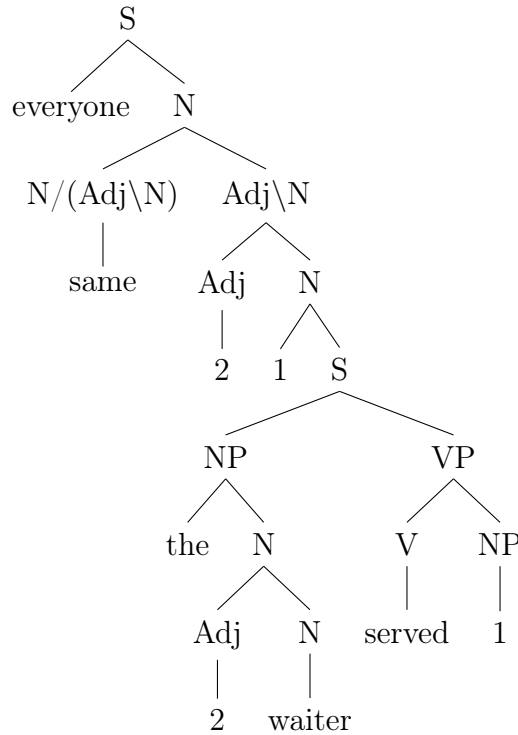
- (293) Two men with the same name are sitting in this room.

Under a deictic reading, there is a specific name like Bernd, and two men with that name are right now sitting in this room. Under the internal reading, the sentence is true as long as there is any name that is shared by two men. The reason Barker calls the internal reading of *same* parasitic scope is that it ‘depends’ on the existence of another scope-taking element in the sentence (absent in e.g. (292a)). In other words, *same* can only take scope if another element also takes scope. In the theoretical implementation, the type conflict induced by scope-taking *same* in its base position can only be resolved after an element like *everyone* has undergone quantifier raising.⁴⁷ The following trees illustrate (Barker 2007:430)



⁴⁷At least in the Heim-and-Kratzer-style sketch of an analysis that Barker provides. He ultimately models the semantics of *same* within a variant of Type-Logical Grammar.

(295)



Same tucks in between the quantifier and the landing site index that triggers λ -abstraction. There the type-clash can be resolved, resulting in the intended meaning. The type Barker proposes for *same* is arrived at as follows. In this system, attributive adjectives are of type N/N , i.e. they take a noun as an argument to their right and then become an element of type N themselves. He uses LIFT to arrive at a higher-order type for *same*, namely $N/((N/N)\backslash N)$, turning an adjective into a ‘parasitic’ scope-taking adjective. This is parallel to LIFTING a proper name of type NP to the type of a generalized quantifier. The NP *John* that can serve as argument to an intransitive verb of type $NP\backslash S$ (give me an NP , then I become a sentence) becomes $S/(NP\backslash S)$ where the functor-argument relation is now reversed. It takes something that has the type of an intransitive verb as an argument and becomes a sentence. In the tree above, Adj replaces N/N for readability. In its base position, *same* cannot compose with the noun. *Everyone* also cannot compose in its base position so it QRs. The associated λ -abstraction creates an object of type N ,⁴⁸ a position that *same* uses parasitically to QR to in a tucking-in fashion, creating an object of type $Adj\backslash N$. This creates a type that *same* can compose with, after which *everyone* composes with its argument in standard fashion.

⁴⁸I am glossing over many details here, the interested reader is referred to the original paper.

What this phenomenon has in common with the two parasitic constructions discussed above is that an operation, scope, ‘depends’ on the presence of another scope. It differs from standard scope interactions where different derivational histories lead to different interpretations in that it requires another scope-bearing element to take scope at all. Incidentally, the type required for *same* is also a *higher-order type* compared to a more standard adjective to derive the correct scopal behaviour.⁴⁹

It differs from the other genuine parasitic phenomena in that *same* is grammatical on its own (albeit with a different meaning). It also differs in that it is not an apparent overapplication of a single process: unlike in parasitic gaps where one item corresponds to two gaps or parasitic ellipsis where (at least in the present approach) one ellipsis elides more material than expected, parasitic scope does not result in e.g. an element taking scope in two positions at the same time (whatever that means). *Same* simply takes scope but requires another scope-taking element to be present. In that sense, parasitic scope might possibly more similar to a *fully dependent* process than a truly parasitic one.

The word ‘parasitic’ is also not only used in syntax (parasitic gaps) and in processes that are to varying degrees oriented towards its interfaces (scope in semantics or the *SynSem*-interface and ellipsis and agreement/licensing in syntax or the syntax/morphology/phonology interface) but also e.g. in phonology proper in the form of parasitic harmony (Steriade 1981, Jurgec 2013 and references cited therein). Parasitic harmony describes the case where vowel harmony with respect to some feature does not hold between all vowels but only between vowels that agree in another feature besides the one that the harmony process affects. An example is rounding harmony in Yowlumne Yokuts (e.g. Steriade 1981, Cole 1987). Here, rounding harmony applies only to vowels that have the same height. Since this rounding harmony ‘depends’ on another feature that does not participate in the harmony, the process is called ‘parasitic’. This is of course very different from the way the term ‘parasitic’ is used in this thesis. There is no rule reference or reference to an abstract constellation that is unusually complex compared to the contexts of other processes. There is simply reference to one more phonological feature that is relevant for a process. A simple SPE-style rule can be formulated that requires a non-round high vowel to turn into a round high vowel if preceded by such a vowel. Nothing forbids phonological rules to make reference to more than one feature per rule; many such rules arguably

⁴⁹Since the second half of the dissertation derives parasitic gaps with a sort of higher-order approach as well, this hints at a connection between higher-order structures and parasitism in general - hence the title of this thesis.

exist without requiring fundamental extensions of various frameworks. As an example, there would be nothing extraordinary about a phonological process that affects only voiced stops, but not all stops or all voiced elements. Computational arguments also back up this point: many non-local phonological process such as vowel harmony can be described by the class of Tier-based strictly local languages (TSL, Heinz, Rawal & Tanner 2011). TSL are weaker than, i.e. contained in the class of regular languages. As discussed by Burness (2022) (section 3.2), whether a harmony is ‘parasitic’ or not has no effect on its complexity, meaning that it can be described by the same machinery as non-parasitic harmony.

To conclude, the epithet ‘parasitic’ is used in various places in the literature. Sometimes this word is used as a more poetic synonym for ‘dependent’, as I would argue is the case in phonology. Other phenomena such as parasitic *heel* show striking similarities to parasitic gaps and parasitic ellipsis, such as obviation of otherwise strong restrictions on the host process and specific domain restrictions similar to parasitic gaps. The agree and licensing phenomena mentioned exhibit the core property of unexpected overapplication of the host process, but are not as straightforward to compare to a process like parasitic gaps owed to their very local nature. Better diagnostics are needed for a more comprehensive study of parasitic processes, as well as more in-depth theoretical investigations to see whether they can be derived by *single process* approaches well.

6 3rd-order structures

In the first part of the dissertation, we have seen the introduction of 3rd-order structures. There are instances where one wants to refer to structures of the form $\lambda f.\alpha f(\beta)$ as a single unit that is the target of one operation. This is the case in the interaction of remnant movement and ellipsis in a derivation tree approach. Less dependent on the type of representation, I demonstrated that parasitic ellipsis is another case where what is elided corresponds to a 3rd-order constituent. After a discussion about what place parasitism should have in grammar, I argued that parasitic ellipsis is best described as a single 3rd-order ellipsis operation that can elide a corresponding constituent all at once.

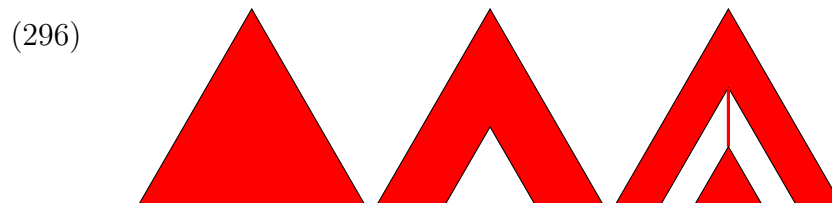
The second part of the dissertation is devoted to exploring the idea that what underlies the other instance of parasitism, parasitic gaps, is similar in nature: a single more powerful, ‘higher-order’ operation that creates the parasitic construction in one stroke. I first develop the idea that what leads to the creation of parasitic gaps is a third structure-building operation besides *merge* and *move* that I call *3rd-merge*. The parallelism to 3rd-order ellipsis (operations) is not quite exact, and the creation of host and parasitic gaps is one step more indirect than the creation of host and parasitic ellipsis; but it turns out this is a feature, not a bug since *3rd-merge* has unexpected further applications and can potentially unify apparently disparate phenomena. In section 6.1 I lay out the reasoning how to arrive at the new operation and which properties it should have. I then formalize the new operation within Minimalist Grammars in section 6.2 and provide an overview over the new rules that are added to the basic system in 2 that govern its behaviour. *3rd-merge* creates a new long-distance dependency between the head that triggers its application and its argument, ‘across’ intervening elements on top of the latter. I argue that this long-distance dependency holds between verbs and their nominal arguments. The effect of this new operation is that an additional root can grow on top of such nominal arguments. Two things can happen now. As detailed in chapter 7, one option for the additional root is to remain independent. I argue that this corresponds to the phenomenon of Horn amalgams. Another option is that the additional root is reintroduced into the main root. If the connecting element in the resulting cyclic structure, the argument of a *3rd-merge* operation, is moved, this leads to the creation of what appears, in a 1D-string, to be two gaps connected to the same element. I argue that this corresponds to the phenomenon of parasitic gaps in chapter 8. Finally I argue that a *3rd-merge* dependency can be proposed as a potential solution to the DP/NP-debate (chapter 9): even in DP-languages, verbs long-distance select for their nouns. Determiners also select for that noun,

but as a secondary root on top of the NP. DPs can therefore be viewed as ‘mini-amalgams’. In such an approach one can say both that D merges with NP *and* that verbs select for nouns. With the postulate that the purpose of *3rd-merge* is to connect the two main spines of natural language, the verbal and the nominal one, the structure of the DP, amalgams and parasitic gap share an underlying core: an operation that leaves ‘enough room’ between both spines allows the growth of a second root on top of nominals in form of a determiner in DP languages. That this root can grow further and either remains independent or is reconnected are natural consequences in such a system. It also follows from this idea that multi-root phenomena always grow on top of nouns which is borne out for parasitic gaps and Horn amalgams.

6.1 ‘3rd-order’ merge

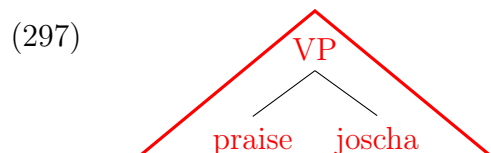
In the following I want to explore what an analogy to higher-order ellipsis would look like for structure building; put differently, where in syntax would one want structure building operations that can refer to or create 1st, 2nd and 3rd-order structures. As a disclaimer, the analogy here is not exact, and a 3rd-order structure building operation might be a misnomer, especially for the actual formal implementation. However, I believe that the analogy is close enough to permit useful comparison. At the very least it provides a good visualization to develop an intuition how such an operation should work.

Remember from the first half of the dissertation that ellipsis can elide a constituent, or a constituent-missing-a-constituent, as is the case in VP-ellipsis and VP-ellipsis with a remnant. Associated with this are 1st and 2nd-order ellipsis operation that refer to the deleted material. I argued that 3rd-order ellipsis operations also exist: sometimes, one elides a constituent, minus a surviving constituent - in which ellipsis continues. To be able to refer to such a 3rd-order constituent at once allows one to describe parasitic ellipsis as a single process. I have used triangles to visualize such (higher-order) constituents.



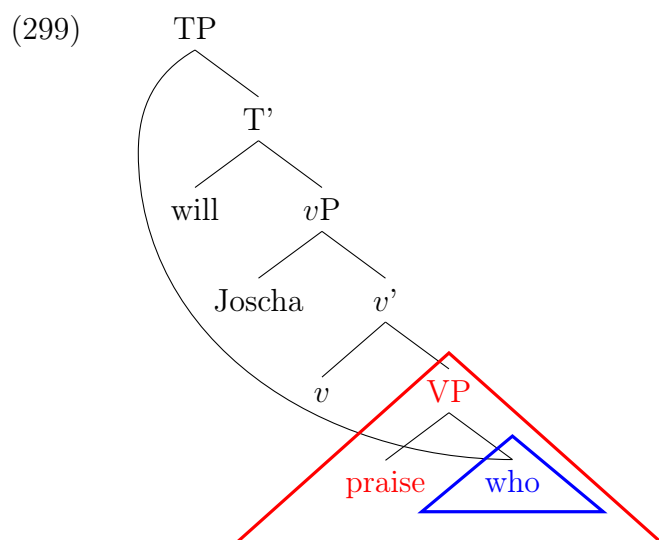
Let us see if a similar logic applies to structure building operations. The operation *merge* takes two (1st-order) constituents and combines them into

a 1st-order constituent. Remember that *merge* in MGs is a function that takes two expressions as arguments and puts out a new expression. One can therefore rewrite the tree representation in a string notation.⁵⁰



- (298) a. `merge(praise,joscha)`
 b. `[VP praise joscha]`

There is also an operation that can, in a sense, turn a constituent into a constituent-missing-a-constituent, i.e. a 2nd-order constituent with a hole (aka a trace), namely *move*:



- (300) a. `λy.move(will, ... merge(praise, y)) (who)`
 b. `[VP praise t ]`

Ellipsis of the above VP would lead to a structure that we have previously notated as `λx.[VP praise x]`. That is, movement creates a constituent with a hole but recombines the cut out element in a higher position.

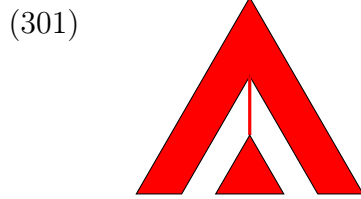
Fittingly, Mark Steedman writes about internal merge aka *move* that “[i]f anything deserves the soubriquet ‘the lost combinator,’ it would be this notional unitary combination of application and abstraction in a single perfect operator” (Steedman 2018:618).⁵¹ Of course, this is only if movement is

⁵⁰Just as one would write *hug(mary, joscha)* to express that *hug* takes these two names as arguments.

⁵¹Combinatory Categorical Grammar (CCG), the brainchild of Steedman, is of course

phrasal and not nested: remnant movement moves 2nd-order constituents, leading e.g. to 3rd-order ellipsis operations as we have seen in 3.1.

The question then is what a structure-building operation would look like that lets us refer to or create a 3rd-order constituent all at once, i.e. whose result can be visually represented like this, with the corresponding λ -expression:



- (302)
- a. $\lambda f.\text{merge-3}(\alpha, f(\beta)) \mid \lambda y.\text{merge}(\delta, y)$
 - b. $\lambda f.[_{VP} V [f(NP)]] \mid \lambda y. [D y]$

An interpretation of such a ‘3rd-order merge’ operation would be a dependency between the selector α and its argument β ‘at-a-distance’, across an intervening 2nd-order constituent in between. One could conceive of this as ‘long-distance in-situ merge’. Such an operation leads to a number of unusual properties, all of which I claim correspond to linguistic phenomena:

- there is long-distance selection by α for β
- β can be subject to further selection (f)
- the top node of f appears unconnected to α

The question is then: where in natural language would one want to state that such a long-distance in-situ selection holds between two elements? A possible application can be found in the context of the DP/NP-debate. The debate revolves around the question what the head of a nominal expression such as ‘*the cat*’ is. Is it the determiner that selects for an NP and therefore projects the category DP ($[_{DP} \text{ the } [_{NP} \text{ cat}]]$), or is the noun the head of a nominal expression, with the determiner as a specifier ($[_{NP} \text{ the}_D [_N \text{ cat}]]$)? An argument by Bruening, Dinh & Kim (2018) against the DP-hypothesis (at the time the standard assumption of the field) is that verbs can select for types of N, but they never select for types of D. In other words, verbs can e.g. require their arguments to be plural (*The cats gathered.* vs **The cat gathered.*) but there is no verb that requires that its argument have a genitive possessor, or that can only select for ‘*every NP*’ but not ‘*some NP*’.

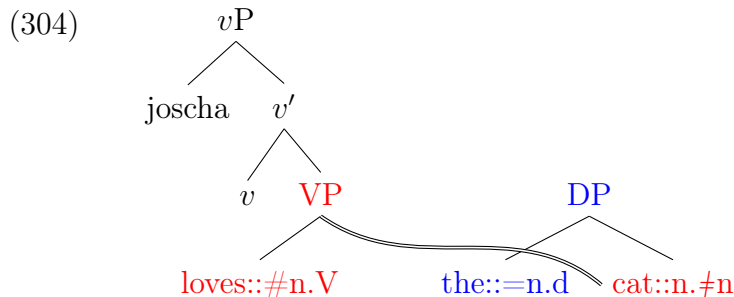
long comfortable with higher-order constituents.

On the other hand, if N-to-D head movement exists (Carstens 2017), this would favour the DP-hypothesis. I cannot do this debate full justice here.⁵² It appears that there are arguments that D is a head selecting N but at the same time there is evidence that there is a dependency between verbs and nouns. However, with the new structure building operation, one can express both at the same time.

Assume that verbs long-distance select for their nominal arguments. Since MGs are a feature-driven formalism, I propose a new positive/negative feature pair for the new operation, $\#f$ and $\neq f$, and that nouns always carry both features ($n.\neq n$). In a sentence such as

(303) Joscha loves the cat.

that means *love* long-distance selects for *cat*.



The new type of dependency between verb and noun is indicated by double lines. In addition to this dependency, we also allow the determiner to select for the noun. In terms of our (rather visual than formal) λ -expression, α and β refer to *love* and *cat*, respectively, while the intervening 2nd-order constituent **f** is the determiner *the*. In such an approach, D selects for NP *and* there is a dependency between verb and noun. As can be seen, two of the properties of the new dependency mentioned above (long-distance selection, allowing for secondary selection) can be put to use. In addition, the resulting tree now has two roots, not one. This implements the third of the desired properties, namely that **f** is connected with β but not α . If we assume that the effect of such an operation is indeed to grow a secondary root, several options present themselves. Besides making room for a determiner in the NP, the secondary root could also grow further into a full clause. It can remain an independent root. I argue that this corresponds to the phenomenon of Horn amalgams (Lakoff 1974):

⁵²For arguments and counter-arguments for and against both sides, see e.g. Glossa's special collection on the DP/NP-debate (Blümel & Holler 2021 and references therein); Salzmann (2020) is also a good overview over the arguments of both sides.

(305) Mary went to [I think it was Golm].

The underlined *Golm* here is the element (sometimes called *pivot*) that is selected both by *went to* and *was*, while the reduced *it*-cleft ‘*I think it was*’ grows on top of *Golm* in the form of an independent, secondary root.

Another possibility is for the secondary root to grow further but to be reconnected into the matrix root at some point in the derivation. This, then, leads to a cyclic graph; if the connecting element at the base of the cyclic structure were to be moved out, this would create the appearance of two gaps in a 1D-string. I argue that this corresponds to parasitic gaps:

(306) Which article₁ did she file *t*₁ [without reading *pg*₁]?

The underlined *which article* is both the argument of *file* and of *reading*. The adjunct *without reading* is the secondary root growing on top of the NP but is at some point in the derivation merged into the main root. Moving the connecting NP creates the appearance of two gaps, both of which, however, refer to the same element. These are exactly the desired properties for the parasitic gap construction. This then mirrors the result in the first part of the dissertation, namely that a more powerful operation can create the host and parasitic phenomenon in one fell swoop.

The above is merely supposed to develop the general intuition for the content of the second part of the dissertation: how I arrive at a new structure building operation beyond *merge* and *move* by employing a (loose) analogy to 1st, 2nd and 3rd-order ellipsis operations and which properties such an operation should have. I have also sketched how this fits into the overarching motif of the thesis, namely parasitism and the claim that parasitism is best viewed as the output of a single operation. In the following, I develop the properties of this additional, third form of *merge*, *3rd-merge*, more precisely and provide a formalization within Minimalist Grammars.

Throughout this thesis I will assume the postulate that only nominals can be the argument of a *3rd-merge* operation. I expand on this point below. Something that is *independent* from this hypothesis is the question whether one applies *3rd-merge* as a solution to the DP/NP-conflict as indicated above. In fact, I will put this issue aside until the very end because the derivation of amalgams and parasitic gaps does not depend on it. I first assume that *3rd-merge* takes as argument that phrase that is the highest projection in the nominal domain, be it NP or DP, depending on the language.⁵³ In the next section I provide a formalization of *3rd-merge* and give an overview over the rules that are used in the following chapters. Actual

⁵³That is one has either elements of the form *d.≠d* or *n.≠n*

in-depth discussion of these rules and the phenomena they are supposed to derive, however, follows in the subsequent chapters. I showcase the rules that deal with amalgams in chapter 7. I then move on to the centrepiece of this dissertation, the application of *3rd-merge* to parasitic gaps in chapter 8. I discuss previous approaches to parasitic gaps, especially in view of the discussion of the properties of parasitism, concluding as well that a *single operation* approach to parasitic gaps is to be preferred over approaches that postulate two distinct operations. Only after that I consider the additional option of applying *3rd-merge* to the DP/NP-conflict as sketched above in chapter 9. I argue that this move would derive the categorial restriction of parasitic gaps and Horn amalgams to nouns, indicating a fundamental connection between these three phenomena. I close with an overview over the results of this thesis and indicate avenues for further research.

6.2 Formalization

Minimalist Grammars are a feature-driven formalism, so to introduce a new type of dependency in form of an additional operation, one requires an additional positive/negative feature pair: one that triggers the operation, and another one that marks an element as the target. I use the features $\#f$ and $\neq f$, ‘plus-equals’ and ‘minus-equals’ for lack of better terms.⁵⁴

Remember the algebraic definition of Minimalist Grammars by Stabler & Keenan (2003) that was introduced in 2.1. The operations *merge* and *move* are defined by rules such as follows:

$$\frac{s :: =f\delta \quad t \cdot f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{ merge1}$$

$$\frac{s : =f\delta, \alpha_1, \dots, \alpha_k \quad t \cdot f, \iota_1, \dots, \iota_\ell}{ts : \delta, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{ merge2}$$

$$\frac{s \cdot =f\delta, \alpha_1, \dots, \alpha_k \quad t \cdot f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{ merge3}$$

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, t : -f, \alpha_{i+1}, \dots, \alpha_k}{ts : \delta, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{ move1}$$

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, t : -f\varepsilon, \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \alpha_1, \dots, \alpha_{i-1}, t : \varepsilon, \alpha_{i+1}, \dots, \alpha_k} \text{ move2}$$

⁵⁴Both signs graphically contain an =, to evoke a similarity to standard selection. $\neq f$ contains a diagonal minus (if one squints long enough), suggesting it is the negative feature, whereas $\#$ is a double-plus of sorts.

One example would be the merger of a verb with its object:

$$\frac{\text{kiss} :: =n.V \quad \text{peter} :: n}{\text{kiss peter} : V} \text{merge1}$$

Since now there is a third type of feature pair, it is important to determine their position in a feature string. For now I propose the following structure of potential lexical items:

$$(307) \quad \{=f, \#f\}^* . f . (\#f) - f^*$$

Positive *3rd-merge* features behave like positive *merge* features in that they can appear any number of times before the category feature. Negative *3rd-merge* features can only occur at most once on a lexical item; if it does, it is directly after the category feature. I also assume the following:

- (308) **Single $\#f$ postulate:** every grammar possesses only a single type of $\#f$ feature, either $\#n$ or $\#d$, depending on what the assumed highest projection in the nominal domain is.

For now this is simply driven by empirical considerations, in order to restrict the phenomenon of Horn amalgams and parasitic gaps to DP/NP. Whether this restriction needs to be relaxed is discussed in later sections. For simplicity I assume for now that N is the head of the nominal domain, as well as that all standard nouns have the feature sequence $n.\#n$ in the lexicon, i.e. they have both the category *and* the negative *3rd-merge* feature.

I define Minimalist Grammars with *3rd-merge* as follows: A Minimalist Grammar with *3rd-merge* $MG_3 = \langle \Sigma, F, Types, Lex, \mathcal{F} \rangle$, with a non-empty alphabet Σ . The set of Features F is the union of the set of features of standard MGs and $\#f$ and $\#f$, i.e. $F_{old} \cup \{\#f\} \cup \{\#f\}$. The $Types = \{::, :\}$, with $::$ for a lexical and $:$ for a derived expression and $\cdot \in \{::, :\}$ as abbreviation. There are Chains $C = \Sigma^* \times Types \times F^*$, and Expressions $E = C^+$, with the lexicon $Lex \subseteq C^+$ as a finite subset of $\Sigma^* \times \{::, :\} \times F^*$. There are three operations $\mathcal{F} = \{merge, move, 3rd-merge\}$. I use $s, t, u, v, w \in \Sigma^*$; $\beta, \gamma \in F^*$; $\delta, \varepsilon, \zeta \in F^+$, chains $\alpha_1, \dots, \alpha_k$ or $\iota_1, \dots, \iota_\ell$ or $\mu_1, \dots, \mu_m$ with $k, \ell, m \geq 0$. The SMC holds for *move*.

I use the symbol ψ for $\#f$. ω is of the form $[t : \psi\gamma, \mu_1, \dots, \mu_m]$; this represents the expression of the pivot in amalgams and parasitic gaps. $\varsigma \in \{n, c\}$, i.e. the subset of F that contains c and the highest projection of the nominal domain (n for now). This is relevant for the permissible categories of secondary roots. The meaning of most of the statements above is spelled out in detail in 2. The additions to the standard definition are elaborated on below.

I now determine how the presence of the new features interacts with the standard rules for *merge* and *move*. Let us assume for the sake of presentation that the head of the nominal domain is n and all nouns have (at least) the feature sequence $n.\neq n$. The most relevant case is the rule *merge4* that governs what happens when a head selects an argument with $\neq n$ after the category feature:

$$\frac{s :: =f\delta \quad t \cdot f\psi\gamma, \alpha_1, \dots, \alpha_k}{s : \delta, [t : \psi\gamma, \alpha_1, \dots, \alpha_k]} \text{merge4}$$

An example for this might be:

$$\frac{\text{was} :: =n.v \quad \text{cats} :: n.\neq n}{\text{was}, [\text{cats} : \neq n]} \text{merge4}$$

A head s selects for its argument t which is of the appropriate category. The feature after the category is $\neq f$. The result of this constellation is that positive and negative merge features are erased, as in a normal *merge* step. Here, however, $\neq f$ triggers that s becomes part of the chains of the resulting expression, akin to merging a moving expression (like the old *merge3* rule). The expression in brackets becomes inaccessible for the rest of the derivation until it is selected by a head from a different root via a *3rd-merge* application. I denote such expressions in brackets ω which take the form $[t : \psi\gamma, \mu_1, \dots, \mu_m]$: an expression that results from a *merge4* rule due to the presence of $\neq f$. It might or might not have movement features (γ) or moving subconstituents ($\mu_1, \dots, \mu_m$).

The purpose of a *merge4* application is to initiate the growth of a *secondary root*. To anticipate its application, it describes e.g. the growth of ‘*I think it was*’ in

(309) Jordan adores [I think it was cats].

The NP *cats* is what is called the pivot of the amalgam and corresponds to the expression denoted as ω . The reason why I call this root secondary is because this root is ‘defective’ in various ways and the root that later selects for the pivot via *3rd-merge* is dominant. This is discussed below in detail, but to mention a few examples: the pivot cannot be moved inside the secondary root, only into the matrix root, and the same goes for subextraction from the pivot which can only proceed into the latter, not the former.

Note that so far I have only allowed the growth of a secondary root on top of an element in complement position: the selector in *merge4* is of lexical type and selects its first argument. There is no corresponding hypothetical

‘*merge5*’ that, in analogy to *merge2*, merges a future pivot into specifier position. As a result, the pivot in a secondary root can only be the most deeply embedded complement of the spine. I discuss this empirical restriction in the chapters devoted to the actual phenomena.

We have not yet defined how *3rd-merge* itself works but I have shown above what happens if there is an interaction with an item that carries $\neq n$ further into the feature string. This lead to the creation of the bracketed expressions denoted ω . It is necessary now to update the former *merge* and *move* rules to account for the potential presence of ω , i.e. for the possibility that one describes a derivation *inside* a secondary root. In what follows, I use ω to denote ‘0 or 1 ω ’ to reduce clutter.

$$\begin{array}{c}
\frac{s :: = f\delta \quad t \cdot f, \omega, \alpha_1, \dots, \alpha_k}{st : \delta, \omega, \alpha_1, \dots, \alpha_k} \text{merge1} \\
\\
\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{merge2} \\
\\
\frac{s :: = f\delta \quad t \cdot f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{merge3} \\
\\
\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{merge3 spec} \\
\\
\frac{s :: = f\delta \quad t \cdot f\psi\gamma, \alpha_1, \dots, \alpha_k}{s : \delta, [t : \psi\gamma, \alpha_1, \dots, \alpha_k]} \text{merge4} \\
\\
\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f, \alpha_{i+1}, \dots, \alpha_k}{us : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{move1} \\
\\
\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f\varepsilon, \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : \varepsilon, \alpha_{i+1}, \dots, \alpha_k} \text{move2}
\end{array}$$

Most of the original rules remain largely unchanged by the potential presence of ω , I have mostly defined at what point in a derivation ω is allowed to be present already. The former *merge1* proceeds as usual, independently of whether the argument contains ω or not. The selector in *merge1* cannot yet contain ω since it is a lexical head. In *merge2*, the selector can contain ω , without any relevant change to its previous version. Note though that I have *not* allowed for the presence of ω *in the selectee*. If this were allowed, this would enable the presence of more than one expression of type ω in a chain. I want to avoid this for now. One reason for this is that it is difficult to keep

track of the original linear position of each ω in the secondary root, at least in the current approach. One would need to mix the phonological string to the left of the type indicator with the presence of elements like ω which still has unchecked features. This is avoided if there is only a single pivot: due to the fact that *merge4* is only defined for complement position, we know that the pivot is always the most deeply embedded complement of the clausal spine.

The ban of ω (or $\neq f$ more generally) in specifier position is quite pervasive. This is also the case for the former *merge3* rules that deal with merging a moving expression. The type of the selector in the original *merge3* rule is given as $\cdot$, as an abbreviation for merging into complement ($::$) or specifier ($::$) position. This makes no difference since the moving expression is not linearized in either place. Here, I have split the two versions of the new *merge3* for precision. In case of a selecting head that comes straight out of the lexicon ($::$), there can of course not be ω already as part of its chain (nor can it have any chain in general). For the selector in a derived expression, this is possible, as I have indicated in *merge3 spec*. I have again not allowed for the *selectee* in the specifier version *merge3 spec* to carry ω since, again, this would lead to potentially multiple ω s, if e.g. the selector also contains an ω .⁵⁵

Lastly, the two *move* rules are updated trivially: ω can be part of the chain but is completely inaccessible; movement proceeds unaffected by the presence of pivots in an expression.

I now turn to the rules that describe a basic application of *3rd-merge*. These are the cases where *no* additional structure has grown on top of the phrase carrying $\neq f$.

$$\frac{s :: \neq f \delta \quad t \cdot f \cdot \neq f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1}$$

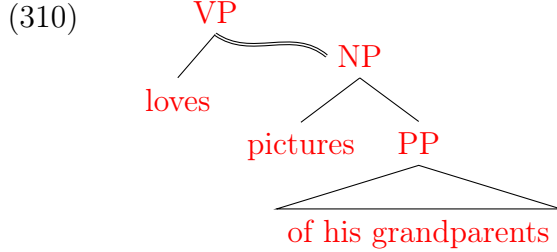
$$\frac{s : \neq f \delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f \cdot \neq f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{3merge-2}$$

$$\frac{s \cdot \neq f \delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f \cdot \neq f \varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{3merge-3}$$

3merge-1 and *3merge-2* describe merging such a phrase into complement and specifier position, respectively. The following is an example:

⁵⁵I have also disallowed ω in the selectee in complement position in *merge3*. Such a rule is undesirable, even if the issue of multiple ω s is dealt with. Such a rule is ‘unphysical’ since it leads to constructions that are to the best of my knowledge not attested and should be excluded for various reasons. I discuss this in chapter 7 under the term ‘distributed remnant movement’.

$$\frac{\text{loves} :: \#n.\#n.v \quad \text{pictures} : n.\#n, \text{ of his grandparents} : -top}{\text{loves pictures} : \#n.v, \text{ of his grandparents} : -top} \text{3merge-1}$$



The NP⁵⁶ *picture of his grandparents* is the designated argument of the verb *love*. The complement of the NP, *of his grandparents*, will be topicalized later in the derivation, so it is part of the chain of the expression on the right side. The verb *love* selects for its nominal argument via $\#n$. I postulated this to always be the case above. As a result, *love* and *pictures* is merged, while the element in the chain of the argument become part of the chain of the resulting expression. Since no additional root has grown on top of the NP in a previous part of the derivation, the outcome appears on the surface to be the same as the standard *merge* step.

Note that both n and $\#n$ are erased in the result. I will address this issue later, but to present a short preview of the reasoning behind it: the feature c of a full clausal secondary root vanishes in a similar manner, to indicate that it is a complete expression at the point of merging it with the matrix root. I treat n here analogously: it is a secondary root, albeit a trivial one since its height is equal to the height of the foot of the root, a null root so to speak. This is different when verbs *3rd-merge* with NP-arguments but determiners grow as additional root, a point I return to in 9. For now it is only relevant that both n and $\#n$ are erased.

3merge-3 is the equivalent of *merge3* in that it describes the processes in *3merge-1* and *3merge-2* above with the additional factor that the argument is a moving expression. The only difference to the other two rules is that it becomes part of the chain of the resulting expression. I have taken the liberty to use *3merge-3* as a unification of two rules, namely whether the merger happens in complement or specifier position. In case of the complement version, the selector (s) can by definition not contain ω or $\alpha_1, \dots, \alpha_k$; this is only possible (but not necessary) in the specifier version.

This concludes the overview of the basic rules of *3rd-merge*. In the rest of this section I provide the remaining rules that govern the behaviour of

⁵⁶Remember that we take NP to be the highest projection in the nominal domain for this and the next sections for simplicity.

amalgams and parasitic gaps. I neither describe or explain them in detail. This is only done in the respective chapters for each of the phenomena. The purpose of the next paragraphs is therefore simply to present all the rules in one place so that the reader can return to this section for a complete overview. I also include a list of the resulting empirical generalizations, i.e. the possibilities and process interactions that the rule system allows and forbids. These also come without justification or examples; a more in-depth discussion is again relegated to the respective sections.

The following rules describe the process of *3rd-merging* complete secondary clauses into complement (*3merge-1'*) and specifier position (*3merge-2'*). This can only take place if the only unchecked feature in the argument, i.e. the whole secondary root, is c (in addition to the unchecked features that reside in ω which is inaccessible for that root). In other words, one can only amalgamate a secondary root if this root is a complete expression save the pivot. The feature c is ‘erased’ in the resulting expression, not as a result of its checking but to indicate that this root does not play a further role for the purposes of this derivation.⁵⁷

$$\frac{s :: \#f\delta \quad t : c, [u : \#f, \alpha_1, \dots, \alpha_k]}{stu : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1'}$$

$$\frac{s : \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : c, [u : \#f, \iota_1, \dots, \iota_\ell]}{tus : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{3merge-2'}$$

The rules allow subextraction out of ω into the matrix root, but there are no rules that allow either extraction of ω itself, nor subextraction out of ω , into the secondary root. Note also that it *is* possible to amalgamate into specifier position (*3merge-2'*) if the secondary root is complete. I have only excluded *active* $\#f$ to occur in specifier position.

The following are the rules that govern parasitic gaps. *3merge-4* describes the case where the head of ω carries a movement feature. The root it is part of

⁵⁷A reviewer (Thomas Graf) comments that the workings of *3rd-merge* could be split into a *merge* and *move* operation; the feature $\#f$ on a lexical item would then instead select for the CP ($=c$) and covertly attract the item carrying $\#f$ via movement ($-f$). This does not work since $=c$ establishes a c-command relation between the selecting item and the selected clause (also setting aside the fact that verbs in question do not otherwise select for a CP and that pivots to do no exhibit movement properties). As discussed in the next chapter, there is ample evidence that no such relationship holds between the selector and the clause, only between the selector and the item carrying $\#f$. It is vitally important that *3rd-merge* does *not* establish any relation to the top of the secondary root. A decomposition into already existing operations is therefore not possible since it would require a new version of *merge* that selects an argument but establishes no syntactic relation otherwise.

must be a complete expression, save either a c or an n/d (abbreviated by ς), to allow for clausal parasitic gap hosts (e.g. relative clauses or adjuncts) and subject parasitic gaps. I leave open the possibility that the head of this root also carries a movement feature (γ), e.g. for potential further extraposition of adjuncts/relative clauses. As a result, a bracketed expression similar but distinct from ω becomes part of the chain of the selector. Note that in contrast to the rules for amalgams, the category feature of this expression is not erased.

$$\frac{s \cdot \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : \varsigma\gamma, [u : \#f\varepsilon, \iota_1, \dots, \iota_\ell]}{s : \delta, \omega, [t : \varsigma\gamma, u : \varepsilon], \iota_1, \dots, \iota_\ell, \alpha_1, \dots, \alpha_k} \text{3merge-4}$$

$$\frac{s : =_\varsigma\delta, \omega, [t : \varsigma, u : \varepsilon], \alpha_1, \dots, \alpha_k}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, u : \varepsilon} \text{chain-merge1}$$

$$\frac{s : =_\varsigma\delta, \omega, [t : \varsigma\varepsilon, u : \zeta], \alpha_1, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, u : \zeta} \text{chain-merge2}$$

$$\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : -f\varepsilon], \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : \varepsilon], \alpha_{i+1}, \dots, \alpha_k} \text{move3}$$

The head of the bracketed expression (i.e. the head that formerly carried $\#f$) can carry more than one movement feature, to allow e.g. for A-movement that precedes the crucial $\bar{A}$ -movement that creates both host and parasitic gap. This is described by *move3*. The rules *chain-merge1/2* describe the process where the matrix root selects for the secondary root that has grown on top of the former ω (for this reason the category of the secondary root has not been erased). This creates the cyclic structure that will lead to the appearance of two gaps in a 1D-string if the connecting element is moved. *chain-merge1* describes merging this root into its final position (i.e. no movement feature after the category), while *chain-merge2* describes merging the root that moves later in the derivation. In both cases, the head formerly carrying $\#f$ becomes a normal member of the chain in the resulting expression, enforcing the anti-c-command property of parasitic gaps since it will move to a position higher than the merge site of its host.

The following generalizations fall out from these rules:

- **complement-only pivots:** the pivot is always the most deeply embedded complement of the spine in the host clause for both amalgams and parasitic gaps

- **ban on $\neq f$ in specifiers** - no expression with active $\neq f$ can occur in a specifier position⁵⁸
- no derivation contains more than one ω
- there is no root-distributed remnant movement
- neither ω nor its subpart can be extracted into the secondary root
- subextraction into the matrix root is possible, while extraction of ω results in parasitic gaps
- single active root: the approach enforces a derivational timing since a secondary root can only be introduced into a structure if it is a complete object save the pivot, and no further operation applies to this secondary root afterwards

This concludes the introduction of Minimalist Grammars with *3rd-merge* and the overview of the rules that will be used in the next chapters. I provide the rules in their totality on the following page.

⁵⁸This does not exclude the possibility of a secondary root in specifier position, however. It is possible via *3merge-2'* to have e.g. subject amalgams.

$$\begin{array}{c}
\frac{s :: = f\delta \quad t \cdot f, \omega, \alpha_1, \dots, \alpha_k}{st : \delta, \omega, \alpha_1, \dots, \alpha_k} \text{ merge1} \\
\\
\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{ merge2} \\
\\
\frac{s :: = f\delta \quad t \cdot f\varepsilon, \alpha_1, \dots, \alpha_k}{s : \delta, \alpha_1, \dots, \alpha_k, t : \varepsilon} \text{ merge3} \\
\\
\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f\varepsilon, \mu_1, \dots, \mu_m}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \mu_1, \dots, \mu_m} \text{ merge3 spec} \\
\\
\frac{s :: = f\delta \quad t \cdot f\psi\gamma, \alpha_1, \dots, \alpha_k}{s : \delta, [t : \psi\gamma, \alpha_1, \dots, \alpha_k]} \text{ merge4} \\
\\
\frac{s : + f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f, \alpha_{i+1}, \dots, \alpha_k}{us : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{ move1} \\
\\
\frac{s : + f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f\varepsilon, \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : \varepsilon, \alpha_{i+1}, \dots, \alpha_k} \text{ move2} \\
\\
\frac{s :: \# f\delta \quad t \cdot f.\#f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{ 3merge-1} \\
\\
\frac{s : \# f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f.\#f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{ 3merge-2} \\
\\
\frac{s \cdot \# f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f.\#f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{ 3merge-3} \\
\\
\frac{s :: \# f\delta \quad t : c, [u : \#f, \alpha_1, \dots, \alpha_k]}{stu : \delta, \alpha_1, \dots, \alpha_k} \text{ 3merge-1'} \\
\\
\frac{s : \# f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : c, [u : \#f, \iota_1, \dots, \iota_\ell]}{tus : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{ 3merge-2'} \\
\\
\frac{s \cdot \# f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : \varsigma\gamma, [u : \#f\varepsilon, \iota_1, \dots, \iota_\ell]}{s : \delta, \omega, [t : \varsigma\gamma, u : \varepsilon], \iota_1, \dots, \iota_\ell, \alpha_1, \dots, \alpha_k} \text{ 3merge-4} \\
\\
\frac{s : = \varsigma\delta, \omega, [t : \varsigma, u : \varepsilon], \alpha_1, \dots, \alpha_k}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, u : \varepsilon} \text{ chain-merge1} \\
\\
\frac{s : = \varsigma\delta, \omega, [t : \varsigma\varepsilon, u : \zeta], \alpha_1, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, u : \zeta} \text{ chain-merge2} \\
\\
\frac{s : + f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : -f\varepsilon], \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : \varepsilon], \alpha_{i+1}, \dots, \alpha_k} \text{ move3}
\end{array}$$

7 Amalgams

I showed above that *3rd-merge* allows for tree structures with multiple roots. I have proposed the following typology: either this additional root⁵⁹ remains independent, or the additional root is reintroduced into what I call the *matrix root* which, via movement of the connecting element, leads to parasitic gaps. This chapter is concerned with the option of independent secondary roots. The main focus of this thesis are parasitic constructions, but I do not want to ignore the analytical possibilities that independent secondary roots open up. I therefore deal with such phenomena here as an excursion before turning to parasitic gaps.

7.1 Empirical landscape

Phenomena that have been claimed to require multi-rooted trees for their description do in fact exist. One example for such a construction are amalgams. Originally described by Lakoff (1974), he claims there are (at least) two different varieties, Horn amalgams and Andrews amalgams, after the people who suggested these constructions to him:

(311) Ville went to [I think it was Golm].

(312) Zvezdana read [I don't know how many queer novels.]

(311) and (312) are examples for Horn amalgams and Andrews amalgams, respectively. The challenge associated with these construction is that the so-called *pivot* (underlined here) appears to play two roles in two different clauses simultaneously. *Golm* is both the argument of *went to* and *was*, i.e. (311) expresses both of the following assertions:

- (313) a. Ville went to Golm.
b. I think it was Golm.

Andrews-amalgams are a bit more complicated. In spirit, a similar thing happens here, but disentangling the sentences in a parallel fashion does not yield completely appropriate ‘sub-sentences’:

- (314) a. #Zvezdana read how many queer novels.
b. I don't know how many queer novels.

Rather, a more accurate depiction might be:

⁵⁹I use the term ‘root’ here as *pars pro toto* for the whole single-rooted subtree that is dominated by that root in a multi-rooted tree.

- (315) a. Zvezdana read a number of queer novels.
 b. I don't know how many queer novels ~~she read~~.

That is, (315a) is not understood as an in-situ or echo question but as an assertion about an unspecified number of novels, while (315b) is understood as containing an elided version of (315a), specifying the pivot more precisely.

This chapter only deals with Horn amalgams. The practical reason for this is that the main focus of this dissertation is the phenomenon of parasitism. Independent secondary roots are a consequence of the operation just outlined, so it is relevant to discuss these data. Horn amalgams are easy to accommodate into the system set up so far. However, Andrews-amalgams differ from Horn amalgams in a number of respects, among them an apparent movement of their pivot inside the secondary root as well as ellipsis of additional material. To argue for a unified account of both types of amalgams would lead too far afield. I therefore restrict myself to Horn amalgams and discuss at the end what extensions to the formalism are necessary to also account for Andrews amalgams.

The most recent and most extensive discussion of amalgams (to my knowledge) can be found in Kluck (2011). In what follows I provide a condensed summary of her overview of Horn amalgams. As indicated above, (311), repeated here, is an example of a Horn amalgam:

- (316) Ville went to [I think it was Golm].

I call the clause '*Ville went to Golm*' the matrix root or matrix sentence, and '*I think it was Golm*' the secondary root (which Kluck calls the interrupting clause, IC). The element connecting both structures, *Golm*, is called the *pivot* (or content kernel in her terminology).

The question that amalgams raise is how the bracketed clause is connected with respect to the rest of the clause. The verb *went to* does not select CPs:

- (317) *Ville went to [_{CP} that it was a nice city].

Instead, *went to* selects nouns, which is also how this sentence is understood: *Golm* is still the object of *went to*. Kluck (2011:ch.3) presents a battery of diagnostics to show two things: 1) the bracketed clause is indeed a root clause, not a subordinated clause and 2) it is distributed as if it had the category of the pivot, i.e. '*I think it was Golm*' appears in those environments where one expects *Golm* on its own to appear, so the former behaves as a whole as if it were an NP-argument to *went to*. In what follows I provide a short overview of some of the diagnostics that Kluck uses. The interested reader is referred to the third chapter of her dissertation for a more in-depth discussion.

One piece of evidence that the secondary root is indeed a root clause is word order in German and Dutch. Declarative main clauses in both languages typically display verb-second (V2) word order while embedded clauses are usually verb-final. The main verb of the secondary root has to be V2 and cannot occur clause-finally (Kluck (2011:52, ex. 6)):

- (318) a. Bill heeft het was waarschijnlijk Bea gekust.
 Bill has it was probably Bea kissed
 ‘Bill kissed it was probably Bea.’
 b. *Bill heeft het waarschijnlijk Bea was gekust.
 Bill has it probably Bea was kissed.

She also argues that the secondary root can have independent illocutionary force. The secondary root allows for speaker-oriented adverbs (Kluck 2011:65, ex. 43):

- (319) Bill stole [{frankly/seriously/honestly}, I thought it was just a dollar] from his grandma.

Matrix and secondary root can also differ in illocutionary force. The following example presumably constitutes a Horn amalgam with an interrogative secondary root and a declarative matrix root (Kluck 2011:68, ex. 56, adapted):

- (320) Bob found [was it a Stradivarius?] in his attic.

Evidence against a relative clause analysis, but for the fact that the secondary root is distributed as if it were of the category of its pivot, are extraposition data (Kluck 2011:71–72, exx. 67–69):

- (321) Ik kan me [_{DP} de film] herinneren.
 I can REFL the movie remember
 ‘I can remember the movie.’
 (322) *Ik kan me herinneren [_{DP} de film].
 I can REFL remember the movie
 (323) a. Bob kan zich [ik geloof dat het Bea was] nog goed
 Bob can REFL I believe that it Bea was still well
 herinneren.
 remember.
 ‘Bob can still remember I believe it was Bea (very) well.’
 b. *Bob kan zich nog goed herinneren [ik geloof dat het Bea
 Bob can REFL still well remember I believe that it Bea

was.]
was

In other words, where *de film* can(not) occur, the secondary root can(not) occur. Note, though, that languages that do allow Heavy NP Shift of eponymous NPs also allow extraposition of heavy NP-pivot+secondary root:

- (324) a. Anton discussed yesterday an extremely long and complicated article on Artificial Intelligence.
b. Anton discussed yesterday [I think it was an extremely long and complicated article on Artificial Intelligence].

Further evidence for the independence of the secondary root comes from binding and thus c-command data. Subjects in the matrix root can bind reflexives in the pivot (Kluck 2011:95, ex. 156):

- (325) De kat₁ joeg op [ik vermoed dat het een schaduw van
the cat chased on I suppose that it a shadow of
zichzelf₁ was].
SE-self was
'The cat chased I suppose it was a shadow of himself.'

This suggests that pronoun and binder are *not* separated by a clause boundary, which might be the case in theories that propose that the bracketed clause is integrated into the main clause as e.g. a relative clause. Other instances of binding yield the same result:

- (326) They₁ compensated [I think you can call it each others'₁ weaknesses].
(327) With this move, she₁ hurt [I honestly think it was herself₁].

Quantifiers yield similar results (Kluck 2011:97, her ex. 163, 164, 166):

- (328) No professor₁ believed the gossip about [I think it was his₁ mistress].
(329) *No professor₁ taught [he₁/his₁ students claimed it was a boring class].
(330) Bea hit [no professor₁ could have known it was his₁ mistress].

The first example shows that the quantifier in the matrix and the pronoun in the pivot can covary. (329) shows that this is not possible across the board. The only part of the secondary root that the quantifier has access to is the pivot; the subject of that clause is invisible. This is difficult to explain under a view where the bracketed portion in (329) is structurally fully integrated. Finally, (330) shows that a pivot can also be bound by an element higher

up in the secondary root. That is, both elements that appear higher in each respective clause can bind the pivot and therefore appear to c-command it, but apparently they do not c-command each other. This follows trivially in an analysis that postulates two independent roots that connect at the pivot.

Condition B shows similar results:

- (331) *Mary₁ loves [I think it was her₁].

The bracketed expression does *not* count as intervening CP which would make a bound reading of *her* possible. Other pronouns in non-pivot position in the secondary root can be bound (Kluck 2011:100, ex. 181):

- (332) The professor₁ was kissing, [the many rumors about him₁ suggested it was Bea].

The following sentences illustrate Condition C effects (Kluck 2011:101, ex. 189-191):

- (333) *He₁ cites [I think it was the professor₁] primarily.
 (334) *The professor kissed, [she₁ claimed it was Bea₁].
 (335) He₁ had been kissing, [the professor₁ finally admitted it was Bea].

The ungrammaticality of (333) and (334) show that the subject of matrix and secondary root clauses both c-command the pivot. (335) on the other hand shows that both elements do not c-command each other.

To summarize these results: there is good evidence that what I call the secondary root indeed behaves as a root clause, as shown by the V2-property and illocutionary independence. The possible positions of the secondary root appear to be determined by the possible positions of the pivot, as shown by the extraposition facts. Matrix root and secondary root appear to be inaccessible for each other *except* at the pivot, as shown by the binding data. The binding data furthermore show that the secondary root, even though it appears linearly between the verb in the matrix root and the pivot, does not count as intervening clause which is evidence for the assumption that this clause is not integrated as a subordinated clause into the matrix root but constitutes its own independent root which is connected to the main root only at the pivot.

7.2 Analysis

I now present an analysis of Horn Amalgams together with the corresponding formalization in Minimalist Grammars with *3rd-merge*. The proposal falls

under the umbrella of what Kluck calls ‘multidominance approaches’ such as e.g. van Riemsdijk (2006) and Guimarães Miranda (2004). They are called multidominance approaches because they follow the basic intuition that in amalgams the pivot is selected by both the verb in the matrix root as well as the one in the secondary root. As such, it has two mothers and is therefore directly dominated by multiple nodes. What is also assumed is that these verbs are each part of a different, independent root. In the following I present a formalization of these approaches within MGs.

Remember that I have postulated that each grammar contains only a single type of $\#f$ -feature, depending on what one assumes as the highest projection in the nominal domain. For the sake of exposition I assume $\#n$ for the rest of the chapter, and therefore the NP-hypothesis. Therefore, all nouns have the feature string $n.\#n$.⁶⁰

To put this assumption about feature strings into words, what $n.\#n$ means is simply that nouns (and only nouns) can ‘play two roles’ in a derivation, and participate in both of those dependencies. There is nothing new about this: $n.-wh$ expresses the idea that a noun can e.g. be selected as an argument, and therefore be part of a *merge*-dependency, but that it also plays a role in the specifier of a higher attracting head where it is linearized and assume additional properties e.g. with respect to binding etc. This means it is also part of a *move*-dependency. Now there is a third way to part of a dependency.

To claim that a phrase is directly dominated by two different nodes is to claim that it plays two different roles for two distinct selectors. I have decided against using the same type of feature for these two different roles for a number of reasons. The main reason is that I want to argue that a *3rd-merge* dependency is a *different type of dependency* than a *merge* dependency. As I demonstrate in this and the coming chapter, there is good reason to assume this. The root connecting with the pivot via n and the root using $\#n$ are not equal (i.e. not ‘parallel’ as e.g. in parallel merge, see Citko 2005). There are various **asymmetries** between both roots in that they allow or disallow different things. I want to argue that the root on top of n is *deficient* compared to the root on top of $\#n$. For this reason I have decided against the option of simply relaxing resource sensitivity, i.e. against the option that the features (the ‘resources’) could be used more than once.⁶¹ Movement is not ‘just’ a relaxation of resource sensitivity in that a merged phrase can be merged again in another position, but a new dependency in its own right, with its own idiosyncratic properties and restrictions. This is

⁶⁰Nothing here hinges on this, one can equally well assume $\#d$ and $d.\#d$ on all D-elements without loss of generality.

⁶¹Sideways movement in the formalization by Stabler (2006) is such a system but with quite different properties from the one presented here.

not to say that there are no similarities between *merge* and *move*. But it is nonetheless insightful to study its properties on their own. For the same reason I treat the ‘new role’ that a phrase can play in the phenomena to be discussed not as just another standard merge step, but as a genuinely new and different dependency. As in the case of *move*, this makes it easier to study its properties and restrictions on their own and to disentangle the possibilities of *merge* from those of *3rd-merge*.

The introduction of the workings of *3rd-merge* in 6.2 started with the rule *merge4*, repeated here.

$$\frac{s :: =f\delta \quad t \cdot f\psi\gamma, \alpha_1, \dots, \alpha_k}{s : \delta, [t : \psi\gamma, \alpha_1, \dots, \alpha_k]} \text{merge4}$$

This rule describes initiation of the growth of the *secondary root* on top of the pivot, e.g.:

$$\frac{\text{was} :: =n.V \quad \text{cats} : n.\neq n}{\text{was} : V, [\text{cats} : \neq n]} \text{merge4}$$

The rule *merge4* states that the growth of the secondary root is triggered by an application of *merge*. Conversely, it will be an application of *3rd-merge* that leads to the integration of this secondary root into the matrix root.

To describe the rule *merge4* in words: the lexical (‘::’, i.e. has not selected any arguments yet) head *s* wants to merge something of category *f* ($=f$) into its complement position. There are other features after $=f$ (δ) in the feature string, namely at least its category. Its argument, *t*, has the appropriate category *f*, that is followed by a feature of type $\neq f$ and 0 or more other features (γ) such as potential movement features. It might also be that the constituent whose head is *t* has moving subexpressions, e.g.:

$$\frac{\text{was} :: =n.V \quad \text{pictures} : n.\neq n, \text{ of his grandparents} : -top}{\text{was} : V, [\text{pictures} : \neq n, \text{ of his grandparents} : -top]} \text{merge4}$$

Here the complement of *pictures* carries a topicalization feature.

In *merge4*, the type of *t* is underspecified ($\cdot$), it can either be $::$ or $:$, depending on whether it has itself selected arguments already or not. The result of *merge4* is that merge and category feature are erased and the expression headed by *t* - the pivot - is bracketed off and becomes part of the chain of the resulting expression (whether it has chains $\alpha_1, \dots, \alpha_k$ itself or not). The pivot and any of its subconstituents are inaccessible for any of the operations applying to this specific root, as will become clear when discussing the rest of the rules.

As mentioned above, *merge4* is only defined for complement position (only type $::$ for s), there is no equivalent *merge5* for s merging such an expression into its specifier position. For Horn amalgams, this appears to be correct. The standard cases of Horn amalgams all appear to involve constructions of the form ‘*It was/appeared to be/... X*’, with X the most deeply embedded complement of the spine.

I have updated all former *merge* and *move* rules in standard MGs to allow for the presence of a pivot ω , i.e. a member of the chain of the form $[t : \psi\gamma, \alpha_1, \dots, \alpha_k]$, with a number of important exceptions in certain configurations. If you remove any mention of ω in the following rules, they collapse to the standard *move* and *merge* rules of original MGs.

$$\frac{s :: = f\delta \quad t \cdot f, \omega, \alpha_1, \dots, \alpha_k}{st : \delta, \omega, \alpha_1, \dots, \alpha_k} \text{merge1}$$

$$\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{merge2}$$

$$\frac{s :: = f\delta \quad t \cdot f\varepsilon, \alpha_1, \dots, \alpha_k}{s : \delta, \alpha_1, \dots, \alpha_k, t : \varepsilon} \text{merge3 comp}$$

$$\frac{s : = f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f\varepsilon, \mu_1, \dots, \mu_m}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \mu_1, \dots, \mu_m} \text{merge3 spec}$$

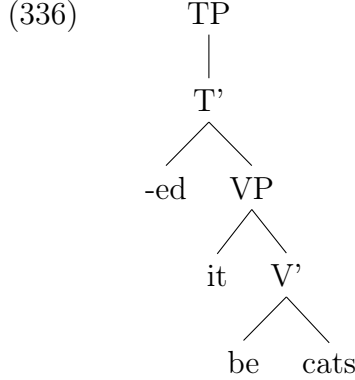
$$\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f, \alpha_{i+1}, \dots, \alpha_k}{us : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{move1}$$

$$\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : -f\varepsilon, \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, u : \varepsilon, \alpha_{i+1}, \dots, \alpha_k} \text{move2}$$

In case any actual application of these rules *does* contain ω , any *merge* or *move* rule describes the construction of a secondary root which contains an inaccessible pivot. Here is a simplified example derivation of the secondary root ‘*it was [cats]*’.⁶² I do not commit to any specific approach to such copular constructions, the only purpose of this example is to demonstrate the growth of the secondary root.

⁶²Glossing over finer details such as a distinction between V and v , where subjects are introduced, the precise make up of verbal and clausal functional projections, head movement, etc. pp.

$$\begin{array}{c}
\frac{\frac{\frac{-ed :: =V.+k.t}{\text{was} : +k.t, [\text{cats} : \neq n], \text{it} : -k} \text{merge1}}{\text{it was} : t, [\text{cats} : \neq n]} \text{move1}}{\frac{\frac{\frac{\text{it} :: n.-k}{\text{be} : =n.V, [\text{cats} : \neq n]} \text{merge3}}{\text{be} :: =n.=n.V \quad \text{cats} : n.\neq n} \text{merge4}}{\text{be} : V, \text{it} : -k, [\text{cats} : \neq n]} \text{merge1}}
\end{array}$$



In this simplified derivation, the copular *be* selects its subject *it* which merges into its specifier. The pronoun happens to also possess a movement feature. Hence, it is an application of the specifier version of the known *merge3* rule, *merge3 spec*. The only difference to the old *merge3* rule is that the phrase headed by *be* contains a pivot. Yet this is immaterial for the application of this rule, since ω cannot be affected. If *it* would not have carried a movement feature, it would have been the argument in an application of *merge2* where the potential presence of ω is equally immaterial. Later the tense head *-ed* merges the result as its complement (*merge1*) and moves *it* into its specifier to assign case (*move1*) - again applications of such rules which happen to contain an ω seemingly without a change.

I want to provide a short overview of the patterns that I do *not* allow before returning to the discussion of the patterns that *are* generated. First, the presence of ω appears as a trivial update to the pre-existing rules in the above discussion. The only non-trivial parts of the updated base rules are the places where ω can *not* occur. Obviously, it can not occur in the selector in *merge1* and the complement version of *merge3* rules, but only because such heads are unable to acquire such substructures yet since they are lexical heads. More importantly, ω is absent in the *selectee* in a *merge2* rule and the specifier version of *merge3*, *merge3 spec*. This is also parallel to the absence of a specifier version of *merge4* (or an already present ω in *merge4*). The overall effect is that it is impossible to merge into specifier position an expression that contains an active $\neq n$. Just as one might want to ban movement out

of specifiers (as in the Specifier Island Condition for MGs, see e.g. Stabler 1999, Gärtner & Michaelis 2005), I prohibit the presence of $\neq f$ in specifiers:

Prohibition on $\neq f$ in Specifiers

No ω can be present in a specifier of any expression.

There are various reasons for this. One is that if this were allowed, one can circumvent the restriction of pivots to complements of the spine introduced by *merge*₄. If growth of a secondary root can only be initiated in complement position, but the phrase containing the pivot can be merged as a specifier within that secondary root, then the secondary root contains a pivot in specifier position, even though locally that pivot is the complement of its selector. However, at least for Horn amalgams pivots seem to be restricted to complement position which is why I want to keep the formalism as restrictive as possible.

Another reason is the potential presence of multiple ω s. If ω -rich versions of the above rules were allowed, the potential presence of ω in *merge*₂ in both selector and selectee would allow for the presence of multiple ω s in the resulting expression. I prohibit this possibility here. In a sense this is a version of the Shortest Move Constraint for $\neq f$.

Single Pivot Constraint

No expression contains more than one ω .

If this were allowed, a single secondary root could provide multiple pivots. I am not aware of constructions that fit this profile. Multiple amalgams exist, but they invariably consist of a single matrix root that contains multiple secondary roots on top of various NPs. Importantly, however, each of these secondary roots is independent, concern only their own unique pivot and do not interact with the matrix root at any other point than at the pivot:

(337) Peter gave [I think it was his sister] [I think it was a present].

These constructions can already be generated. However, a relaxation of the above would allow for one root with multiple ‘docking sites’ in form of pivots:

(338) [I] think it was [Peter]

A matrix root could then select at different points in the derivation for both of these pivots, resulting in something like:

(339) I₁ heard that Mary kissed [_1 think it was Peter].

Here, *kiss* selects for the pivot *Peter* with the additional secondary root, and later *hear* selects the second pivot *I* out of the secondary root as its subject. Even worse, another possibility that this presents is those of sentences where different roots select for each of these pivots respectively, i.e. one matrix root selects for *I* above as pivot and another selects for *Peter* as a pivot, while both share the same secondary root, yet at different places. While this might be due to lack of creativity on my part, I am not aware of any linguistic construction where one can make use of any of those options. I therefore exclude this option before it lays eggs.

As a last point, another possible pattern that I rule out is the case where one merges into complement position a moving phrase that contains a pivot, i.e. *merge3 comp* where ω is allowed in the selectee. Below is a version of how *merge3* would look like in complement position which would at least avoid the issue of multiple ω s (*merge3 comp*), but it is greyed out to indicate that I will not adopt it into the overall rule system.

$$\frac{s ::= f\delta \quad t : f\varepsilon, \omega, \alpha_1, \dots, \alpha_k}{s : \delta, \omega, t : \varepsilon, \alpha_1, \dots, \alpha_k} \text{merge3 comp}$$

Imagine the type of construction this would induce: an expression of the form $t : f\varepsilon, \omega, \alpha_1, \dots, \alpha_k$ is a secondary root since it contains ω as the result of an application of an earlier *merge4*. Later, a head that selects for this expression containing a pivot carries a movement feature. Any movement of this phrase excludes its subconstituent ω since it is not affected by operations in the secondary root and will be linearized according to the demands of the matrix root in a *3rd-merge* step. This then leads to what could be called ‘root-distributed remnant movement’: t moves as a 2nd-order constituent, however *without* prior extraction of ω . Such a sentence would look roughly as follows:

- (340) *Joscha liebt [[t_1 gewesen]₂ war es glaub ich die Katze₁].
 Joscha loves been was it think I the cat
 ‘Joscha loves I think it was the cat.’

The PP [*die Katze gewesen*] is topicalized to the initial position of the secondary root; the inert and inaccessible pivot *Katze* is still part of the chains of this secondary root and not linearized with its VP. It is only a subsequent step of *3rd-merge* that determines its position later. I am not aware of any such phenomenon and therefore exclude this option. For completeness’ sake, another version of *merge3 comp* in which pivot and its moving selector are treated as one unit and moved as such also leads to ungrammatical results:

- (341) *Joscha liebt [[die Katze gewesen] war es glaub ich].
 Joscha loves the cat been was it think I
 ‘Joscha loves I think it was the cat.’

After this excursion into excluded eccentricities let me return to the patterns that the rule system can and should generate. The rules encountered so far allow the initiation of the growth of a secondary root (*merge4*), and the further business as usual growth of a tree, whether it is a secondary root (i.e. contains ω) or not, with the restrictions discussed above. What we have not seen so far is an actual application of *3rd-merge* where secondary roots are involved. *3rd-merge* combines the selector with its argument, with an additional root on top or without. Below are the cases, discussed in section 6.2, where *3rd-merge* applies e.g. to a verb and its argument NP, but without additional structure on top of the noun:

$$\frac{s :: \#f\delta \quad t \cdot f.\#f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1}$$

$$\frac{s : \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f.\#f, \iota_1, \dots, \iota_\ell}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{3merge-2}$$

$$\frac{s \cdot \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t \cdot f.\#f\varepsilon, \iota_1, \dots, \iota_\ell}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, \iota_1, \dots, \iota_\ell} \text{3merge-3}$$

Horn amalgams are those cases where an additional secondary root has grown on top of such an NP. The following rules describe these cases:

$$\frac{s :: \#f\delta \quad t : c, [u : \#f, \alpha_1, \dots, \alpha_k]}{stu : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1'}$$

$$\frac{s : \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : c, [u : \#f, \iota_1, \dots, \iota_\ell]}{tus : \delta, \omega, \alpha_1, \dots, \alpha_k, \iota_1, \dots, \iota_\ell} \text{3merge-2'}$$

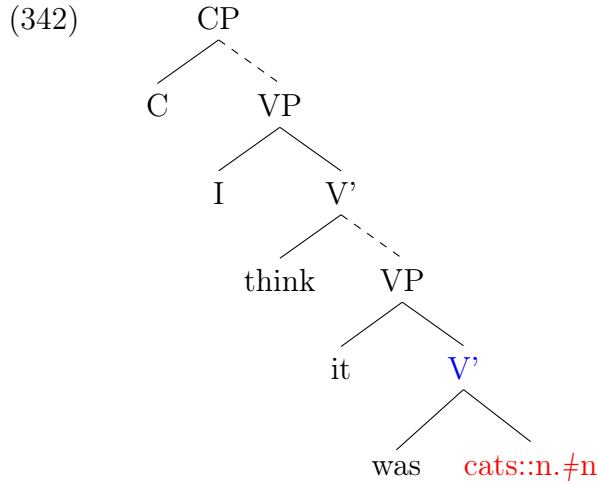
3merge-1'/2' are the counterparts to *3merge-1/2* in that they *3rd-merge* an argument ($u : \#f$) into complement (*3merge-1'*) and specifier (*3merge-2'*) position, with the difference that a secondary root ($t : c$) has grown on top of it. First, a simplified derivation of a secondary root:

$$\frac{\text{was} :: =n.=n.V \quad \text{cats} : n.\#n}{\text{was} : =n.V, [\text{cats} : \#n]} \text{merge4}$$

$$\frac{\text{it} :: n}{\text{it was} : V, [\text{cats} : \#n]} \text{merge2}$$

$$\vdots$$

$$\text{I think it was} : c, [\text{cats} : \#n]$$

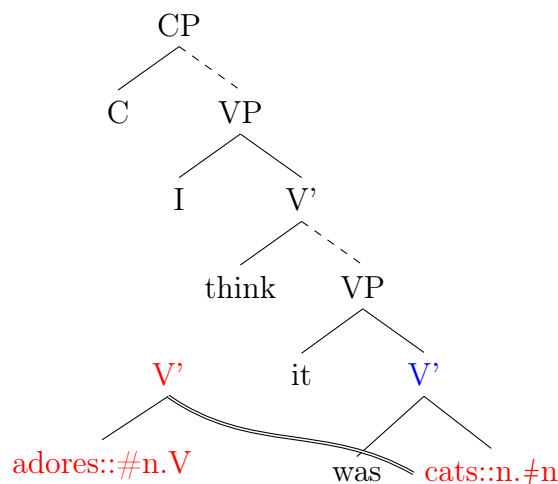


The item *was* selects for *cats* in an application of *merge₄*, turning the latter into a pivot and initiating growth of a secondary root on top of that pivot. The derivation then proceeds normally, the CP containing *it was cats* is build (omitted here) and then selected as complement by *think*. This structure then also turns into a full CP. This sentence would be a complete syntactic object were it not for the last unchecked feature $\#n$ on *cats* (besides the unchecked C of the root, of course).

This $\#n$ can only be checked if it is selected via *3rd-merge* by an element out of a different root. This step is illustrated here:

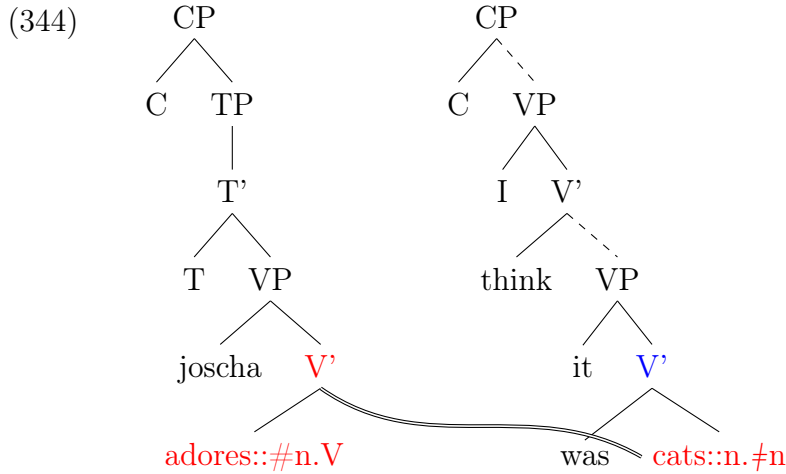
$$\begin{array}{c}
 \frac{\text{was} :: =n.=n.v \quad \text{cats} : n.\#n}{\text{was} : =n.v, [\text{cats} : \#n]} \text{merge}_4 \\
 \frac{\text{it} :: n}{\text{it was} : v, [\text{cats} : \#n]} \text{merge}_2 \\
 \vdots \\
 \frac{\text{adore} : \#n.V \quad \text{I think it was} : c, [\text{cats} : \#n]}{\text{adore I think it was cats} : \#n.v} \text{3merge-1}'
 \end{array}$$

(343)



The rule *3merge-1'* requires that its head s selects for the pivot u inside the root headed by t and linearizes them as $s-t-u$, i.e. merging the complement pivot to the right of the head s , with the amalgamated clause on top of the pivot ‘squeezed in between’. This yields the correct linearization. The pivot is linearized relative to that side of the selecting head that this head requires. The amalgam is in a position relative to the pivot that the verb in the secondary root demands. In the verb-initial language English, this results in the order *verb-secondary root-pivot*. This also correctly derives the generalization mentioned in 7.1 that the secondary root is distributed as if it were of the same type as just its pivot, since the unit $t-u$ appears exactly where u on its own would occur. As a result, the remaining feature of the pivot is erased and the secondary root is therefore syntactically inactive. Note that technically, the secondary root still possesses the unchecked c on its highest head. This is not an issue, however: previously, a syntactic object was complete if its only remaining feature is c on its root. In a multi-root object one now has to enforce that *all* its roots carry one unchecked c to count as complete. The rule above ensures that and the c can be ignored for the remaining derivation. I expand on this point below shortly.

For the current derivation, the only thing left now is to grow the matrix root into a full CP so the derivation can terminate:



Note that the existence of *3merge-2'* does allow for amalgams in specifier position, as is needed e.g. for indirect objects. The only thing that is disallowed is the presence of *unchecked* $\neq f$ -features in specifiers, not the merging of a secondary root into said position. In the same fashion, a secondary root can *itself* merge an argument with a secondary root on top. Below is such an admittedly artificial-sounding example of ‘amalgams-in-amalgams’.

- (345) ?Peter hates [I assume with [I think it can be called extreme certainty]
it is the Axiom of Choice].

If one wants to exclude recursive amalgams, one can simply forbid the presence of ω in *3merge-2'*. Since the above sentence is not obviously completely ungrammatical, I leave that option open for now.

The rules above are unusual in that they use not only variables for features but reference *a specific category*, c . This is less construction-specific than it appears at first glance. The rules as they are set up enforce the following properties: a pivot plus secondary root can only be introduced into the matrix root if (apart from the pivot itself) no other features in the secondary root are unchecked save a single c , the category of the head of that tree. This means that there can be no unfinished material in this tree, and it has to be a complete clause. Remember that an MG derivation is complete if the only feature left unchecked is the start feature c , i.e. if the tree is a CP. Since *3rd-merge* introduces multi-rooted trees, one must now determine at what point such a multi-rooted tree is complete. The natural assumption is that every single non-trivial root must be a CP. The above rules simply enforce this, hence the mention of a specific category. Note also that the feature c vanishes in the result. This is *not* due to feature checking but simply a notational way of expressing that this root is not active any more. Otherwise one would

also have to introduce feature symbols into what is standardly simply the phonological string. This is independently undesirable since as mentioned above I want to enforce that only *completed* roots can be introduced into the matrix root. I call this the *single active root condition*:

Single Active Root

A secondary root can only be introduced into the matrix root if it is a complete expression except for the features inside the pivot. At any point in the derivation, there can only be a single root that is active.

This not only avoids notational difficulties in the algebraic definition used here but also potential non-determinism that results from multiple active roots.

A further consequence of the above rules is also that they introduce an **asymmetry**. The selector in a *3rd-merge* dependency is always part of what will be the matrix root, while the root on top of the selectee is always the secondary root. On top of a difference in the operation that jump-started the growth (*merge* in secondary root, *3rd-merge* for matrix root), there are a number of empirical asymmetries that I discuss below.

The first is an asymmetry in subextraction. Note that the rules above allow for extraction of subconstituents of the pivot into the matrix root, but not into the secondary root. This is empirically justified (this subextraction possibility was to my knowledge first mentioned by van Riemsdijk 2006):

(346) Of his grandparents, Mary collected I think it was [pictures *t*].

Throughout a derivation of the secondary root, the pivot with potential sub-movers is isolated since no movement rule can access the latter. This also appears to be empirically accurate:

(347) *Mary collected [Of his grandparents, I think it was pictures *t*].

Submovers are ‘released’ from the pivot into the resulting expression after an application of *3rd-merge*, which ensures that they can only be extracted into the matrix root.

Another possibility that is excluded in the present system is that of extraction of the pivot itself (this was already noted in Kluck 2011, ch.3.4).

(348) *Chicago, Peter went to [I think it was *t*].

Since this pattern is not attested I do not want the current system to be able to produce it. Structurally, such a sentence would also be unusual. Both

Peter went to and *I think it was* are independent clauses, joined at *Chicago*, but the linking element is moved out of its connecting position, breaking its connection with the secondary root. Metaphorically speaking one could then visualize the secondary root as a piece of unconnected structure. One could say that such a structure is forbidden by positing a **ban on unconnected structure**.^{63,64} To anticipate the next chapter, I want to reserve cases of moving pivots for parasitic gaps (where unconnected trees are avoided since the parasitic gap host clause is always reintroduced into the matrix root).

I have also not formulated any rules that allow for movement of the pivot on its own within the secondary root:⁶⁵

- (349) *Joscha liebt [[die Katze] glaub ich, dass es gewesen war *t*.]
 joscha loves the cat think I that it been was
 ‘Joscha loves the cat, I think it was.’

This simplifies the system since one does not have to state different movement-features on a single item for different roots.⁶⁶

The exclusion of rules that allow movement of either a constituent containing the pivot or the pivot itself leads to the exclusion of the possibility that the pivot appears in any other position besides its base position in the secondary root. This can be rephrased as the following empirical claim:

Single secondary root position:

An element shared by two roots appears in only a single position inside the secondary root, namely its base position.

⁶³Of course the structure is not literally unconnected, especially not in a derivation tree.

⁶⁴The question is if there are some structures that could reasonably be assigned such a structure. One option might be parentheticals. To explore this, however, is beyond the scope of this thesis. One issue that would need to be addressed is how such detached structures are linearized with respect to the matrix root.

⁶⁵One could also say that movement of the pivot within the secondary root is excluded because all movement features are by postulate ordered after $\neq n$ and are therefore not accessible until the point of amalgamation. Yet that adds little explanatory depth.

⁶⁶As only possible exception to this, Kluck mentions topicalization within the secondary root in Dutch even though she claims the sentence is marginal (Kluck 2011:91, ex. 140):

- (i) Bob heeft [*Bea* dacht ik dat het was _] ten huwelijk gevraagd.
 Bob has Bea thought I that it was to marriage asked
 ‘Bob proposed to Bea I thought it was.’

The German equivalent sounds quite degraded to my ear. A possible explanation for this might be that the sentence above is rather a parenthetical. One would have to check if this works with other verbs than *I thought*, and if it underlies the same restrictions as secondary roots, e.g. if it can appear sentence-initially which secondary roots cannot. I have nothing more insightful to say about this for now and leave it as a potential counterexample.

This appears to be true at least as a strong tendency. For now, I leave an investigation into possible counterexamples, and the consequences for the formal system to accommodate such data, for further research.

To summarize, there is an asymmetry between matrix root and secondary root (i.e. the reason why I call them this way): subextraction out of the pivot is only possible into the matrix root; extraction of the pivot itself is prohibited for both options in amalgams, but in the chapter on parasitic gaps we see that in this construction it is possible, but only into the matrix root.

There is a last option which I want to mention briefly: movement of the pivot plus the secondary root as a single unit. As an example, say an NP is scrambled to a position higher in the Mittelfeld. Whether an NP can be scrambled should at least at first sight not be affected by the presence or absence of a secondary root on top of it. What one would want to say then is that the NP pivot ‘pied-pipes’ the whole secondary root. However, such a rule has a strong overlap with the rules introduced to deal with parasitic gaps; most importantly, they look very similar to rules with pivots that have left-over movement features but differ in small respects. In order to preserve the order of presentation of the two phenomenon, I delay the explanation of this rule to the chapter on parasitic gaps. For completeness, however, I list the rule here:

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, [t : c, u : -f], \alpha_{i+1}, \dots, \alpha_k}{tus : \delta, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{move1},$$

After discussing at length all the various interactions between amalgams and other grammatical operations that are *not* possible, let us close with a summary of things that the formalism presented so far *does* derive. One can grow a secondary root on top of an element carrying $\neq f$, but only in complement position (*merge4*). The growth of such a secondary root proceeds ordinarily, with the only restriction that the pivot ω is inaccessible for any structure-building operation in the secondary root. This derives a number of **asymmetries** between secondary and matrix root: movement of the pivot or one of its subconstituents is impossible into the secondary root; subextraction is only possible into the matrix root (extraction of the pivot itself into the matrix root is reserved for parasitic gaps).

3rd-merge can combine a head with its nominal argument. This also works if a secondary root has grown on top of this argument (*3merge-1’/2’*). The rules as they are formulated enforce that this can only happen if, apart from the pivot itself, the secondary root is a complete clause with no unchecked features left except the highest c . This enforces a derivational timing: any operation that needs to occur in this root needs to be completed

before it is *3rd-merged* into the matrix root. After that, the root is not active any more. As a consequence, there is only ever a single root that is active at any point in the derivation (**single active root**). In addition to this there is the additional restriction that every root can only contain a **single pivot**.

Secondary root plus pivot are linearized as a single unit to the position that the selecting head requires, in the same manner as the position of nominal arguments without a secondary root is determined. This derives the correct linearization facts as well as the observation that the secondary root is distributed as if it were of the same category as its pivot. The fact that both roots are only connected at the pivot and are structurally unrelated otherwise derives the binding data discussed in section 7.1. The system as presented derives the basic cases of Horn amalgams.

7.3 Further issues

In what follows I discuss a couple of further issues, such as the categorial restriction for pivots and a short comparison of the analysis in this formalism with previous approaches.

7.3.1 Categorial restriction

At first sight, Horn amalgams appear to be restricted to nominal pivots. Kluck (2011) cites predicative and attributive APs as pivots as degraded (ibid:74, ex. 76 & 78):

(350) ?*Bea is [I think it's blond].

(351) ?*an [I think it was [_{AP} blond]] woman.

She relates this to the fact that predicative pivots are ruled out in *it*-clefts (which she takes to underlie Horn amalgams).

However, she cites van Riemsdijk (2006:35, ex. 39) who claims the relative acceptability of Horn amalgams with attributive kernels (even though she and her informants find them marginal):

(352) Dit is een, [ik denk dat je het zou mogen noemen
 this is a I think that you it would may call
 eenvoudig-*(e)] oplossing.
 simple solution
 'This is a, I think you might call it simple solution.'

This is only possible if the adjective is exceptionally extraposed in the secondary root (which is otherwise not possible). Default word order leads

to ungrammaticality; equally ungrammatical is the Horn amalgam with the standard copular construction instead of the verb *to call*.

She later claims that predicative adjective pivots are ameliorated somewhat if they are contrastively focussed (Kluck 2011:133, ex. 86):

(353) ?Bea is I think it's BLOND (, but it could be GINGER).

For German, I cannot detect any difference, a contrastively focussed version is as (un)grammatical as its unmarked counterpart:

(354) ??Bea ist ich glaub es war BLOND/blond.

Lastly, she reports that in English, PP-pivots are dispreferred but possible (Kluck 2011:208, her ex. 169):

(355) ?Dickens wrote his books [I think it was with a feather].

She reports German PP-pivots as markedly more dispreferred but still possible (her ex. 173):

(356) ??Bob dekorierte sein Zimmer [ich glaube es war mit einem Bild
 Bob decorated his room I believe it was with a poster
 von Marx].
 of Marx
 'Bob decorated his room I believe it was with a poster of Marx.'

To conclude this short overview, the default category of pivots in Horn amalgams appears to be NP. AP and PP remnants appear to be marginally acceptable in sometimes exceptional circumstances. So far I have postulated that only a single category has a corresponding $\neq f$ -feature, namely *n*. To accommodate the cited data, one option would be to allow for $\neq a$ and $\neq n$ in the grammar. The other option would be to conserve the restriction and postulate analyses for predicative adjectives and PPs in which their 'nouniness' is retained. Predicative adjectives could contain a silent NP core which survives under category-preserving adjunction in APs. The accessibility of the NP contained in PPs might be derived in an appropriate theory such that one could claim that the secondary root grows on top the NP contained in the PP. This might correlate with the possibility of pied-piping; after all, the examples in English, a pied-piping language, are reportedly better than the counterparts in German which does not have this property. At present, however, I have nothing more of substance to add. Even though the restriction to a single $\neq n$ might need to be relaxed, it is at least true that NP-pivots are the default case and cover most uses. We will see parallels in a similar

discussion about parasitic gaps: NP-gaps are the default, but sometimes AP and PP-gaps are reported as marginally possible.

7.3.2 Previous approaches

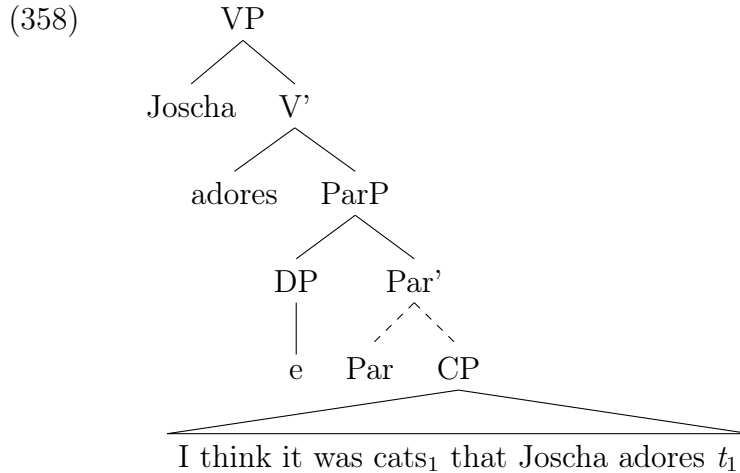
Let me start comparing this approach to the approach to amalgams that it is most dissimilar to, namely Kluck's own. Kluck wants to avoid a multidominance approach to (Horn) amalgams where the pivot has two different mothers. For that she invokes a different kind of structure-building operation called *par-Merge* (based on de Vries 2007's *b-merge*). To summarize the idea behind this operation shortly: in normal merge, merger of two elements A and B leads to a phrase C where the node C dominates A and B; as a result, A also c-commands B (and vice versa). *Par*-merging two objects does not lead to a dominance relation and hence no c-command relation. This avoids the inevitable consequence of subordination in *merge* and enables the implementation of parataxis. This allows one to model properties of parentheticals that appear to be structurally integrated in the clause but at the same time inaccessible for many syntactic relations such as binding. Amalgams share a lot of these properties with parentheticals, as demonstrated in 7.1. Kluck's own proposal is that amalgams are sluiced parentheticals. The basic idea for Horn amalgams is that the secondary root is a cleft-construction that contains elided material:

(357) She went to [I think it was Chicago₁ ~~that she went to~~ _{t₁}]

The secondary root consists of (at least) an *it*-cleft whose pivot is specified by a relative clause; the cleft-construction itself might be further embedded by e.g. *verba dicendi*. Kluck argues that the construction in the secondary root involves sluicing, i.e. the ellipsis of the relative clause CP. The pivot actually originates in the lower relative clause. She proposes a raising analysis for these cleft relative clauses where the head noun moves into a higher CP-layer CP₁ while sluicing obligatorily elides a lower CP-layer CP₂.

The way interrupting clauses such as '*I think it was Chicago*' are integrated into the matrix root is by way of a head *Par* that takes the former clause as its complement via an application of *par-merge*. Crucial to her account is that *Par* not only takes a complement via *par-merge* but also a specifier via normal merge. This specifier is a null correlate (*e* in the tree) that is of the category that the verb in the matrix root selects. *Par* is categorially underspecified and takes on the category *of its specifier*. The verb (e.g. *went to* in the above example) then selects the *ParP* as its argument since it is of category DP. The complement of *Par* is not accessible for c-command relations since it is connected via *par-merge* which does not lead

to a dominance relation.



There are a couple of issues with this approach. The first is, as Kluck notes, that ellipsis is obligatory. Standard cases of ellipsis such as gapping, VP-ellipsis or crucially sluicing are optional. Obligatory ellipsis appears to be rather a theoretical device where one wants to express that the pivot plays a role in two positions at the same time, i.e. both the cleft and the matrix root are predicated of the pivot. Instead of saying this directly as in a multidominance account, this account states that both cleft and what the matrix sentence expresses are present in the secondary root. The former is never pronounced (pronunciation leads to ungrammaticality), for which the assumption of obligatory ellipsis is evoked. Nor can any other material distinct from the matrix root content be present. The fact that syntactically the pivot behaves as if it were present in its base position in the matrix root,⁶⁷ Kluck attributes to reconstruction effects of sluicing remnants: the pivot which in her account originates in the relative clause in the secondary root is reconstructed into the obligatorily elided CP *as if it had never moved*. The latter fact is derived without stipulation in an account in which the pivot does not move in the first place. However, reconstruction is not identical to not having moved; there are environments that block reconstructions and conversely, there are infelicitous overt configurations that are ameliorated by ellipsis (such as vehicle change effects, see Fiengo & May 1994. An extensive investigation to determine whether the pivot in amalgams behaves as a reconstructed ellipsis remnant or base generated element is beyond the scope of this chapter. Yet it appears that crossover effects remain unaffected even though a rescuing constellation would in principle be possible:

⁶⁷It is necessary to address this because the pivot *cats* both does *not* c-command e.g. *joscha* in the matrix, and *does* c-command it in the obligatorily elided material.

- (359) a. *His₁ mother adores [I think it was Joscha₁ ~~that his₁ mother~~
adores.]
b. ?I think it was Joscha₁ that his₁ mother adores.

One argument that Kluck mentions are case mismatches where the verbs in matrix and secondary root require different cases. According to her, multidominance accounts predict strict case matching. I disagree with this argument. A multidominance account is logically independent from whatever approach to case theory is implemented in it. Any account of case that determines what happens when multiple cases are assigned to a single element (strict matching, first/last wins, ...) can be adopted in such a framework.

Another issue pertains to the null correlate. Some mechanism needs to ensure that the pivot of the cleft-construction is always coreferent with the null correlate in the specifier of the ParP and not some other element (which could turn any language into a radical pro-drop language). This is similar to an approach to parasitic gaps with null operators without a theory of its coindexation.

The binding facts detailed in 7.1 also follow in a different way than they would in a multidominance theory. She goes into detail for Condition C effects (Kluck 2011:299, ex. 70):

- (360) *He₁ cited [I think it was the professor₁].

In her approach, the ungrammaticality results not from the first *he* in the sentence c-commanding a coreferential NP (*the professor*). This is because this *he* only c-commands the null correlate, but not the *par*-merged interrupting clause. Instead it follows from the fact that it is the *he* that is elided in the secondary root that c-commands the professor. The full structure of the clause in brackets is, in her analysis, ‘*I think it was the professor that he cited*’. This is unexpected under an ellipsis approach. Deletion of offending material often leads to amelioration via vehicle change. See e.g. the sentences cited in Hunter & Yoshida (2016:562, ex. 4):

- (361) A: Someone said that John₁ left.
B: Yeah, but not he₁/him₁.

Here, a pronoun c-commands an elided coreferential proper name. In contrast to an ellipsis approach it appears instead that amalgams behave as if the material is present only *once* but as if both roots are predicated of the pivot at the same time. Doubling the material and obligatorily eliding one copy again superficially leads to the same result (with more assumptions though). But none of the additional possibilities enabled by ellipsis appear to occur.

Kluck also offers the criticism that multidominance approaches cannot deal with the obligatory intensionality of the secondary root. The pivot is the daughter of two verbs. As the daughter of the verb in the secondary root, it is within the scope of the intensional operator, but as the object of the matrix verb, it is supposedly outside its scope and for this reason a wrong inference follows (Kluck 2011:246, ex. 63)

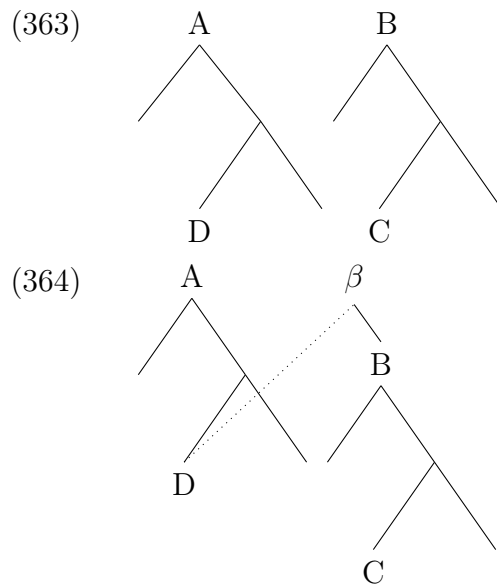
- (362) Bob found [I think it was a Stradivarius].
 $\nrightarrow$ Bob found a Stradivarius.

The above sentence does not necessarily entail the indicated proposition. As in the case with case mismatches, I disagree with the conclusion that multidominance accounts cannot derive this. Nothing a priori in a multidominance account predicts how the two roots are composed together. The development of an account of the precise semantics of amalgams is an important goal but it is unfortunately beyond the scope of this thesis. A simple sketch of a solution offers itself, however, from the way I set up the rule system: the secondary root needs to be build - and therefore most likely also interpreted - in full first before it is integrated into the matrix root. This way it should be possible to let the pivot ‘pick up’ whatever intensionality is generated in the secondary root (e.g. that the pivot is something of which the speaker thinks it is a stradivarius, which as a whole is then the object of the matrix verb)). This could rather be taken as evidence that derivation of the secondary root must be prior to that of the primary one. Conversely, there is no reason why her approach should not suffer from the same issues. After all, the null correlate that is coreferent to the pivot in her approach is accessible to the matrix verb as well. An account that relies on the fact that the intensionality of the pivot in the secondary root carries over to the matrix root by timing of the construction can be implemented in a multidominance account as well.

To summarize, the arguments against multidominance accounts are not as problematic as they appear. In case of intensionality and case clashes, they are orthogonal to the underlying account and can be solved by an appropriate implementation of a case account or by derivational timing. Other properties are equally problematic for both and require stipulation. This is e.g. the case for the intensionality of the secondary root. Kluck implements this by stating that Par selects a complement that is specified for [+SD], i.e. for Speaker Deixis. Similar stipulations can be proposed for the properties of secondary roots in the rules that integrate them which would have the same explanatory depth. On the other hand, to introduce an obligatory ellipsis operation into the analysis makes it less parsimonious. For interpretative reasons the pivot

is obligatorily interpreted in its reconstructed position (i.e. as if it had never moved) and it appears that none of the *additional* interpretative options that are enabled by ellipsis are allowed. Such an approach wants to derive the empirical fact that one element is connected to two otherwise independent clauses by doubling it and then deleting the double while ensuring their identity, yet all consequences of deletion need to be avoided.

The account that the approach to Horn amalgams presented here is most similar to is by van Riemsdijk (2006), van Riemsdijk (2010) in terms of *grafts*. van Riemsdijk motivates grafts by looking at potential versions of *merge*. There is external merge that merges two trees at their root; there is internal merge (move) that merges a subconstituent of a tree with the root of that same tree. Then ‘external-internal merge’ is also a possibility where a node internal to a tree (i.e. not its root) merges another external tree as its sister (van Riemsdijk 2006:20, exx. 5&6):



To translate this to the terminology in this thesis, D would be the pivot with the secondary root headed by A that is grafted into the matrix root now headed by β .

While van Riemsdijk focuses mostly on transparent free relatives and constructions such as ‘*a far from simple matter*’ he also applies the operation *graft* to Horn amalgams. Besides the formal implementation within MGs, there are number of smaller differences between the way van Riemsdijk sets up *graft* and the operations defined here. One of them is that *graft* must ap-

ply before the complete secondary root is built for phase reasons.⁶⁸ For the way *3rd-merge* is set up here it is crucial that the complete secondary root is built so as to avoid multiple active roots in a single derivation. Another small difference regards the asymmetry between both roots. van Riemsdijk notes some empirical asymmetries between both roots but maintains that “all trees are equal” (van Riemsdijk 2006:fn.8) and that only at some point in the derivation it is decided which of the trees takes on primary status. For *3rd-merge*, the asymmetry between the two roots is baked into the definition. Whatever grows on top of a pivot via a normal merge-operation is the secondary root, whereas whatever phrase has a positive *3rd-merge* feature will be a part of the primary root (conversely, whatever carries $\neq f$ in its chain is the secondary root). Overall, however, the differences between the two approaches are minor when compared to markedly different proposals such as the one by Kluck (2011), so that *3rd-merge* implements closely the idea behind *grafts*.

7.4 Conclusion

I want to conclude this chapter by providing a short summary of the core mechanisms proposed here to deal with Horn amalgams. The core idea is that nouns carry two types of features, normal *merge*-features and the new *3rd-merge*-features. This allows them to be doubly-selected where the standard merge-feature can lead to the growth of a secondary root on top of the noun (then called the pivot). After this secondary root is built completely, with all features satisfied save the features on the pivot, an item from another tree can select for the pivot inside the other root via an application of *3rd-merge*. This then leads to the merge of the selector and its argument together with the whole secondary root on top of that argument, creating a multi-rooted tree. The rules set up so far derive the basic cases of Horn amalgams. There are a number of theory-internal and empirical properties that are to varying degrees crucial to the formalism. I want to list them to make it clear what the predictions of the current approach are and where its potential shortcomings are.

Theoretically I have proposed that any secondary root is always built completely before it is integrated into another root, i.e. this enforces a derivational timing where there is only ever a **single active root**. Also, no derivation contains more than one active pivot. I have also prohibited the negative feature in a *3rd-merge* operation to appear in specifier position.

⁶⁸van Riemsdijk 2006:27–28: “I conclude that external merge of any kind, that is including graft, can apply to a pair of nodes that each belong to their respective trees, where every one of these trees can be at some intermediate stage of its growth.”

Besides avoiding issues with the theoretical apparatus, this is also related to a stronger empirical claim.

The empirical claims about Horn amalgams I made in this chapter are the following. I have claimed that the pivot is always the most deeply embedded complement of the spine of the secondary root. In other words, neither is it a specifier in the secondary root, nor is it a complement inside a specifier of the verbal spine.

The second claim is about asymmetries between both roots. I have claimed that subextraction from the pivot is only possible into the matrix root, but not into the secondary root. There is also no movement of the pivot inside the secondary root. That is, I claim that the pivot appears in no other position inside the secondary root aside from its base-generated position. This appears to hold for English and German. The only potential counterexample comes from topicalization in the secondary root in Dutch. Further research is needed how robust and pervasive these counterexamples are and to what extent the formalism would need to be amended. For now I want to maintain the stronger and more easily falsifiable claim.

I have also decided to put aside a more in-depth discussion of Andrews amalgams such as ‘*He invited [you’ll never guess who]*’. For these constructions it is *prima facie* more plausible that they contain *wh*-movement of the pivot plus sluicing. Note though that Andrews amalgams slightly differ from Horn amalgams in a number of properties (it is e.g. possible to topicalize the pivot inside the secondary root even in English Andrews amalgams even though this is ungrammatical for Horn amalgams, ‘**Bob kissed [how many girls you can’t even begin to imagine!] at his party*’, Kluck 2011:91, ex. 138). It might be necessary to extend the formalism at hand to cover Andrews amalgams as well. But since parasitic constructions, not amalgams are the focus of this thesis, a detailed investigation into Andrews amalgams and closely related constructions such as transparent free relatives are beyond the scope of this work.⁶⁹ I also do not attempt to use *3rd-merge* to derive

⁶⁹There might also be reason to doubt that free relatives are the movement counterpart to Horn amalgams with their in-situ pivots, as a reviewer (Thomas Graf) suggests (though see van Riemsdijk 2006). One relevant question is whether the relative clause is also an independent root. Kluck (2011) notes that while reflexives can also be bound across an apparent clause boundary similarly to Horn amalgams, bound variable readings are indeed available in transparent free relatives, in alleged contrast to Horn amalgams:

- (i) Every student₁ gave what he₁ considered a great presentation.
- (ii) #Every student₁ gave I think he₁ called it an amazing lecture.

It is therefore unclear to what extent the current mechanisms can carry over without modification to free relatives. A more elaborate combination of multi-dominated pivots

the superficially similar phenomenon ATB-movement since it differs in quite a number of respects from the centrally relevant parasitic gaps, as shown in Postal (1993).

The last empirical generalization concerns the categorial nature of the pivot. The prototypical pivot in a Horn amalgam is an NP. I have indicated in 7.3.1 that there are AP and PP candidates for pivots as well. Either the assumption needs to be given up that there is only one type of $\neq f$ in a lexicon, or one attempts to unify apparent counterexamples with standard NP pivots. Either way, it is interesting to note that parasitic gaps share this profile in that prototypical uses involve moved NPs, but claims of exceptions in the form of APs and PPs in the relevant position go back to Engdahl (1983). We turn to this construction now in the next chapter.

and re-merged additional roots might be at play here.

8 Parasitic Gaps

This chapter deals with the second instance of parasitism, parasitic gaps. After providing a short introduction in 8.1, I discuss the relevant rules for parasitic gaps in 8.2, together with detailed example derivations. Since Horn amalgams and parasitic gaps are derived via similar mechanisms, namely multi-rooted trees, I list a number of empirical parallels between those constructions in 8.3 that justify this similar treatment. In 8.4 I discuss how the analysis proposed here fares with respect to the desiderata for theories that derive parasitic phenomena from chapter 3. I shortly compare this approach to others from the literature in 8.5 and close with an outlook over issues that need to be addressed in future work (8.6).

8.1 Introduction

I have discussed the parasitic gap construction already in 4.3, but I want to provide a brief introduction and overview of this phenomenon here as well to make this chapter self-contained. Parasitic gaps are the phenomenon where one gap (or long-distance dependency) can occur in the presence of another gap (long-distance dependency). The reason this gap is called parasitic is that in many cases the gap on its own would be ungrammatical. The classic example is from Engdahl (1983):

- (365) a. [Which article]₁ did you file t_1 without reading it?
 b. *[Which article]₁ did you file it without reading t_1 ?
 c. ?[Which article]₁ did you file t_1 without reading pg_1

In (365a), there is a run-of-the-mill *wh*-extraction of the object of *file*. (365b) is ungrammatical because of a *wh*-extraction out of the adjunct *without reading t*. The crucial case is (365c): there is a licit long-distance dependency between the fronted *wh*-phrase and the now empty object slot of *file*, yet in this context the otherwise ungrammatical long-distance dependency into the adjunct is surprisingly grammatical. Crucially, the empty object slot of *reading*, aka the parasitic gap *pg*, is interpreted as identical to the filler for the empty object slot (the real gap *t*) of *file*. That is, the interpretation of (365c) is the question ‘*Which article did you file without reading that same article?*’. As such, parasitic gaps are a one-to-many filler-gap construction.

The literature has settled on a number of core properties (see e.g. Culicover 2001): one is already illustrated above, namely that parasitic gaps can correspond to its filler across one island boundary. Another property is that parasitic gaps are only licensed by (overt) $\bar{A}$ -movement as in the sentence

above, but not e.g. by A-movement (Engdahl 1983:13):

(366) *John₁ was killed t_1 by a tree falling on pg_1

The host gap must not c-command the parasitic gap, but both are c-commanded by their filler (*anti-c-command condition*). The first condition is not fulfilled in the following sentence (Engdahl 1983:20):

(367) *Who₁ t_1 sent a picture of pg_1 ?

There is also a categorial restriction to nominals, i.e. parasitic gaps can only correspond to NPs, not other types of phrases.

(368) *Smart₁, his mother is t_1 without wanting her son to be pg_1 .

There are a number of other properties that Culicover (2001) lists as the ‘Common Consensus Position’. The interested reader is referred to this work for a more in-depth overview of this phenomenon. However, it is not the case that all these properties are uncontested across the board. Especially the categorial restriction is claimed to be non-total in Engdahl’s original paper where she mentions the marginal possibility of AP and PP fillers for parasitic gaps in Swedish. However, the above-mentioned properties are definitely the prototypical cases of parasitic gaps and I therefore treat them as the default case of this construction, putting off a discussion of potential exceptions.

Nissenbaum (2000) provides a sketch of the landscape of theories for parasitic gaps which fall into two major classes: there is the class of ‘shared antecedent proposals’ that assume that host and parasitic gap are bound by the same antecedent. There is also the class of what Nissenbaum calls ‘separate antecedent proposals’ where host and parasitic gap have distinct antecedents. The account I propose for the derivation of parasitic gaps with the current extension to MGs falls firmly into the first class. This account is detailed in the next section.

8.2 Analysis

I derived Horn amalgams in the previous chapter by making use of the additional possibilities that come from the new structure-building operation *3rd-merge*, and the feature sequence proposed for nominals ($n.\neq n$). This way, nominals can be selected by two verbs: once by a step of normal *merge* which initiates growth of a secondary root, and once by a verb via an application of *3rd-merge* which results in the connection of the secondary root with the primary root at the pivot. In other words, there is a system where one can derive trees with multiple roots that are conjoined at a nominal.

The approach to parasitic gaps here will also make use of the idea of multiple roots with nominals that are shared between them, but with a twist. The idea is the following. In a system that allows the growth of an additional root, there are two possibilities: either the additional root remains independent, leading to the phenomenon of Horn amalgams; or the additional root is reconnected into the matrix root at some point during the derivation. In a derived tree perspective, this leads to a cyclic structure. If the nominal that is part of both roots is moved out, however, the connecting element that lead to the cyclic structure in the first place is pronounced elsewhere. What results is what appears to be two distinct gaps in a 1D-string, referring to the same element: there is a gap after the matrix verb (*file* in (365)) and a gap after the verb from the host clause (*reading* in (365)). From a tree-perspective, however, the appearance of two distinct gaps is illusory: there is only a single element that has moved that corresponds to seemingly two gaps at the same time. In other words, the option of reconnecting the additional root leads to the phenomenon of parasitic gaps. This idea of parasitic gaps as the result of multidominance structures was first proposed in Kasai (2010). The analysis proposed here follows this basic insight. In what follows I introduce the rules in MGs with *3rd-merge* that determine the behaviour of derivations in which additional roots are reconnected.

As in the case with amalgams, the derivation of what ultimately will turn out to be a parasitic gap construction starts with the growth of a secondary root, via *merge4*:

$$\frac{s :: = f\delta \quad t \cdot f\psi\gamma, \alpha_1, \dots, \alpha_k}{s : \delta, [t : \psi\gamma, \alpha_1, \dots, \alpha_k]} \text{merge4}$$

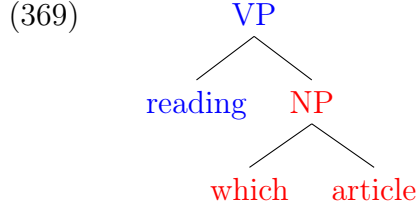
Let us take the classic examples about our filed articles for illustration:

$$\frac{\text{reading} :: = n.V \quad \text{which article} : n.\#n.-wh}{\text{reading} : V, [\text{which article} : \#n.-wh]} \text{merge4}$$

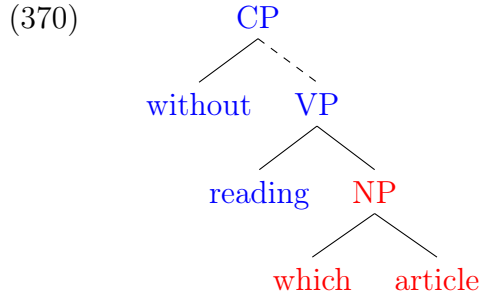
I significantly simplify the syntactic structure of the secondary root, e.g. I leave open how *read* requires the *-ing* suffix and many other issues. The only relevant point here is that some form of *read* selects *which article* as its object. Later head movements and the like do not change this fact. The crucial difference is that the element that will later be shared by both roots carries a movement feature (*-wh*).⁷⁰ For now, this fact is not yet relevant,

⁷⁰As in the last chapter, I use the NP-hypothesis for illustrative purposes and therefore have to assume here that the *-wh* somehow ends up on the whole NP. Nothing much hinges on this, I could as well assume that *which* has the feature sequence *d.\#d.-wh*. I keep to the NP structure simply for consistency.

the *merge*₄ rules allow for that (as can be seen in the feature string $f\psi\gamma$, i.e. 0 or more instances of features after $\#f$).



Let us now say that we have grown the secondary root until it is a full sentential adjunct (without reading : c , [which article : $\#n.-wh$]).



Now that this is a complete secondary root except for the features on the pivot, an element out of another root can select for that pivot via an application of *3rd-merge*. *3rd-merge*₁' and *3rd-merge*₂' dealt with cases where the pivot carries only a $\#n$ -feature but nothing else (resulting in Horn amalgams). This pivot, however, also carries a movement feature in which case another rule applies, *3merge*₄:

$$\frac{s \cdot \#f\delta, \omega, \alpha_1, \dots, \alpha_k \quad t : \varsigma\gamma, [u : \#f\varepsilon, \iota_1, \dots, \iota_\ell]}{s : \delta, \omega, [t : \varsigma\gamma, u : \varepsilon], \iota_1, \dots, \iota_\ell, \alpha_1, \dots, \alpha_k} \text{3merge-4}$$

*3merge*₄ is short-hand for two rules, one rule that merges a secondary root into complement position, in which case there are no moving parts in the expression of the function ($s :: \#f\delta$), and one rule that merges into specifier position in which case all elements after δ might be present. The argument in the rule is a tree that is headed by t , what will be the host of the parasitic gap; the head t has category feature ς which is a shorthand for c or n (the latter is to allow parasitic gaps inside subject NPs). Additionally, it can also carry a movement feature (γ); I have left open this possibility e.g. for rightward dislocation of an adjunct host of the parasitic gap. The connecting element or pivot, u , carries the negative *3rd-merge* feature which marks it as the argument of the selector s , and it crucially carries a movement feature

(ε). There might also be moving elements inside the pivot ($\iota_1, \dots, \iota_\ell$). In the resulting expression, both *3rd-merge* features are erased and the secondary root becomes a chain, now in brackets ($[t : \varsigma\gamma, u : \varepsilon]$). Submovers of the pivot are also released into the chains ($\iota_1, \dots, \iota_\ell$).

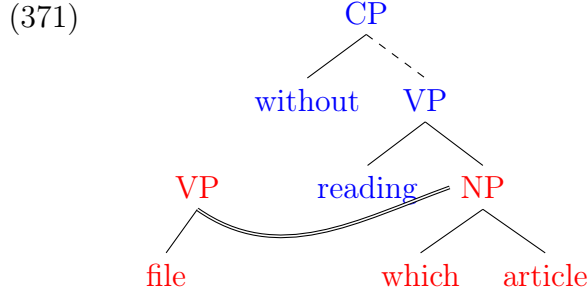
Crucially, the category feature of the head of the secondary root is *not* erased. This was the case in the rules that result in Horn amalgams where I used this as a notational ploy to express that the secondary root is an independent CP which is syntactically a complete single root to which no other operations apply any more. Here, however, the root head remains active since it needs to be re-connected to the main structure.

The above rule covers a lot of special cases, so let us start with the simplest case: a verb selects a nominal as its complement via an application of *3rd-merge*; this nominal is one that has an additional root on top of it and also a movement feature:

$$\frac{s :: \#f\delta \quad t : \varsigma, [u : \#f\varepsilon]}{s : \delta, [t : \varsigma, u : \varepsilon]} \text{3merge-4 (vanilla)}$$

One such case happens when *file* selects for its object in our example:

$$\frac{\text{file} :: \#n.V \quad \text{without reading} : c, [\text{which article} : \#n.-wh]}{\text{file} : V, [\text{without reading} : c, \text{which article} : -wh]} \text{3merge-4}$$



file selects for its object *which article* via an application of *3rd-merge*. The object *which article* so happens to carry additional structure on top of it in form of a secondary root (*without reading : c*). The object also carries a movement feature, so it cannot be linearized yet. As a result of an application of *3merge-4*, $\#n$ and $\#n$ are erased and the secondary root becomes a chain in the resulting expression where it is represented inside brackets.

We now have two entities that are enclosed in square brackets: previously, only pivots (ω) with an unchecked $\#f$ were represented with brackets; now there is the structure that results from the application of *3merge-4* (*without $\#f$*). This is a distinct object whose behaviour is determined by the rules

that I discuss next, but it is similar enough, and I want to keep this notation to express that this bracketed expression can only be operated on as a single unit.

Three rules govern the behaviour of this object. The first rule governs the behaviour of non-final movement of the pivot, the last two rules govern the reconnection of the secondary root into the matrix root. The movement rule discusses a special but relatively mundane case:

$$\frac{s : +f\delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : -f\varepsilon], \alpha_{i+1}, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_{i-1}, [t : \varsigma\gamma, u : \varepsilon], \alpha_{i+1}, \dots, \alpha_k} \text{move3}$$

The above rule talks about the case where the pivot has multiple movement features. This can come about e.g. if the pivot is subject to scrambling before the final $\bar{A}$ -movement that leads to the creation of the parasitic gap. For this reason I have made non-final movement features in bracketed expressions accessible for attractors on the head of the expression (s). The effect of this rule is movement of the pivot+secondary root as one unit to that position without being linearized yet, while leaving the overall bracketed structure unaffected in all other respects. This can be viewed as the pivot ‘pied-piping’ the secondary root along with it until the latter is merged to the matrix root.

As an aside, in the section about amalgams I have alluded to the fact that such a rule might also be necessary for amalgams. Such a rule would of course differ from the parasitic gap case in that the head of the secondary root will remain independent and unconnected.

$$\frac{s : +f\delta, \alpha_1, \dots, \alpha_{i-1}, [t : c, u : -f], \alpha_{i+1}, \dots, \alpha_k}{tus : \delta, \alpha_1, \dots, \alpha_{i-1}, \alpha_{i+1}, \dots, \alpha_k} \text{move1},$$

An expression of this kind would come about via a previous application of *3merge-4* since a pivot with a movement feature is involved. The difference between amalgams and parasitic gap constructions then simply comes down to whether the head of the secondary root is reconnected to the matrix before the final movement feature on the pivot is erased ($\rightarrow$ parasitic gaps) or if the final movement step of the pivot is executed with the root still independent in which case an amalgam will result.

After this detour, I now return to more crucial issues of the current system, namely the rules that govern the reconnection of the secondary root to the matrix root:

$$\frac{s : =\varsigma\delta, \omega, [t : \varsigma, u : \varepsilon], \alpha_1, \dots, \alpha_k}{ts : \delta, \omega, \alpha_1, \dots, \alpha_k, u : \varepsilon} \text{chain-merge1}$$

$$\frac{s : =_{\varsigma}\delta, \omega, [t : \varsigma\varepsilon, u : \zeta], \alpha_1, \dots, \alpha_k}{s : \delta, \omega, \alpha_1, \dots, \alpha_k, t : \varepsilon, u : \zeta} \text{ chain-merge2}$$

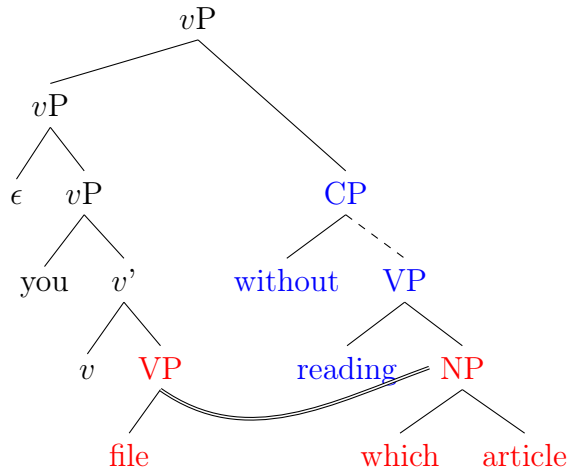
The pivot, i.e. the mover that is the antecedent for both host and parasitic gap only becomes accessible after the secondary root is re-connected to the matrix root. The rules *chain-merge1/2* describe this point. These rules are akin to movement rules in that they are unary rules: there is a chain as part of the expression that has an unsatisfied feature. The crucial difference to the movement rules is that the unsatisfied feature is a *merge feature*. If the head of the current tree selects for a category feature ($=_{\varsigma}$) that is matched by the category feature on the head of the secondary root (ς) inside square brackets, then the matrix root head can select this structure as its argument. This root (t) is then merged with the head (s) if the secondary root does not carry a further movement feature (*chain-merge1*). If the secondary root does carry a further movement feature ($t : \varsigma\varepsilon$), then it becomes part of the movers (*chain-merge2*). In both cases, however, the pivot becomes part of the movers ($u : \varepsilon$ or $u : \zeta$) and is accessible for extraction. This automatically ensures the anti-c-command property of parasitic gaps: the host gap does not c-command the parasitic gap since the host of the latter is merged at a position higher than the host gap; movement of the filler becomes available only after the secondary root is reconnected to the matrix root and as a result its movement will be to a position higher than both host and parasitic gap, thereby c-commanding them.

Here is an example for the application of such a rule. To proceed with the example I have used throughout this exposition, let us say we are at the point where the adjunct *without reading* is adjoined at some point in the verbal domain. I abstract away from the intricacies of adjunction and simply treat it as the result of selection by a silent adjunction head ($\epsilon : =v.=c.v$, selecting vP , then a CP-adjunct and becoming a vP).⁷¹

$$\frac{\epsilon \text{ you file} : =c.v, [\text{without reading} : c, \text{which article} : -wh]}{\epsilon \text{ you file without reading} : v, \text{which article} : -wh} \text{ chain-merge1}$$

⁷¹I also ignore the question how the adjunct is linearized to the right which is an orthogonal issue independent from the ones discussed here.

(372)



At the vP level we want to adjoin the adjunct *without reading* which happens to be a tree that is not independent but already connected with the current root at the pivot. The top of the secondary root is merged to the vP and the pivot at which it is connected to the matrix root is released for movement in the resulting expression where it becomes just a regular moving chain (which article : $-wh$).

Both in the rules *chain-merge1/2* and in *3merge-4*, I have again indicated the possible presence of an ω . This would mean that the expression of the selector on the left is *itself* a secondary root since it contains a pivot. This would describe a case where we are generating a parasitic gap construction in a tree that itself contains a pivot; this tree would then later serve as a secondary root which is connected to a matrix root at that pivot, resulting either in a Horn amalgam or yet another parasitic gap. These ‘embedded parasitic gaps’ are similar to cases of recursive amalgams mentioned in the last chapter. It is beyond the scope of this chapter to discuss whether this inclusion of ω can find a reasonable empirical application. I have therefore greyed it out in the three rules mentioned.

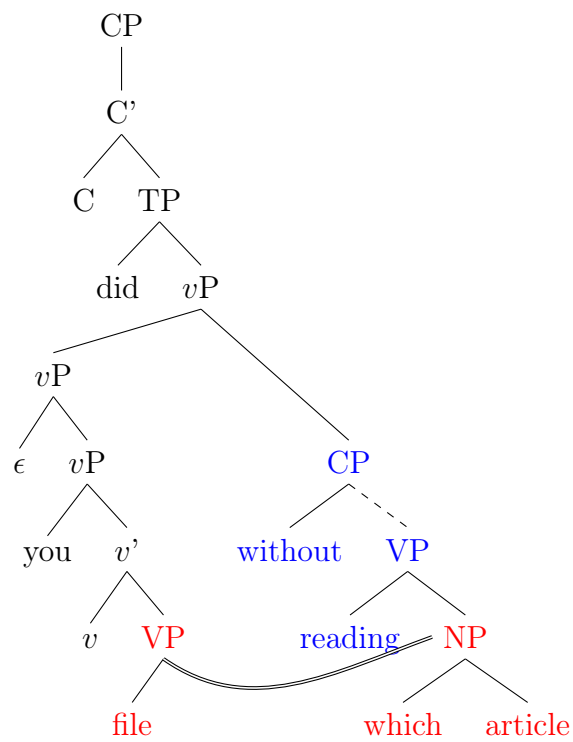
There is a last point that I want to discuss with respect to the presence of those bracketed structures with unchecked merge features. It is in principle possible to allow for multiple such bracketed structures, even though in the rules above I have indicated that there is only one (assuming that α or ι only ever denote movers). This would correspond to a structure that contains multiple pivots which at some point later will each correspond to both a host and a parasitic gap, i.e. multiple parasitic gap constructions with distinct host clauses. Besides the empirical properties of these constructions, there is also a more abstract point: what are the consequences of allowing more than one such brackets? If there are multiple CPs at some point that can be merged by an application of *chain-merge*, the derivation is non-deterministic.

This is of course also the case where verbs choose a nominal while multiple nominals are available in the ‘numeration’. The relevant issue is rather if the availability of multiple targets has similar effects as the availability of multiple movers. MGs enforce the Shortest Move Condition, i.e. that in an expression no chain starts with the same type of movement feature. The question is therefore whether one should enforce an SMC equivalent for such bracketed chains. An investigation of the above issues, however, is beyond the scope of this thesis and I continue for now with the simplifying assumption that there is only ever one such bracketed chain.

Let us now return to the derivation of the exemplar parasitic gap sentence. Above we arrived at that point in the derivation where the top of the secondary root is selected by the head of the matrix root, turning the multi-rooted tree into a single-rooted tree again. It now contains a cyclic structure but the connecting element, the pivot *which article*, contains a movement feature. After building the rest of the matrix clause, C attracts this *wh*-word into its specifier:

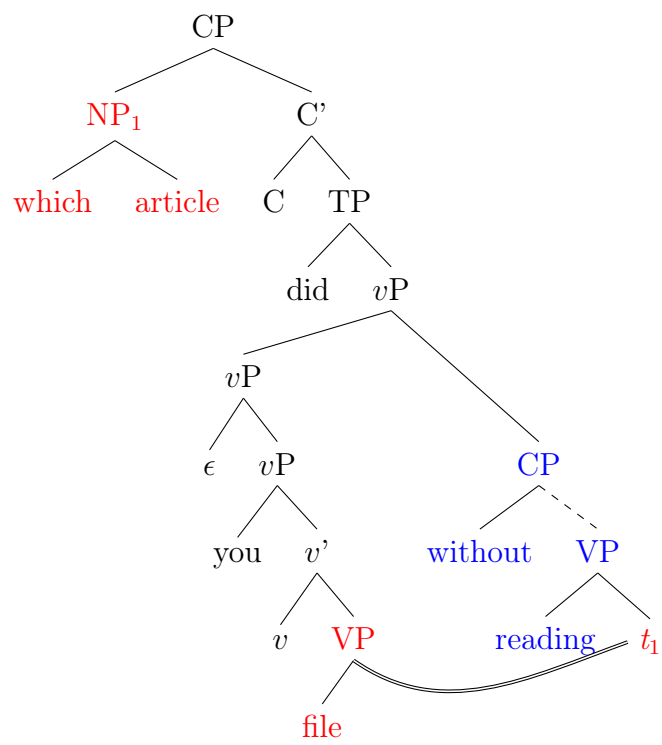
$$\begin{array}{c}
\frac{\text{reading} :: =n.V \quad \text{which article} : n.\neq n.-wh}{\text{reading} : V, [\text{which article} : \neq n.-wh]} \text{merge4} \\
\vdots \\
\frac{\text{file} :: \#n.V \quad \text{without reading} : c, [\text{which article} : \neq n.-wh]}{\text{file} : V, [\text{without reading} : c, \text{which article} : -wh]} \text{3merge-4} \\
\vdots \\
\frac{\epsilon \text{ you file} : =c.v, [\text{without reading} : c, \text{which article} : -wh]}{\text{you file without reading} : v, \text{which article} : -wh} \text{chain-merge1} \\
\vdots \\
\frac{\text{did you file without reading} : +wh.c, \text{which article} : -wh}{\text{which article did you file without reading} : c} \text{move1}
\end{array}$$

(373)



Translated into a derived tree, this would look as follows:

(374)



To walk through the whole derivation again: the construction of this sentence starts by building the adjunct clause. The verb *read* selects its argument *which article* via an application of standard merge, thereby triggering the growth of a secondary root (*merge₄*) because its argument carries a $\neq n$ feature. The pivot *which article* is isolated for the rest of the derivation of the adjunct and hence does not play a further syntactic role inside its host. After the adjunct is built fully, *file* selects its argument via an application of *3rd-merge*. The pivot *which article* inside the adjunct is a suitable target since it carries a $\neq n$ feature; but it is special in the sense that it is inside an additional root. It is a defining feature of *3rd-merge*, however, that it can select an argument from inside an independent root. As a result, *which article* also becomes the object of *file*, and the two roots, the adjunct and the root headed by *file* are conjoined at the pivot *which article*. Crucially, however, no other syntactic relationship is established between the two roots. *file* selected and therefore c-commands *which article*, just as *read* selected and c-commands the latter; but no relationship is established between the other elements of the adjunct and the matrix root. The matrix root grows further (e.g. *file* selects its subject). At some point at the *vP* level, an adjunct clause is to be selected (which I do here via an empty adjunction head). Instead of an adjunct that is built separately, the adjunction head can also select the rooted (sub)tree with which it is connected at the pivot, as long as the correct subcategorization requirements are met (*chain-merge₁*). As a result, the formerly independent root is now reconnected to the matrix root and the sentence only contains a single root again. This results in a cyclic structure from a tree perspective. The tree is constructed further until the CP; there, the C head attracts the pivot which carries a $-wh$ -feature, breaking up the cyclic structure in a derived-tree perspective. This leads to the creation of what appear to be two distinct gaps in a 1D-string: a gap after *reading*, and a gap after *file*, both corresponding to *which article*. In the underlying tree, however, there is only a single element that caused the creation of two gaps at the same time. MGs with *3rd-merge* therefore achieve the main goal of this chapter, namely deriving the basic patterns of parasitic gap constructions, doing so with a single movement step.

Before closing this section, I also want to provide an example for more elaborate cases of *3merge-4* for completeness' sake. In the general rule I have allowed for the presence of submovers inside the pivot, and I have also indicated that they are released into the matrix root at the point of merging the pivot. This seems to be justified. Take the following example from German:

- (375) ?[Welche Bücher t_1]₂ hat [über Potsdam]₁ jeder gekauft t_2
 which books has about potsdam everyone bought
 ohne je zu lesen pg_2 ?
 without ever to read
 ‘Which books about Potsdam did everyone buy without ever reading?’

Here, *which books about Potsdam* is the antecedent for both real and parasitic gap. Its complement *about Potsdam*, however, can be scrambled outside of it, crucially to a position that is lower than the landing site of its selector. It is therefore necessary to make $t_1, \dots, t_\ell$ already accessible to movement operations at this point:

$$\frac{\text{gekauft} :: \#n.V \quad \text{ohne zu lesen} : c, [\text{w. bücher} : \#n.-wh, \text{über potsdam} : -scr]}{\text{gekauft} : \#n.V, [\text{ohne zu lesen} : c, \text{w. bücher} : -wh], \text{über potsdam} : -scr} \text{3merge-4}$$

Crucially, however, subextraction into the secondary root is not possible, just as in the case of Horn amalgams (see (347)):

- (376) *[Welche Bücher t_1]₂ hat jeder gekauft t_2 [ohne [über
 which books has everyone bought without about
 Potsdam]₁ je zu lesen pg_2]?
 potsdam ever to read
 ‘Which books about Potsdam did everyone buy without ever reading?’

That is, both apparently unrelated constructions share a fundamental *asymmetry between the roots*. This can be taken as an indicator that a treatment of both constructions with the same mechanism is justified.

I want to close this section by providing an overview of what MGs with *3rd-merge* achieve. *3rd-merge* extends MGs by allowing the growth of trees with additional roots that are conjoined to the matrix root at an element inside it. If you grow an additional root, there are two things that can happen: either the root remains independent, or it is selected and reconnected to the matrix root again. I claimed that there are two empirical phenomena that correspond to these options: Horn amalgams for the former, and parasitic gaps for the latter. In case the root is reconnected, I have made the additional assumption that this always happens in cases where the connecting element is a mover so that the resulting cyclic structure is always ‘broken up’ by its displacement. One might view this as an avoidance of otherwise unresolvable

linearization requirements.⁷²

Allowing additional roots to grow and reconnect to the main root while moving out the connecting pivot, leads to a construction with a number of properties. It creates a 1-to-2 filler-gap dependency where a single item acts as the filler to two apparently distinct gaps which nonetheless obligatorily refer to the same element. The host gap cannot c-command the real gap, and the phrase containing the parasitic gap c-commands the host gap, i.e. it fulfils the anti-c-command requirement. The connecting element is later moved out of its position in the matrix root which leads to the possibility of what looks like movement out of at least one island, e.g. a long-distance dependency into an adjunct island.

8.3 Parallels between Horn amalgams and parasitic gaps

From the perspective outlined above, Horn amalgams and parasitic gaps are related constructions. Both are multi-root constructions, at least at some point in the derivation. In both constructions there is an NP that is the argument of two verbs at the same time that come from different roots. In Horn amalgams the pivot stays in-situ (plus an unconnected root) whereas in parasitic gaps that pivot moves out (and roots are conjoined again). There are further parallels between the two constructions that lend credibility to an approach that postulates that they are derived by similar mechanisms.

Both share the categorial restriction to nominals. Prototypical pivots in Horn amalgams and parasitic gaps are nominals. Both constructions also share claims about exceptions to this rule. As discussed in the chapter on amalgams, AP and PP pivots might be marginally possible for some speakers. Engdahl (1983:17) provides examples of Swedish parasitic gaps with AP and PP pivots, though English counterparts are quite unacceptable to most speakers. To repeat, prototypical pivots in both cases are nominals. For this reason I have postulated that there is only one instance of *3rd-merge* features, namely $\#n$. Postulating this restriction does of course not derive this property. The cost of this assumption is slightly decreased, however, by the fact that all constructions derived by *3rd-merge* share this property. If there are exceptions, then the most promising candidates are APs and PPs. Either one develops theoretical solutions where even in those cases it is actually the nominal core of these phrases with which both roots connect; or one needs to allow for $\#a$ and $\#p$ as well, presumably for both constructions together.

⁷²Another option is to say that phenomena corresponding to this structure do exist; a candidate that comes to mind is right-node raising. An exploration of this possibility is beyond the scope of this thesis, however.

Both constructions also share certain *asymmetries* of the two roots, unlike the superficially similar ATB-construction, in that the secondary root is **defective** in some way with less possibilities than the primary root. It appears generally to be the case that neither the pivot nor a constituent containing the pivot can move inside the secondary root in Horn amalgams.⁷³ Similarly, the final landing site of the pivot in parasitic gap constructions cannot be within the secondary root. The same also holds for subextraction: one cannot scramble e.g. the complement of the pivot into the secondary root in either Horn amalgams or parasitic gaps, only into the primary root.

Another property of both constructions is the position of the pivot inside the secondary root. In Horn amalgams, the pivot appears in complement position. Prototypically, the situation is the same in parasitic gap hosts. The default parasitic gap cases are those where the position of the parasitic gap appears in the direct object position of e.g. the adjunct clause. Parasitic gaps in subject position are claimed to be either ungrammatical or degraded compared to their object counterpart (see e.g. a discussion in García Mayo 1997 and works cited therein):⁷⁴

(377) *The person they insulted *t* after *pg* left the room

Parasitic gaps in indirect object position appear to be at least worse than their direct object counterparts.

(378) ?*Welchem Patienten half der Arzt *t* ohne *pg* ein
 which patient helped the doctor without an
 Antibiotikum zu geben?
 antibiotics to give?
 ‘Which patient did the doctor help without giving antibiotics?’

Postal (1993) claims that parasitic gaps are ungrammatical if they correspond to an indirect object (which he attributes to Cinque 1990, Postal 1993:752, fn. 14):

(379) *Who₂ did they praise *t*₂ after giving *pg*₂ the prize?

At the very least there seems to be a strong preference for complement position in both constructions though it does not appear to be total in parasitic gaps. For this case one would need to allow growth of a secondary root on top of a moving pivot in specifier position, i.e. a specifier-version of *merge*₄.

⁷³With one potential exception in Dutch as discussed in 7

⁷⁴This ban could be explained in two ways: either it is about the non-complement status, or it is about the fact that subjects typically move. If movement is generally forbidden inside the secondary root, subjects are doubly excluded.

To keep the system simple, I remain with the stronger and more easily falsifiable claim that pivots occur only in complement position (even the stronger claim that pivots are the most deeply embedded complement of the clausal spine).

In conclusion, there are a number of restrictions that need to be built into a formalism that generates multi-rooted trees. But these restrictions hold for both the phenomena that can be derived with it and do not need to be implemented for each construction specifically.

8.4 Parasitic gaps as a single process

Since the overarching theme of this thesis is an investigation of parasitism, this is a good point to consider whether the approach to parasitic gaps presented here meets the requirements for a solution to parasitic phenomena that I have proposed in 3.

As a quick reminder, I discussed in 4.3 that parasitism exhibits unexpected properties when viewed as the outcome of two separate but dependent processes. Descriptively, the parasitic process appears to refer rather to the host process itself than the context it creates. It also appears that parasitic relationships exist only between processes of the same type. Lastly, if the parasitic process is a separate operation of its own, one would expect it to underlie the restrictions that hold for operations of this type; in stark contrast to that, parasitic processes underlie at first sight unexpected ad hoc restrictions, but on the other hand they are also more permissive in that they can violate otherwise fundamental restrictions.

As a first point, in the current approach parasitic gaps are not produced by two operations that depend on each other; instead, two gaps are produced by one single movement of an item that simultaneously belongs to two different roots. In this case, producing the parasitic construction is more indirect compared to the parasitic ellipsis case where the 3rd-order ellipsis operation produced said construction directly. Here, the ‘more powerful’ operation *3rd-merge* does not lead to the creation of two gaps directly but rather the more general operation of allowing growth of secondary roots. However, this allows describing two apparently distinct phenomena with the same structure-building operation. It is a single operation (movement) applied to a possible structure that *3rd-merge* creates that generates host and parasitic gap at the same time. This achieves that reference to the presence of other processes is excluded.

The parasitism of likes observation is also derived trivially in this approach. A gap must be parasitic on another gap since in this analysis they are actually the same gap. It is only in a 1D-string that they create missing

material in two positions.

One idiosyncratic restriction on parasitic gaps is also derived under the current analysis. The anti-c-command property of parasitic gaps follows from the derivational timing that is enforced: first build the secondary root completely, then introduce it into the matrix root, as was the case for Horn-amalgams; then the secondary root can optionally be reconnected to the matrix root at a later stage. From this it follows that the parasitic gap is not c-commanded by the host gap but that the moved item c-commands both gaps.

Another idiosyncratic restriction remains simply stipulated for now, namely the restriction of parasitic gaps to nominals. This is achieved by allowing only one single $\#f$ -feature in the grammar, namely $\#n$. This assumption is also necessary for Horn amalgams, so while stipulated it is an assumption that is shared by more than one phenomenon. In the next chapter I suggest that the actual purpose of *3rd-merge* is to let verbs select their nominal argument across the determiner, as a potential solution to the DP/NP-issue. The D-layer of the nominal domain is then a miniature secondary root. The possibility to grow that root on top of this phrase further and potentially reconnect it is then a mere consequence. The fact that Horn amalgams and parasitic gaps show categorial restrictions then follows from an apparently unrelated phenomenon that would share a common core.

Turning from restrictions to additional freedom, this approach also derives why parasitic gaps can seemingly violate island restrictions. In a derivation containing a parasitic gap construction, the pivot is not moved within e.g. the adjunct clause across the island barrier, but only along the spine of the matrix root. In other words, the path of the mover does not cross an island.

What this analysis does not derive as a consequence so far is the restriction to *at most* one island. This is not an insurmountable problem. One option to implement island restrictions into MGs would be an algorithm along the following lines: a derivation does not yet crash at a point where a structure is headed by an element that is a potential island barrier (say an adjunct C head); at this point, if there are moving chains in the expression, those items could still move to its specifier. At the point where yet another head selects this tree as its argument, any moving items would need to cross the island, hence ungrammaticality results. In other words, whenever merging with a tree headed by an island head, that tree must not contain any moving elements.

To apply this to the current context, when growing a secondary root on top of an element that also contains movement features, such a secondary root does not yet lead to ungrammaticality if its highest head is a potential island barrier. Only further growth would yield an island violation. But

if at the previous point the pivot plus secondary root is *3rd-merged* into the primary root, no island violation results since the secondary root is now a chain in the expression of the matrix root, ensuring that any movement occurs within that root (which means that only islands within the matrix root are relevant for the pivot now).⁷⁵

To conclude, this approach, like the approach to parasitic ellipsis presented in the first half, generates the parasitic phenomenon in one fell swoop and avoids potential pathologies of communicating operations. The parasitism of likes generalization also falls out naturally. Some of the idiosyncratic properties of parasitic gaps can also be derived with the system proposed here.

8.5 Other approaches

Nissenbaum (2000) proposes the following classification for approaches to parasitic gaps: there are *shared antecedent* proposals and *separate antecedent* proposals. In his words, the first class of theories accepts “the premise that the relationship between the PG and the overt antecedent is what needs to be explained. [...] Implicit in these proposals is an assumption that both gaps (parasitic and main) are bound by the same antecedent.” (2000:24–25). Examples for this class are the empty pronominal approach in Chomsky (1982) and the sideways movement approach in Nunes (1995).

The other class is the *separate antecedent* family of proposals which “reject[...] the premise that there is a relationship between the PG and the overt antecedent that needs to be explained” (ibid.). “There is no dependency between the PG and the antecedent of the licensing movement because the PG is the trace of a separate movement that doesn’t cross the island boundary”

⁷⁵An alternative approach would be to deny that parasitic gaps are restricted to crossing only one island. One can find examples for this even in literature that maintains the island restrictions, e.g. in Nissenbaum (2000), in the context of multiple parasitic gaps (2000:112, ex. 32d):

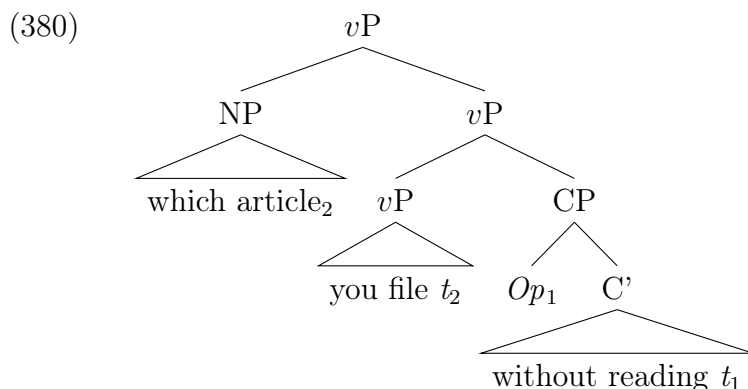
- (i) ?Which actor did they assign which role to _ [without even asking _ if he wanted _].

In his approach, this would require movement of an empty operator corresponding to *which role* to the edge of the *without*-clause; this would require extraction out of the *if*-clause which is itself an island. This might be restricted only to secondary parasitic gaps. In the absence of a modified theory, however, it would be necessary to allow empty operator movement across an island boundary, contradicting the putative predictions of empty operator theories regarding islands. A simple modification where every operator only moves to the edge of its island would have the same outcome of allowing multiple island violations.

(2000:27). As examples of this type he cites the empty operator theory in Chomsky (1986); his own work also belongs to this class.

My approach obviously falls into the first category. I do not necessarily agree with all of Nissenbaum's criteria for these classes,⁷⁶ but the overall classification is clear and intuitive enough for a comparison of various approaches. In what follows I discuss a number of properties of Nissenbaum's approach (as the most well-known recent representative of the other class of parasitic gap proposals) in light of my discussion of parasitism; I also point out a number of smaller differences between my approach and others from the same class.

Nissenbaum (2000), following Chomsky (1986), postulates the existence of empty operators Op that an element can take as a nominal argument. An empty operator $\bar{A}$ -moves to the edge of the clauses that hosts it. Doing so triggers predicate abstraction, similar to the way relative pronouns do in Heim & Kratzer (1998), thereby creating e.g. a vP -adjunct of type $\langle e, t \rangle$. Nissenbaum assumes that $\bar{A}$ -movement proceeds successive-cyclically, also through the specifier of vP ; this intermediate movement also leads to predicate abstraction and therefore to a phrase of type $\langle e, t \rangle$ below the landing-site of the mover. In contrast to Op , the mover can compose with its sister node to satisfy its individual variable to result again in an expression of type $\langle t \rangle$. Nissenbaum proposes that a vP -adjunct can late-merge countercyclically to the vP -node of type $\langle e, t \rangle$, i.e. the node below the mover. Since both sisters are of type $\langle e, t \rangle$, they can then combine via Predicate Modification which results in the unification of both variables created by movement of the antecedent and movement of Op , respectively. Composition with the next-higher mover then ensures that it is this element that constitutes the argument to the open slots in this function.



In a sense, then, parasitic gaps are an interface phenomenon in this approach.

⁷⁶In my approach e.g. the movement does not technically cross the island of the pg host.

The existence of moving empty operators and the independently necessary composition mechanism Predicate Modification conspire to produce a configuration in which one moved element semantically acts as the filler for two gaps in the sentence.

The appeal of such an approach is that parasitic gaps might be explained without additional machinery that needs to be postulated for parasitic gaps. This approach ties the coreference between host and parasitic gap to the independently necessary semantic composition mechanism Predicate Modification. The anti-c-command condition and path containment follow as a consequence from this approach.

A potential worry might be that too many things need to conspire to result in parasitic gap constructions (and only those): empty operators, the requirement that all parasitic gap hosts are exclusively attached exactly to the *vP* level, late merger of those hosts and the fact that semantics (blind to what a parasitic gap is) needs to help out crucially to generate the necessary conditions for what at first sight appears to be a syntactic phenomenon.

A number of assumptions underlie empty-operator theories. In order to explain a phenomenon where there appears an additional gap, akin to a trace, in the context of $\bar{A}$ -movement, that can only correspond to DPs, empty operator theories postulate that which ideally should be derived, namely the existence of a silent element that ultimately has almost exactly the properties of an additional trace.

Op has no phonetic realization and no overt counterpart. It can only correspond to category DP. It is not like a pronoun in that it could also be potentially free, but it always needs to be ‘bound’. It has no content, and the predicate abstraction that it triggers is of the type that it only re-opens the variable, *Op* cannot satisfy it again - in contrast to the predicate abstraction triggered by the intermediate movement step that it is accidentally parasitic on. *Op* is also an item that always *has* to move, it cannot stay in-situ, i.e. it is necessarily equipped with the correct $\bar{A}$ -movement feature. Furthermore, its movement and its landing site in almost all cases do not have any overt counterpart; e.g. neither English nor German allow movement of an NP to the specifier of a *without*-clause:

(381) *Peter went home [the articles₁ without reading *t*₁]

These assumptions allow a derivation which results in a construction where there is an additional silent DP trace that corresponds to another trace and its filler, potentially across an island. Many of these properties, however, needed to be baked into the properties of the postulated empty operator (phonologically null, the categorial restriction and obligatory movement).

The free availability of empty operators and predicate modification might also lead to unwanted overgeneration. It should in principle be possible to solve any type-mismatch between sisters of type $\langle e, t \rangle$ and $\langle e \rangle$ by postulating empty operator movement in the latter. Since predicate modification is blind as to how phrases of type $\langle e, t \rangle$ were created, it is entirely possible that this type is not the result of movement.^{77,78} In a sense then, the dependence on movement is accidental. All this comes on top of the necessary assumption that all relevant adjuncts are adjoined at vP to create the required structural configuration.

To conclude, the empty operator approach is a theory of parasitic gaps in which a lot of heavy machinery needs to be employed to ensure the complete but also accidental dependence of a freely available empty operator and a moving item. Many syntactic properties of the construction are postulated as features of the empty operator. The syntax then produces via countercyclic late merge just the right outcome so that the semantics module can ensure the identity of two gaps. In my view that amounts to a number of conspiracies that need to come together to produce accidentally what is actually a defining property of the phenomenon, namely the parasitic relationship.

Another approach is to assume that parasitism is a genuine phenomenon in its own right. If one wants to assume fewer conspiring factors, then most likely a new mechanism is necessary that can generate the structures on purpose. One such option is sideways movement (Nunes 1995, Nunes 2001, for a formalisation within MGs see Stabler 2006). The general idea is to allow an additional kind of movement, namely movement between trees that are not yet connected. To give an example, one can first start generating the adjunct clause *without reading which article*. When building the matrix clause headed by *file*, then one can copy or *sideward move which article* into the object position of *file*, creating a gap in the adjunct clause.

Later movement of the object of *file* will create a standard $\bar{A}$ -gap. The *wh*-word then corresponds to both gaps because it has been present in their place during some point in the derivation. The creation of the additional, unusual gap is due to the more powerful movement option *sideward movement*. In this account, the parasitic phenomenon is also in a sense the outcome of a ‘more powerful’ version of the operation that leads to the creation of the host phenomenon. A major difference between sideways movement and

⁷⁷This might make it possible to combine an adjective and a phrase with empty operator movement, without overt movement in the former. Such examples might only be avoided because the required selection criteria never arise, another necessary coincidence.

⁷⁸Furthermore, parasitic gaps inside NPs already create issues for semantic types since they create a phrase of type $\langle e, e \rangle$; Nissenbaum & Schwarz (2010) propose another composition mechanism COMPOSE to solve this issue.

3rd-merge is that the former does not lead to multi-rooted trees. As such sideways movement cannot generate Horn amalgams as a sort of in-situ counterpart to parasitic gaps.

As mentioned above, such a multidominance account (employing parallel merge by Citko 2005) to parasitic gaps has first been proposed by Kasai (2010). The account in this chapter follows the same underlying ideas. There are some differences, however. In Kasai (2010), the pivot does move within the adjunct, to the edge of the CP where it is then merged with the matrix verb. The root of the adjunct clause can also grow after that point and the pivot can move independently before the secondary root is reconnected. These options are forbidden in the system set up in this chapter. Apart from that, however, this account proposes that the parasitic gap host clause is at one point a secondary root, and that a single pivot corresponds to both host and parasitic gap.

For the reasons outlined throughout this thesis I prefer solutions from this class of theories that take the apparent parasitic relationship as the central explanandum of this construction. A multidominance approach to parasitic gaps also meets the requirements I have set myself that both host and parasitic phenomenon are created in one fell swoop by a single operation.

8.6 Further issues

To close this chapter on the main result of this half of the dissertation, I want to point out issues that require further work. One issue that I have left aside in this thesis is the empirical fact that A-movement does not license parasitic gaps. This does not follow from the theory proposed here. More work is necessary to derive this property (ideally other than just stipulating it).

As a more theoretical point, it is quite obvious that the current approach contains a huge number of rules. Ideally, these rules can be simplified in future work. As it stands, one of the reasons for the number of rules might also be reason for hope. Many of those rules underlyingly express the same idea: merge an amalgam into complement position, merge an amalgam into specifier position, merge what is going to be a parasitic gap host into specifier position, and the same for complement position and so on. If the algebraic system here would not account for linearization as well and instead leave that for another (sub)module, maybe the number of rules can be cut in half. Right now, connected and unconnected roots require distinct rules, even though there are not very different.⁷⁹ In a more suitable notation where

⁷⁹A reviewer (Thomas Graf) comments that this does not provide a unified operation in

the root category symbol does not need to be erased to avoid havoc in the notation, the difference between a connected or unconnected root is left to whether an item chooses to merge with it or not without needing separate rules for both cases beforehand.

In the end, what this extension of MGs encodes are the following ideas: under certain conditions, growth of a secondary root can be triggered (*merge4*); after that root is built completely, it can be selected for at the pivot to be connected with another root (*3merge-1,2,3,4* and its variations); that additional root can also be subject to *merge* (*chain-internal merge1/2*). In this dissertation I have tried to err on the side of formal precision. The precise formalization that covers as many subcases as possible is what led to the blow-up of the number of rules. Yet only a small number of core ideas underlies them. Future work ideally finds a more suitable and more concise way to represent them.

the strict sense since different rules are required for amalgams and parasitic gaps. Whereas the number of rules obscures the fact that this is underlyingly the same operation, this is not different from the definition of *merge* which requires three rules to define *merge* into specifier or complement position and *merge* with a moving element.

9 The DP/NP issue

In this chapter I take a closer look at the postulate that only nouns are selected via an application of *3rd-merge*. Here I propose that this can be utilized as a potential solution to the theoretical issue whether N or D is the head of the nominal domain. In fact I want to suggest that the *only* purpose of *3rd-merge* is to connect verbs and nouns, the core elements of the two major spines in syntax: the clausal and the nominal one. The possibility of further growth on top of the nominal root is then a consequence of the relationship between verb and its nominal argument.

Throughout this dissertation I have made the simplifying assumption that the NP is the highest projection in the nominal domain (i.e. I have assumed the NP-hypothesis). This was mostly immaterial for the issues discussed,⁸⁰ I might as well have assumed the DP-hypothesis; the only change this requires is to assume that the only *3rd-merge* feature is $\neq d$ (and the features sequence of determiners therefore at least $d.\neq d$).

The core function of *3rd-merge* is long-distance in-situ selection across intervening material. In the previous chapters, this additional material has been the secondary root in form of an Horn amalgam or a parasitic gap host clause. The idea is now that the intervening material is actually smaller. My proposal is that negative *3rd-merge* features are *always* on N, even in the presence of a D-layer as in the DP-hypothesis. Verbs then select - via *3rd-merge* - for the nominal core which allows one to state that verbs (can) place restrictions on the type of noun they select, but never the type of determiner (see Bruening, Dinh & Kim 2018 for a discussion of this claim). However, determiners *also* select for the NP, albeit via an application of *merge*. The D-layer can then be understood as a *mini-amalgam*. Every DP argument of a verb is then a small secondary root, with the NP core as pivot; that the secondary root can grow even further and result in Horn-amalgams and parasitic gaps is just a natural consequence.

In this approach, there is both a selection relationship between verb and noun *and* the DP is the highest projection in the nominal domain, without contradiction. This could then potentially reconcile the arguments for both sides of the theoretical divide.

The main focus of this dissertation are parasitic phenomena. It is therefore far beyond the scope of this thesis to do full justice to the literature on the structure of the nominal domain, or to discuss all consequences or complications that this proposal might imply. The modest goal of this chapter is

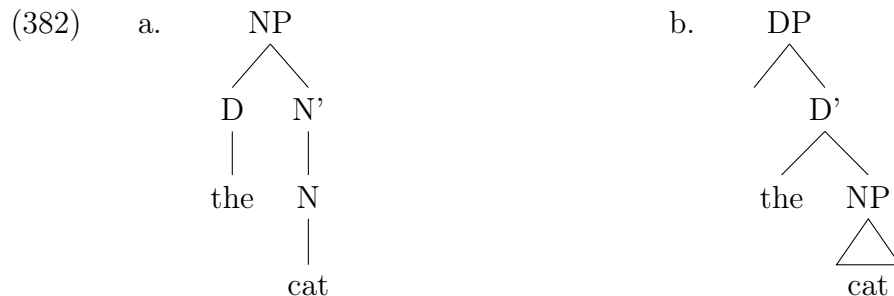
⁸⁰The only exception was whether parasitic ellipsis after nouns is a counterexample to the locality restriction.

to provide a potential analysis of the DP/NP phenomenon in its most basic form and discuss some of its consequences for different kinds of nominals, especially in the context of Postal’s 1993 anti-pronominal condition.

The structure of this chapter is as follows: after a short overview of the theoretical debate regarding the structure of the nominal domain, I showcase the updated formal system of rules that implements the basic idea discussed above (9.1). I also sketch how a more elaborate structure of the DP can be accommodated in this basic system (9.2). I then discuss the observation that verbs that cannot select for (certain) pronouns also cannot host a parasitic gap (9.3). Adopting certain assumptions about the structure of pronouns from previous literature, this fact follows as a consequence from the current proposal. I then close this dissertation with a discussion of a number of issues and further research avenues that follows from this ansatz (9.4).

9.1 DPs as secondary roots

Phrases like *the cat* have traditionally been assumed to be NPs. This changed at the latest with Abney (1987) who proposed that such phrases are better described as DPs,⁸¹ i.e. that they are headed by a determiner that selects an NP as complement:



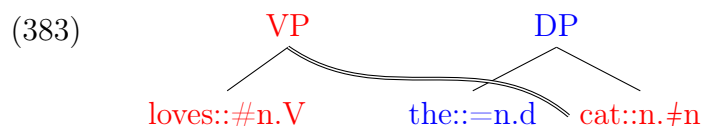
Recently, the DP-hypothesis has most prominently been challenged by Bruening (2009), Bruening, Dinh & Kim (2018). Chief among their arguments are generalizations from selection: verbs select for types of nouns, but not for types of determiners. This is unexpected under the view where D heads the nominal domain. This has lead to a re-examination of the underlying assumptions and arguments for the DP-hypothesis and a debate between both positions (see e.g. the Glossa special collection ‘New Perspectives on the NP/DP debate’ introduced in Blümel & Holler 2021). For a good overview and critical

⁸¹For a list of earlier works that suggest the DP hypothesis, as well as early counterarguments, consult Bruening (2009).

examination of claims both for/against the DP- or NP-hypothesis, see Salzmann (2020) and references therein. He points out that several arguments in favour of the DP-hypothesis are either theory-internal or conceptual. These are for example arguments that D as specifier would require multiple specifiers, something that is more commonly accepted today, or that D would be the only category that does not project (also not a priori problematic). Another conceptual argument is the proposed parallelism between the clause and the nominal domain so that in a sense C and D are the respective functional categories that head their domain at the top of the spine starting with either V or N. Salzmann (2020) also points out that phenomena that appear to require head movement from N to D which is impossible under the NP-hypothesis might receive alternative analyses in the latter that do without such a mechanism.

As mentioned above, I cannot do this debate full justice here; the interested reader is referred to the above-cited works where these issues are discussed better and in more depth than I could do here. For the purposes of this chapter, let us assume that arguments for or against any of the two positions are inconclusive, though some generalizations might be more succinctly or elegantly expressed under one or the other. On the one hand, one would like to be able to express in some instances that there is a connection or relationship between the verb and the noun; on the other hand, some analyses are much more natural if D selects NP (e.g. in N-to-D movement analyses, Carstens 2017).

I want to propose that with *3rd-merge* one can assume both points at the same time. This would be the case when $\neq f$ is always located on nouns even in a ‘DP-language’, i.e. a language in which D is the highest projection in the nominal domain. For simplicity I will concentrate on those languages, without taking a stance on whether all languages are DP-languages or if there are also in addition NP-languages (see e.g. Bošković 2008 for such a claim). *3rd-merge* can then long-distance in-situ select for a noun while still allowing more structure, namely the D-layer, to grow on top of it. The following tree serves as illustration:



In this chapter I want to suggest that this is the core purpose of *3rd-merge*. Everything else discussed in previous chapters simply follows as possible consequences from this basic configuration. In short: The purpose of *3rd-merge* is to allow long-distance in-situ selection of nominal arguments. It is there-

fore the mechanism that connects the two main spines of natural language syntax: the nominal and the verbal one.

This means that a new type of long-distance dependency holds between verbs and nouns that allows for enough room to grow more (functional) structure in the nominal domain. If this is the case, the existence of other multi-root phenomena is expected.

This approach to combining the verb with a nominal is similar to the one pursued in Thoms (2019). He proposes a *layering* approach (citing Sportiche 2005 as a spiritual predecessor) where the verb selects the *nP* and D is ‘layered’ on top of *nP* via parallel merge/sideways movement/external remerge. The resulting DP-root is reintroduced into the tree e.g. at SpecTP via movement. Thoms (2019) uses this to derive anti-reconstruction effects of arguments that are introduced this way via the invisibility of the *full* DP. One difference between the approach in Thoms (2019) and mine is that here the new operation (*3rd-merge*) underlies the relationship between verb and noun, and the relationship between D and N is standard (*merge*), whereas in his approach it is the addition of the determiner that requires an additional operation.

To make the analysis as proposed above work requires modifications to the rule system set up so far. Let us start with the simple case where a verb such as *pet* selects the DP *the cat*. A DP in this approach will be a secondary root - a *mini-amalgam* so to speak. I maintain the previous assumption that secondary roots must be built first before they can be connected to the primary root. Therefore the determiner needs to select the noun first, similar to the *merge4* step in the previous sections.

Where it differs from its previous incarnation is that here this step does not trigger the growth of a secondary *clausal* root but it completes the minimum element that can be selected by a verb. In the previous sections I have treated NPs as ‘trivial’ secondary roots, in the case of *3merge-1*:

$$\frac{s :: \#f\delta \quad t \cdot f.\#f, \alpha_1, \dots, \alpha_k}{st : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1}$$

$$\frac{\text{adore} :: \#n.V \quad \text{cats} :: n.\#n}{\text{adore cats} : V} \text{3merge-1}$$

It is a trivial root since there is no new material above the NP that selects the NP but is *not* selected by the verb, as would be the case when an amalgamated clause of the type ‘*I think it was*’ grows on top of the NP. It is still a root since its category feature is not checked but simply vanishes in the output to represent the fact that it is syntactically inert, similar to the

case where a whole CP on top of a pivot NP was connected to a verb; there, C was erased from the representation to express the fact that a linguistic expression is complete if the only unchecked features on *all* its roots are C. Previously, I have implicitly treated *3merge-1* as an operation that connects two roots even though it did not lead to a detectable difference compared to a standard *merge* application.

What one needs to say now is that a linguistic expression is complete if the only unchecked features in a tree are C *or* D on all its roots. In other words, *both* C and D are now treated as start categories in MGs. This establishes a parallelism between CP and DP that is viewed as desirable under the DP-hypothesis.

Therefore, I propose a new rule (*DP-merge*) that leads to the completion of the smallest possible ‘trivial’ root, and not to the growth of an extended (i.e. *clausal*) root:

$$\frac{s :: =n.d\gamma \quad t \cdot n.\#n, \alpha_1, \dots, \alpha_k}{[s : d\gamma, t : \#n, \alpha_1, \dots, \alpha_k]} \text{DP-merge}$$

An example for illustration is the following:

$$\frac{the :: =n.d \quad cat :: n.\#n}{[the : d, cat : \#n]} \text{DP-merge}$$

In this rule, an element of category *d* selects as its complement an NP (*=n*); it might or might not have movement features (γ).⁸² It selects an NP that, as stipulated, has the features *n.#n* and might or might not contain moving elements ($\alpha_1, \dots, \alpha_k$). An application of *DP-merge* therefore creates the bracketed structure that I have previously called ω , a pivot; the new ω now differs slightly from the old one, it contains both determiner and the noun that needs to be selected via a later application of *3rd-merge*. This means that in the updated rule system, ω now reads as $[s : d\gamma, t : \#n, \alpha_1, \dots, \alpha_k]$.

Next I describe the step where a verb selects for its nominal argument (without a clausal root on top) via *3rd-merge*. This was done in the previous system via *3merge-1*. It is now replaced by a new version that reflects that the verb selects for the NP across an intervening D-head:

$$\frac{s :: \#n\delta \quad [t : d, u : \#n, \alpha_1, \dots, \alpha_k]}{stu : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1}$$

Again, an example illustrates, with the corresponding tree:

⁸²To reduce clutter I have abstracted away from the possibility that the determiner might also select for a specifier. The full rule would then start with ‘*s :: =n\beta d\gamma ...*’

$$(384) \quad \frac{\text{pet} :: \#n.V \quad [\text{the} : d, \text{cat} : \#n]}{\text{pet the cat} : V} \text{3merge-1}$$

There is also of course the versions that correspond to *3rd-merging* a specifier and a moving element:

$$\frac{s : \#n\delta \quad [t : d, u : \#n, \alpha_1, \dots, \alpha_k]}{tus : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-2}$$

$$\frac{s \cdot \#n\delta \quad [t : d\varepsilon, u : \#n, \alpha_1, \dots, \alpha_k]}{s \cdot \delta, tu : \varepsilon, \alpha_1, \dots, \alpha_k} \text{3merge-3}$$

The rules above are now more similar to the rules that described amalgams in the previous system (i.e. *3merge1'/2'*) in that pivot (u) and its (mini-) amalgam (t) are linearized together while simultaneously erasing the category of the top node of the amalgamated secondary root (d in this case). Functionally, however, they achieve the same as the old *3merge-1/2/3* rules under the NP-hypothesis in that it connects the selector with its nominal argument, with the crucial difference now that the argument carries a non-zero root on top.

The rule that initiated the growth of a non-trivial, clausal root on top of an NP/DP, *merge4*, now also receives an update:

$$\frac{s :: =d\beta \quad [t : d\gamma, u : \#n, \alpha_1, \dots, \alpha_k]}{s : \beta, [tu : \#n\gamma, \alpha_1, \dots, \alpha_k]} \text{merge4}$$

Growth of a clausal secondary root is initiated if an element selects for an ω , i.e. a bracketed DP that contains a negative *3rd-merge* feature. In the process, the category feature on D is erased. Again an example that initiates what will become a Horn amalgam illustrates:

$$\frac{\text{was} :: =d.V \quad [\text{the} : d, \text{cat} : \#n]}{\text{was} : V, [\text{the cat} : \#n]} \text{merge4}$$

For simplicity I combine the phonological realization of D with that of the NP so as to avoid clutter and items with a phonology but without features in the notation, in case it does not have an additional movement feature. Nothing much hinges on this, but it avoids having to write two different rules. I want

to stress, though, that a later application of *3rd-merge* selects for $\neq n$ that is still on NP, ignoring the D. This aesthetic simplification therefore should not be mistaken for a syntactic feature percolation analysis.⁸³

The rules that select a secondary root or a parasitic gap host into complement or specifier position (*3merge-1'/2'/4*) remain unchanged since (due to the notational simplification above) pivots look the same in both systems once they are part of a clausal secondary root:

$$\frac{s :: \#f\delta \quad t : c, [u : \neq f, \alpha_1, \dots, \alpha_k]}{stu : \delta, \alpha_1, \dots, \alpha_k} \text{3merge-1' old/new}$$

$$\frac{\text{pets} :: \#n.V \quad \text{I think it was} : c, [\text{the cat} : \neq n]}{\text{pets I think it was the cat} : V} \text{3merge-1'}$$

As an intermediate summary, this is what I have proposed: verbs select for their nominal arguments via an application of *3rd-merge*. NPs generally carry both a category feature and a negative *3rd-merge* feature $\neq n$ (the only such feature in the grammar). That is, verbs select for the NP core of their argument directly, allowing one to express selectional restrictions on the type of noun. However, determiners also select for NPs, but via standard *merge* ($=n$); NPs even must be selected by determiners. This is enforced indirectly by stating the allowed start category symbols of a linguistic expression. The single root axiom needs to be abandoned for Horn amalgams anyway; I have also proposed to make the parallelism between the nominal and clausal domain more exact by stating that the only permitted unchecked features at every root is *either* C or D. Therefore an additional root needs to grow to at least DP.

If it is postulated that this is how verbs and their nominal arguments are connected, then the categorial restriction to DP for any multi-root phenomenon does not need to be stated separately any more but follows as a consequence. A connection between the verbal and nominal spine always leads to a secondary root. Any further growth of a secondary root therefore occurs on top of a nominal, deriving the categorial restriction in the analysis I have provided for Horn amalgams and parasitic gaps.

Under this view, the structure of the nominal domain, Horn amalgams and parasitic gaps show a surprising common core: the structure of the

⁸³This also obviates the need for a feature percolation mechanism that I have temporarily assumed in the previous chapter on parasitic gaps where I assumed the NP-hypothesis and noted that the *-wh* on *which* in *which article* somehow percolates up to the whole NP. In the final version of the rule system used here, *-wh* is on D and ultimately leads to movement of the whole pivot while the verb still selects for $\neq n$ on NP.

nominal domain is what underlies the latter two phenomena since it provides the seed for their growth. This is the main result of this chapter.

The specific solution to the structure of the nominal domain proposed here makes it possible to say two things at the same time. On the one hand, there is a direct connection between the verb and the noun which makes it possible to derive generalizations where the noun can select for and is sensitive to features of the NP. On the other hand, D is simultaneously also the highest projection in the nominal domain. This allows for analyses that appear to be more succinct when D selects for N, such as N-to-D head movement.

A full exploration of the consequences of this proposal for the various generalizations and analyses that exist regarding the nominal domain is beyond the scope of this thesis. Future work needs to determine where it allows to unify apparently contradictory generalizations and where it runs into problems. The goal of this chapter is simply to demonstrate that this new structure-building operation can be applied to this theoretical issue and that doing so reveals an unexpected common core underlying the phenomena that have been discussed in this half of the dissertation.

Before continuing with the next section, let me review an issue that arose in the first half of the dissertation with respect to the parasitic ellipsis generalization (3.2). Parasitic ellipsis is possible after the head of a gapping remnant, e.g. in a PP remnant, the complement of P can be elided. The only potential counterexample were nominal remnants where what can be elided is the complement of the noun. Under the DP-hypothesis, this is unexpected since D is the head of the nominal domain and NP should be parasitically elided, contrary to fact. In that chapter I assumed the NP-hypothesis to make the generalization total. With the approach to the structure of the DP outlined in this chapter it is possible to maintain the parasitic ellipsis generalization *and* the DP-hypothesis. Nominals are DPs, but the phrase that is connected to the clausal spine that is elided is the NP. Gapping remnants are arguments of an elided head; ellipsis can continue parasitically after the head of the remnant. In parasitic ellipsis in CPs, an elided verb selects its CP argument whose head in turn selects an elided complement. This constellation is now identical in nominal remnants: an elided verb selects (via *3rd-merge*) its NP argument whose head in turn selects an elided complement. The D in form of a secondary root is invisible for these purposes and therefore survives as well. This also predicts that larger clausal roots are similarly unaffected which appears to be borne out:

- (385) ?Joscha las einen ARTIKEL über seinen Helden und Jonas ich
 Joscha read an article about his hero and Jonas I

glaub es war ein BUCH.
 think it was a book
 ‘Joscha read an article about his hero and Jonas I think it was a
 book.’

9.2 More complex DPs

Above I have assumed that there is no intervening material between the determiner and the NP.⁸⁴ This assumption is not universally shared. There are also other complications that might arise. In this section, I want to address some of those.

There is more material that can occur in the nominal domain such as adjectives and numerals. The question then is the structural configuration in which they occur and what that means for the current system of rules. If the noun selects for, say, adjectives, i.e. if those are the daughters of an N’ node, then nothing changes. The determiner can select the NP carrying $n.\neq n$, as described in *DP-merge*. Matters become a bit more involved if the NP is the complement of the AP (the AP-over-NP analysis, already assumed in Abney 1987). One would then need at least two rules: one describing the step of selecting an NP but without thereby turning it into the smallest possible trivial root, a DP; and one that describes the first D that selects such an intermediate object (while making sure that no later D selecting a constituent containing such a pivot has a similar effect). Such rules can become quite convoluted. There are several ways around that. One option is to say that there is a functional category, say n , that takes the whole nominal complex except the determiner as complement, and it is only this head on which $n.\neq n$ resides.⁸⁵ Another option is that of category-preserving adjunction. For a proposal to implement adjunction in MGs, see Frey & Gärtner (2002) (for further discussion, see e.g. Fowlie 2013, Graf 2014). Roughly, in their model there is a new operation *adjoin* where one element can adjoin to another without erasing the category feature of the latter. For example, the adjective *black* :: $\approx n$ can adjoin to *cat*:: n . As a result, the adjoin-feature ($\approx n$) is erased, but not the category feature of *cat* (‘asymmetric feature checking’), resulting in *black cat*:: n . As long as the feature sequence of the phrase that was adjoined to remains the feature sequence on the head of the resulting phrase, nothing about the above rule *DP-merge* needs to change since the crucial $n.\neq n$ sequence remains that is in the end selected by D.

⁸⁴To muse: in the default case, then, *3rd-merge* ‘skips’ over exactly one head, similar to the locality condition in parasitic ellipsis.

⁸⁵A similar nP-shell analysis is suggested in Bruening (2009), with the obvious difference that he also proposes determiners to be specifiers of nP.

To conclude, the exact structure of the nominal domain might require a more elaborate version of the rules above, but I have also indicated ways that might not make this necessary.

Another issue is extraction from DP. Extraction of e.g. the complement of the noun is already accounted for (the $\alpha_1, \dots, \alpha_k$ in the respective rules). I am concerned here with extraction of a phrase that contains part of the nominal spine. An example of this is split topicalization in German:

- (386) Italienische Weine hat er nur billige getrunken.
 Italian wines has he only cheap drank.
 $\approx$ ‘As for Italian wines, he only drank cheap ones.’

One might assume that the unmarked version of this sentence where the object has not been topicalized is as follows:

- (387) Er hat nur billige italienische Weine getrunken.
 He has only cheap Italian wines drank.
 ‘He only drank cheap Italian wines.’

Then it would follow that part of the nominal spine has been moved. In an AP-over-NP structure this would look as follows:

- (388) [_{AP} Italienische Weine]₁ ... [_{DP} [_{AP} billige t_1]]

Under the NP-over-AP structure, however, a non-maximal projection would move. This can be circumvented if APs are adjoined to NP.

This leads to the question where $\neq n$ is located and/or how it behaves under adjunction. If AP adjoins to NP, then one cannot uphold the previous assumption that the new NP inherits *all* the features of the previous phrase (such as $n.\neq n$), since one NP appears to have a movement feature (on [italienische Weine]) yet the NP [billige italienische Weine] does not inherit it. It should not be the adjective that contributes the movement feature since it is not considered the head in this approach. Either one needs to propose a pied-piping analysis that is regulated in some way, or one needs to change the way adjunction works. Adjunction might be able to choose whether to inherit movement features or not. Alternatively, one might propose different modes of adjunction: one, where the adjoined to element remains the head; and one where the adjoiner becomes the head (imitating an AP-over-NP vs. NP-over-AP analysis) and accumulates the features of its complement on its own head. Either way, an adjunction analysis makes it possible that not only n but also $\neq n$ remain the features on the head of the phrase that is the outcome of adjunction. This avoids the issues that might arise by distinguishing between intermediate and final versions of *DP-merge*.

On top of that, it is unclear whether split topicalization actually does involve a (naive) movement dependency. For a critical discussion of arguments for and against this position, see Fanselow & Ćavar (2002) and Ott (2015).⁸⁶ For these reasons I leave a more thorough investigation of the consequences of partial NP movement for the current approach to future research and maintain the simpler system above.

9.3 Deriving the anti-pronominal condition of parasitic gaps

In two seminal papers discussing properties of parasitic gaps and how they differ from other superficially related constructions, Postal (1993, 1994) makes the following observation. There are a number of verbs or constructions that do not allow pronouns like *it* to occur in a specific position, namely naming verbs and existential *there*-constructions, change-of-colour verbs, as well as the impossibility of parasitic gaps to correspond to predicate nominals (for a more extensive discussion, see also Poole 2017):

- (389) He named his cat Mikesch/*it.
- (390) There are three articles/*them in the drawer.
- (391) She painted the red door black/*it.
- (392) She became an astronaut/*it.

Intriguingly, these verbs/constructions also forbid a parasitic gap to occur in this position:

- (393) *Which name₁ did he grow to love *t*₁ after naming his cat *pg*₁?
- (394) *Which magazines₁ did he order *t*₁ even though there were *pg*₁ in the drawer?
- (395) *Which colour₁ did she learn to hate *t*₁ after painting the door *pg*₁?
- (396) *What kind of teacher₁ did she despise *t*₁ before becoming *pg*₁?

This is not due to unavailability of the host extraction:

- (397) What₁ did he name his cat *t*₁?
- (398) Which articles₁ are there *t*₁ in the drawer?
- (399) Which colour₁ did she paint the door *t*₁?

⁸⁶Although both authors propose a movement/copy-*like* analysis, neither is of the naive type as in (388).

(400) What₁ did she become *t*₁?

In what follows I will attempt to derive this pattern from the proposal about the structure of the DP made in this chapter.

Something that I have so far not addressed in the previous chapters is the **issue of variable selection**: in e.g. a parasitic gap construction, one verb (the one that occurs with the host gap) selects its argument via $\#n$, the other verb in the secondary root via $=n$ – or now $=d$ as of this chapter. To make this work, one has to assume that both options are available to all verbs. At first, this seems to be an unnecessary complication of the theory. I want to argue, however, that one can derive certain properties from this assumption that can be expressed directly via this distinction.

The proposal that there are two different ways of selecting a nominal-like element would be strengthened if there were some verbs in which one mode or the other is missing. I propose that this is what underlies (at least some of) the anti-pronominal contexts above.⁸⁷ This can be given a syntactic/lexical account by saying that for the position in question, the above verbs miss the option $=d$, i.e. to select something of category D. Together with independently proposed approaches to the structure of pronouns, it follows that for this position, the selection of a definite pronoun like *it* is impossible.

Postal (1966) already argued that pronouns are more similar to determiners than other noun-like elements, given that it is impossible to combine determiners with pronouns, in contrast to nouns or proper names; and that it is possible to combine pronouns with nouns in certain contexts:

- (401) a. *the she that I know
 b. the Dana that I know
 c. the woman that I know
- (402) a. us linguists
 b. Ich Trottel
 I fool

One possible assumption, therefore, is that pronouns are bare D-elements without an NP complement (see also Abney 1987, Stanton 2016). This means that besides nouns like *cat* :: $n.\#n$, pronouns like *she* :: d are also part of the lexicon. Then it is independently necessary to propose that verbs can either select pronouns, or ‘full’ DPs (via their lexical core using *3rd-merge*), quite apart from any matters related to Horn amalgams or parasitic gaps. If pronouns lack an NP with a putative $n.\#n$ feature bundle, then it is impossible to select ‘into’ a pronoun via *3rd-merge*, and selection via

⁸⁷For other accounts of anti-pronominality, see e.g. Poole (2017) or Stanton (2016).

$=d$ is necessary.⁸⁸ Since most verbs do not care about the distinction, this difference is usually not noticed (even though it is then made use of for ‘secondary selection’ in Horn amalgams and parasitic gaps); i.e. most verbs are of the form $s :: \#n/=d.V$. Yet if some verbs *do* make this distinction (such as $name :: \#n.V$), patterns as above with anti-pronominal contexts are one expected outcome. In other words, in order to accommodate an independently necessary distinction between nominals and pronouns, one has to postulate two ways of selecting the two different noun-like elements. This difference then makes it possible to state the impossibility of some verbs to select pronouns syntactically via feature bundles on lexical items.

Once these assumptions are in place, however, the fact that ‘anti-pronominal verbs’ (i.e. verbs without the option $=d$) cannot host parasitic gaps is *predicted*. You can only grow a secondary root that is the host for a parasitic gap via an application of *merge*, namely $=d$. If a verb does *not* have the option of selecting its argument in this way, the verb in question is unable to host a parasitic gap in this position. In this approach, this fact follows simply from selectional restrictions.⁸⁹

I also want to bring attention to the fact that this captures the underlying intuition in some approaches that the parasitic gap corresponds to a kind of pronoun (see e.g. Postal 1993:744, and for mention/discussion of this idea Stanton 2016, e.g. her footnotes 17 & 20). For the verb that initiated growth of a secondary root, the parasitic gap indeed *is* a pronoun since for all it knows it selected a determiner-like element ($=d$) as an object. In that sense there is some partial overlap to empty operator theories because what is selected in the parasitic gap position is a pronoun without lexical content.

I want to point out that it is not possible to just select a ‘full D’, i.e. a DP with NP-content via $=d$ only. Such a derivation would not converge since the $\#n$ -feature on the NP remains unchecked. That is, there are not

⁸⁸As noted already in Postal (1993, 1994) and expanded on in Poole (2017), not every pronoun is incapable to occur in these anti-pronominal positions. It is e.g. possible to have *that* in some of these contexts. Poole (2017:ch.4) discusses at length which class of elements is forbidden there, namely ‘anaphoric ($=$ strong) definite descriptions’. I cannot do full justice to the extensive discussion there. For now, I restrict myself to pronouns that behave parallel to *it* and assume that it is those that have the structural description given above. Usage of the word ‘pronoun’ then is understood as reference to this restricted set.

⁸⁹Ideally one would like to find evidence of anti-pronominality in Horn amalgams as well. This is difficult to achieve due to the nature of the construction since it is restricted to the cleft-expression. None of the above types of verbs can appear in the amalgamated clause in the relevant position. A weak pronoun like *it* appears odd in Horn amalgams (*?*Susan talked about I think it was it.*); but this might simply be due to the cleft-expression itself (*?*I think it was it.*).

two equally available ways to select a DP in a matrix root. Selecting for D is only possible if $\#n$ is already checked by another verb, or if there is no NP to begin with.⁹⁰

To conclude, one can view the fact that anti-pronominal verbs are unable to host parasitic gaps as evidence that the ‘parasitic’ verb selects for the pivot as well (as it does in a multidominance analysis) and that the mode of selection differs from that of the matrix verb (the distinction between $\#n$ and $=d$).

9.4 Further issues and applications

More work needs to be spent on teasing out the consequences and predictions of the account in this chapter, as well as its compatibility with various proposals about the structure and properties of the DP/NP. One example might be different kinds of pronouns and the various structures they have been assigned (Wiltschko 1998, Cardinaletti & Starke 1999, Déchaine & Wiltschko 2002, Patel-Grosz & Grosz 2017). Another source of theoretical variation with various consequences is the overall structure of the nominal domain: AP-over-NP vs. NP-over-AP, different modes of adjunction etc. The approach in this chapter depends in part on the choice of these theoretical assumptions. Investigating this web of interdependence is beyond the scope of this chapter.

There is one aspect that I left implicit so far that I want to discuss here in more detail, namely the workings of movement, c-command and search under *3rd-merge*. In the following rule, the DP moves due to its movement feature (ε). This might seem innocuous but it runs counter to how search and c-command usually works, obscured by the rule notation.

$$(403) \quad \frac{s \cdot \#n\delta \quad [t : d\varepsilon, u : \#n, \alpha_1, \dots, \alpha_k]}{s \cdot \delta, tu : \varepsilon, \alpha_1, \dots, \alpha_k} \text{3merge-3}$$

The verb *3rd-merges* the NP, tagging the D-shell along as a secondary root. Naively, one might expect that therefore the verb only c-commands the NP,

⁹⁰Another analytical option for the structure of pronouns would be to say that they do contain a silent NP (see e.g. Déchaine & Wiltschko 2002) but that they are semantically/syntactically bleached and hence only carry n as category feature without the accompanying $\#n$. This would achieve the same result.

not the D-part of the nominal complex. Viewed over a tree-perspective, then, an attracting movement feature $+f$ scans downward into its c-command domain, but once it crosses a *3rd-merge* path (represented by double lines) it needs to search *upwards* to find the movement feature $-f$ located on D. However, in case that more than a D-layer occurs on top of an NP, namely an interrupting clause or a parasitic gap host, we do not want the upward search to extend indefinitely but to stop within the (bracketed) pivot. In other words, one needs to state that an application of *3merge1* (and related rules) leads to a c-command relation between selector and pivot (and pivot only). Rephrased in terms of search paths, one would need to state something along the following lines in addition to the usual search of c-command domains:

- (404) When search crosses a path established by *3rd-merge*, it can probe ‘upwards’ up until the first start category, i.e.
it can probe all elements that are dominated by the phrase of the closest start category (i.e. DP) that dominates the NP that was *3rd-merged*

A probing movement feature needs to be able to see the whole DP that came along with the NP that was *3rd-merged*. The fact that it must not see further is in line with the special, isolated status of the pivot in the secondary root that was a recurrent factor in the previous chapters (such as invisibility for (sub)extraction into the secondary root).

Such paths might not only introduce more elaborate definitions but might also find possible applications. One such application of a *down-then-up* mechanism might be determiner sharing (McCawley 1993, as discussed in 5.3), especially as envisioned in a system like Ackema & Szendrői (2002).

- (405) Every composer likes Beethoven and every painter likes Götz.

To simplify, they claim that an ellipsis like gapping is licensed by an elliptical head (say C) that itself can optionally determine the non-pronunciation of the head of its complement (and so on in a transitive fashion). This derives variable sizes of gapping remnants (*try to begin to write a novel, begin to write a novel, a novel,...*). As discussed in 5.3, parasitic ellipsis would be an instance where it is possible to skip a single head and then continue eliding afterwards. In parasitic ellipsis proper, the noun is skipped and deletion continues in its complement (as sketched in 9.1). The possibility of determiner sharing might also be a consequence of the more involved, multi-root structure of the nominal domain. If in a DP (*every picture of his grandmother*), N (*picture*) is marked as an ellipsis remnant (as the head selected by V), in contrast to all other phrases there are now two options for parasitic el-

lipis: either continue downwards (*of his grandmother*) or continue upwards (*every*). The definition of search needed to be updated for DPs to allow for both downward and limited upward probing starting from N, and parasitic deletion of additional material in both directions is possible in that same phrase.

One might also reconceptualize agreement/concord within the DP with downward-upward paths. Bruening (2009) observes that form determination in the clausal domain is usually *downward* (C determines I, auxiliaries select the form the auxiliary they select) whereas in the nominal domain, form determination is *upward*, depending on the head noun: the form of adjectives/determiners can be determined based on the number, gender, mass/count distinction of the noun. This itself is not indicative of anything yet, but it might be when considering properties that are not inherent to the noun but assigned by the verb such as case. If the verb assigns, say, accusative to the NP via the *3rd-merge* path ('downward') then some mechanism needs to ensure that this information travels right up to the D-layer ('upward'), i.e. the same space that a head with a positive movement feature needs to probe to find the corresponding feature on the head of D.

To conclude, either one simply defines *3merge-1* such that it leads to c-command between selector and D; or one takes the consequences of the structure literally and attempts to find phenomena that might correspond to such upward-then-downward paths. Determiner sharing might be a suitable candidate.

Turning to the more general consequences of the approach set up here, another phenomenon that might be modelled with nominals as secondary roots is quantifier raising (QR) and its often covert nature, as proposed in e.g. Johnson (2012) and Poole (2017). If one proposes that quantifiers have a different category than determiner-like elements, then the issue arises that sentences containing QPs are not complete syntactic objects. Remember that in the new system, a syntactic object is complete if the only unchecked features are either C or D on each root. Q does not fit the bill. This then necessitates that some operation (say a rule like *chain-merge*) reconnects the secondary QP roots into the main root, leading to potentially various different scopes depending on re-attachment site.⁹¹

Of course these are quite preliminary notes that would need to be worked out into a specific theory of these phenomena.

⁹¹To muse, it is then quite fitting that an operation that was arrived at via imagining what a '3rd-order' structure-building operation might look like is responsible for DPs and QPs which are also higher-order in their semantics. This provides a closer match between syntax and semantics.

9.5 Conclusion

To conclude, in this chapter I have proposed that the core purpose of *3rd-merge* is to establish a dependency between verbs and their nominal arguments, thereby connecting the two major spines of natural language syntax while leaving enough room for functional structure to grow on top of the noun. In other words, verbs long-distance select their argument NPs which carry on top a D-layer as a secondary root. This allows one to state the main claims of both DP- and NP-hypothesis without contradiction: a direct connection between verb and noun, and functional structure selecting NP.

If every DP containing an NP is an additional root, this opens up the possibility of growing this root further. In the previous two chapters I have argued that two linguistic phenomena correspond to two theoretical options, namely either letting the additional large root remain independent, or reconnecting it while moving the pivot.

In this approach, the structure of the DP is the reason that Horn amalgams and parasitic gaps can exist in the first place, revealing an unexpected common core underlying the three phenomena. If the above-mentioned purpose of *3rd-merge* is postulated, one predicts that multi-root phenomena are restricted to occur on top of DPs. Such a restriction is indeed what one finds (at least in the form of a strong tendency) which provides further credibility to the claim that those phenomena are fundamentally related.

Once one makes any claim about the structure of the nominal domain, one is confronted with the vast literature on that topic, as well as related phenomena. It is impossible to do all this justice in a single chapter of a dissertation. Instead, what I hope to have achieved here is to develop the core idea that works for the simple case, indicating at various points which unintuitive assumptions are made necessary or which empirical phenomena would need to be accommodated in the future for a more complete theory. I believe that this is an approach worth taking if it means that a unification of the three phenomena discussed in this part of the thesis is possible.

10 Conclusion & Outlook

The leitmotif of this dissertation is an investigation into parasitism in syntax, in ellipsis in the first half, and in the creation of gaps in the second half of this thesis – even though this investigation lead to sometimes unexpected connections to other phenomena in both parts.

One of the main claims in this thesis is that parasitism is best viewed as a single phenomenon, not the outcome of two separate but somehow dependent processes. I have argued at length in chapter 3.2 that when viewed as a separate operation, parasitic processes exhibit unexpected properties. They are at the same time subject to additional requirements and restrictions *and* enjoy more freedom than their host counterparts. Such parasitic dependencies also seem to be restricted to operations of the same type.

I have argued that allowing parasitic dependencies between operations in the literal sense harbours dangers of overgeneration. An alternative is to deny the parasitic status of the construction and aim to derive it from independently available processes that align in the right way to produce this construction. I have argued that this represents a conspiracy of sorts since it reduces the parasitic construction to an accidental outcome of the interaction of unrelated processes while often at the same time requiring a number of stipulations that ensure that exactly the required conditions are reliably provided.

In this thesis, I have pursued a different approach to parasitism: viewing parasitism as a genuine notion in its own right and treating it as such. I have assumed that parasitic dependencies are sufficiently distinct from other process interactions like feeding to warrant their own mechanisms. While ensuring that parasitic phenomena are generated ‘on purpose’ and not as the accidental confluence of various factors, at the same time I have also attempted to avoid the issues of operations that refer to each other. Instead I have argued that a parasitic phenomenon is the outcome of a *single* but more powerful operation; in other words, the appearance of a separate parasitic process is illusory.

The underlying idea was that a succession of ever more powerful operations ultimately leads to the creation of parasitic processes. These operations refer to or create increasingly complex syntactic structures. I have claimed that 3rd-order constituents and parasitism are intimately linked and that ‘3rd-order operations’ are the ones that generate the parasitic phenomenon in question (even though this is more transparent and direct in ellipsis than in parasitic gaps).

For this reason I have proposed two new operations: a 3rd-order ellipsis operation, and a new structure-building operation born out of a similar idea.

In the first half of the dissertation I have shown that parasitic ellipsis is a much more widespread and general phenomenon than originally assumed and that this phenomenon can be regarded as a *single* ellipsis if one allows not only 1st- and 2nd-order ellipsis operations but also 3rd-order ones. Parasitic ellipsis as a single process due to a 3rd-order ellipsis operation derives the indifference of parasitic ellipsis regarding the category of the elided structure and the ‘skipping’ over a focussed head. This also derives the parasitism of likes generalization: turning a 2nd-order ellipsis operation into a 3rd-order one cannot but result in anything other than more ellipsis.

In the second half of the dissertation I have attempted to apply the same reasoning to parasitic gaps: if parasitic ellipsis can be modelled as a single ellipsis by a higher-order ellipsis operation, can parasitic gaps result from a more powerful, ‘higher-order’ operation as well? While the connection is more indirect in that case, this approach turned out to be fruitful nonetheless. The resulting new structure-building operation resembled a form of long-distance in-situ merge. I made the case that this operation *3rd-merge* can be applied to reconcile conflicting evidence for the DP- and NP-hypothesis regarding the structure of the nominal domain. I argued that this new type of long-distance dependency holds between verbs and the NP part of their nominal arguments. *3rd-merge* functions in such a way that the argument in this relation becomes a secondary root, leading to multi-rooted trees. This allows functional layers to merge on top of the NP without interrupting the relationship between verb and noun. Two apparently conflicting properties can be stated without contradiction: a direct connection between verb and the type of noun that it selects, and the fact that functional projections select for NP.

In an approach where all DPs are secondary roots, it is possible to grow this secondary root still further, into a full clause. I have argued that this results in two different phenomena, depending on whether the new root remains independent, or if it is reconnected into the main root at some point during the derivation. In 7, I have shown that Horn amalgams correspond closely to a secondary root on top of an NP. However, when the secondary root is remerged into the primary root, a cyclic structure results. But if the connecting element is moved and the resulting structure linearized, this creates the appearance of two different gaps corresponding to the same element in a 1-dimensional string. This corresponds to the phenomenon of parasitic gaps, as detailed in 8.

Overall, this approach claims an underlying common core between the three phenomena. Multi-root phenomena are restricted to occur on top of DPs, and two logical possibilities of independent and reconnected roots result in two attested phenomena with a number of desired properties. I have taken

the fact that Horn amalgams and parasitic gaps appear to be restricted to DPs as evidence for the unified approach of these phenomena. This summarizes the main results of this dissertation.

In the rest of this chapter I want to lay out future avenues for the proposals made here. One question is not only whether one *can* solve certain phenomena with more powerful phenomena, but whether one *should*. The above proposals amount to the claim that 3rd-order structures are indeed part of natural language and that grammar must be able to refer to them. One answer to the question why parasitic phenomena exist is then to say that syntax does not stop at 2nd-order structures.

This leads to the next question whether there are other phenomena that are best described as a single 3rd-order structure, and if they are always connected to parasitic dependencies. Reversely one can ask whether all apparently parasitic phenomena share the same properties as parasitic gaps/ellipsis as detailed in 3.2 and require similar mechanisms. I have mentioned a number of constructions in 5.4 that can be compared with the phenomena discussed here.

When it comes to the formalism, it would be a welcome result if the extension to MGs proposed here could be significantly simplified. Another question involves the complexity and expressive power of these operations and structures, an issue that I have ignored so far. Ideally one can demonstrate that *3rd-merge* indeed does not increase the weak generative capacity of MGs.⁹² Since the formalism allows only for a single active root and the presence of a single pivot (ω), the maximum number of chains in an expression is still bounded. Only *3merge-4* needs to be restricted so as to not allow for an unbounded number of selectable secondary roots. But as long as the number of chains is bounded, I conjecture this extension of MG to have the same weak generative capacity as the vanilla version.

3rd-merge also leads to another collection of research avenues to pursue in the future. **It is an underappreciated fact that proposing a new structure-building operation means to propose a new type of dependency.** One case I know of where this is explicitly recognized is van Riemsdijk (2006) where he mentions properties common to constructions that he proposes to describe with *grafts*, and points out the need to investigate restrictions on grafts just as is done with movement dependencies.

MGs usually know two syntactic dependencies, one brought about by

⁹²It is at least interesting to note that another way to create 3rd-order structures in e.g. ellipsis is remnant movement, the mechanism that grants MGs their mild context-sensitivity (Kobele 2010). Maybe operations that create 3rd-order structures can achieve the same even in the absence of remnant movement.

merge, and one brought about by *move*. These dependencies fall into different clusters of properties that can also be used as diagnostic criteria to show their presence. By showing that various constructions involving long-distance dependencies pattern together with *wh*-movement in question formation, Chomsky (1977) proposed that what underlies them is the same type of dependency, *A*-movement. Even though relativization, *wh*-movement and topicalization might differ in detail, the number of non-trivial properties they have in common justifies the unification, viewing them as instantiations of the same underlying dependency.

This leads to the question: can the same be done for *3rd-merge* (or grafts, sideways movement, parallel merge)? This means asking not only: can and should one describe a number of phenomena with this operation, but: are there independent empirical properties that these phenomena share that provide evidence for treating these phenomena as a single class?

Are there arguments to propose such a *Third Dependency*? There are various structural configurations that multi-rooted trees due to *3rd-merge* can produce, so the unity of various phenomena might be difficult to establish. I have proposed that this new long-distance dependency holds between verbs and their nominal arguments, with Horn amalgams and parasitic gaps as indirect consequences of this dependency. At the very least then, a hallmark of this dependency should be a categorial restriction to DP which is borne out in the latter two phenomena. Indirectly, other arguments for a new class of phenomena (and therefore other potential criteria) have already been made, starting with the discovery in Postal (1993, 1994) that ATB-movement and parasitic gaps differ with respect to a number of diagnostic criteria, among them the anti-pronominal contexts discussed above. Later, Postal (1998) applies these and other contexts (with a precursor in Lasnik & Stowell 1991) to various types of leftward extractions and argues that they fall into two distinct classes, named A-extractions and B-extractions, with B-extractions those that are subject to anti-pronominality. This interaction of antipronominality and movement types is also used in Poole (2017) (who claims, however, that the ability to change scope is a relevant factor, cross-cutting these movement types) and in Stanton (2016), who takes up this idea (although with a slightly different partition such as a subdivision of B- into B₁-type and B₂-type extractions depending on whether they lead to Condition C violations) to explain certain pronoun-stranding patterns in English.

It might be worthwhile to investigate other members that are in the same class as parasitic gaps, such as gapped degree-phrases or *tough*-movement. I illustrate the latter here:

(406) Perfectionism₁ is difficult to defeat t_1 .

tough-movement prototypically involves nominals and disallows stranding of certain pronouns which is connected to anti-pronominality in Stanton’s 2016 approach. Since *tough*-constructions appear to lack the possibility of reconstruction, it has been argued that a different dependency than standard $\bar{A}$ -movement underlies them; they have also received an empty-operator analysis in various approaches (for a recent paper discussing these matters, see Poole, Mendia & Keine 2023 and references therein).

Just as with parasitic gaps, it is possible to reformulate an empty operator theory in an approach that makes use multi-rooted trees. A possible analysis within the current system would be to state that minimal secondary roots, i.e. DPs, remain active upon being *3rd-merged*, at least within a finite CP. A *tough*-predicate always selects its subject via $=d$, i.e. a pronoun(-like) element. Besides merging an actual pronoun, this can also occur via an application of *chain-merge*, i.e. merging the DP-layer of a more deeply embedded pivot into subject-position, leading to the relocation of the pivot without movement. In fact, a similar analysis has been proposed by Chung (2023) who proposes that *tough*-constructions involve merging of a lower NP that raises higher to merge with a D (an idea similar to the general approach in Sportiche 2005). In the approach proposed here, *tough*-constructions would also be a multi-root phenomenon and their shared properties with parasitic gaps unsurprising. However, the properties of other ‘B-type extractions’ might not in all cases be direct diagnostics, since they also address extraction in addition to a putative third dependency. Other phenomena to look out for are those that have already received multi-root analyses such as free relatives and constructions like ‘*a far from simple matter*’ (van Riemsdijk 2001).

If the dependency brought about by *3rd-merge* as proposed here exists, it results in at least three potential points of interaction involved in a given construction: first, the original dependency between verb and nominal argument; second, the selection relationship between a verb out of a secondary root and that argument; and third, the point of connection or lack thereof between primary and secondary root. Other operations such as movement might interact with these factors in specific ways.

At least for the first two points of contact I have proposed a number of characteristic properties. The hallmark property of the core *3rd-merge* dependency is the categorial restriction to nouns. Further selection of the pivot from a secondary root also underlies selectional restrictions since the verb in question can only select the D-shell of that noun. In 9.3, I have claimed that this is what underlies anti-pronominal contexts. The selectional

deficiency of certain verbs can be noticed in these contexts. It also appears that the secondary root is deficient in further ways, such as the inability to subextract from the pivot, move the pivot itself and possibly positional restrictions of the pivot within the secondary root.

Horn amalgams and parasitic gaps share the first property of a categorial restriction. The latter exhibits the anti-pronominal condition, a consequence of selectional deficiency. For Horn amalgams this is more difficult to establish since verbs in the secondary root are restricted to cleft-like constructions (and hence naming verbs and the like are impossible to test). However, Horn amalgams might share the related property that predicate nominals resist becoming a pivot (see Kluck 2011:ch.4.3, though this might simply due to the semantics of cleft-constructions in general), just as predicate nominals cannot correspond to parasitic gaps as discussed by Postal in connection with anti-pronominal contexts. Both constructions also share the root asymmetries mentioned above.

To conclude, a search for further diagnostic criteria shared by various constructions that can be derived via multi-rooted trees is an important research avenue to provide stronger justification for the claim that using such a mechanism makes the right analytical cut.

This thesis is intended as a contribution towards showing the necessity of additional grammatical operations to satisfactorily explain parasitic phenomena. Future work needs to investigate the consequences of these operations and search for further evidence for their existence.

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